Tools for the Optimal Operation and Planning of Power Systems

Lecture 3: Operational Variability

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Solutions to the proposed exercises

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

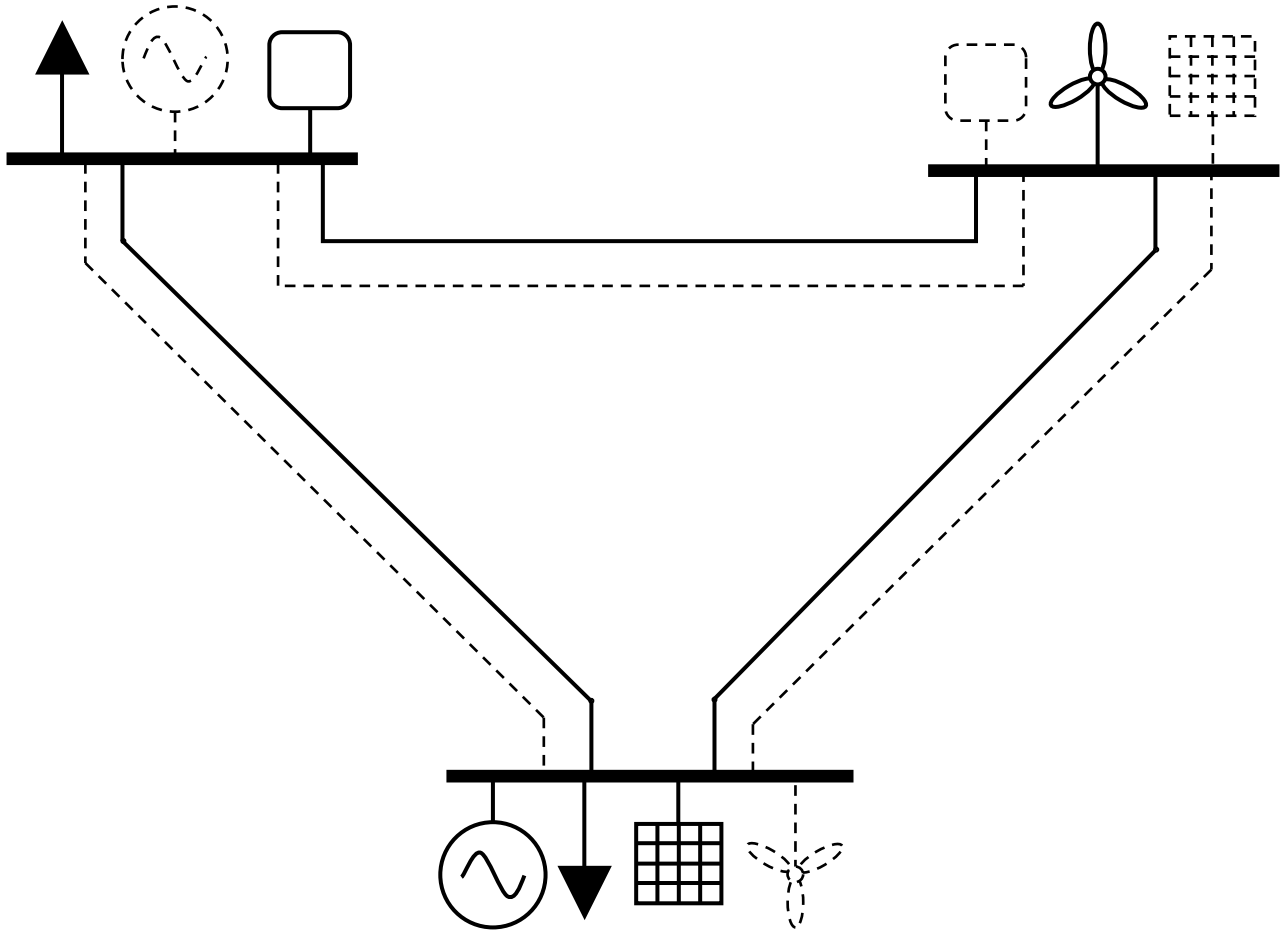


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1. Introduction
2. Representative time periods
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4. Inter-day dynamics
5. Extreme conditions
6. Advanced aggregation techniques
7. Alternative aggregation techniques

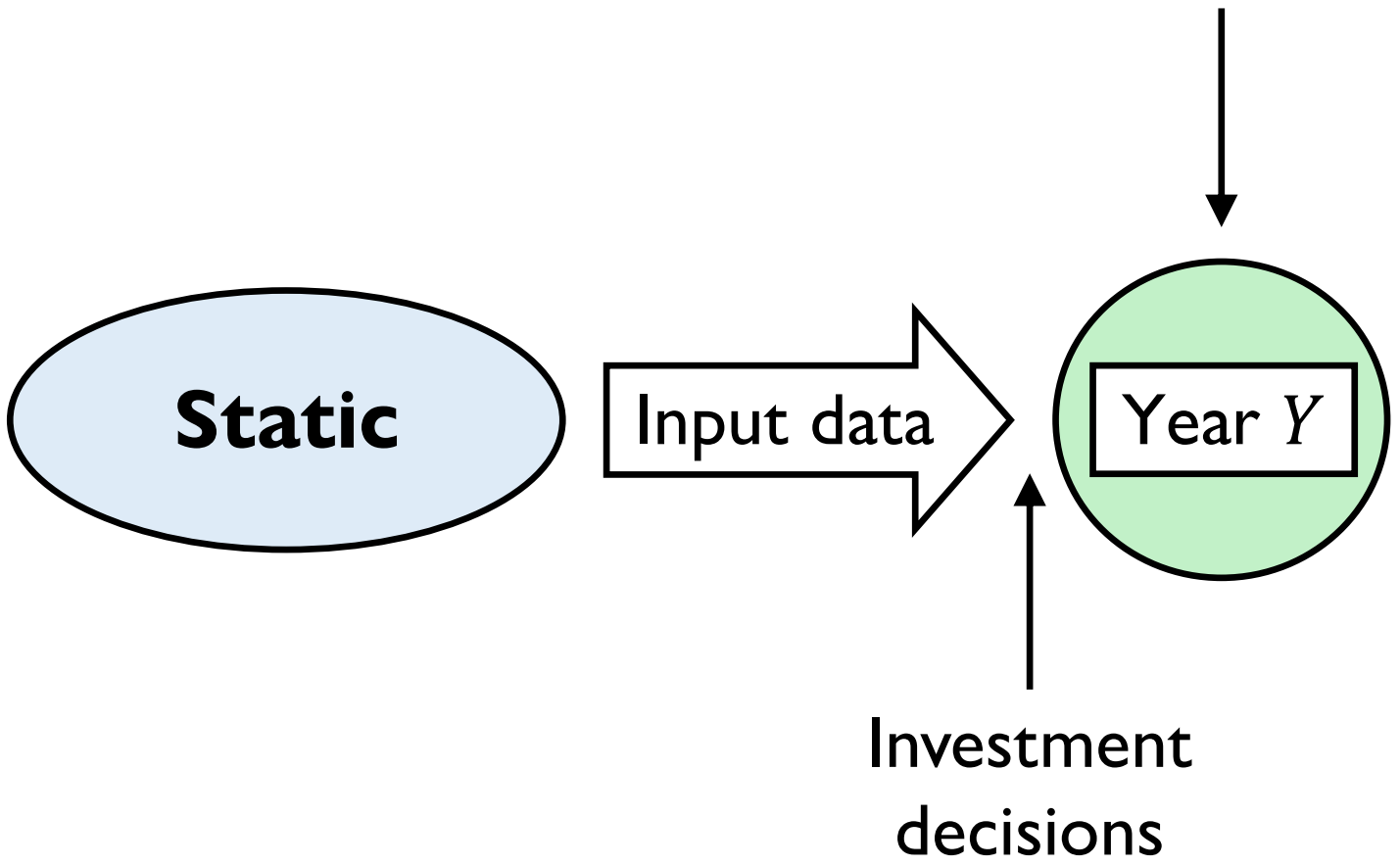
1. Introduction

Long-term planning problems



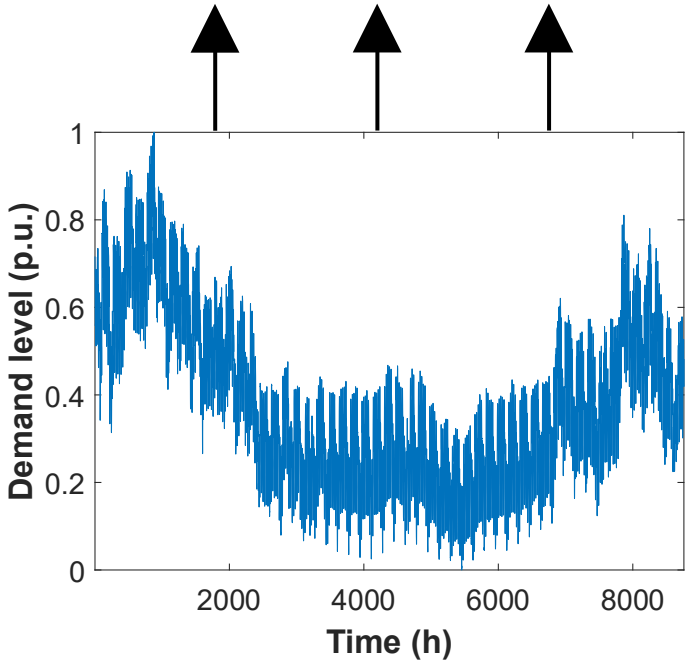
Planning

We focus on the end of the planning horizon

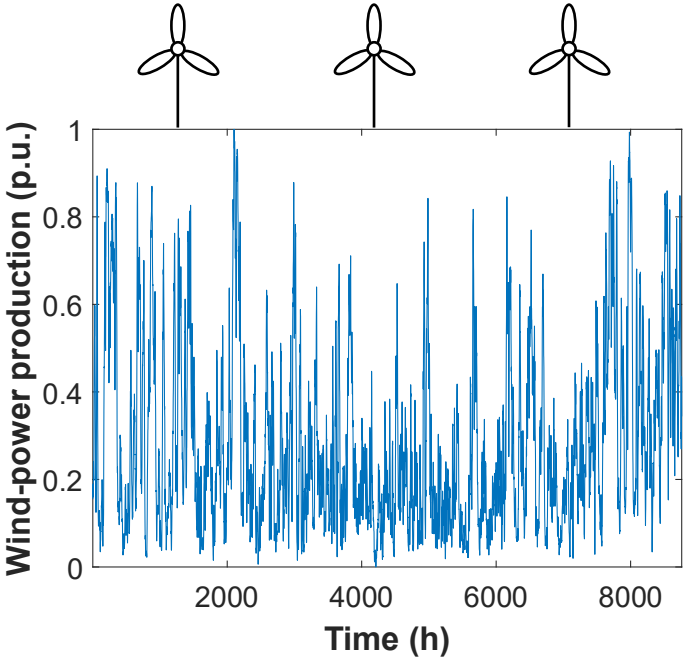


Historical data

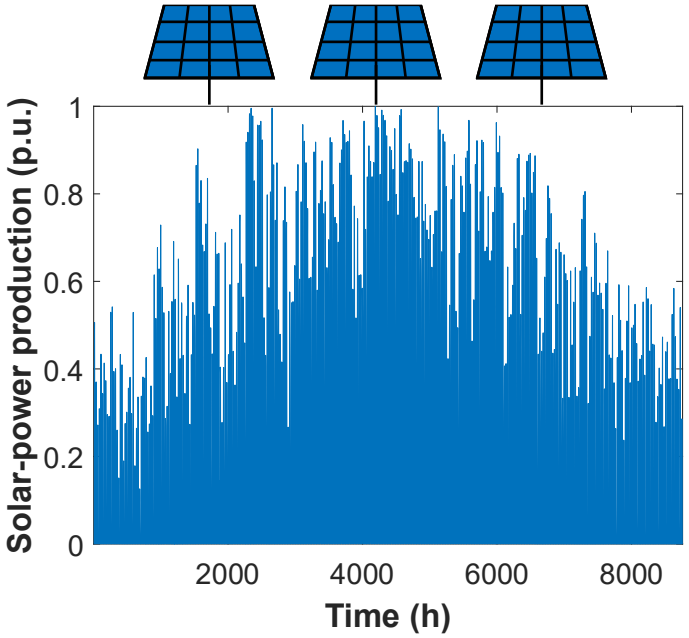
Electrical demand



Wind-power production



Solar-power production



Computational issues

Static approach → 8760 hourly time periods (1 year).

Computational issues may arise due to the **high number of variables and equations**.

Modeling operational variability

Use of representative time periods:

- Each representative time period t is associated with a value of the weight ρ_t^{T} and the parameters that vary with t, such as the electrical demand P_t^{D}.

- $\sum_{t \in \Omega^{\mathrm{T}}} \rho_t^{\mathrm{T}} = 8760$ h.

Modeling operational variability

$$\min_{\Phi} \sum_{t \in \Omega^{T}} \rho_t^{T} \cdot \text{Operating costs} + \text{Annualized investment costs}$$

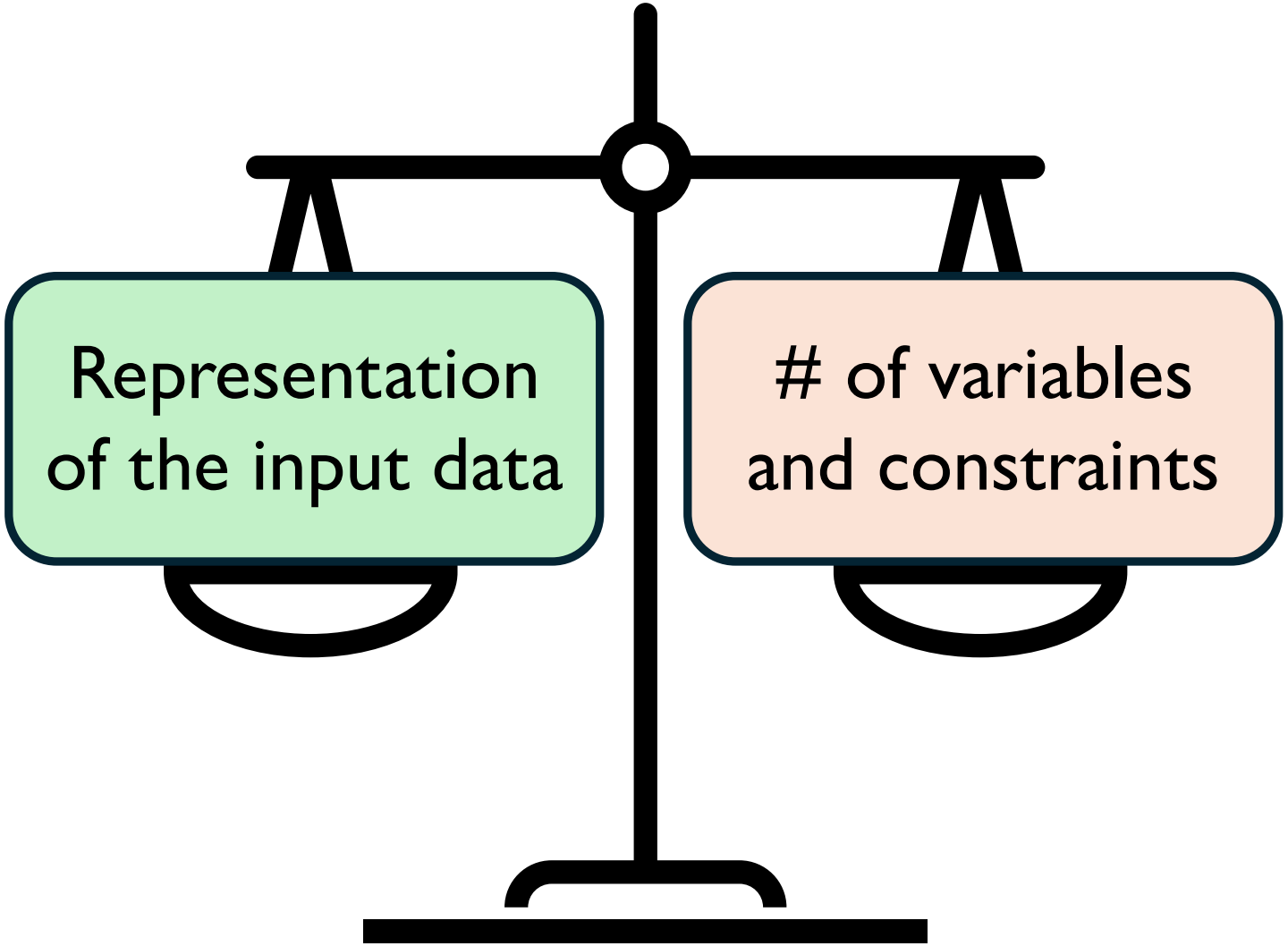
Weight of representative time period t (input data)

Modeling operational variability

Optimization variables associated with the operation:

$$variable_t$$

Modeling operational variability



Modeling operational variability

Operational Variability

Short-Term Uncertainty

Modeling operational variability

Operational Variability
is **NOT**
Short-Term Uncertainty

Modeling operational variability

Operational variability:

- **All** representative time periods **will occur**.
- Each representative time period is linked to a **weight**.

Short-term uncertainty (lecture 4):

- **Only one** of the scenarios **will occur**.
- Each scenario is associated with a **probability**.

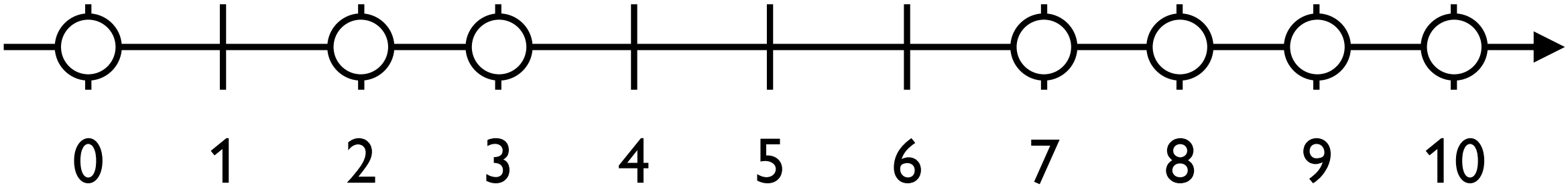
What will you learn?

- To apply popular clustering techniques (K-means and K-medoids).
- To generate representative time periods, representative days, and representative weeks using MATLAB.
- To solve planning optimization problems using representative input data.
- Advanced aggregation techniques.

2. Representative time periods (RTPs)

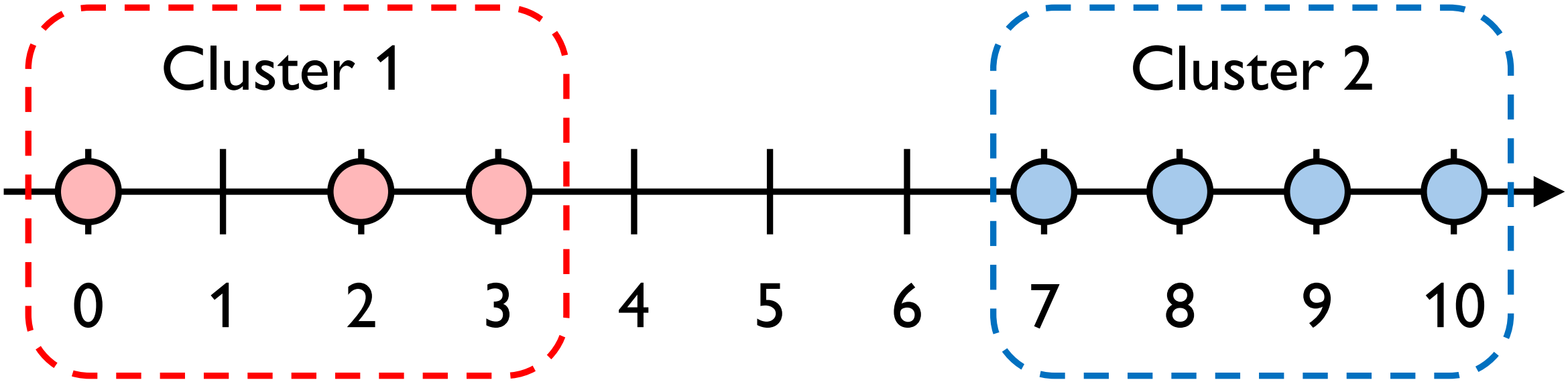
RTPs – Definitions

We are going to aggregate the input data into $K = 2$ groups (**clusters**).



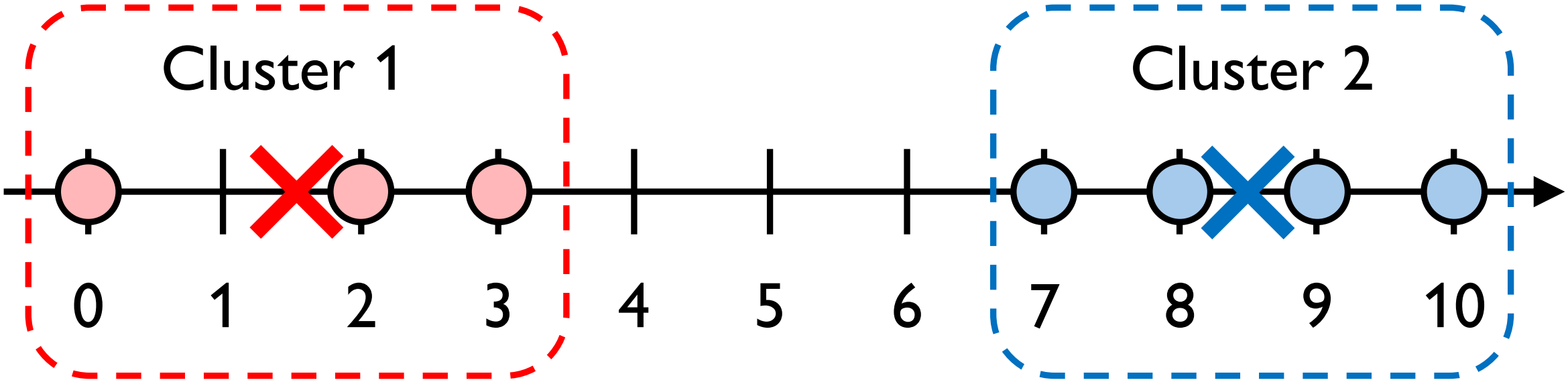
RTPs – Definitions

We are going to aggregate the input data into $K = 2$ groups (**clusters**).



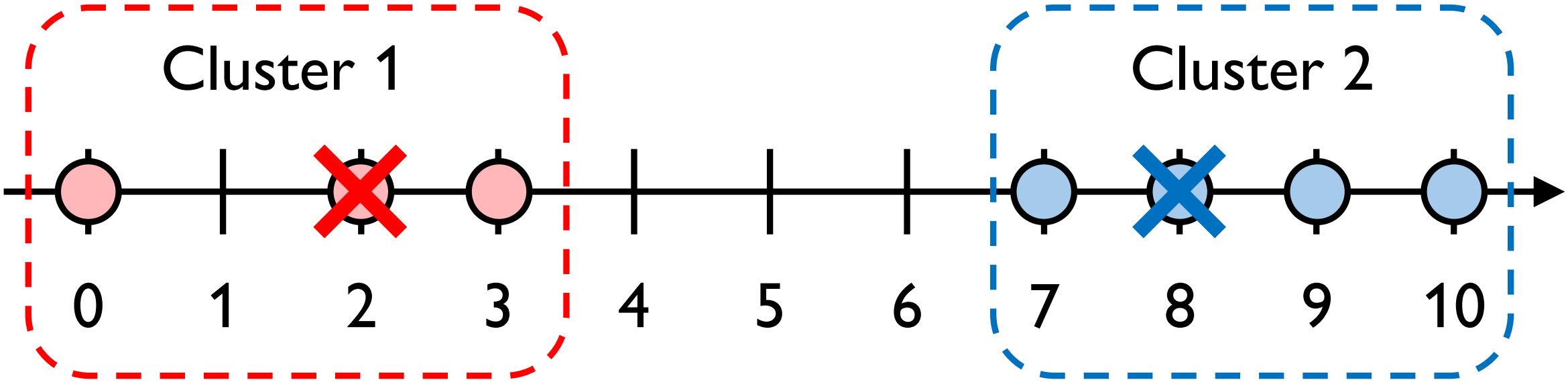
RTPs – Definitions

RTPs are generally identified with **centroids** or medoids.



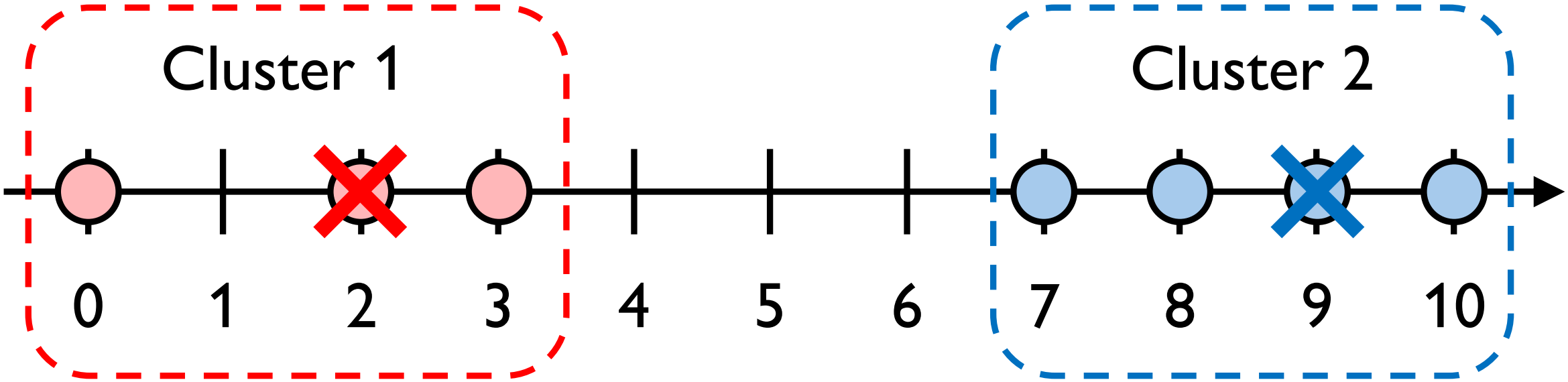
RTPs – Definitions

RTPs are generally identified with centroids or **medoids**.



RTPs – Definitions

RTPs are generally identified with centroids or **medoids**.

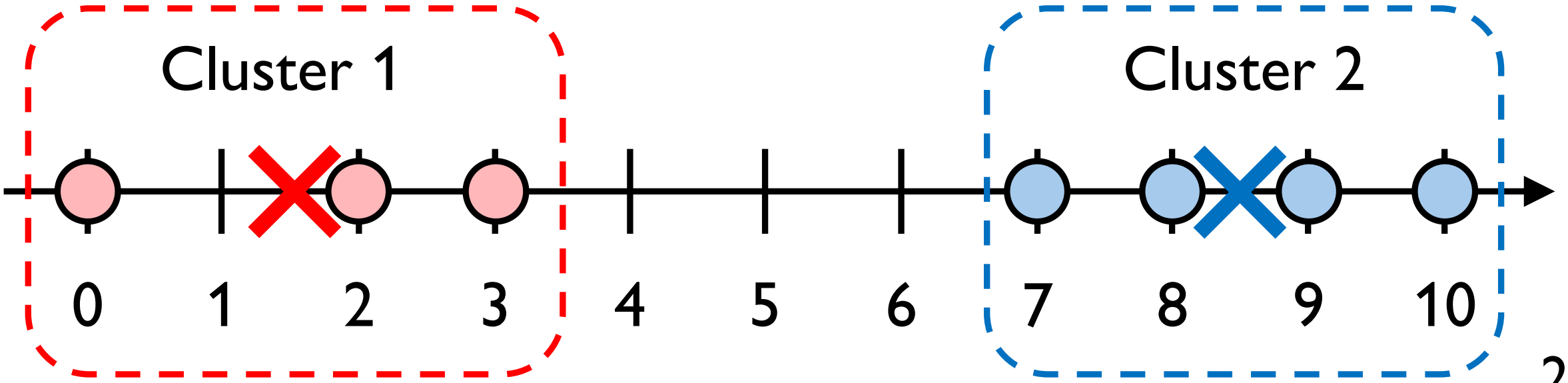


RTPs – Definitions

Each RTP is associated with a value of the **parameter** and a **weight**.

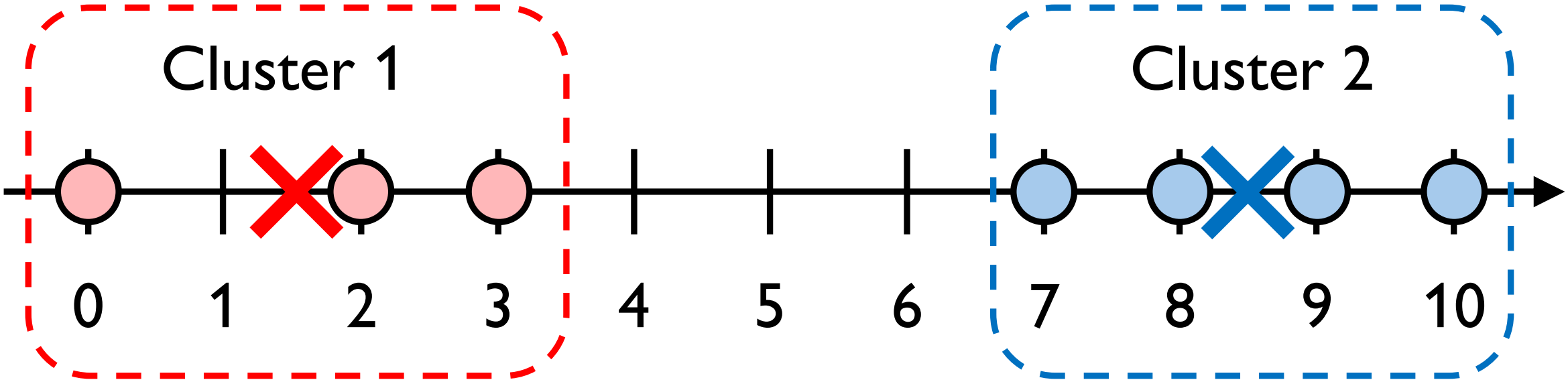
Value = 1.67
Weight = 3

Value = 8.5
Weight = 4



RTPs – Definitions

The sum of the weights must be equal to the number of input data of the aggregation/clustering procedure.



RTPs – Definitions

Other names for RTPs in the literature:

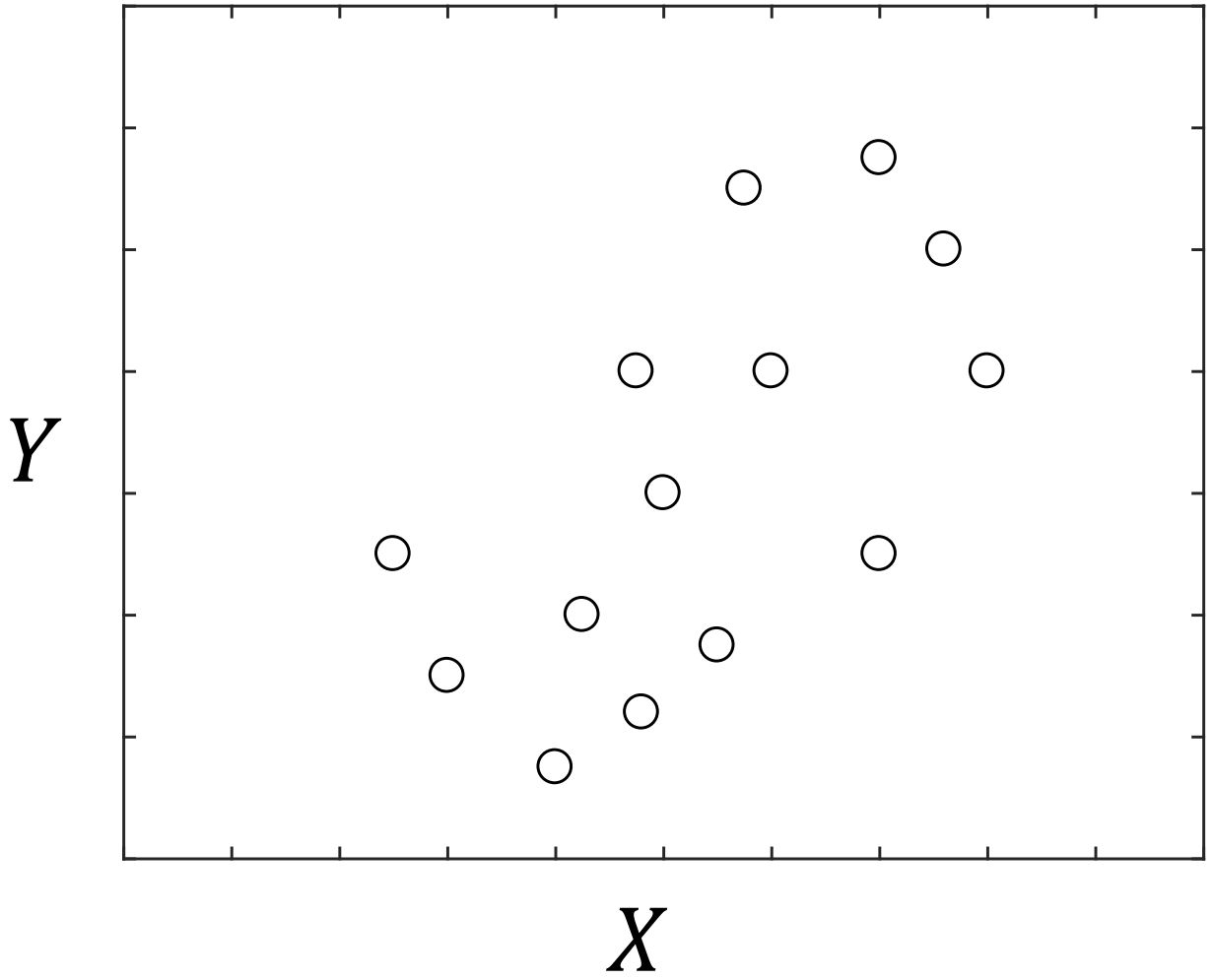
- Operating conditions.
- Snapshots.
- Time slices.
- System states.

RTPs – Popular algorithms

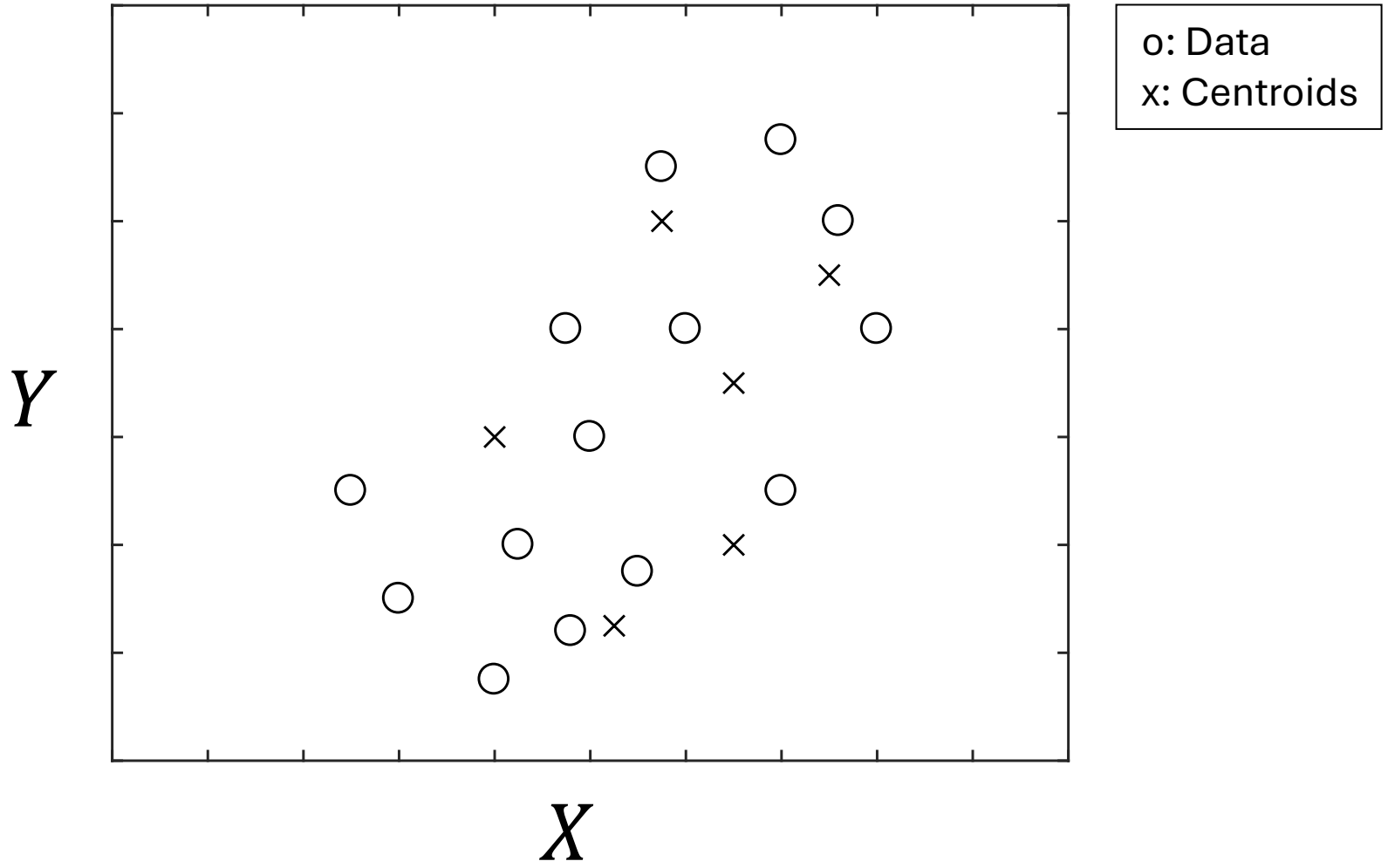
K-means algorithm → Centroids

K-medoids algorithm → Medoids

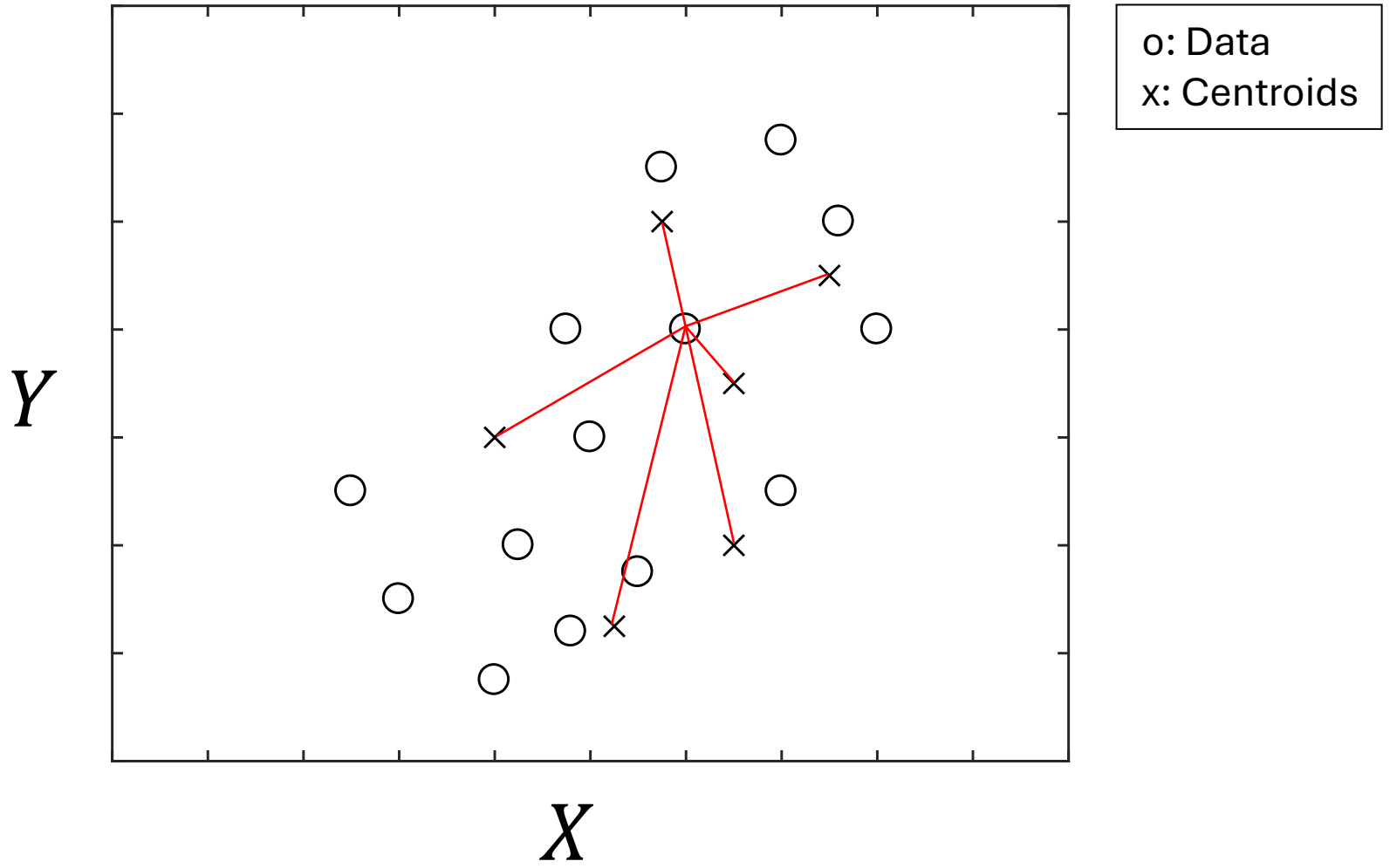
RTPs – K-means algorithm



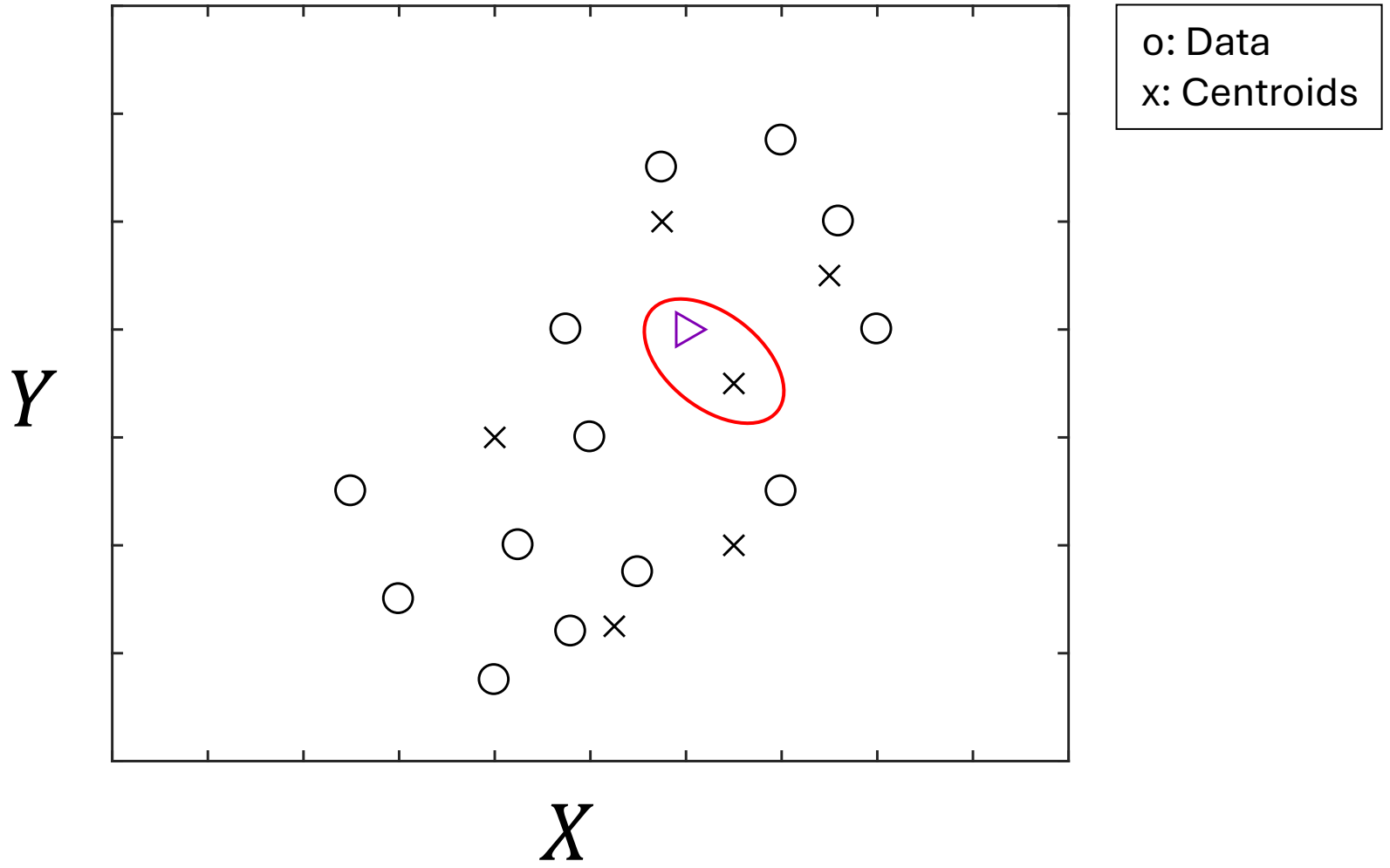
RTPs – K-means algorithm



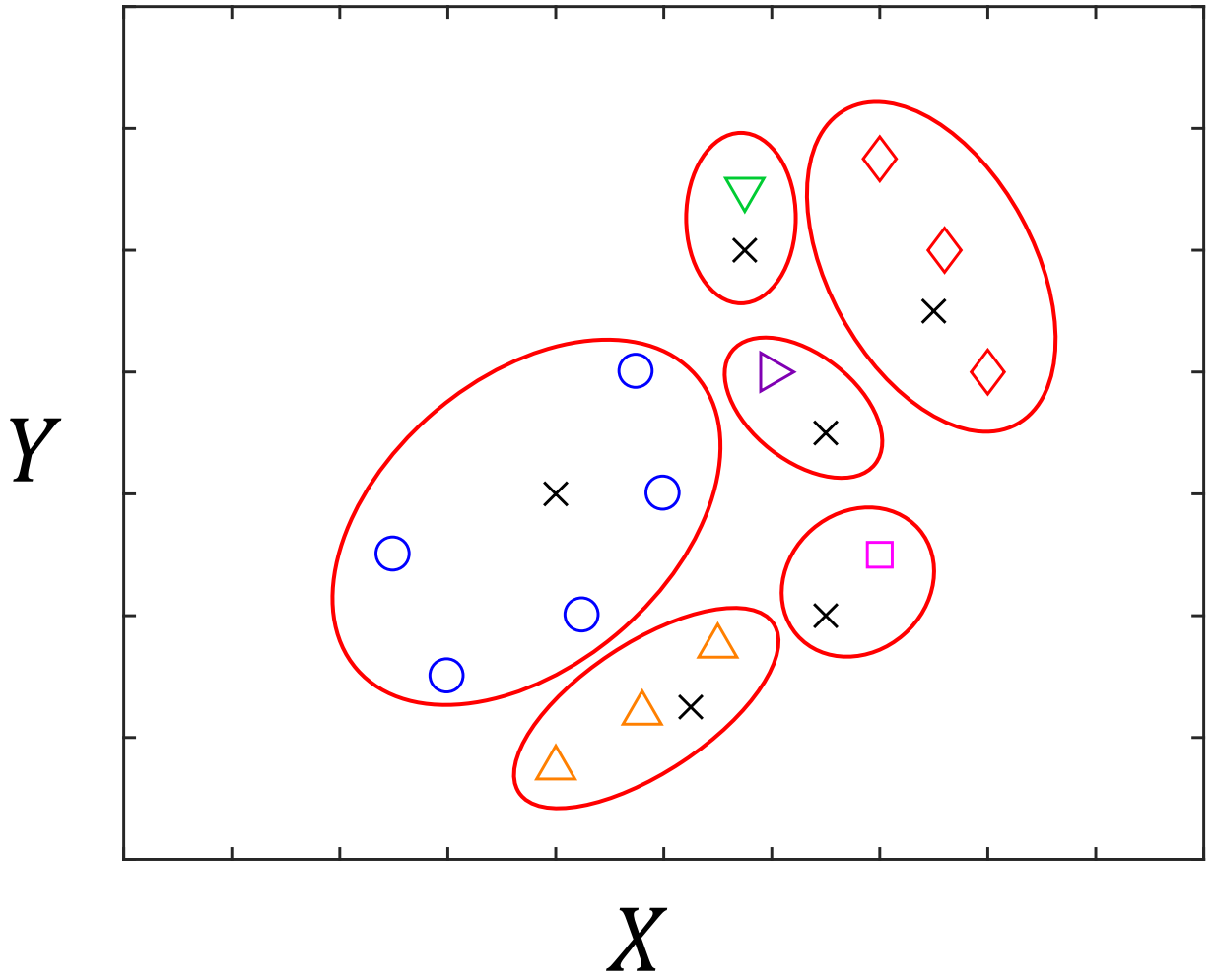
RTPs – K-means algorithm



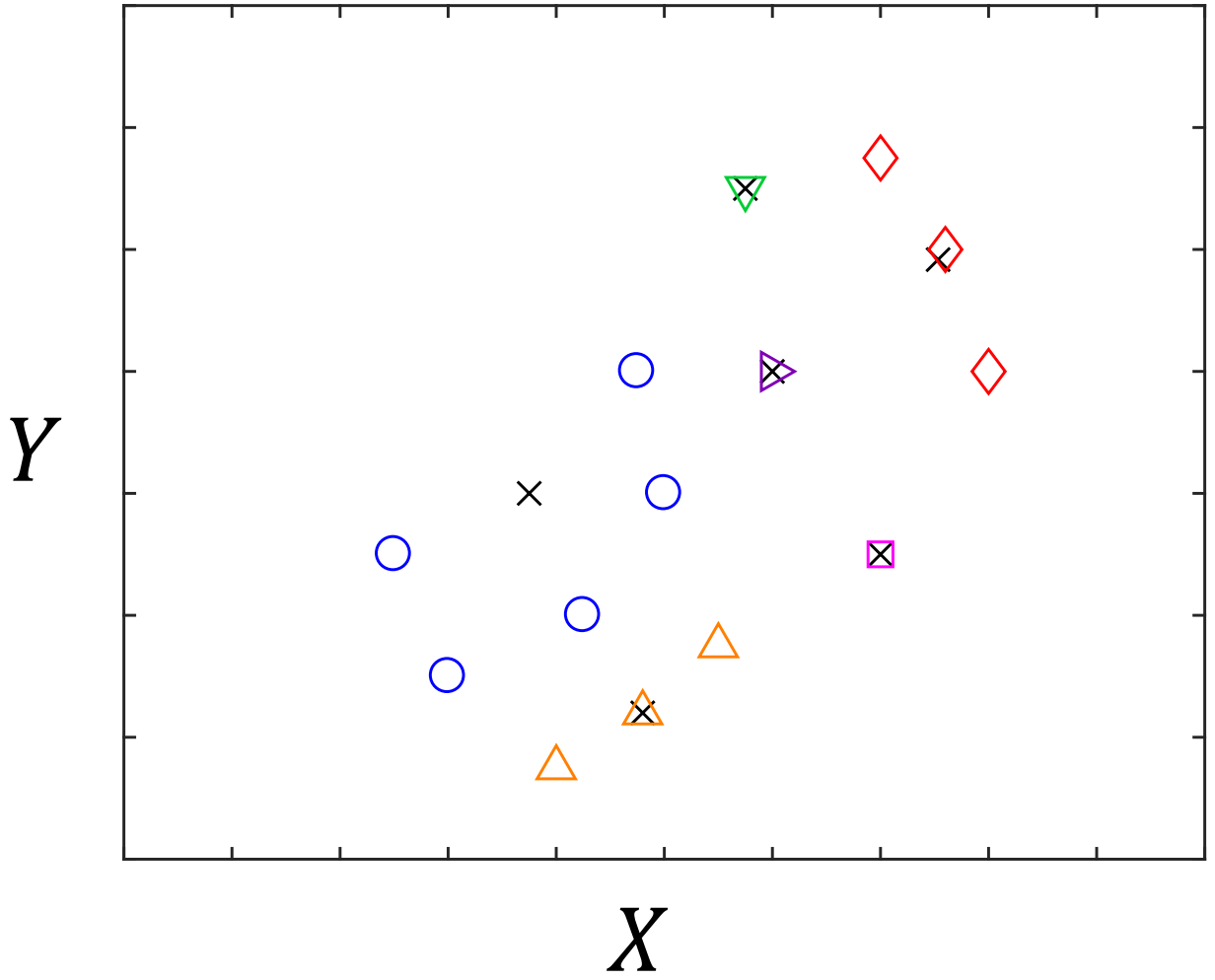
RTPs – K-means algorithm



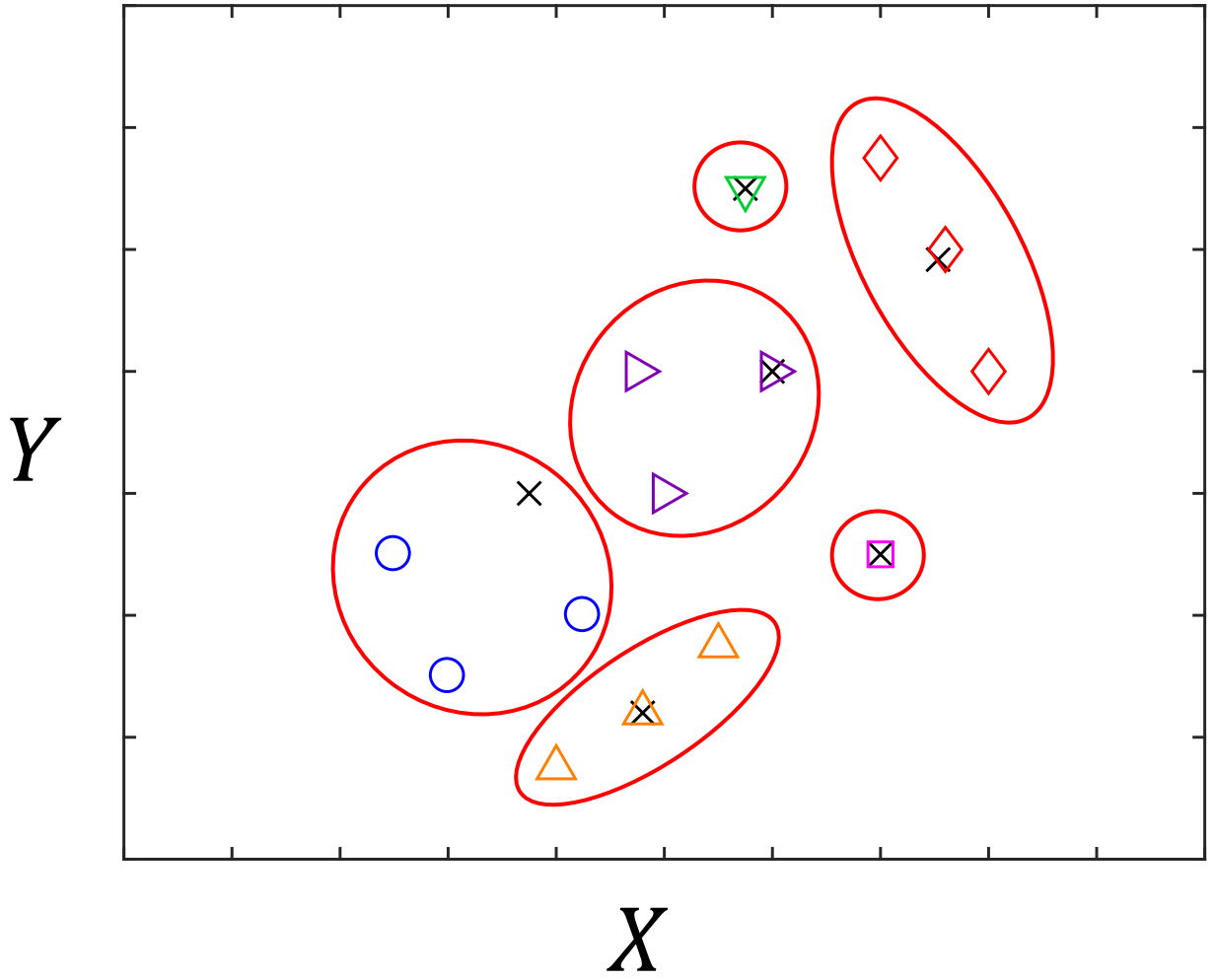
RTPs – K-means algorithm



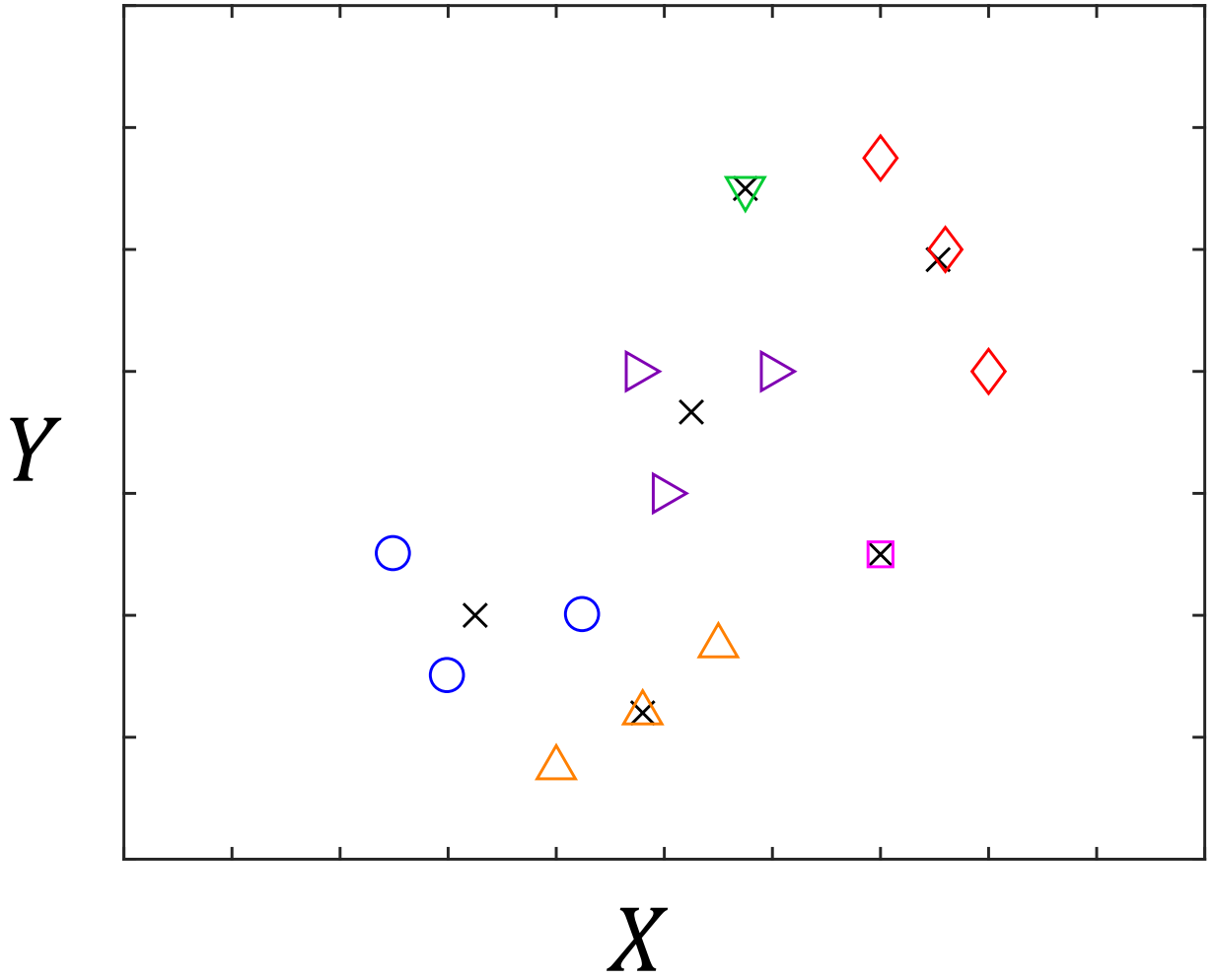
RTPs – K-means algorithm



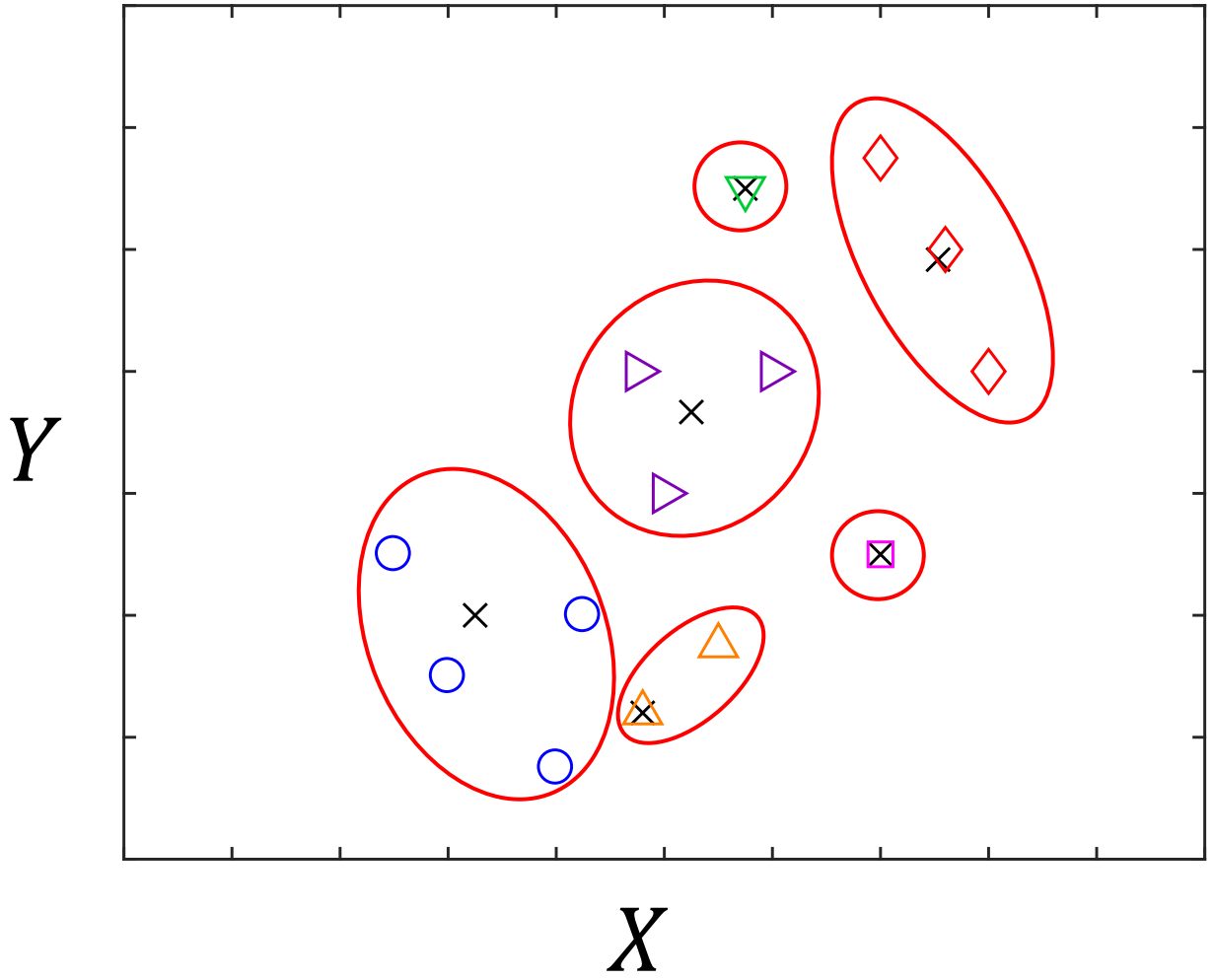
RTPs – K-means algorithm



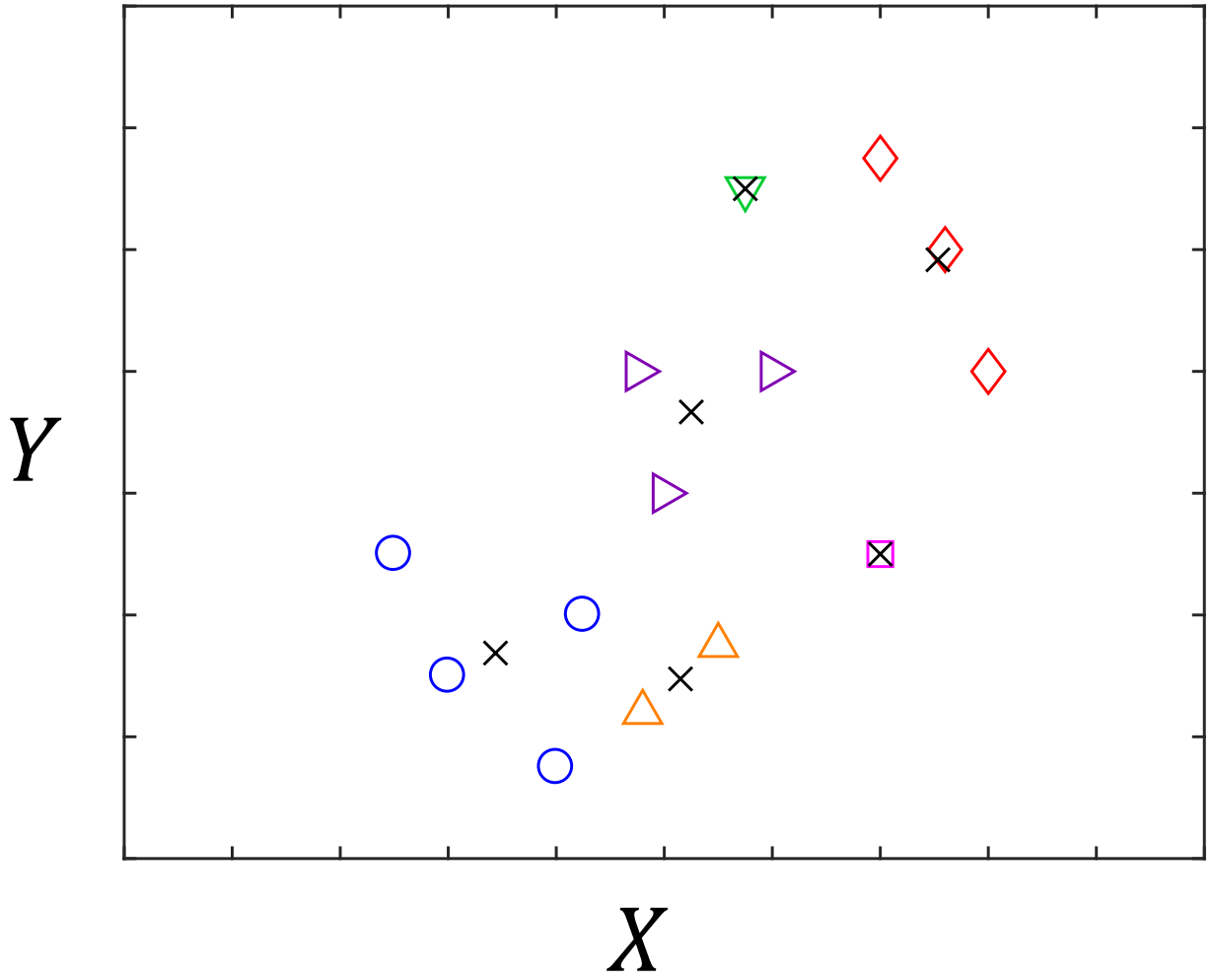
RTPs – K-means algorithm



RTPs – K-means algorithm



RTPs – K-means algorithm



RTPs – Popular algorithms

K-means algorithm → Centroids

K-medoids algorithm → Medoids

RTPs – K-medoids algorithm

K-medoids algorithm → Medoids are used rather than centroids.

Medoid-based algorithms are useful when the input data involve parameters with integer nature.

For example, consider input data related to tariff periods (1, 2, 3, etc.) of electricity consumers. The use of centroid-based algorithms may yield an RTP associated with tariff 1.67, which lacks practical meaning in the context of the problem.

RTPs – Exercise 1

Apply the K-means algorithm (kmeans MATLAB's function) considering the following input data and number of clusters:

- Input data: 0, 2, 7, 10, 3, 8, 9.
- Number of clusters: 2.

Solution:

- RTP1: value (centroid) = 1.67, weight = 3.
- RTP2: value (centroid) = 8.5, weight = 4.

RTPs – Exercise 1

Parameter	Cluster
0	1
2	1
7	2
10	2
3	1
8	2
9	2

RTPs		
Cluster	Centroid	Weight
1	1.67	3
2	8.5	4

RTPs – Exercise 2

Apply the K-medoids algorithm (kmedoids MATLAB's function) considering the following input data and number of clusters:

- Input data: 0, 2, 7, 10, 3, 8, 9.
- Number of clusters: 2.

Solution:

- RTP1: value (medoid) = 2, weight = 3.
- RTP2: value (medoid) = 9, weight = 4.

RTPs – Exercise 3

Apply the K-medoids algorithm (kmedoids MATLAB's function) considering the following input data and number of clusters:

- Input data:
  - Parameter 1: 0, 2, 7, 10, 3, 8, 9.
  - Parameter 2: 100, 300, 150, 225.6, 500, 800, 679.
- Number of clusters: 2.

Solution:

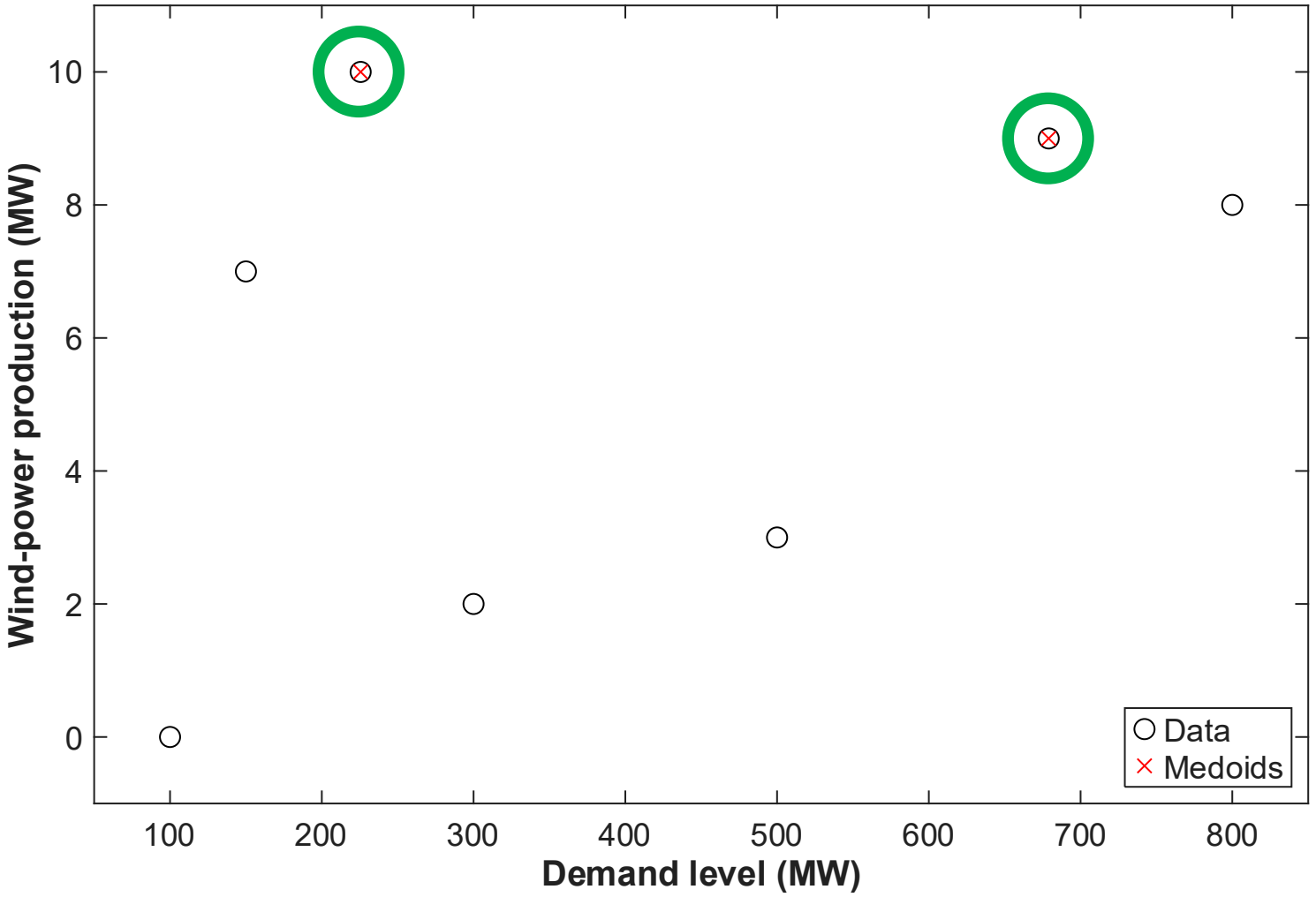
- RTP1: parameter 1 = 9, parameter 2 = 679, weight = 3.
- RTP2: parameter 1 = 10, parameter 2 = 225.6, weight = 4.

RTPs – Exercise 3

Input data		
Parameter 1	Parameter 2	Cluster
0	100	2
2	300	2
7	150	2
10	225.6	2
3	500	1
8	800	1
9	679	1

RTPs			
Cluster	Par. 1	Par. 2	Weight
1	9	679	3
2	10	225.6	4

RTPs – Exercise 3



RTPs – Normalization

Min-max normalization:

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

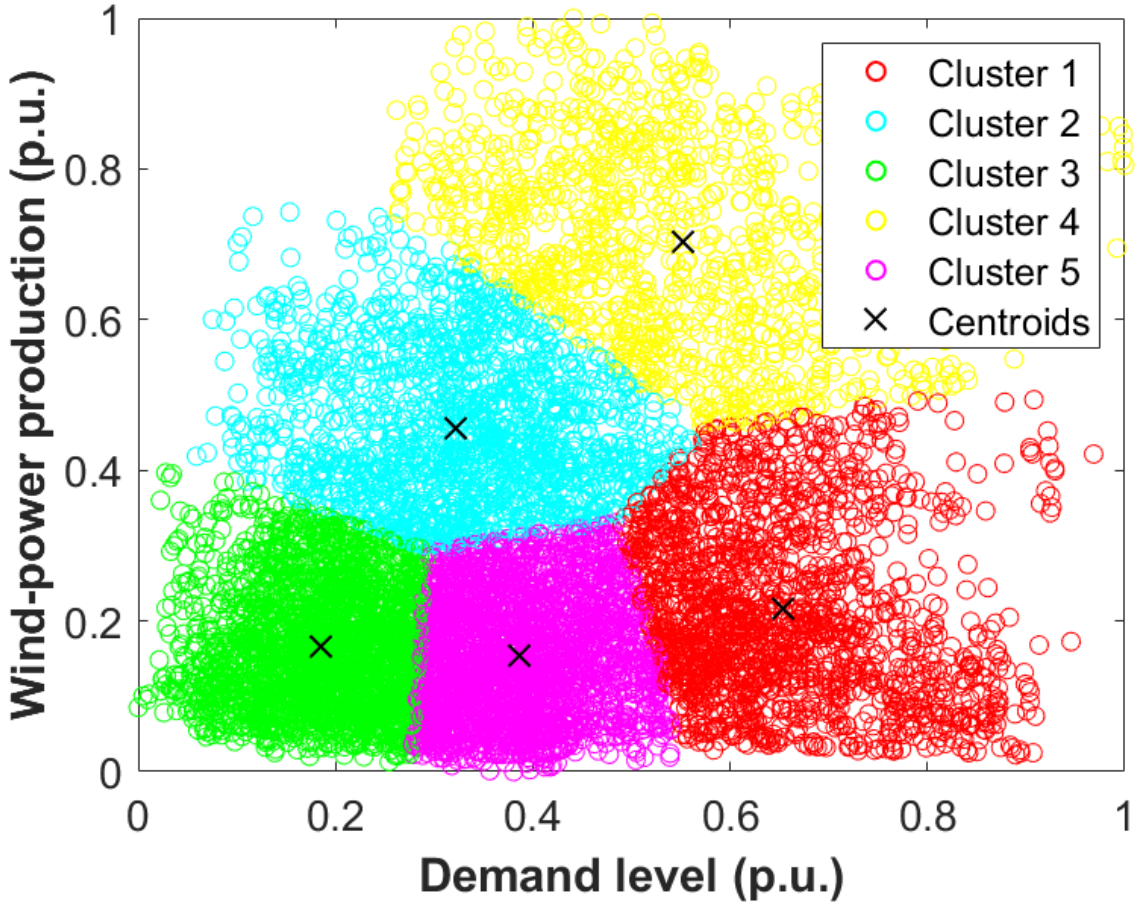
M. Hoffmann, L. Kotzur, D. Stolten, and M. Robinius, "A review on time series aggregation methods for energy system models", *Energies*, vol. 13, no. 3, art no. 641, Feb. 2020. DOI: 10.3390/en13030641.

RTPs – Normalization

Once the normalized representative value $x_{rep,norm}$ (centroid or medoid) has been obtained, the representative value is computed as follows:

$$x_{rep} = x_{rep,norm}(\max(x) - \min(x)) + \min(x)$$

RTPs – Normalization



Centroid	Weight (h)
1	1576
2	1492
3	2220
4	1084
5	2388
Total	8760

RTPs – Exercise 4

Apply the K-medoids algorithm (kmedoids MATLAB's function) using the min-max normalization and considering the following input data and number of clusters:

- Input data:
  - Parameter 1: 0, 2, 7, 10, 3, 8, 9.
  - Parameter 2: 100, 300, 150, 225.6, 500, 800, 679.
- Number of clusters: 2.

Solution:

- RTP1: parameter 1 = 9, parameter 2 = 679, weight = 3.
- RTP2: parameter 1 = 2, parameter 2 = 300, weight = 4.

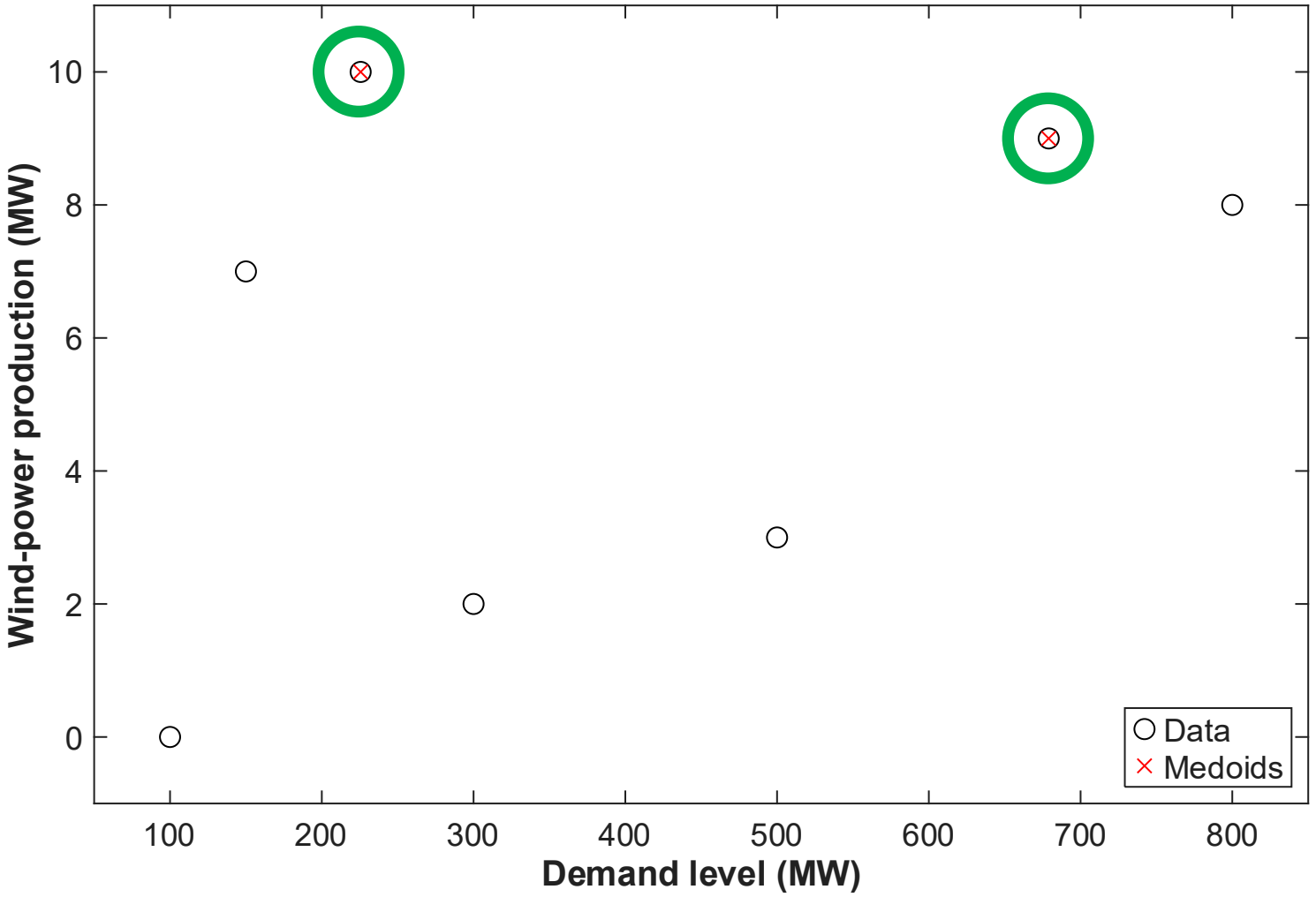
RTPs – Exercise 4

Input data		
Parameter 1	Parameter 2	Cluster
0	100	2
2	300	2
7	150	2
10	225.6	1
3	500	2
8	800	1
9	679	1

RTPs			
Cluster	Par. 1	Par. 2	Weight
1	9	679	3
2	**2**	**300**	4

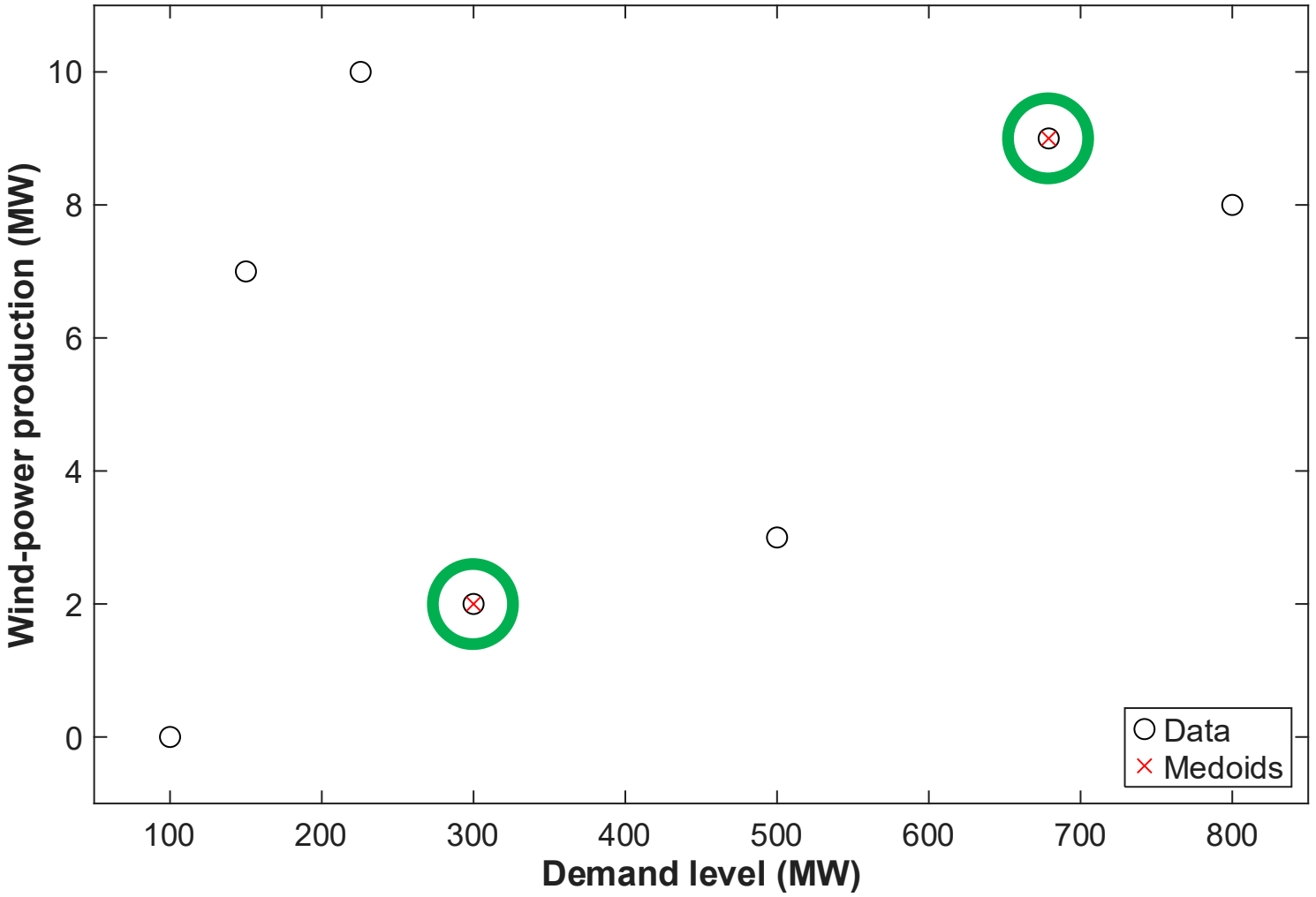
RTPs – Exercise 3

Without Normalization



RTPs – Exercise 4

With Normalization



RTPs – Other techniques

Duration curves:

L. Baringo and A. J. Conejo, "Correlated wind-power production and electric load scenarios for investment decisions," *Applied Energy*, vol. 101, pp. 475–482, Jan. 2013. DOI: 10.1016/j.apenergy.2012.06.002.

Hierarchical clustering (agglomerative and divisive):

L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. Chichester, UK: John Wiley & Sons, Inc., 1990. DOI: 10.1002/9780470316801.

RTPs – Transition matrix

S. Wogrin, P. Dueñas, A. Delgadillo, and J. Reneses, "A new approach to model load levels in electric power systems with high renewable penetration", *IEEE Transactions on Power Systems*, vol. 29, no. 5, pp. 2210–2218, Sep. 2014. DOI: 10.1109/TPWRS.2014.2300697.

Sonja Wogrin: https://orcid.org/0000-0002-3889-7197

3. Representative days (RDs)

RDs – Motivation

Inter-temporal dynamics that should be modeled:

- Start-up cost.
- Shut-down cost.
- Minimum down time.
- Ramping limits.
- Evolution of the energy stored in batteries.

RDs – Definitions

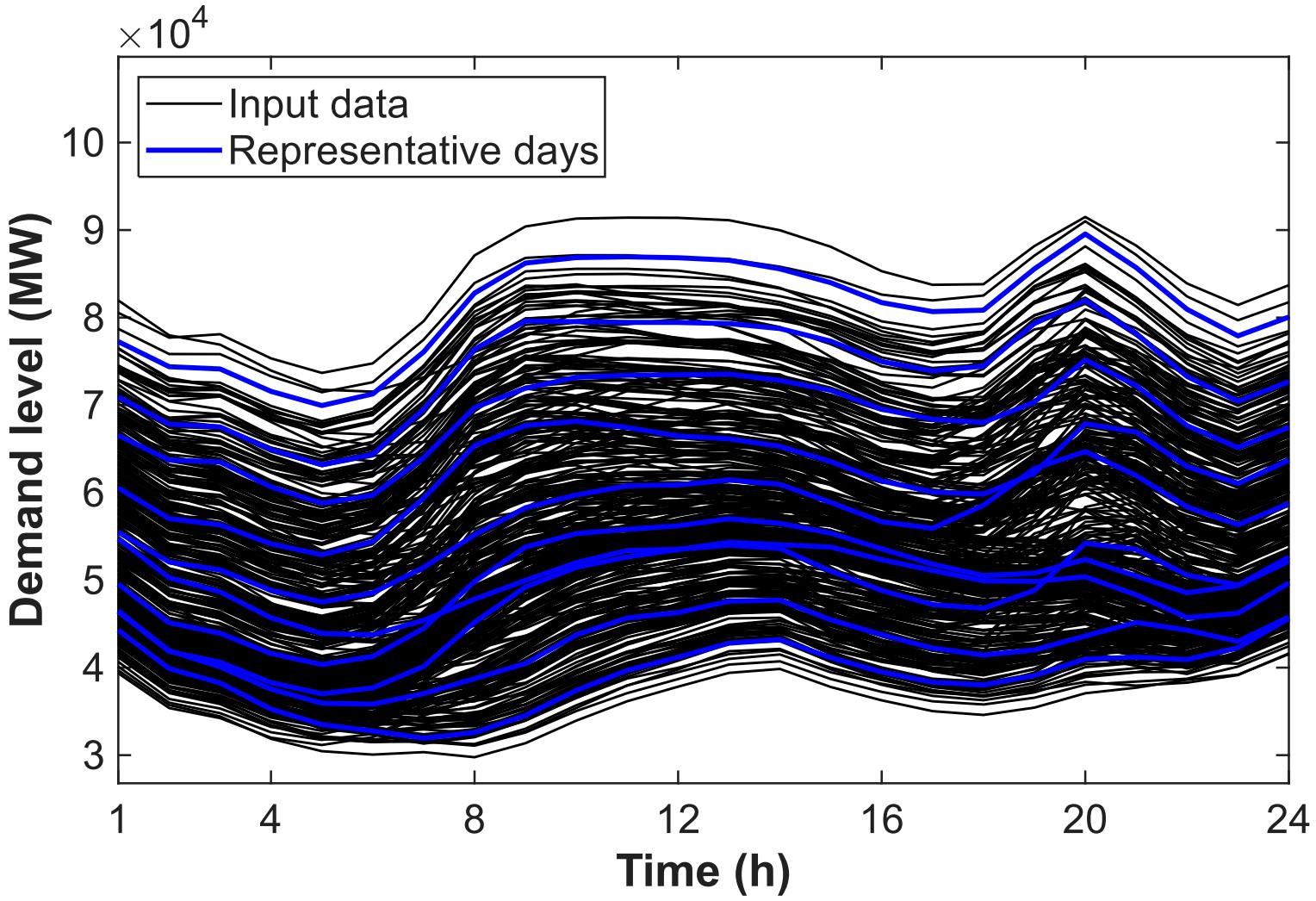
I will consider hourly input data.

Then, each RD is composed by 24 hourly time periods.

Each RD is associated with:

- 24 values of each parameter from the input data (demand, wind-power availability, solar-power availability, etc.).
- Weight = Number of days from the input data within the cluster.

RDs – Example


×10^4
Input data
Representative days
10
9
8
7
6
5
4
3
Demand level (MW)
1
4
8
12
16
20
24
Time (h)

RDs – Formulation

$$\min_{\Phi} \sum_{d \in \Omega^{\mathrm{D}}} \rho_d^{\mathrm{D}} \sum_{h \in \Omega^{\mathrm{H}}} \text{Operating costs} + \text{Annualized investment costs}$$

RDs – Formulation

$$\min_{\Phi} \sum_{d \in \Omega^{\mathrm{D}}} \rho_d^{\mathrm{D}} \sum_{h \in \Omega^{\mathrm{H}}} \text{Operating costs} + \text{Annualized investment costs}$$

RDs

RDs – Formulation

Weight of RD d (number of historical days)

$$\min_{\Phi} \sum_{d \in \Omega^{\mathrm{D}}} \rho_d^{\mathrm{D}} \sum_{h \in \Omega^{\mathrm{H}}} \text{Operating costs} + \text{Annualized investment costs}$$

RDs – Formulation

$$\min_{\Phi} \sum_{d \in \Omega^{\mathrm{D}}} \rho_d^{\mathrm{D}} \sum_{h \in \Omega^{\mathrm{H}}} \text{Operating costs} + \text{Annualized investment costs}$$

Hours

RDs – Formulation

Optimization variables associated with the operation:

$$variable_{dh}$$

RDs – Formulation

If the input data involve 8760 hourly time periods, then:

$$\sum_{d \in \Omega^{\mathrm{D}}} \rho_d^{\mathrm{D}} = 365$$

RDs – Different techniques

Some researchers use:

- One RD per season.
- One RD per month.

These methods neglect the possible similarities among historical days from different seasons and months.

RDs – Different techniques

Typical method used by researchers:

- K-means algorithm.
- K-medoids algorithm.
- Hierarchical clustering (agglomerative and divisive).

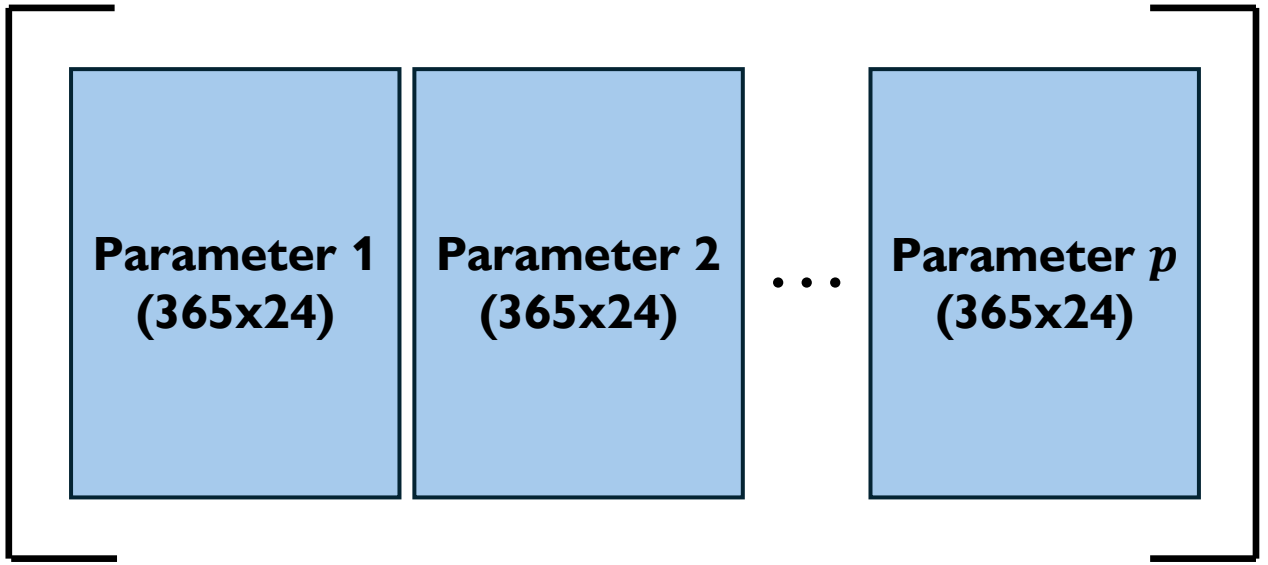
RDs – Input and output data

Input data:

- 8760 hourly time periods and p parameters.
- Input matrix: 365x24p.

Clustering considering K RDs.

Output matrix: Kx24p.

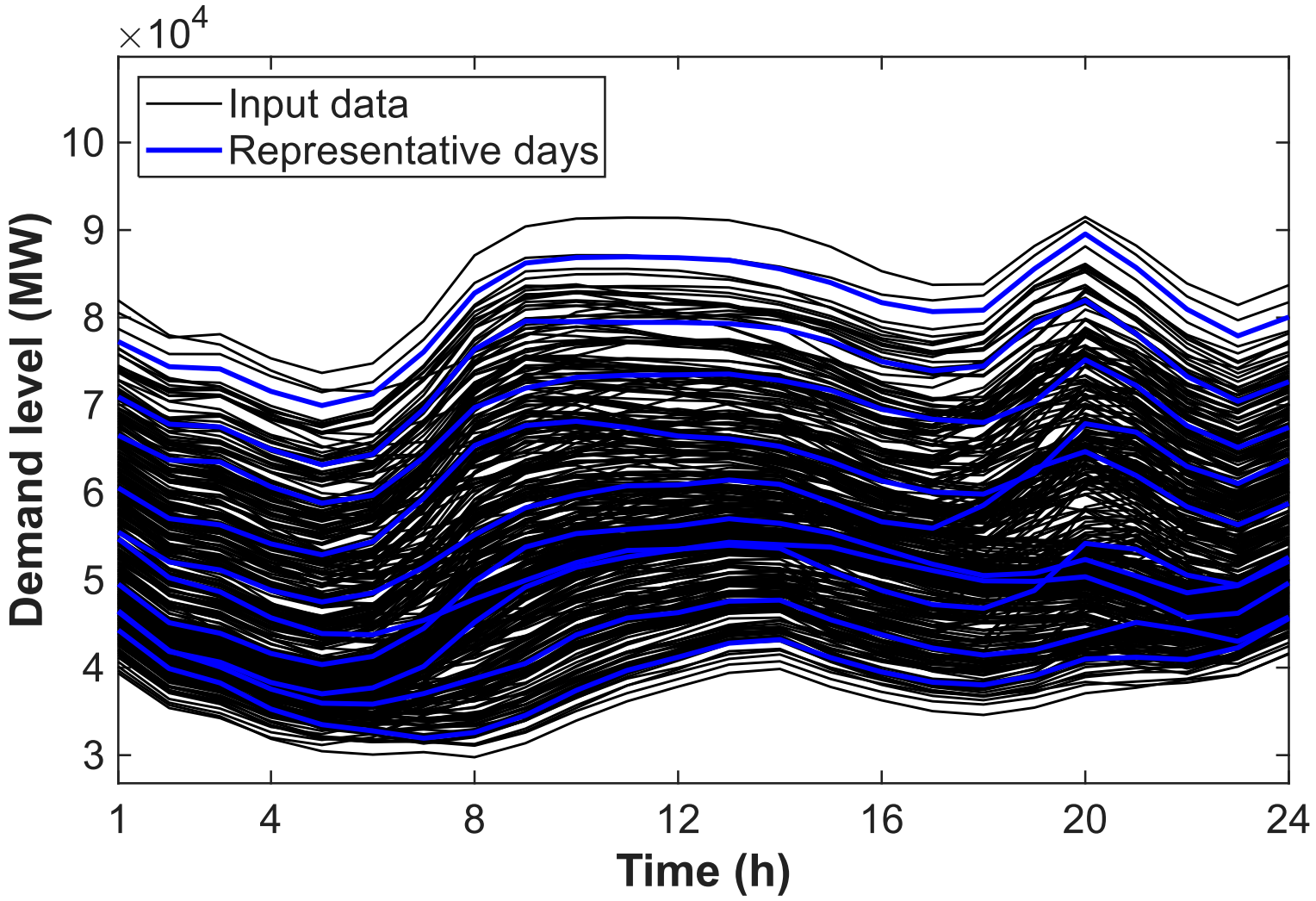


RDs – Exercise 5

Apply the K-medoids algorithm (kmedoids MATLAB's function) considering 10 RDs and the input data from the following link (exercise3_05.m, 2015_f_load.xls):

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

RDs – Exercise 5

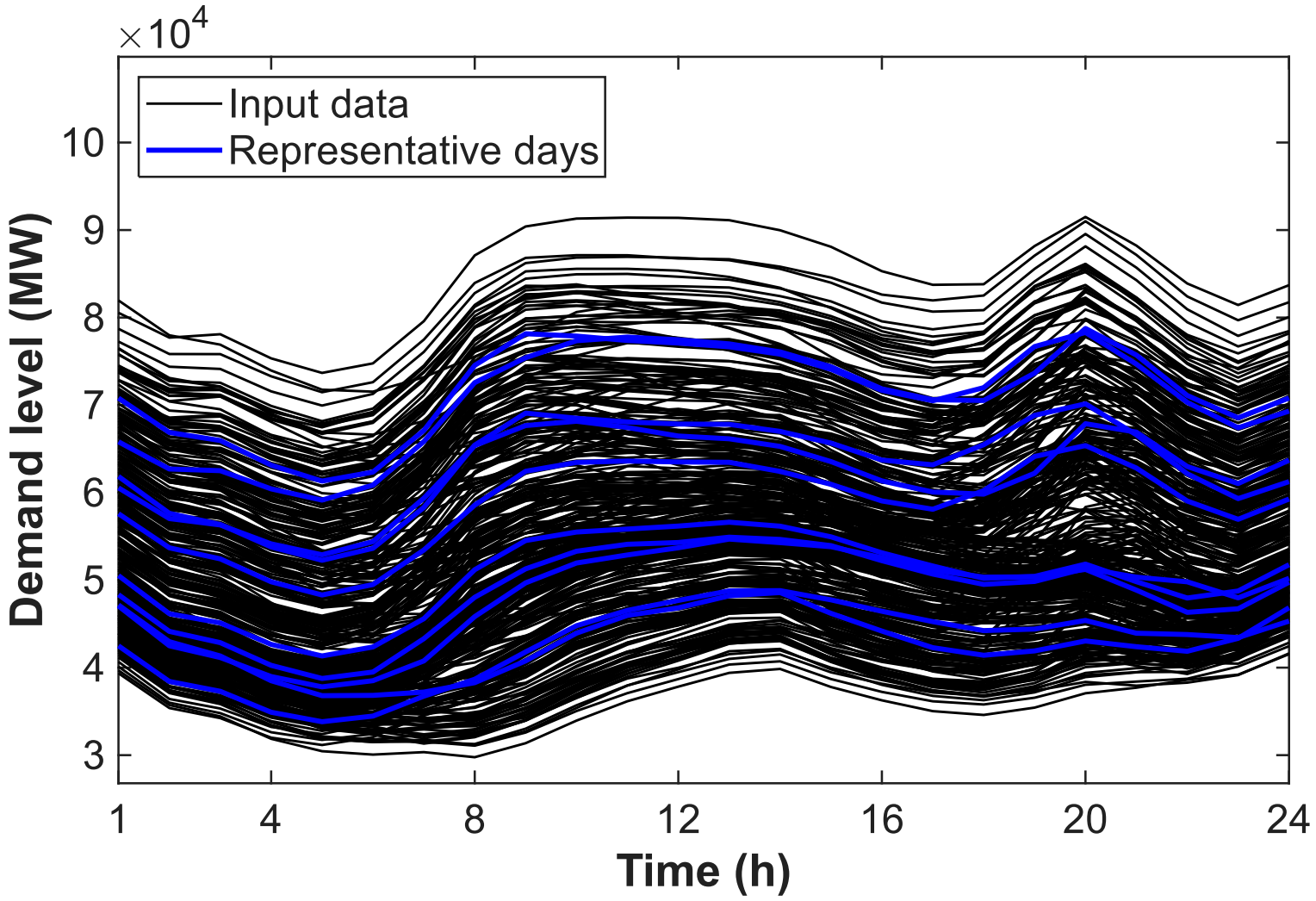


RDs – Exercise 6

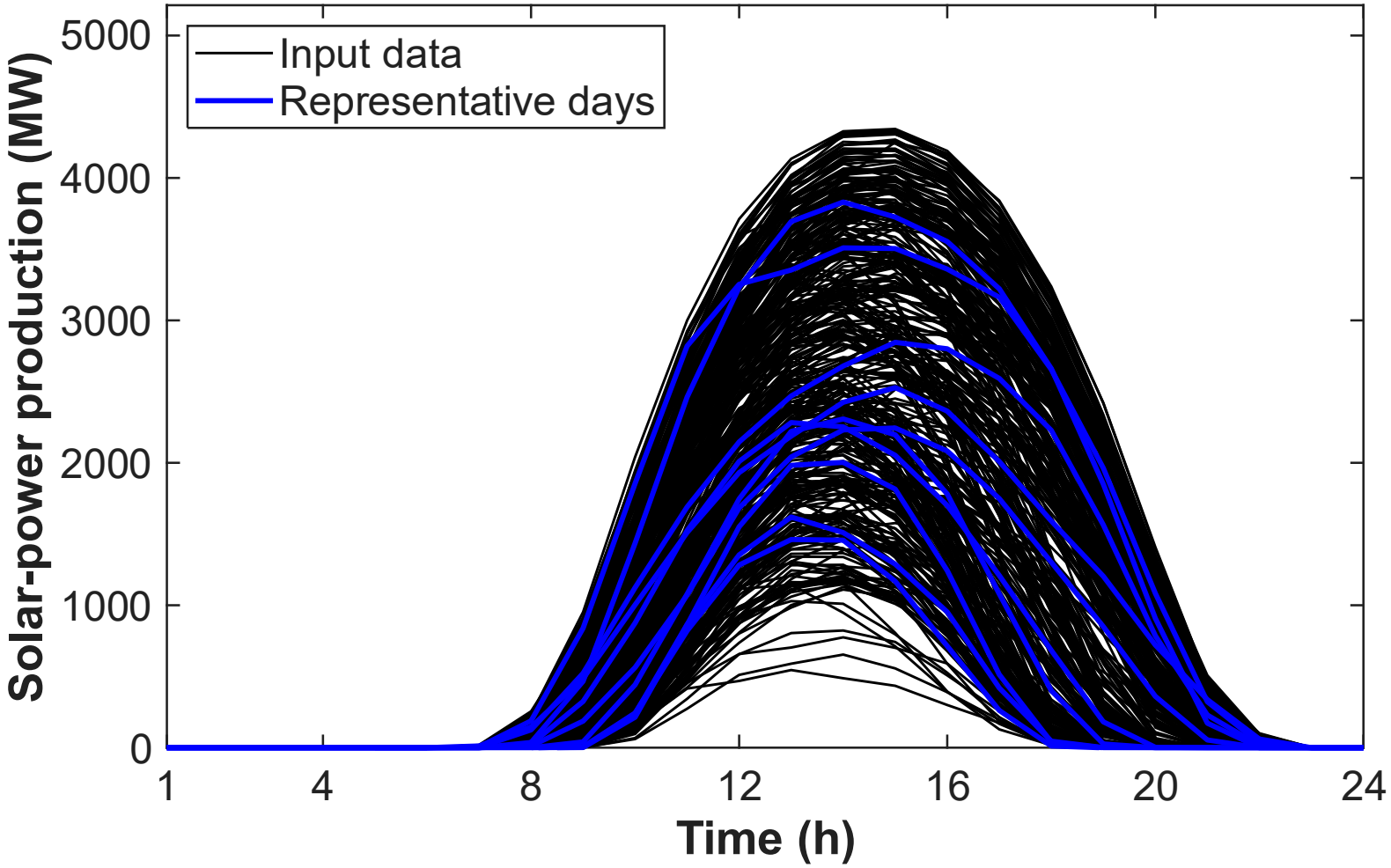
Apply the K-medoids algorithm (kmedoids MATLAB's function) considering 10 RDs and the input data from the following link (exercise3_06.m, 2015_f_load.xls, 2015_f_pv.xls, 2015_f_wind.xls):

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

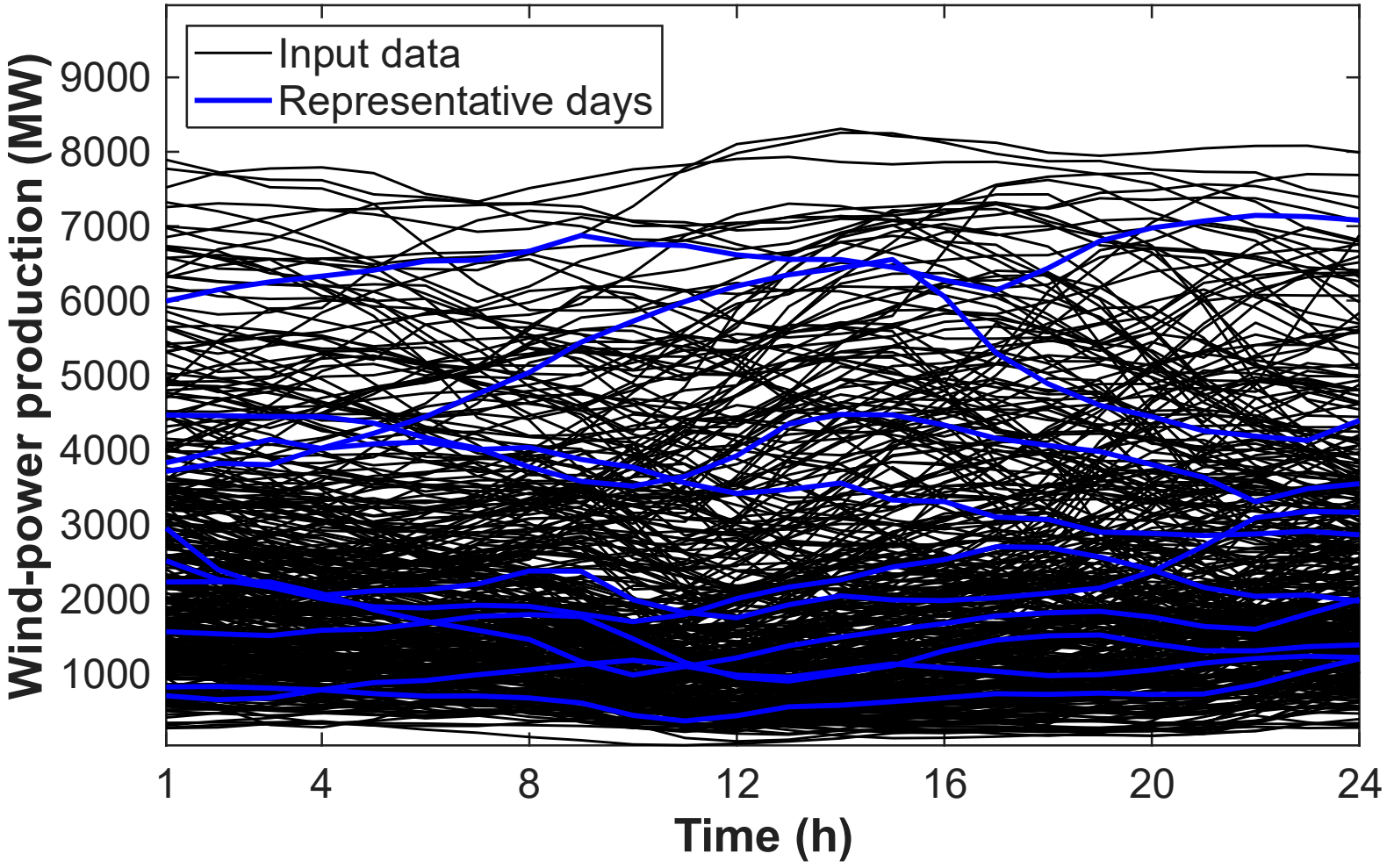
RDs – Exercise 6



RDs – Exercise 6



RDs – Exercise 6



RDs – Exercise 7

Export the results from exercise 6 to GMS files.

Sets d and h used for representative days and hours, respectively.

Demand → Parameter PD(d,h).

Solar-power production → Parameter APV(d,h).

Wind-power production → Parameter AW(d,h).

RDs – Exercise 8

Solve the centralized GEP problem using the data, including 10 RDs, from the following link (exercise3_08.gms, dataRho_3_08.gms, dataPD_3_08.gms):

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

Solution:

Objective function value = M€54.88.

Capacity built of the generating unit g_4 = 50 MW.

Capacity built of the generating unit g_5 = 250 MW.

Capacity built of the generating unit g_6 = 0 MW.

RDs – Exercise 9

Solve the centralized GEP problem using the data, including 60 RDs, from the following link (exercise3_09.gms, dataRho_3_09.gms, dataPD_3_09.gms):

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

Solution:

Objective function value = M€56.86.

Capacity built of the generating unit g_4 = 150 MW.

Capacity built of the generating unit g_5 = 150 MW.

Capacity built of the generating unit g_6 = 50 MW.

RDs – Exercise 10

Solve the centralized GEP problem using the data, including 365 RDs, from the following link (exercise3_10.gms, dataRho_3_10.gms, dataPD_3_10.gms):

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%203

Solution:

Objective function value = M€56.96.

Capacity built of the generating unit g_4 = 150 MW.

Capacity built of the generating unit g_5 = 150 MW.

Capacity built of the generating unit g_6 = 50 MW.

RDs – Exercises 8–10

Representative days	**10**	**60**	**365**
Objective function value (M€)	54.88	56.86	56.96
Capacity built of the generating unit g_4 (MW)	50	150	150
Capacity built of the generating unit g_5 (MW)	250	150	150
Capacity built of the generating unit g_6 (MW)	0	50	50
Computational time (s)	< 0.001	2	40

RDs – Exercises 8–10

Representative days	**10**	**60**	**365**
Objective function value (M€)	54.88	56.86	56.96
Capacity built of the generating unit g_4 (MW)	50	150	150
Capacity built of the generating unit g_5 (MW)	250	150	150
Capacity built of the generating unit g_6 (MW)	0	50	50
Computational time (s)	< 0.001	2	40

RDs – Exercises 8–10

Optimal investment plan

Representative days	**10**	**60**	**365**
Objective function value (M€)	54.88	56.86	56.96
Capacity built of the generating unit g_4 (MW)	50	150	150
Capacity built of the generating unit g_5 (MW)	250	150	150
Capacity built of the generating unit g_6 (MW)	0	50	50
Computational time (s)	< 0.001	2	40

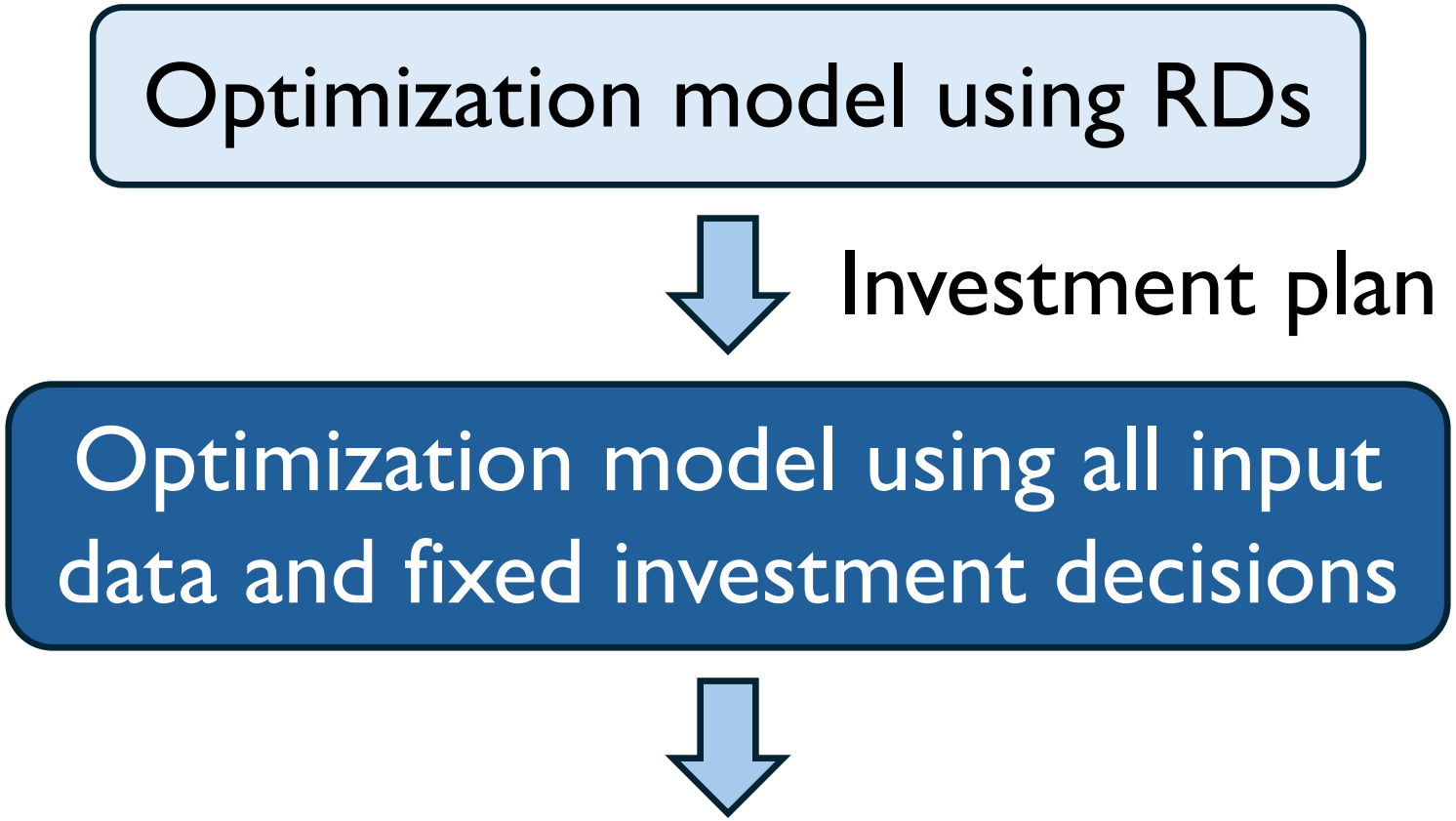
RDs – Exercises 8–10

Lowest costs
(minimization problem)

Best solution?

Representative days	**10**	**60**	**365**
Objective function value (M€)	54.88	56.86	56.96
Capacity built of the generating unit g_4 (MW)	50	150	150
Capacity built of the generating unit g_5 (MW)	250	150	150
Capacity built of the generating unit g_6 (MW)	0	50	50
Computational time (s)	< 0.001	2	40

RDs – Comparison procedure



RDs – Exercises 8–10

Representative days	**10**	**60**	**365**
Objective function value **using all input data and fixed investment decisions** (M€)	57.41	56.96	56.96
Capacity built of the generating unit g_4 (MW)	50	150	150
Capacity built of the generating unit g_5 (MW)	250	150	150
Capacity built of the generating unit g_6 (MW)	0	50	50

4. Inter-day dynamics

Inter-day dynamics – Motivation

Independent RDs → Inter-day dynamics cannot be modeled.

Dunkelflaute → Lack of wind and sunlight → Null or low renewable production → Inter-day storage become relevant.

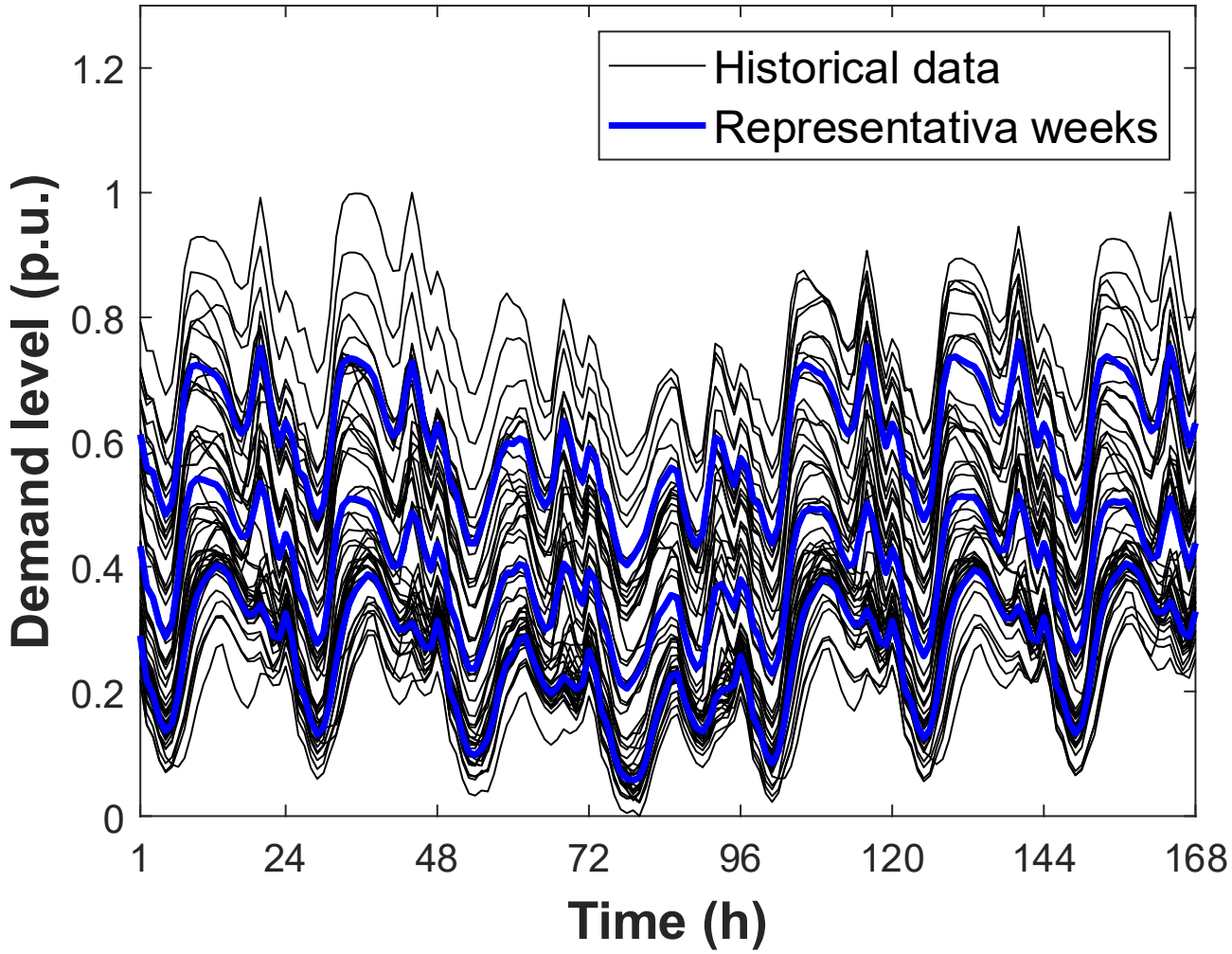
Inter-day dynamics – Solutions

Possible solutions:

- Representative weeks (RWs).

Input matrix

- RDs → 24 columns per parameter.
- RWs → 168 columns per parameter.



Inter-day dynamics – Solutions

Possible solutions:

- RDs with inter-day dynamics:
  - RDs and transition matrix.
  - The technical limits of the storage assets are not imposed for all time periods → Practical solution may be infeasible.
  - If these constraints are imposed for all time periods → → Significant increase in the computational time.

D. A. Tejada-Arango, M. Domeshek, S. Wogrin, and E. Centeno, "Enhanced representative days and system states modeling for energy storage investment analysis", *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6534–6544, Nov. 2018.

Inter-day dynamics – Solutions

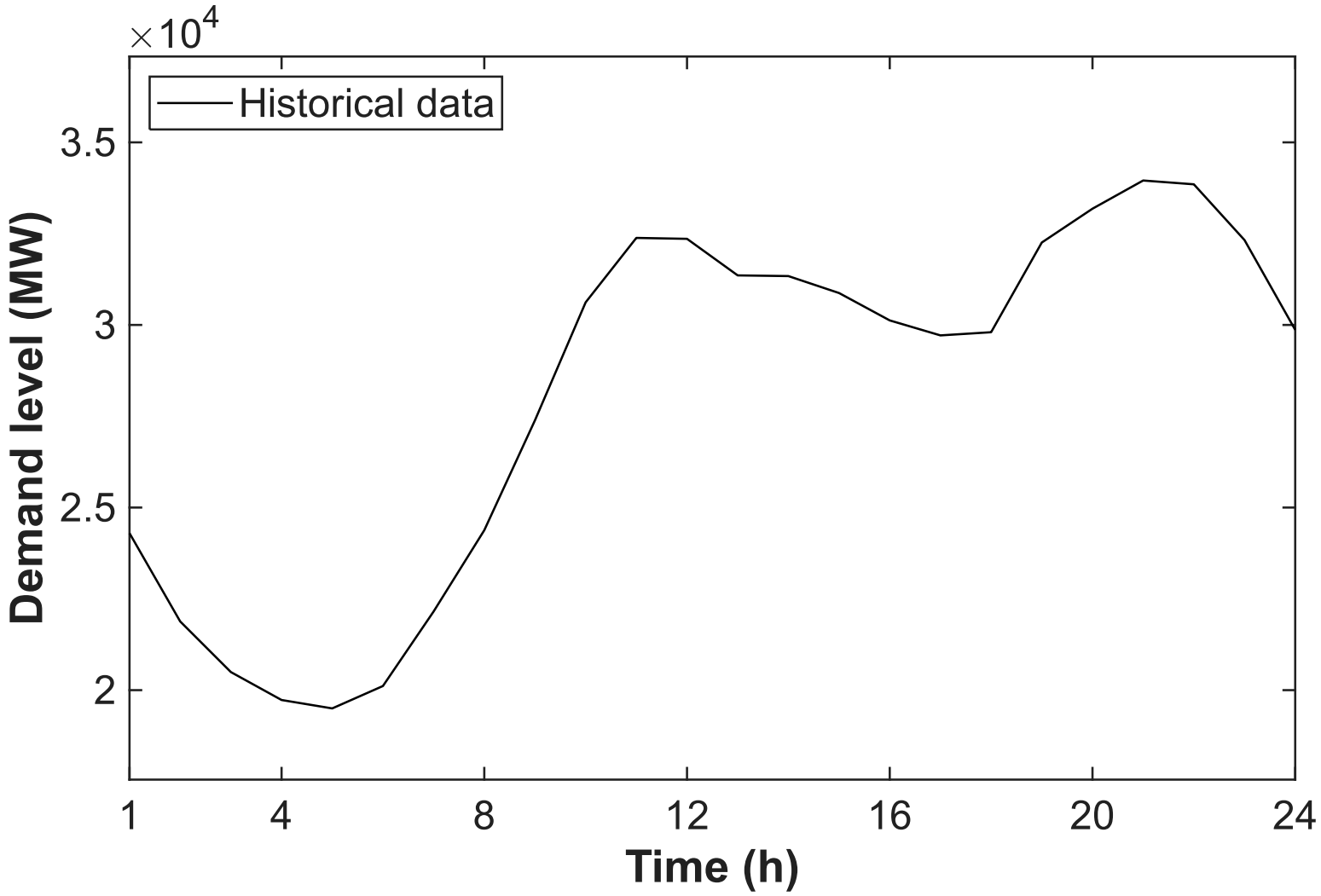
Possible solutions:

- RTPs with inter-day dynamics throughout the entire time horizon:
  - Chronological Time-Period Clustering (**CTPC**).
  - The agglomerative hierarchical clustering is applied for **adjacent** time periods.

S. Pineda and J. M. Morales, "Chronological time-period clustering for optimal capacity expansion planning with storage", *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 7162–7170, Nov. 2018.

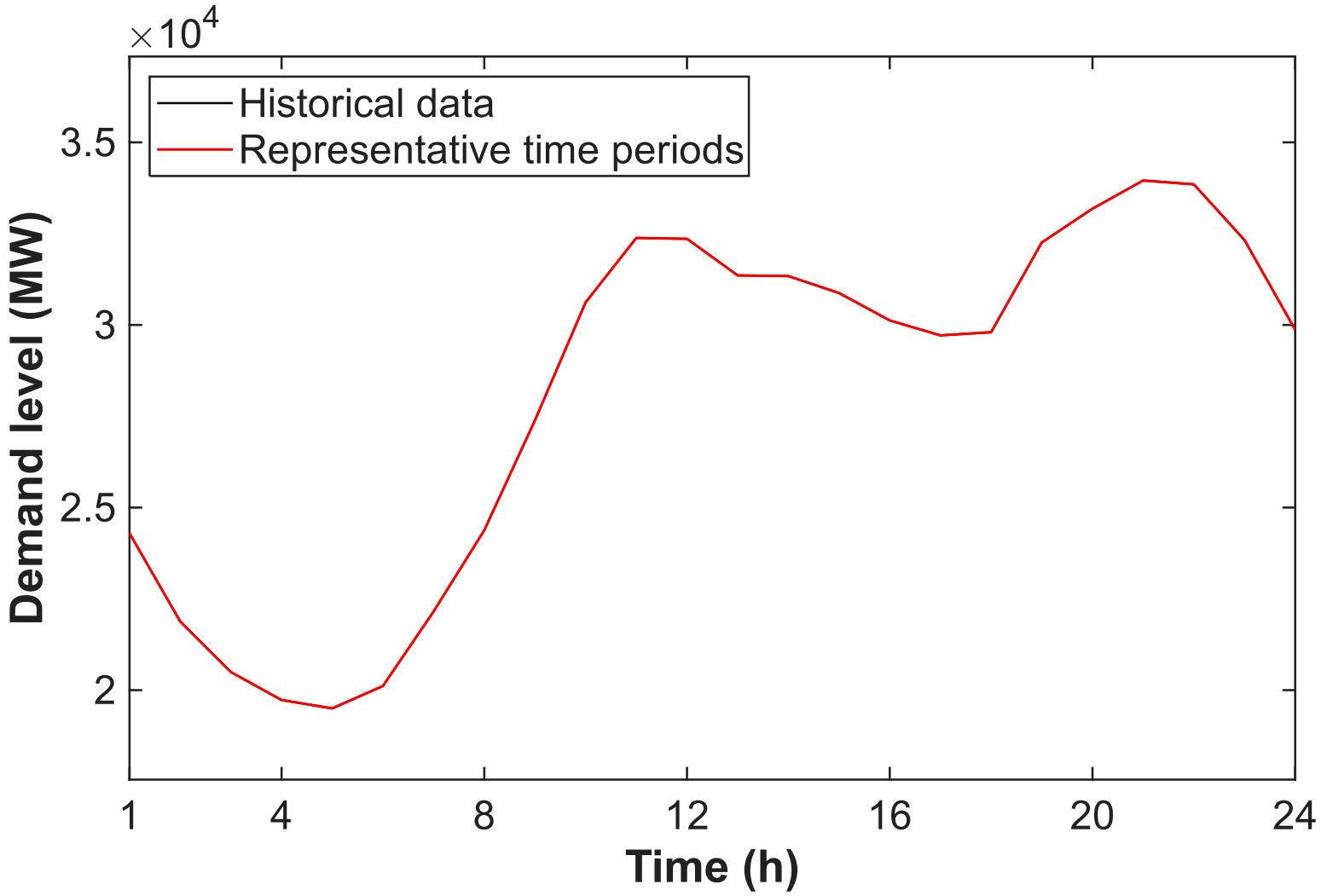
CTPC

Input data: 24



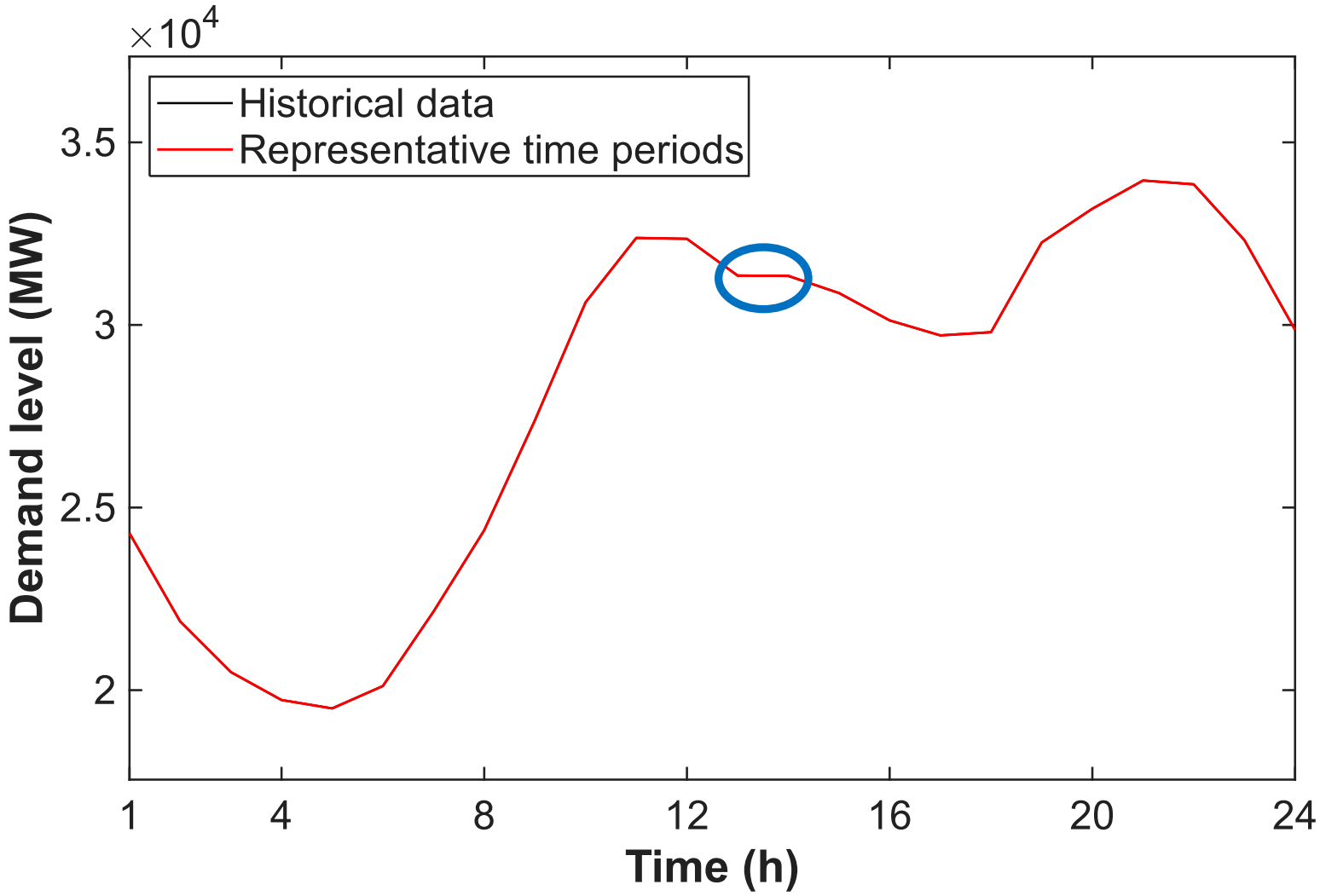
CTPC

Input data: 24
RTPs: 24



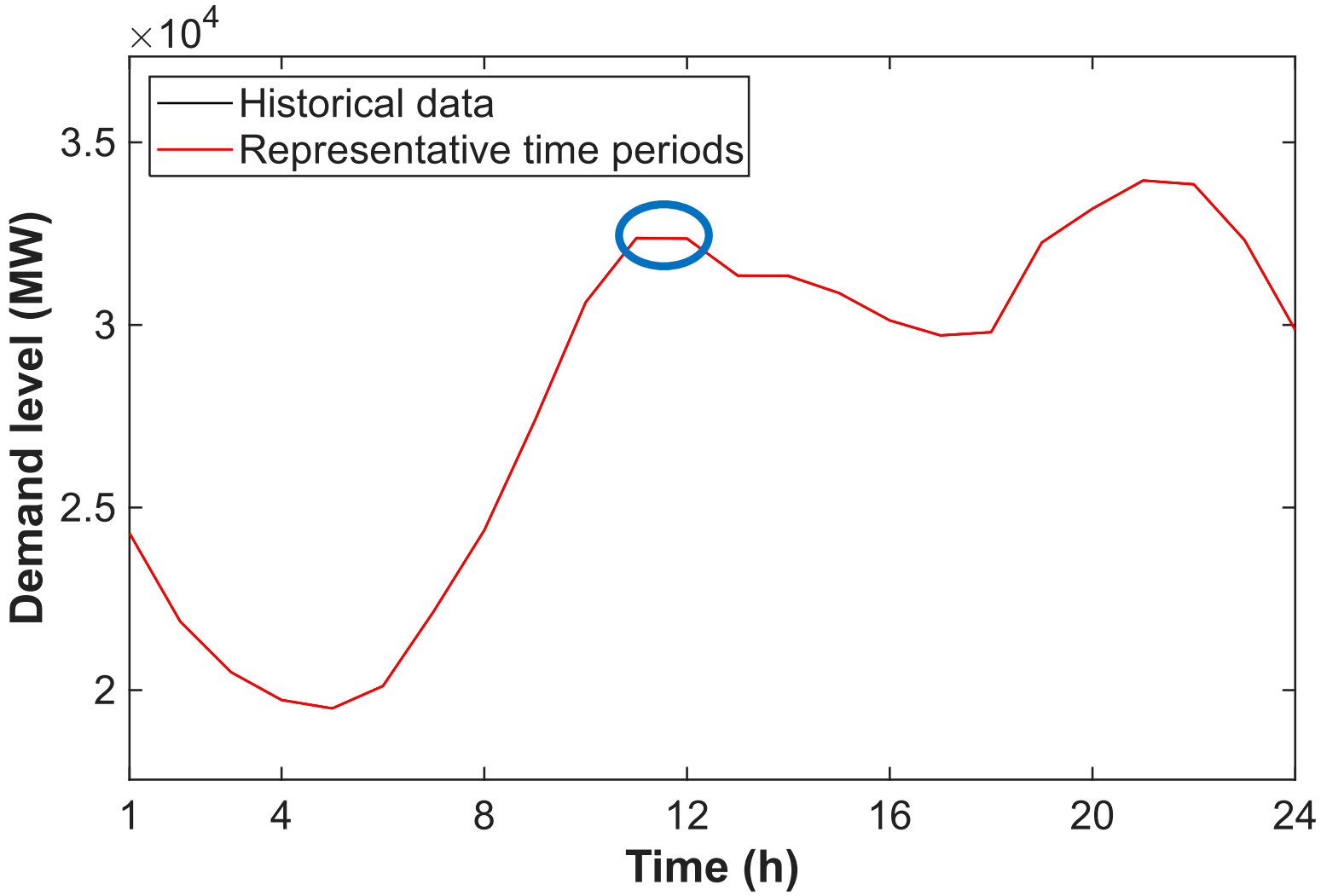
CTPC

Input data: 24
RTPs: 23



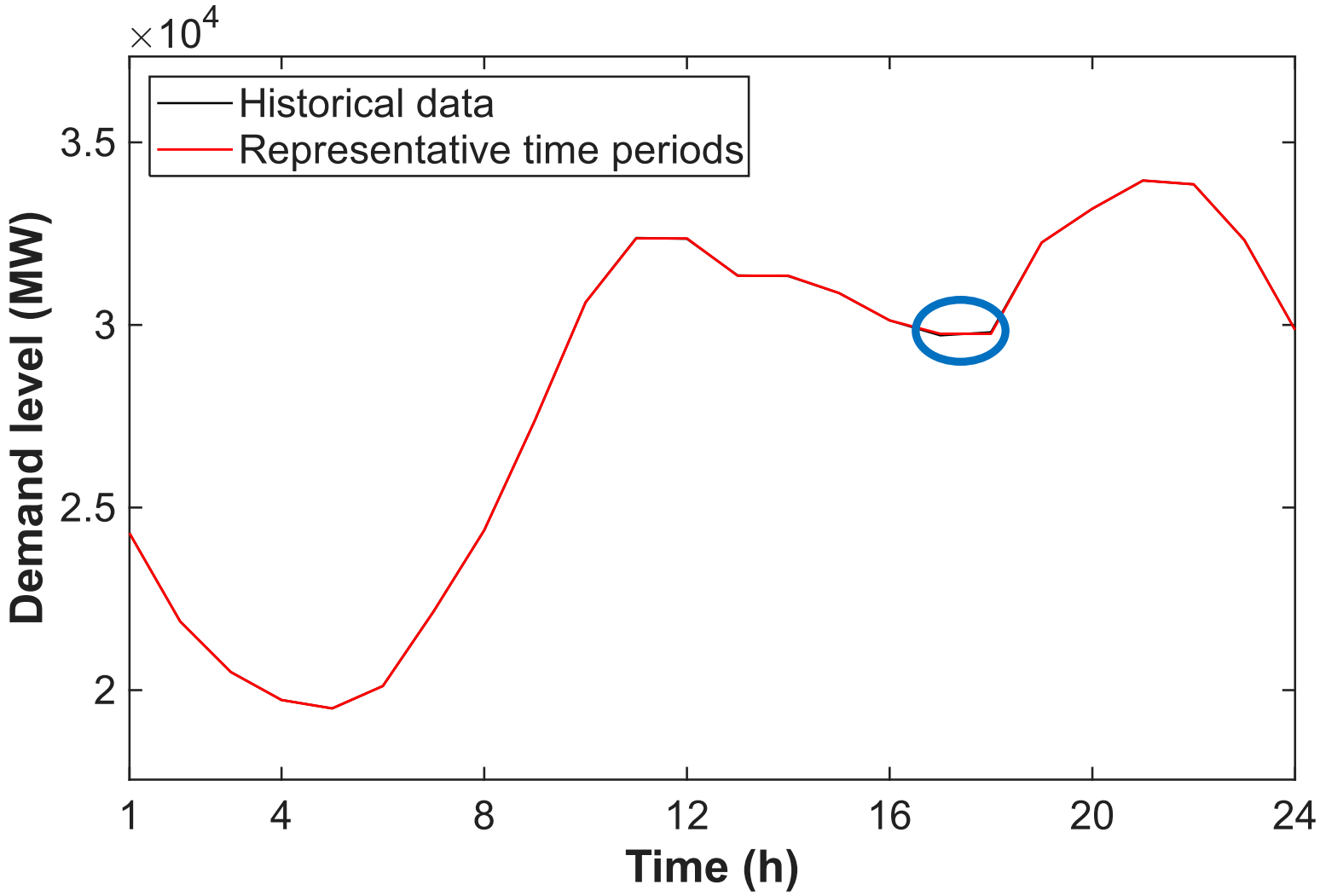
CTPC

Input data: 24
RTPs: 22



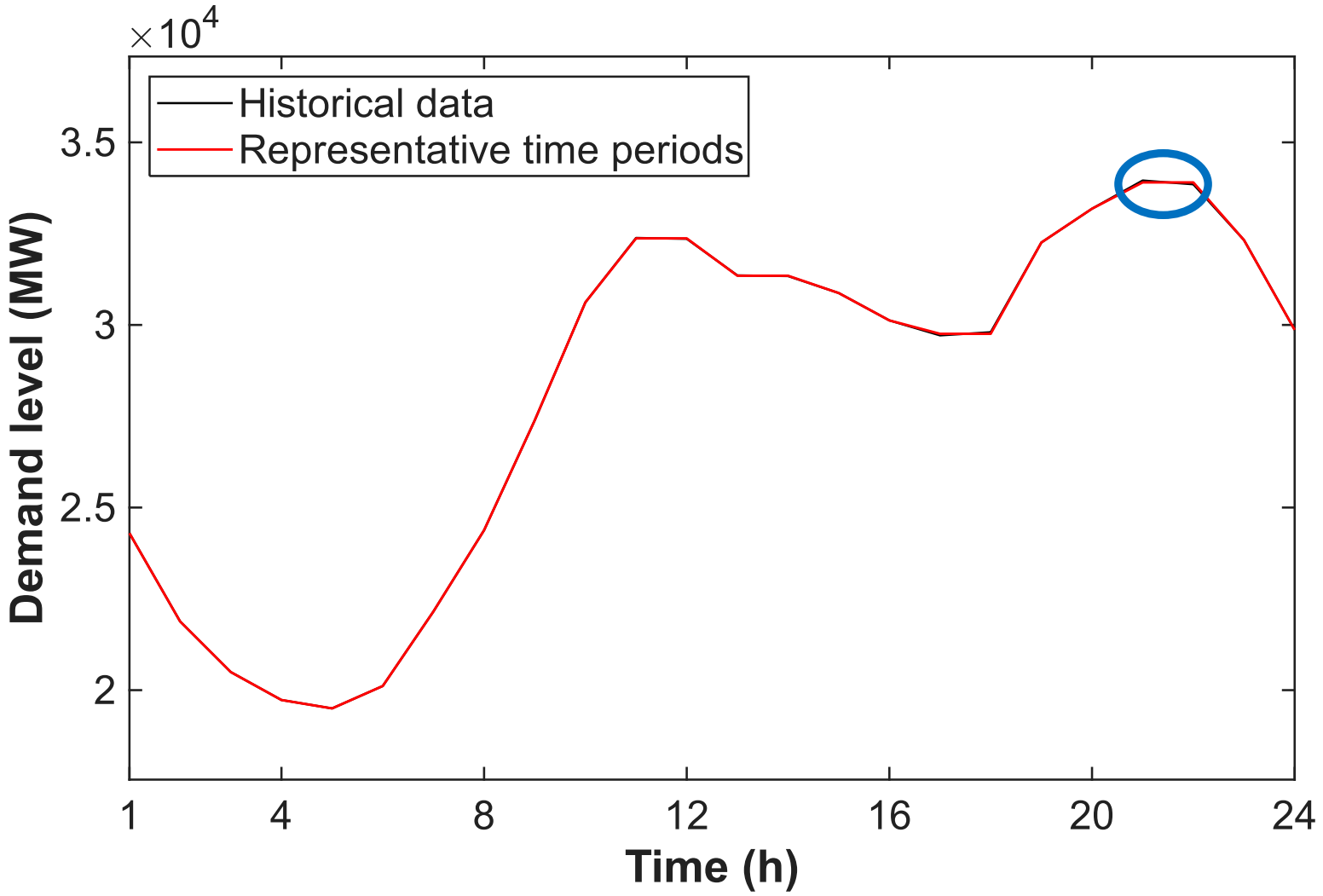
CTPC

Input data: 24
RTPs: 21



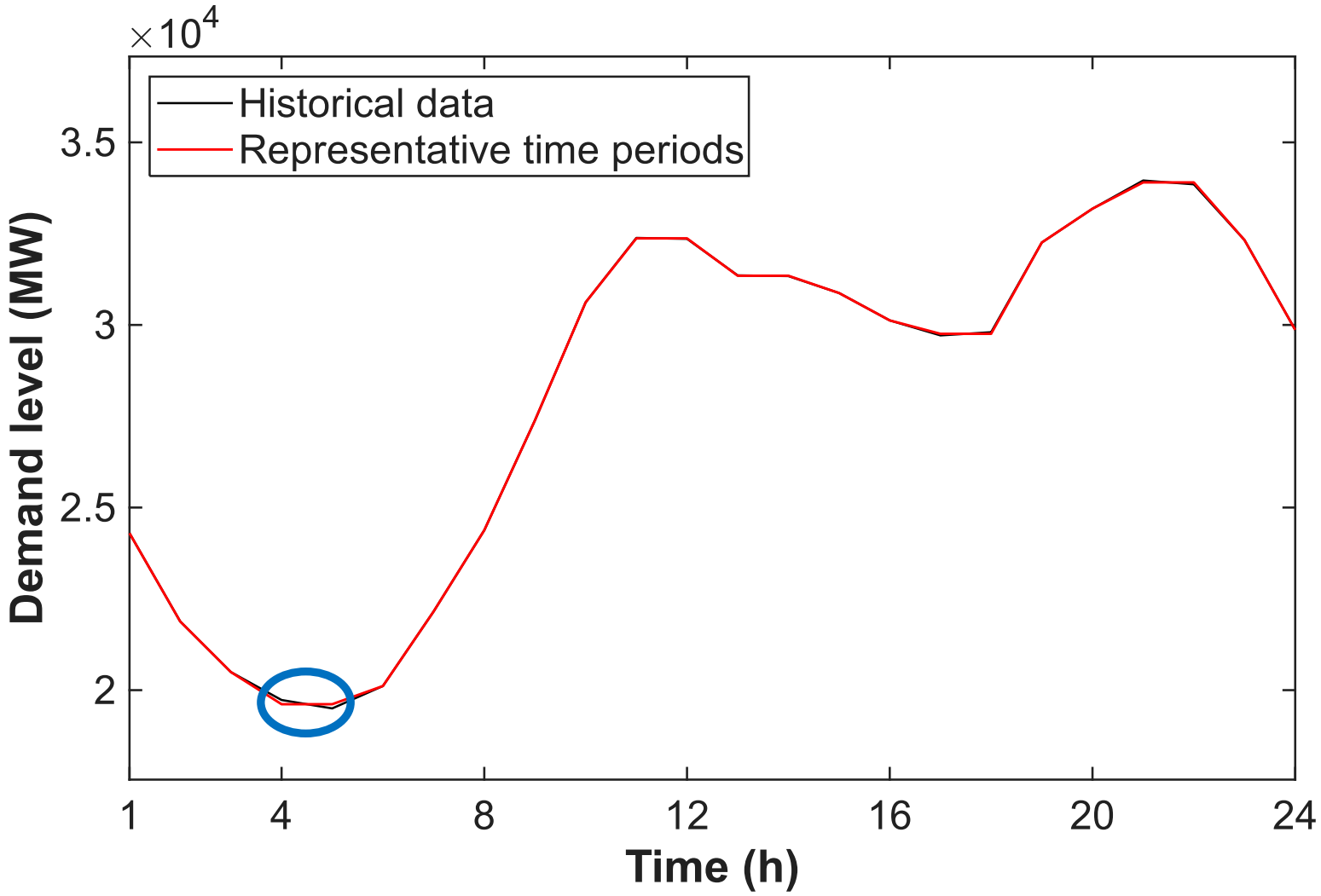
CTPC

Input data: 24
RTPs: 20



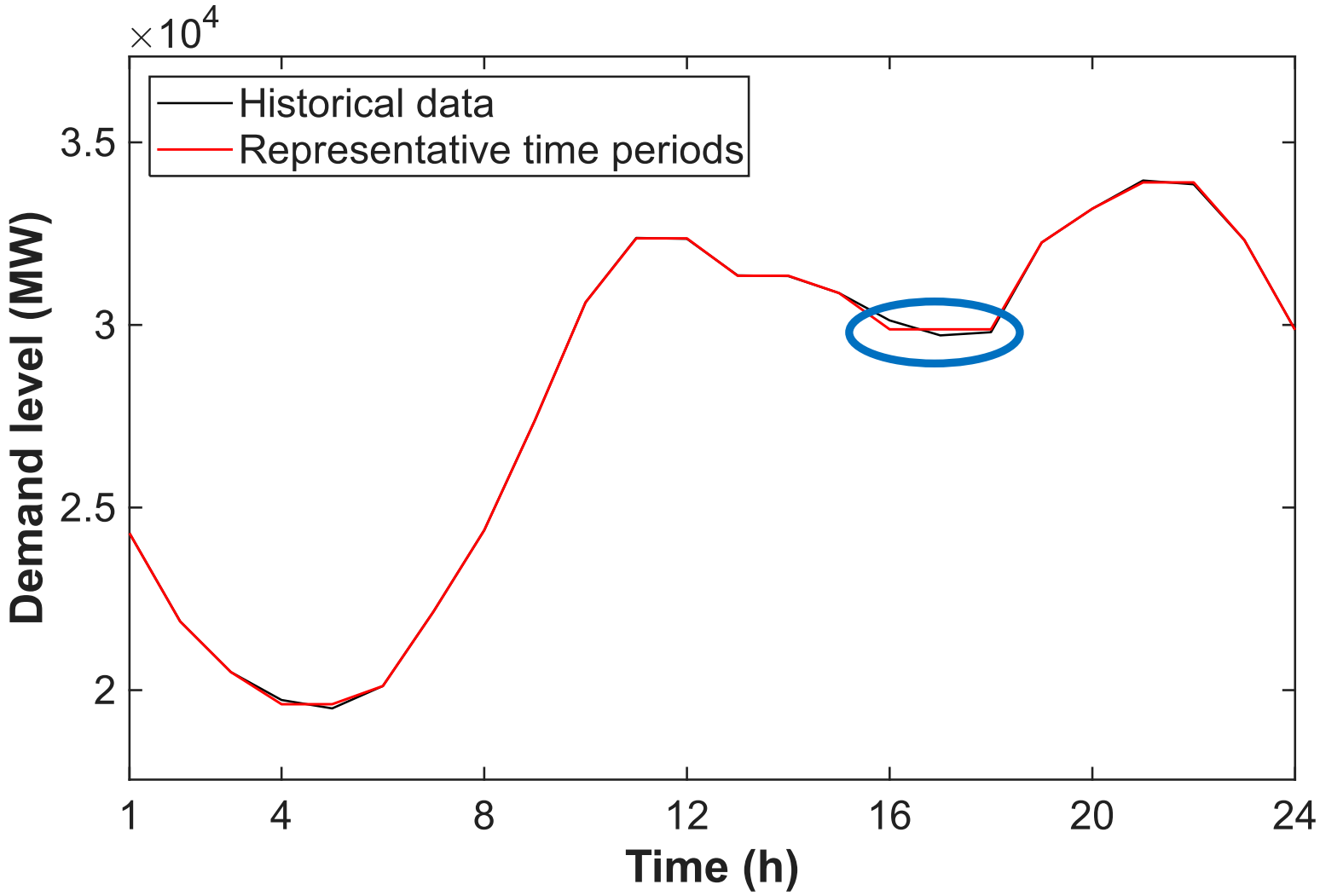
CTPC

Input data: 24
RTPs: 19



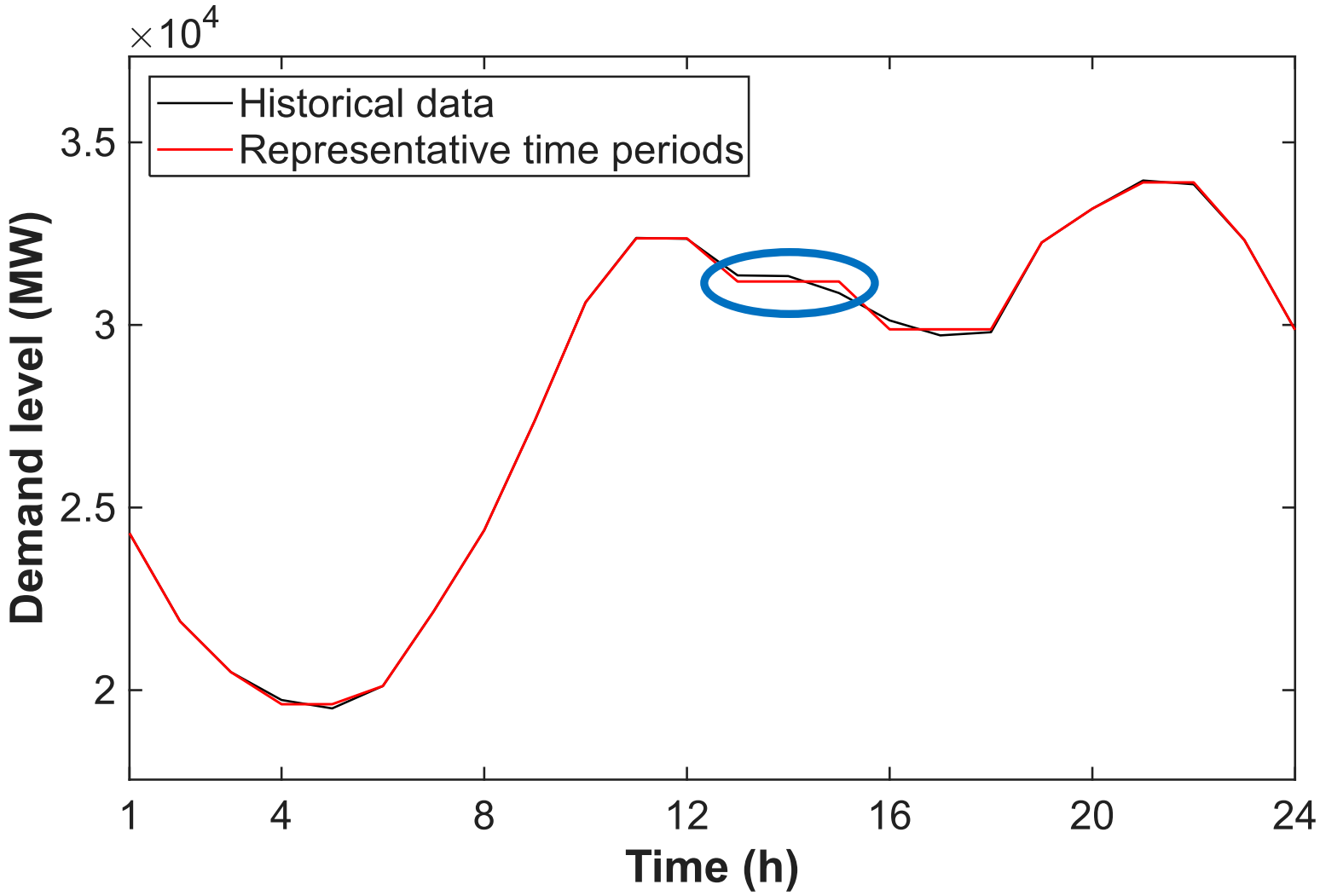
CTPC

Input data: 24
RTPs: 18



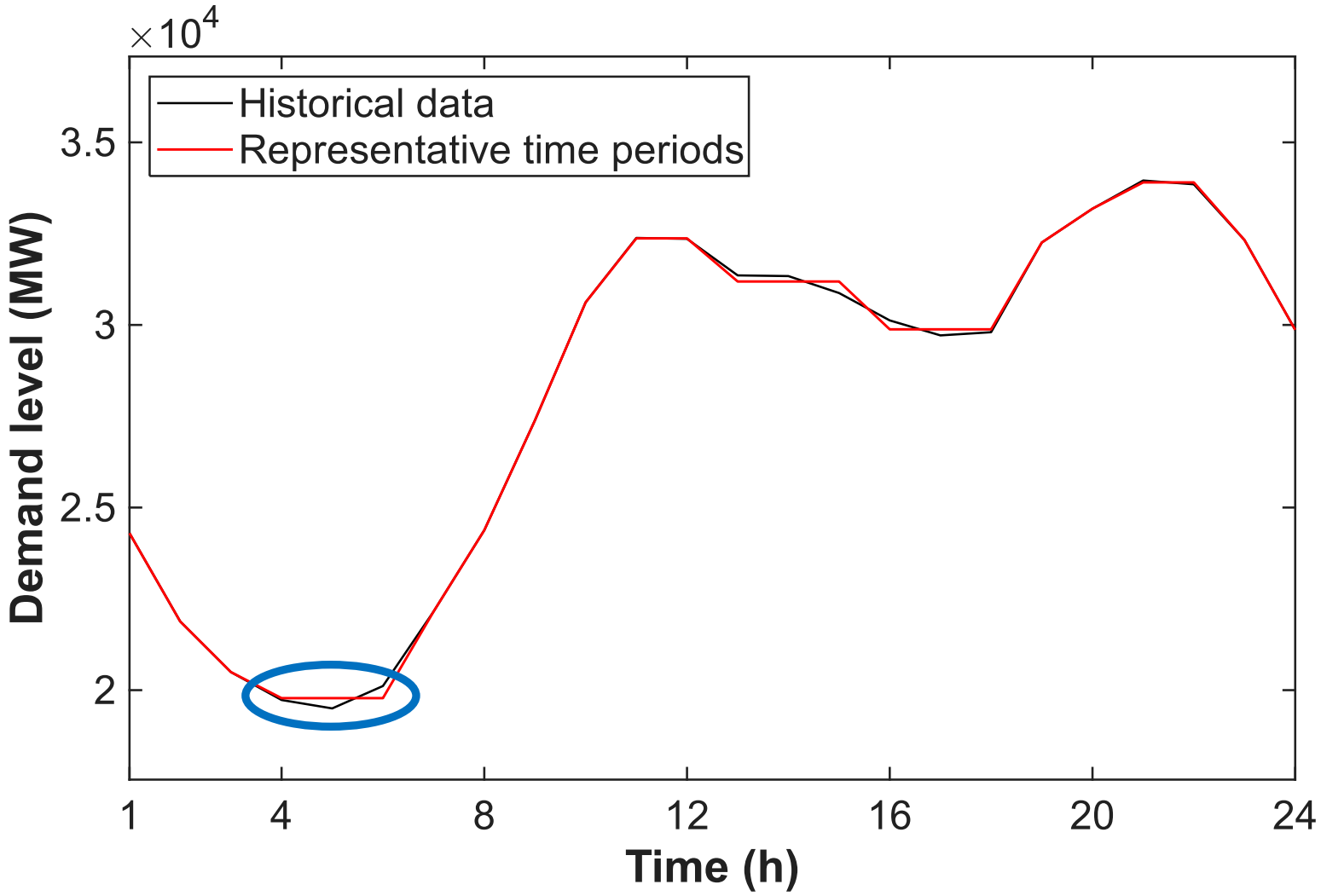
CTPC

Input data: 24
RTPs: 17



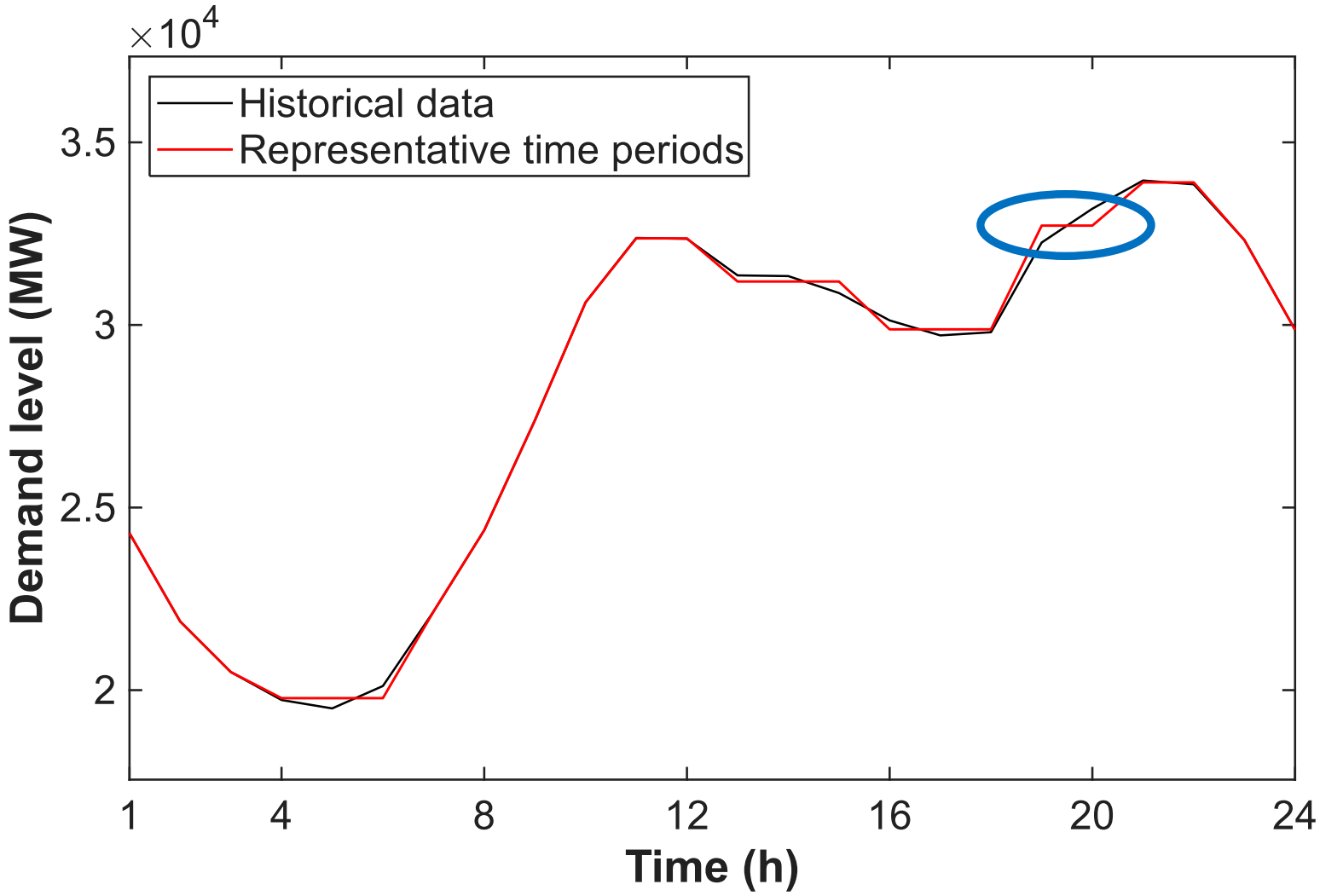
CTPC

Input data: 24
RTPs: 16



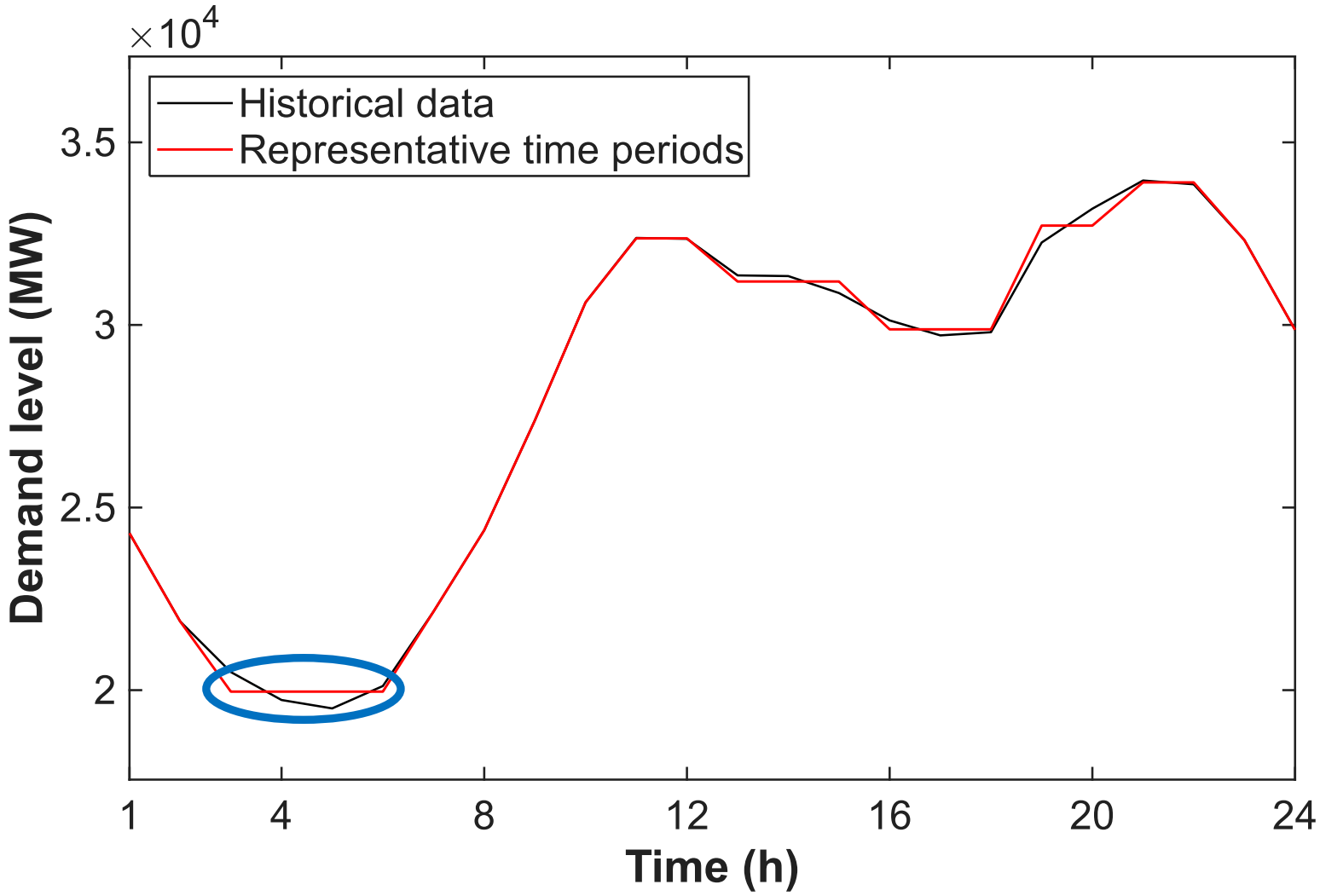
CTPC

Input data: 24
RTPs: 15



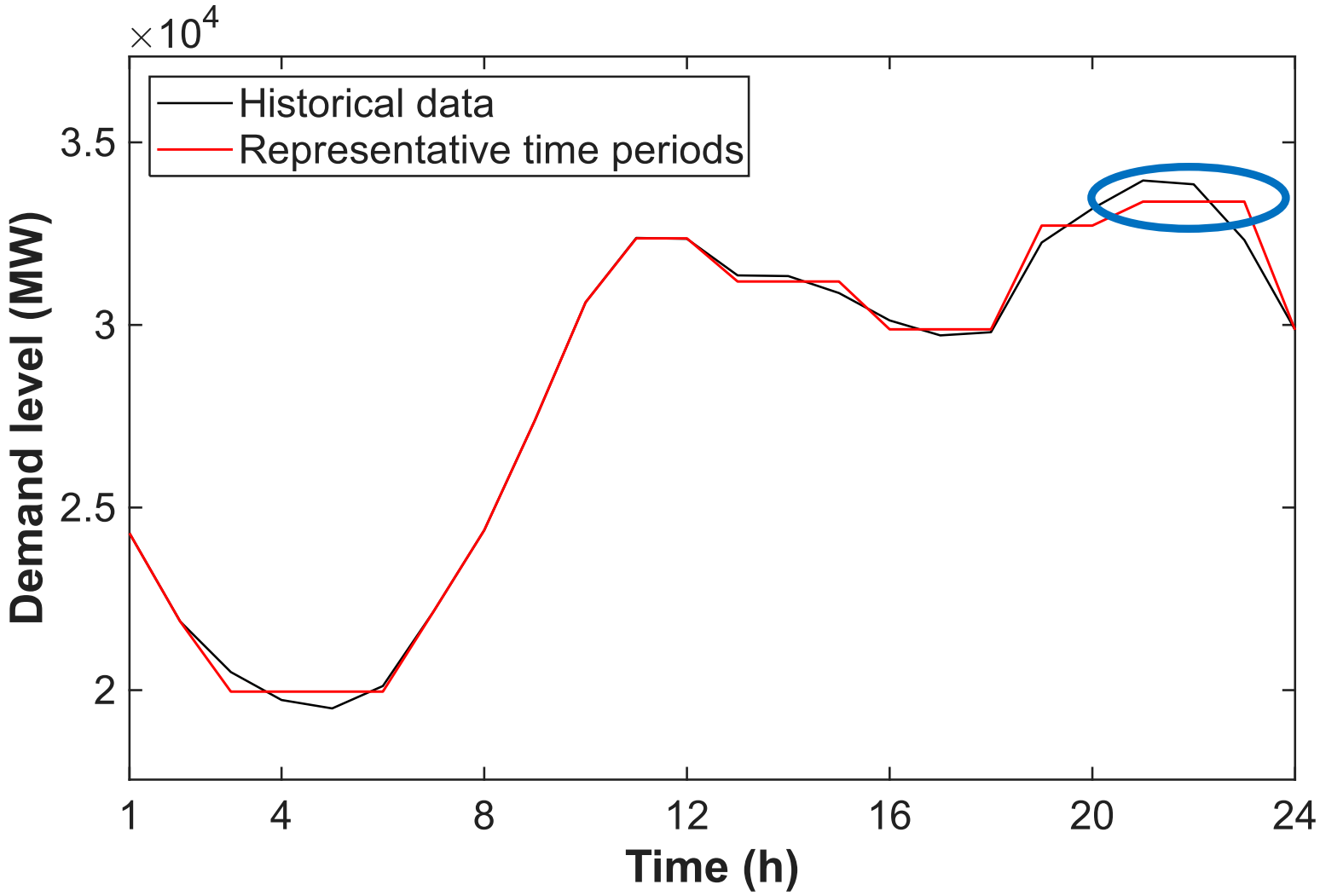
CTPC

Input data: 24
RTPs: 14



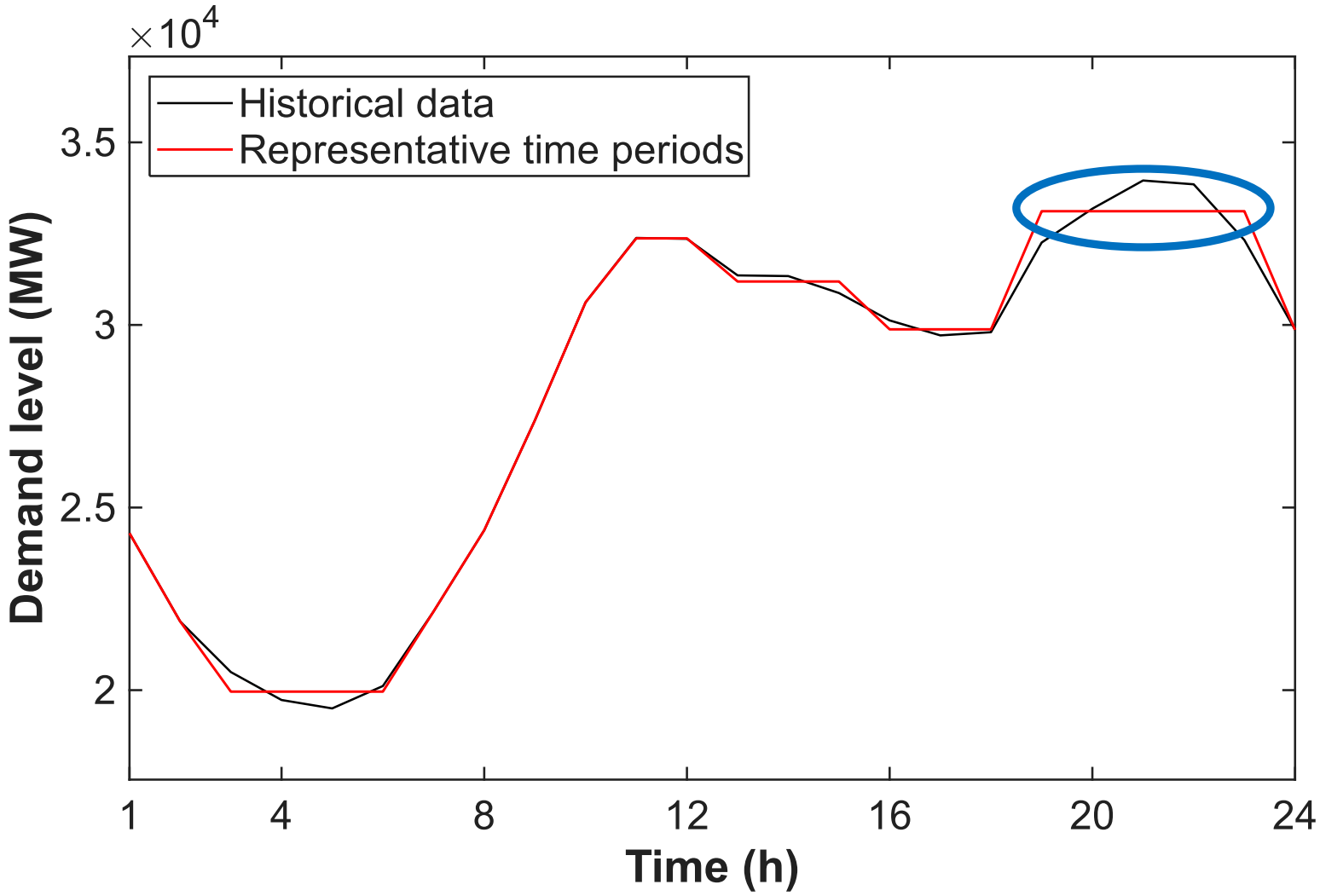
CTPC

Input data: 24
RTPs: 13



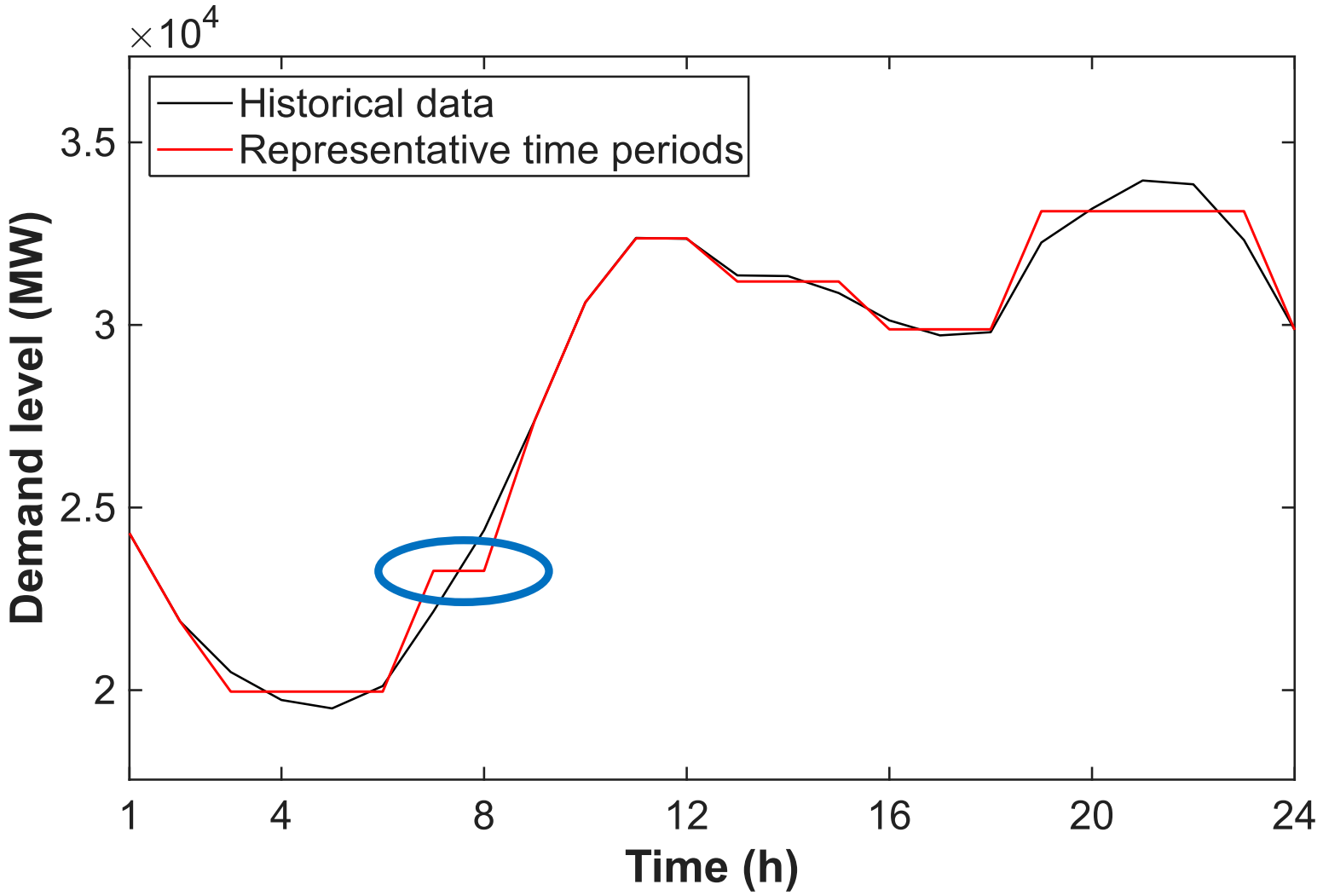
CTPC

Input data: 24
RTPs: 12



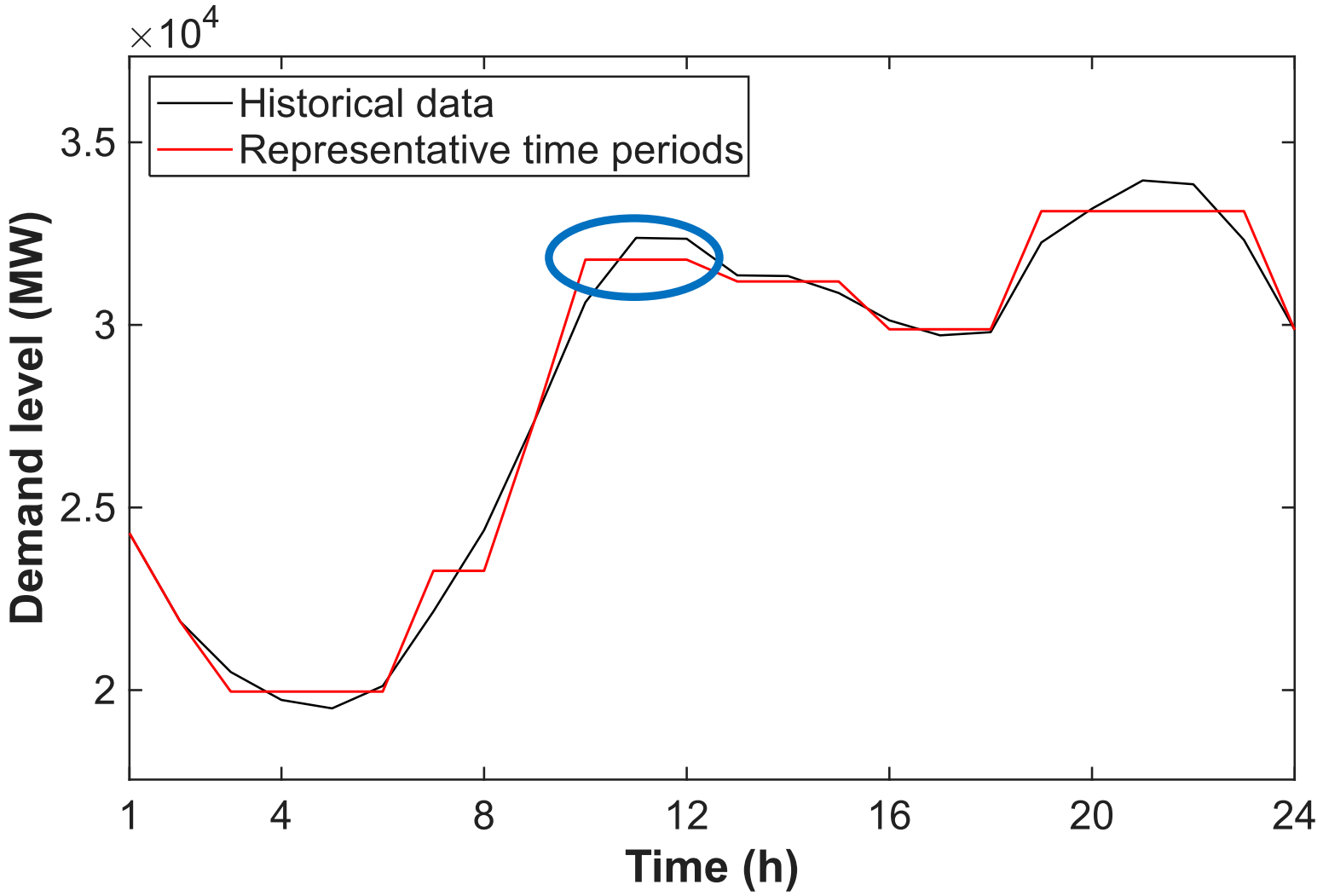
CTPC

Input data: 24
RTPs: 11



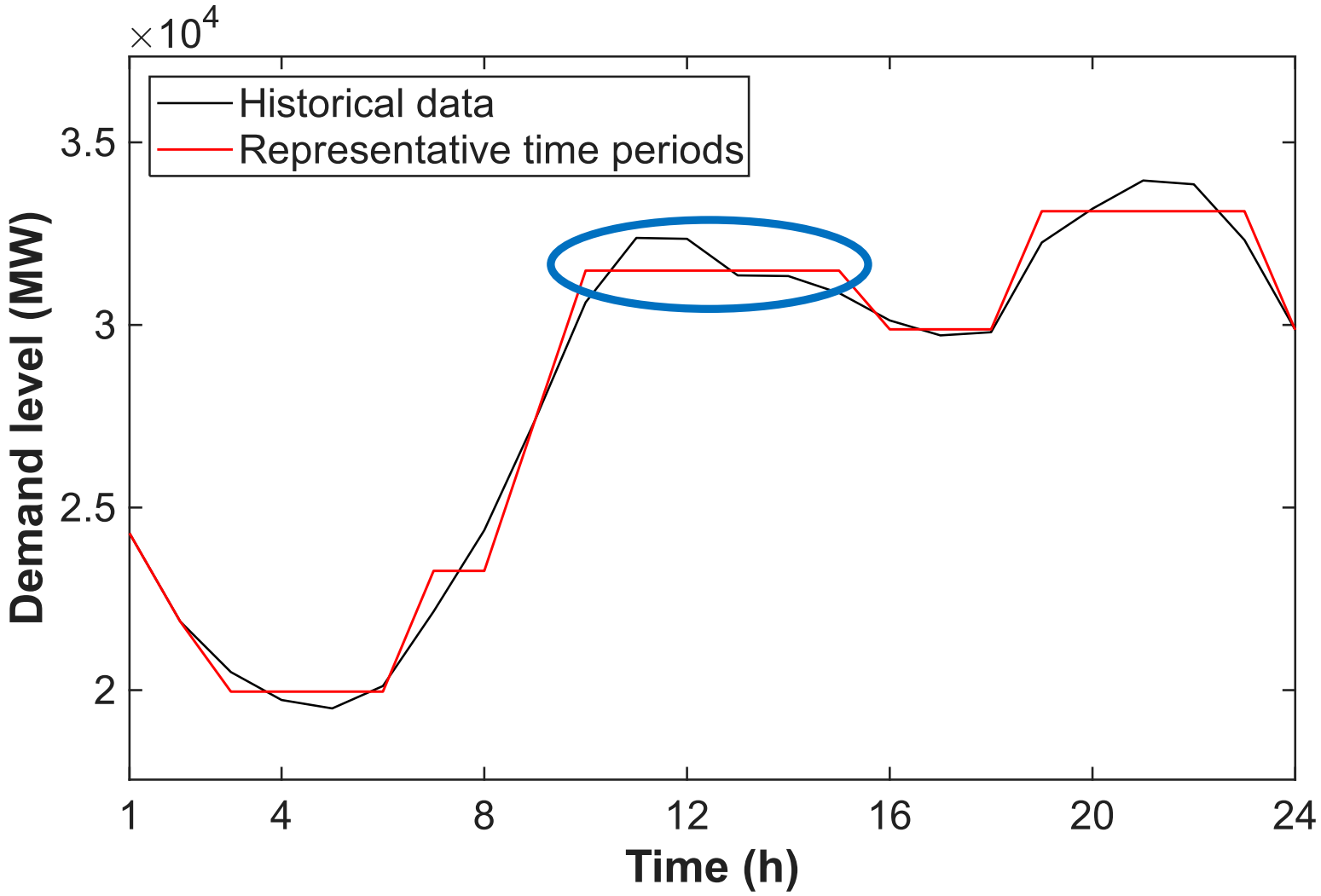
CTPC

Input data: 24
RTPs: 10



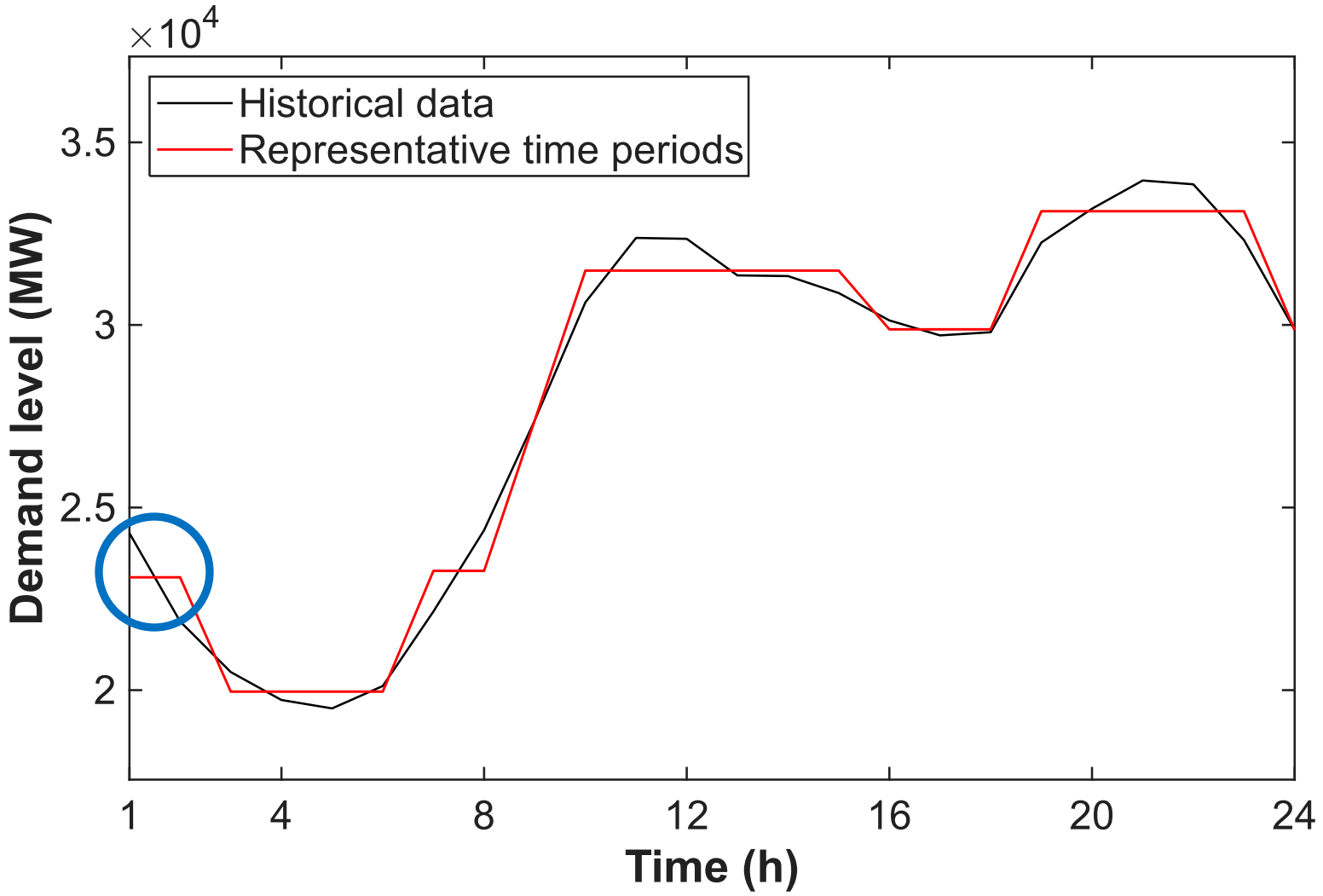
CTPC

Input data: 24
RTPs: 9



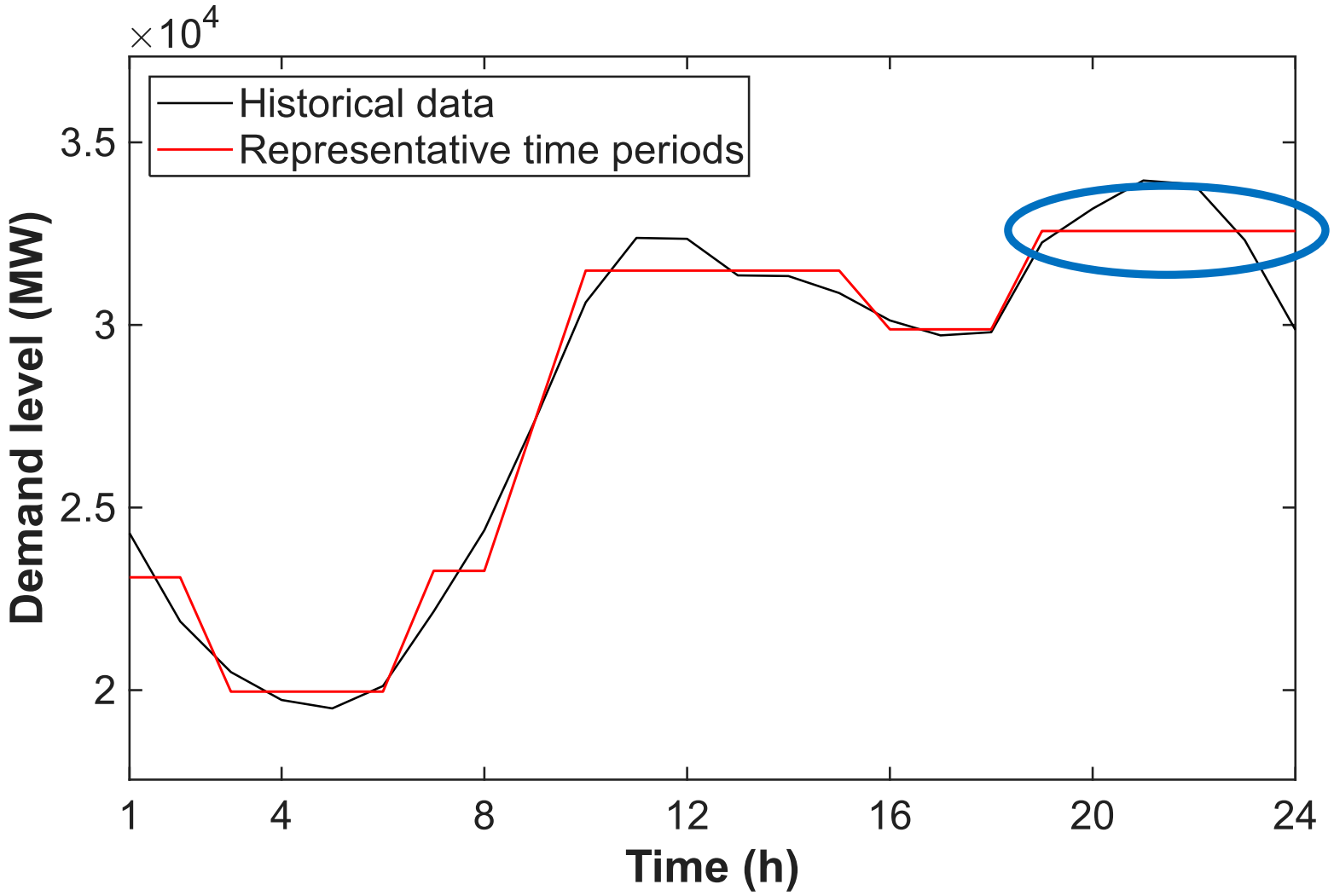
CTPC

Input data: 24
RTPs: 8



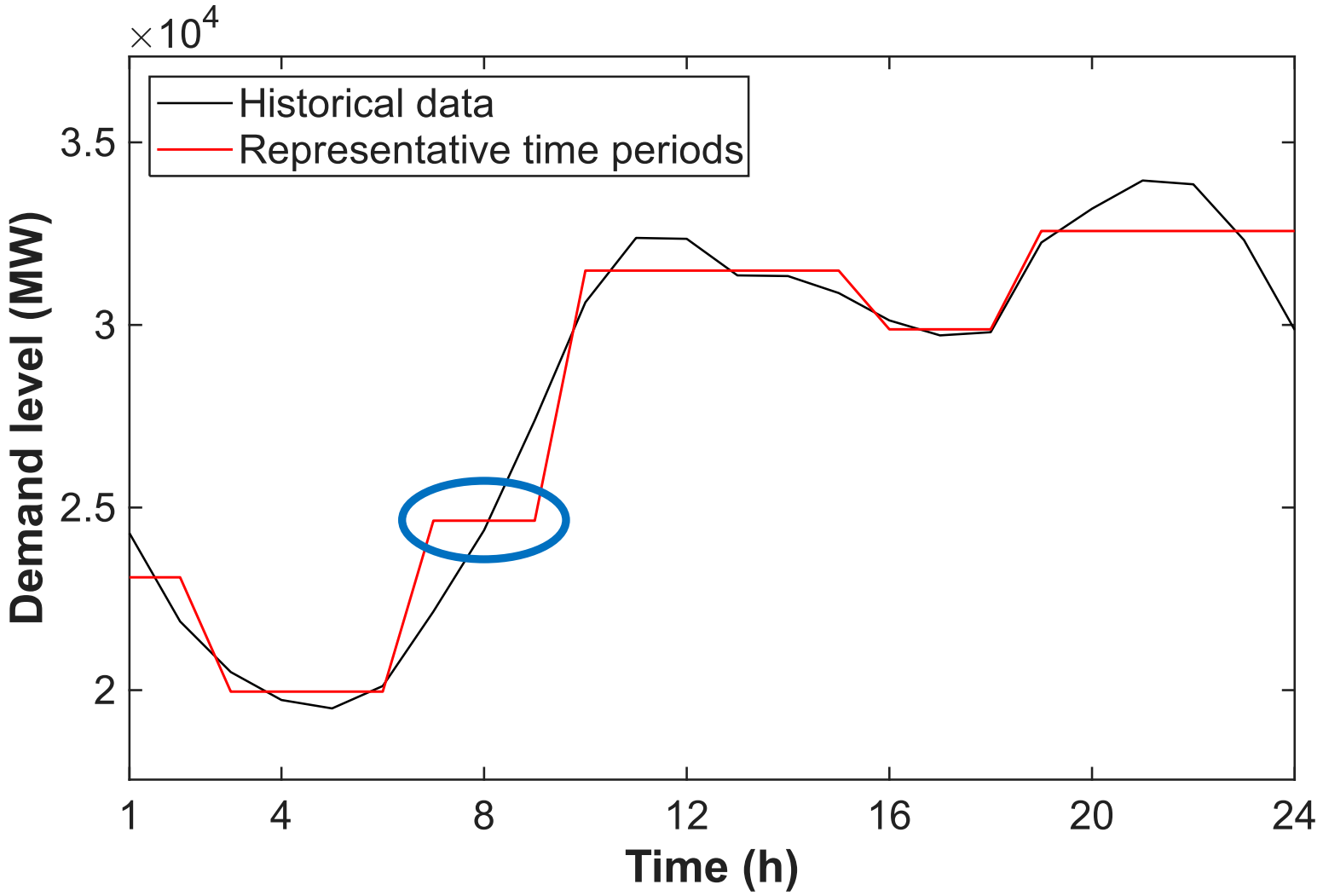
CTPC

Input data: 24
RTPs: 7



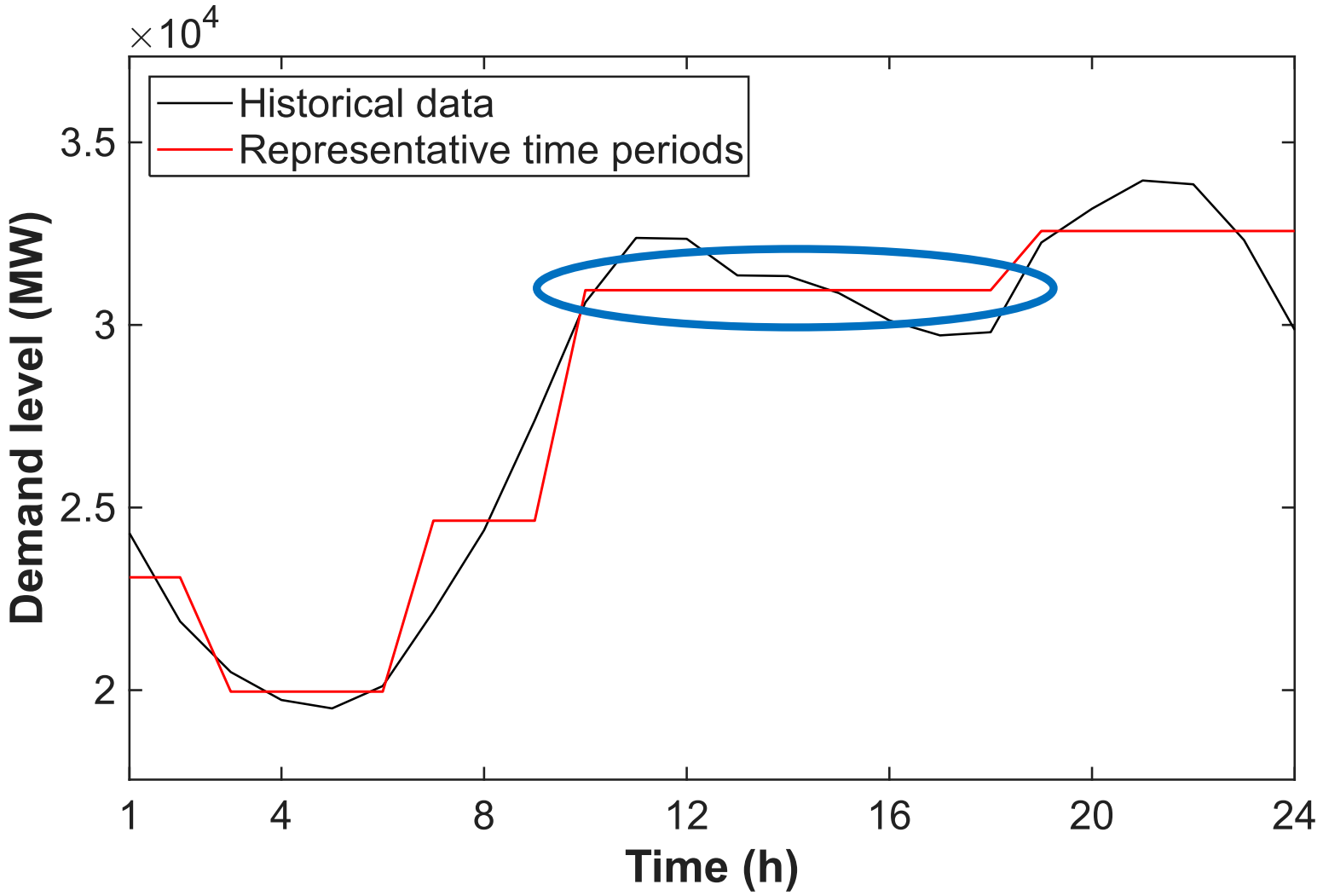
CTPC

Input data: 24
RTPs: 6



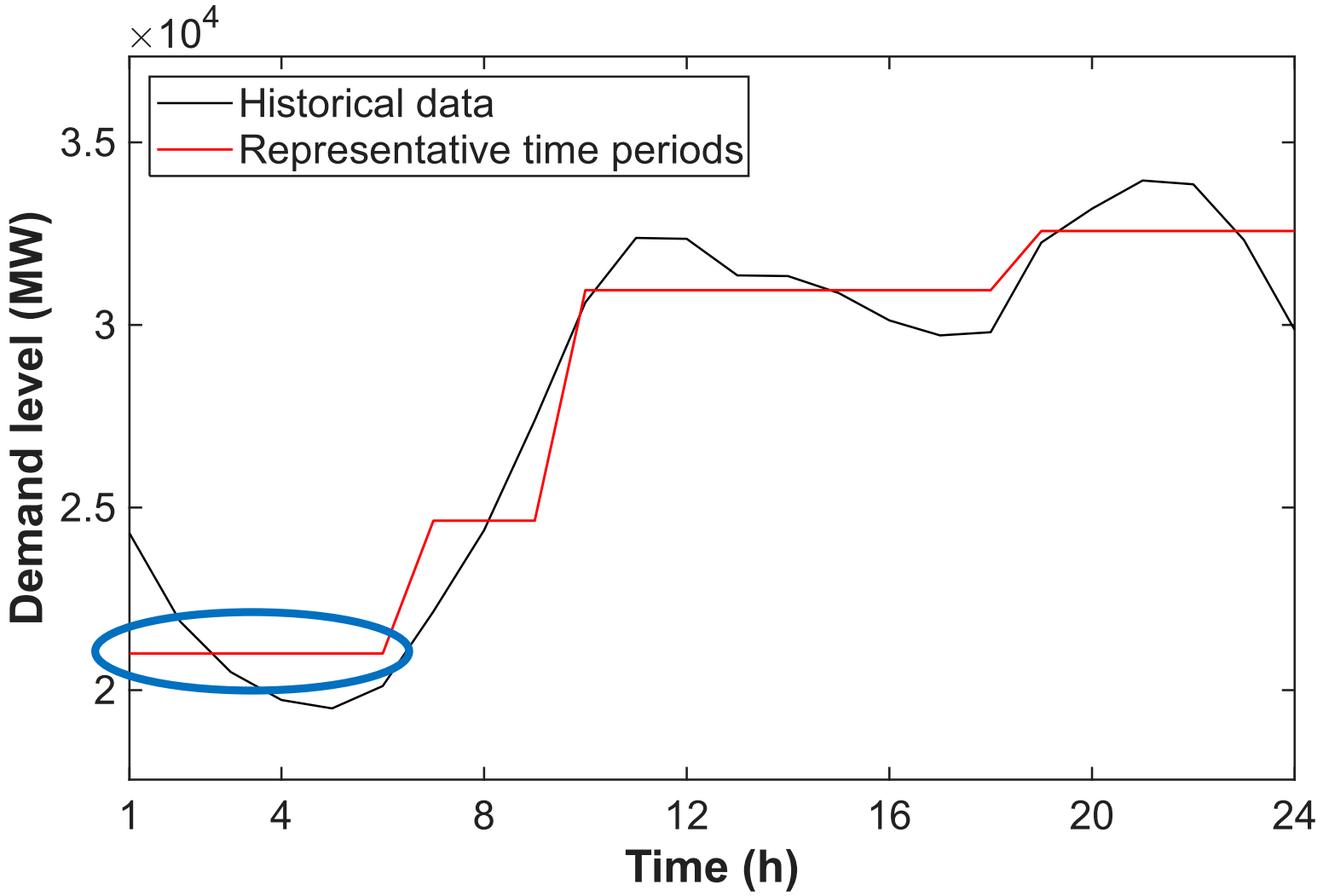
CTPC

Input data: 24
RTPs: 5



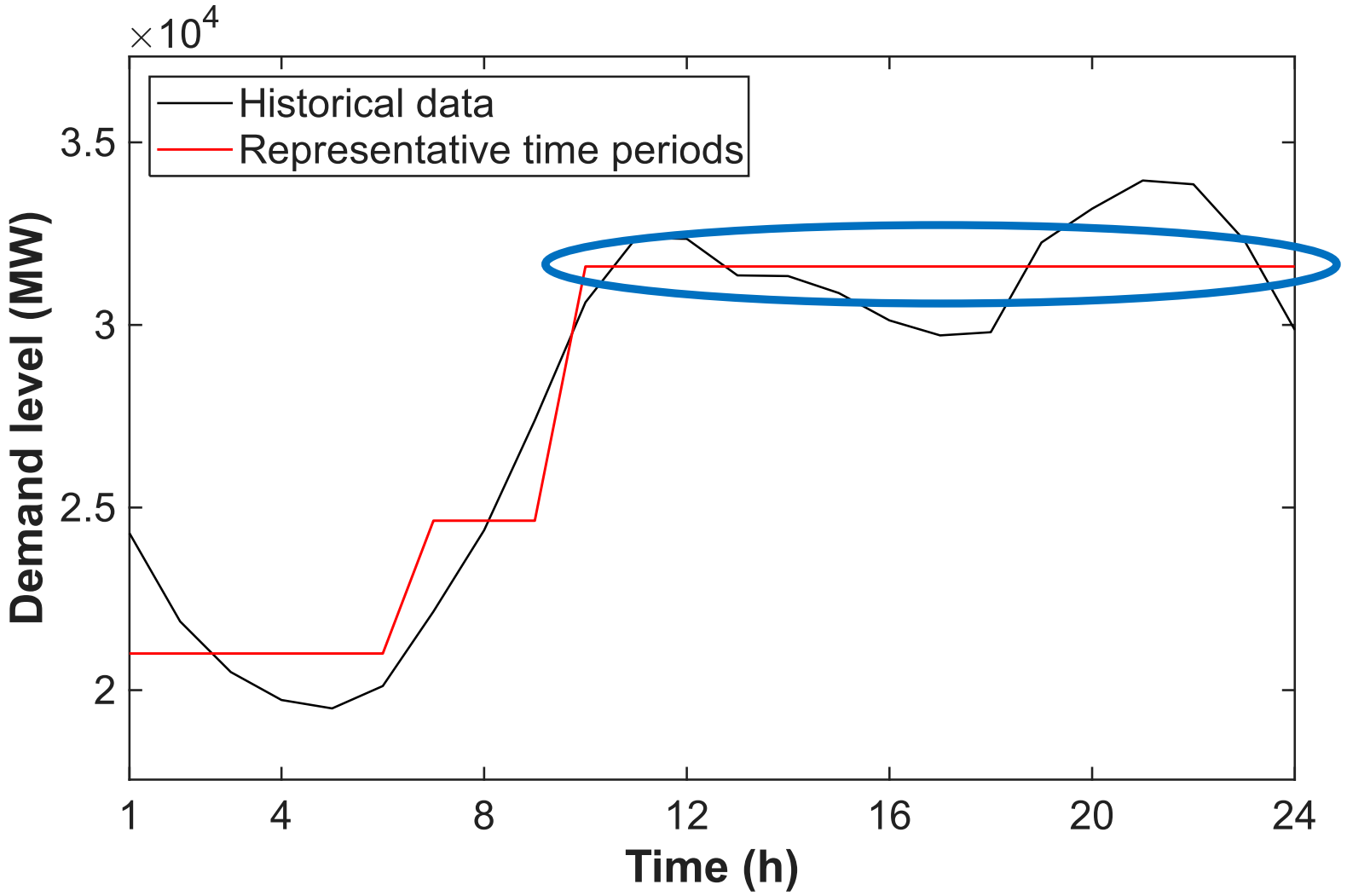
CTPC

Input data: 24
RTPs: 4



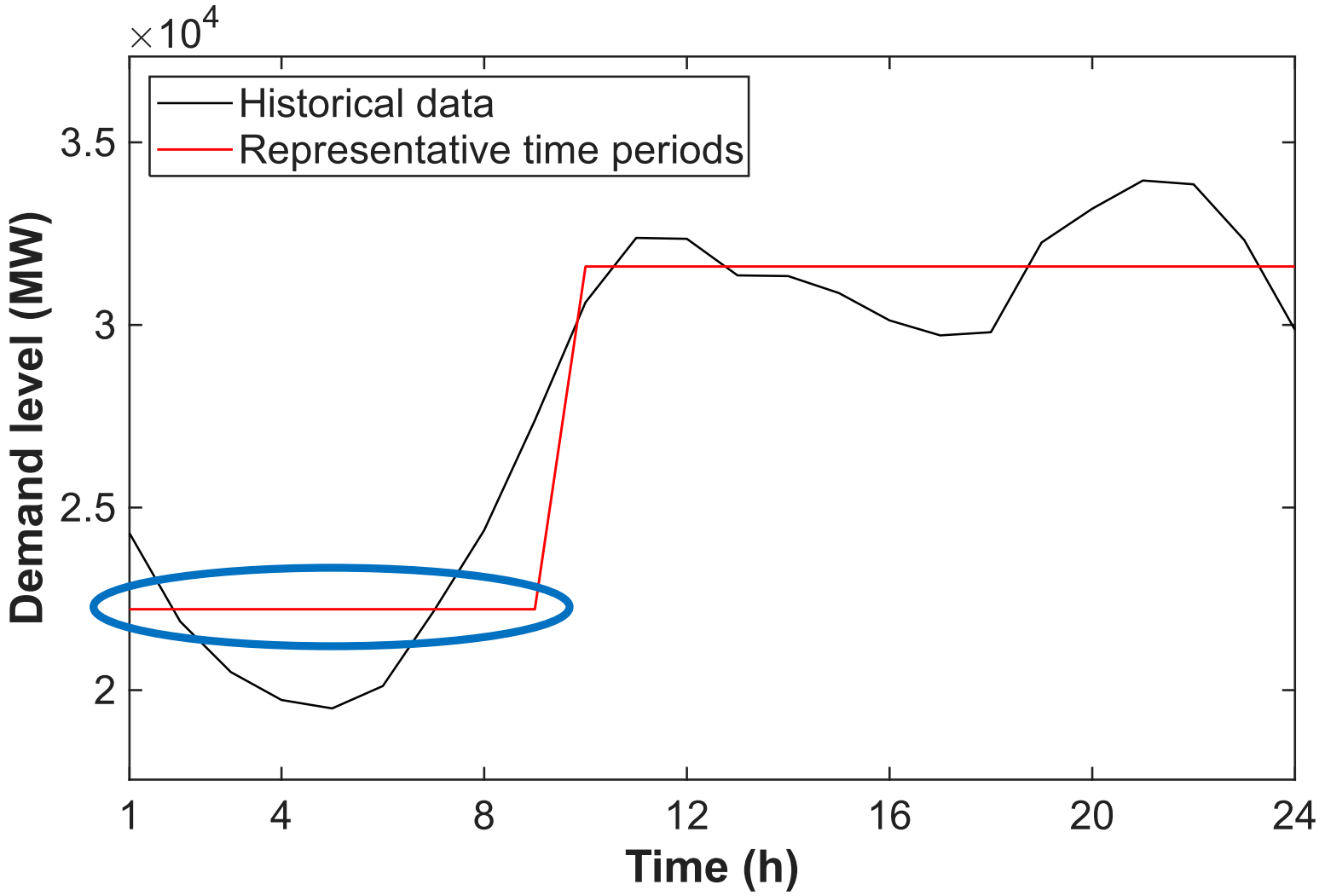
CTPC

Input data: 24
RTPs: 3



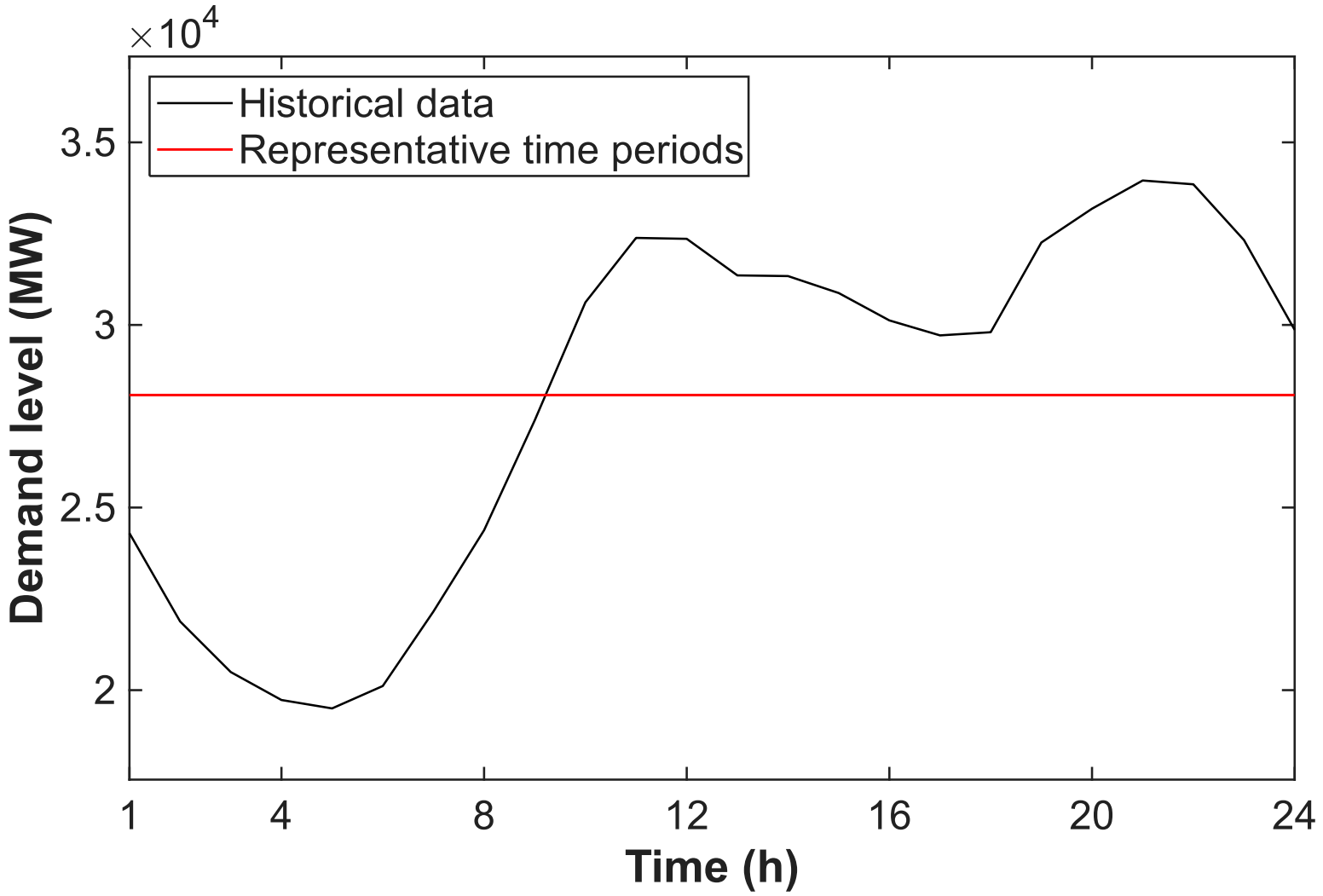
CTPC

Input data: 24
RTPs: 2



CTPC

Input data: 24
RTPs: 1



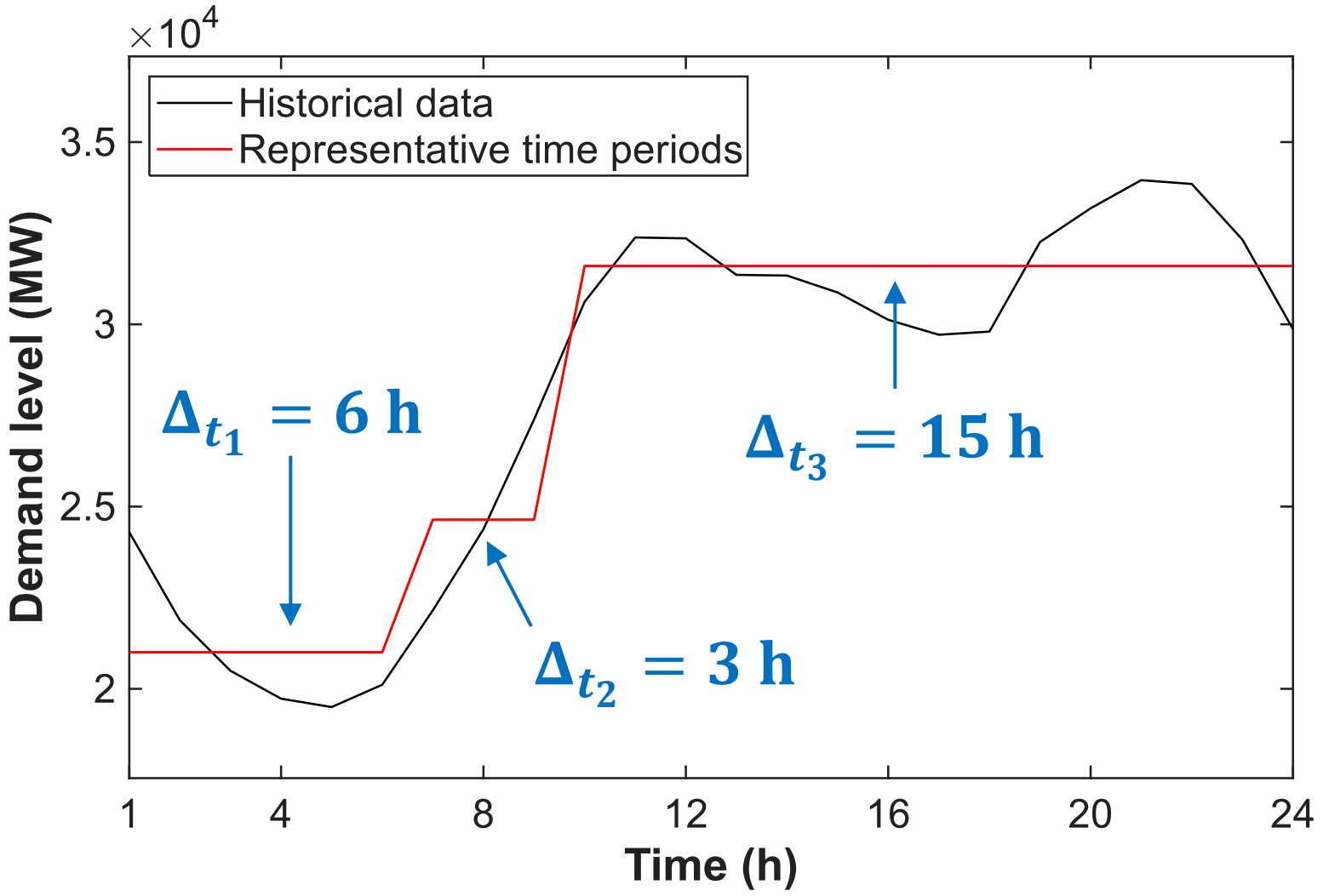
CTPC

The durations of the RTPs must be considered in the formulation of the problem: objective function, energy balance in storage assets, ramping limits, etc.

$$\Delta_t$$

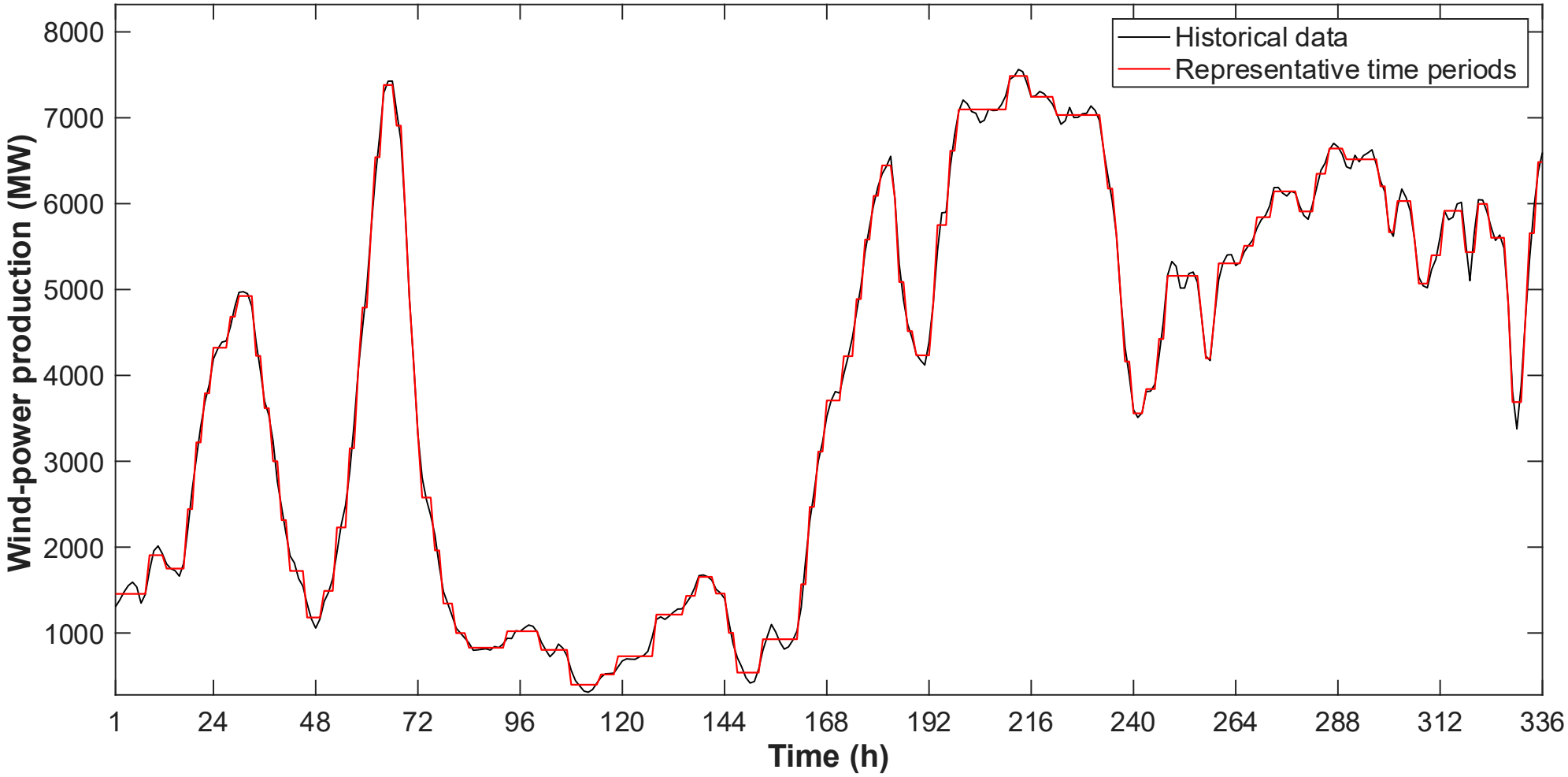
CTPC

Input data: 24
RTPs: 3



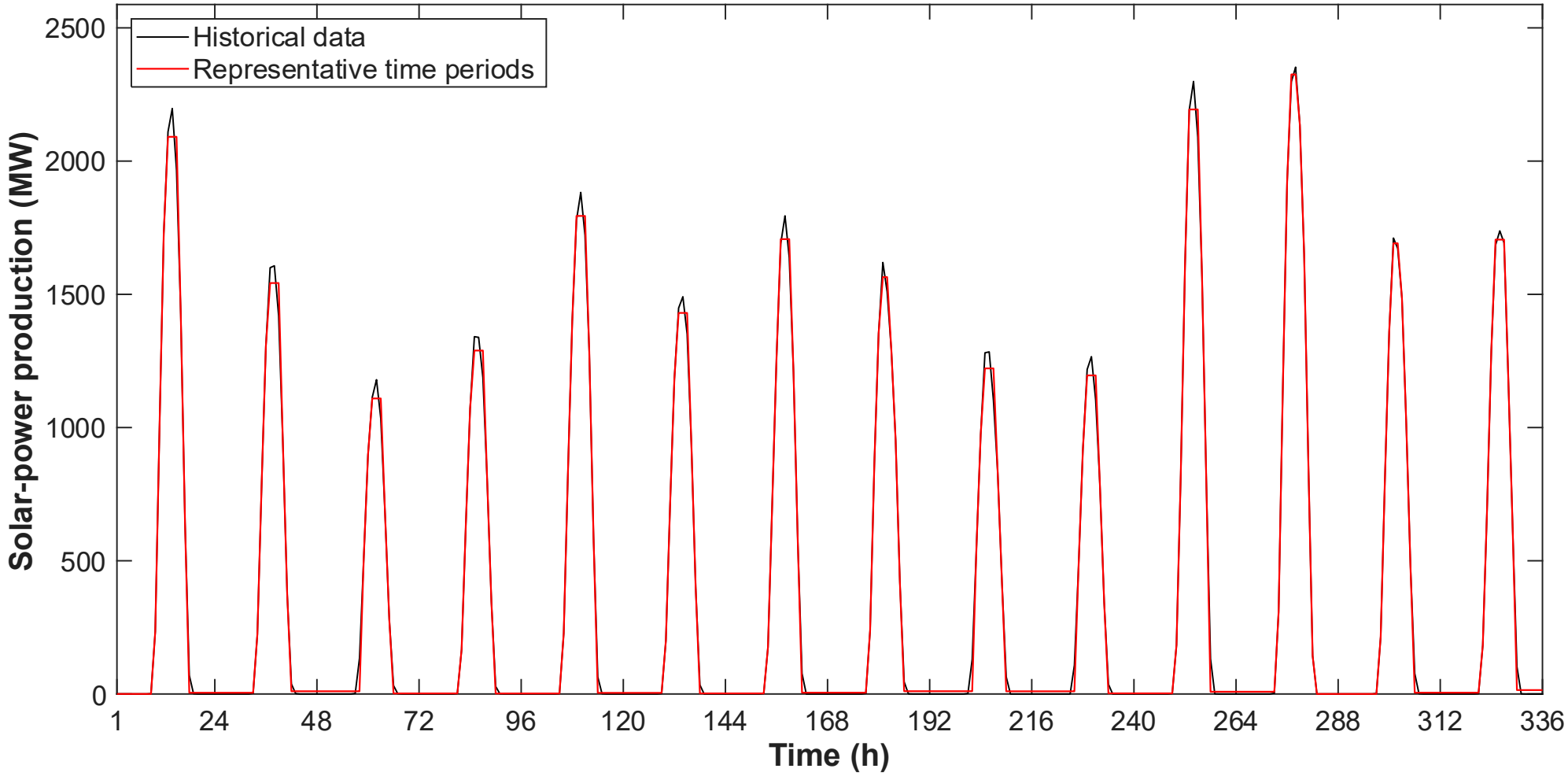
CTPC

Input data: 336
RTPs: 100



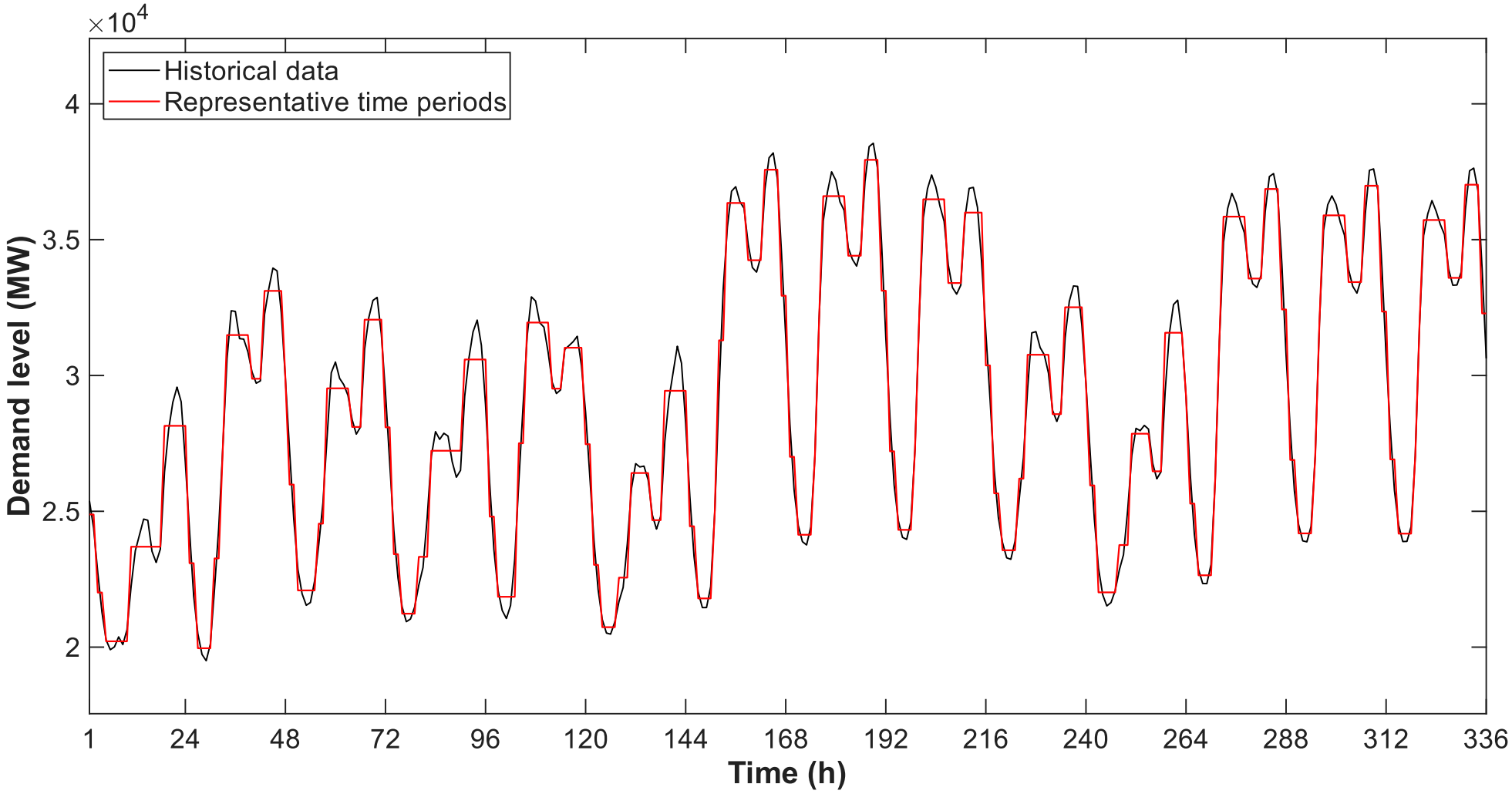
CTPC

Input data: 336
RTPs: 100



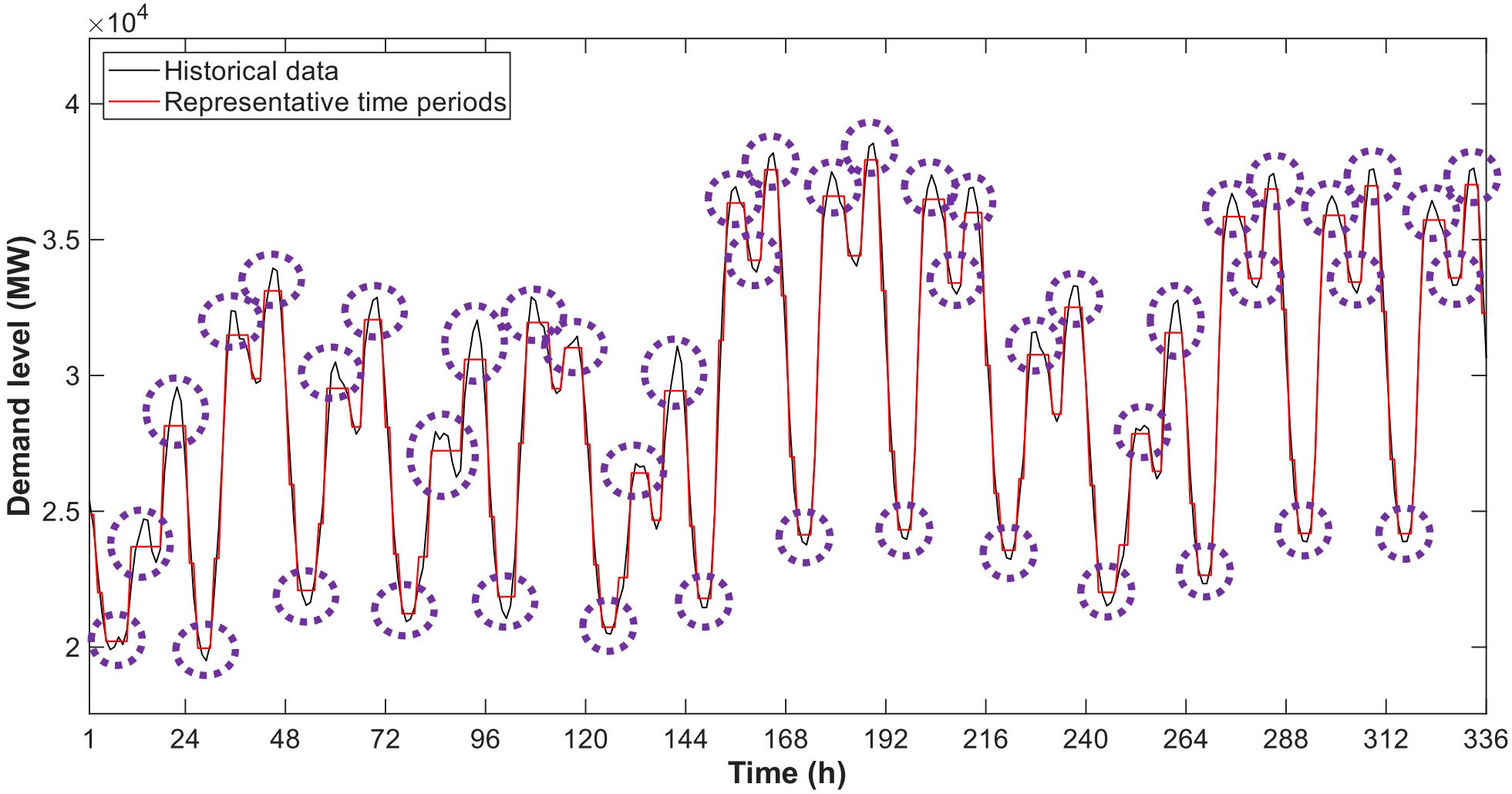
CTPC

Input data: 336
RTPs: 100



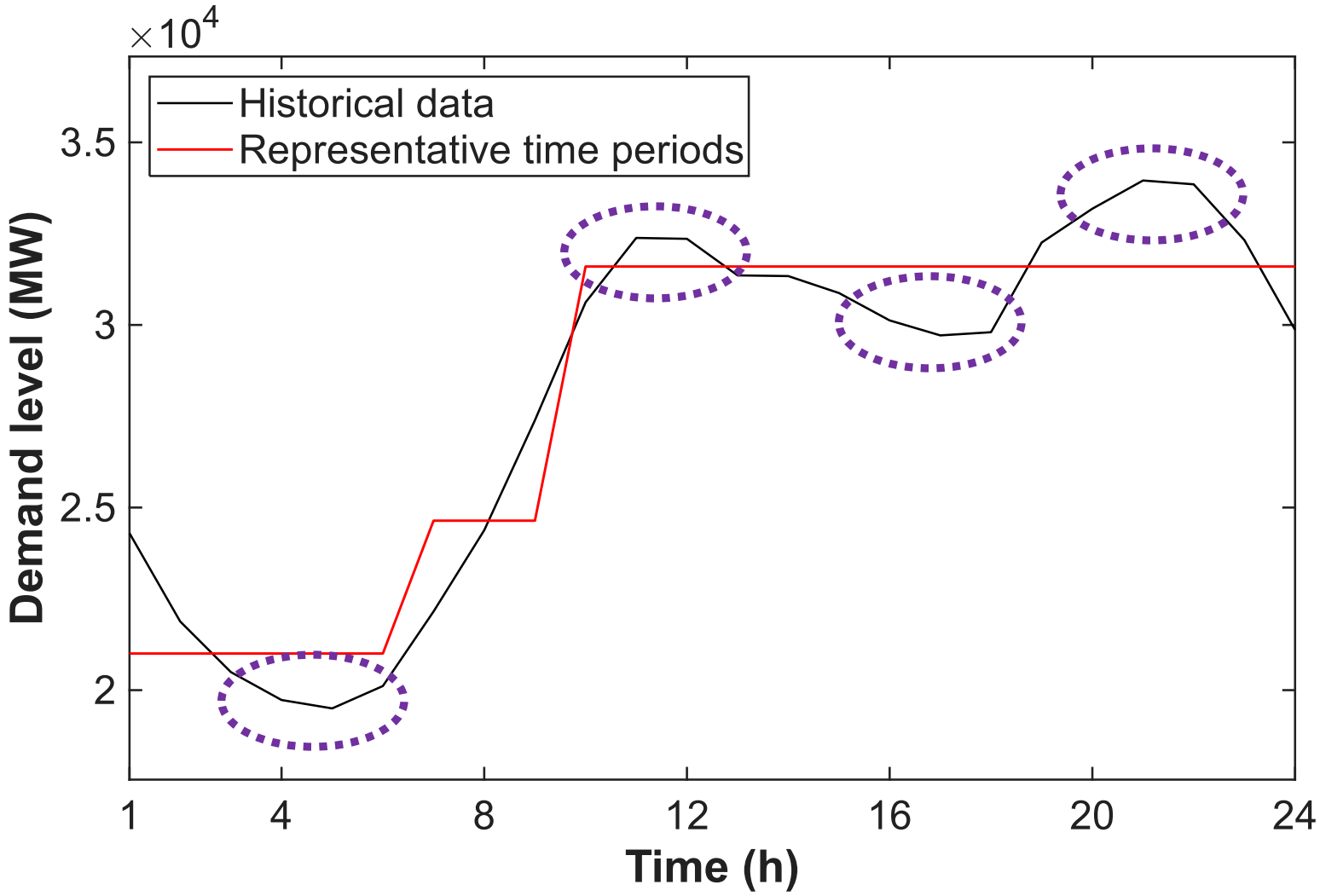
CTPC

Input data: 336
RTPs: 100



CTPC

Input data: 24
RTPs: 3



5. Extreme conditions

Extreme conditions

- Peak demand levels and low renewable production availability.

- Previous approaches:
  - Maximum, average, and minimum values.
  - Extreme values with unitary weights + clustering the remaining input data.
  - Replace the centroid/medoid of the cluster with the maximum/minimum value without changing the weight.

– Francisco Umbral, Spanish journalist and writer

“He venido a hablar de mi libro”.

“I came here to talk about my book”.

– Francisco Umbral, Spanish journalist and writer

6. Advanced aggregation techniques

Advanced aggregation techniques

Works resulting from my Bachelor's Final Project and PhD Thesis:

- Modified K-means Method (MKM).
- Modified Maximum Dissimilarity Algorithm (MMDA).
- Priority Chronological Time-Period Clustering (PCTPC).
- Combination of previous techniques (MMDA + PCTPC).

Advanced aggregation techniques

Works resulting from my Bachelor's Final Project and PhD Thesis:

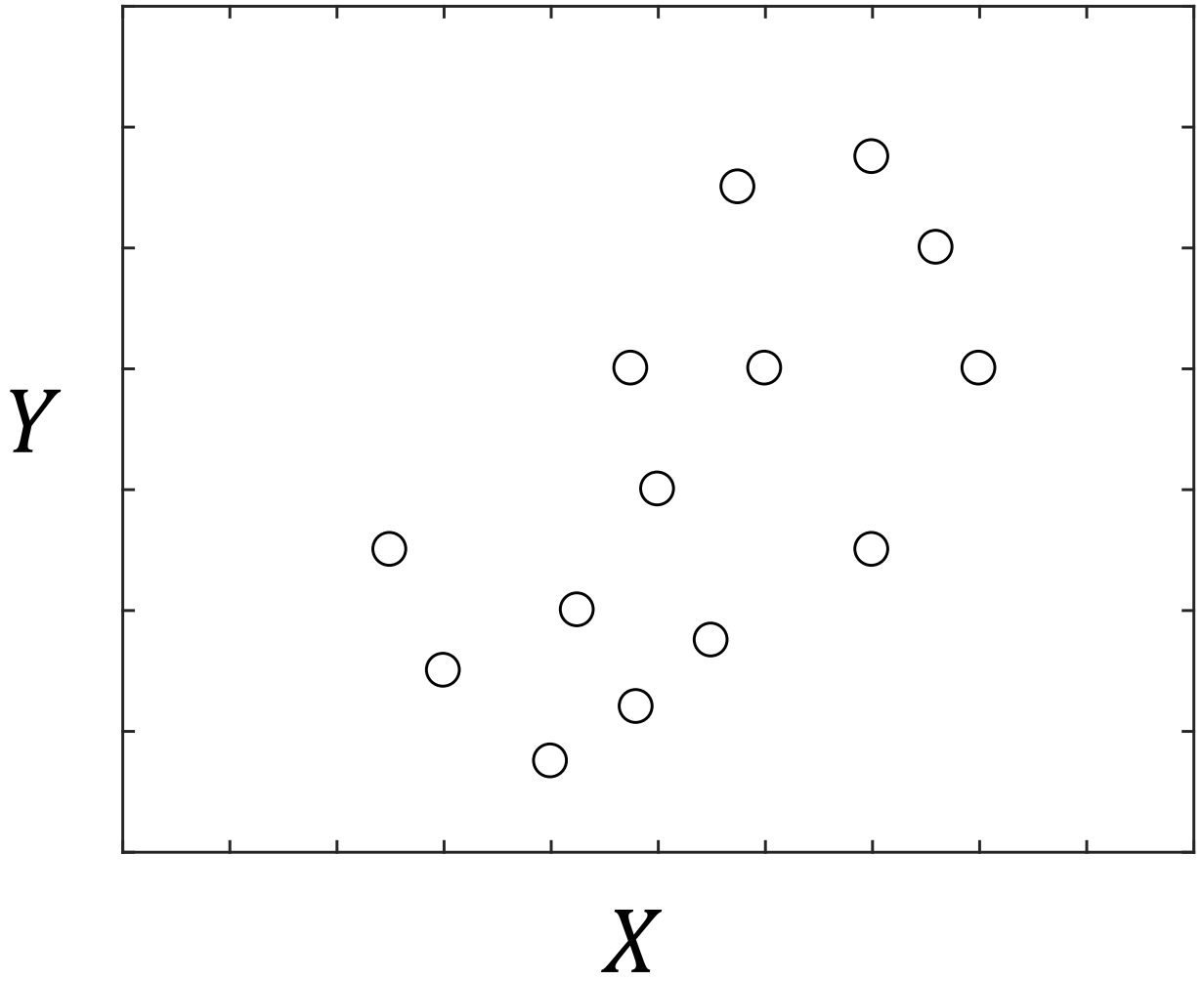
- Modified K-means Method (MKM).
- Modified Maximum Dissimilarity Algorithm (MMDA).
- Priority Chronological Time-Period Clustering (PCTPC).
- Combination of previous techniques (MMDA + PCTPC).

Modified K-means Method (MKM)

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Motivation

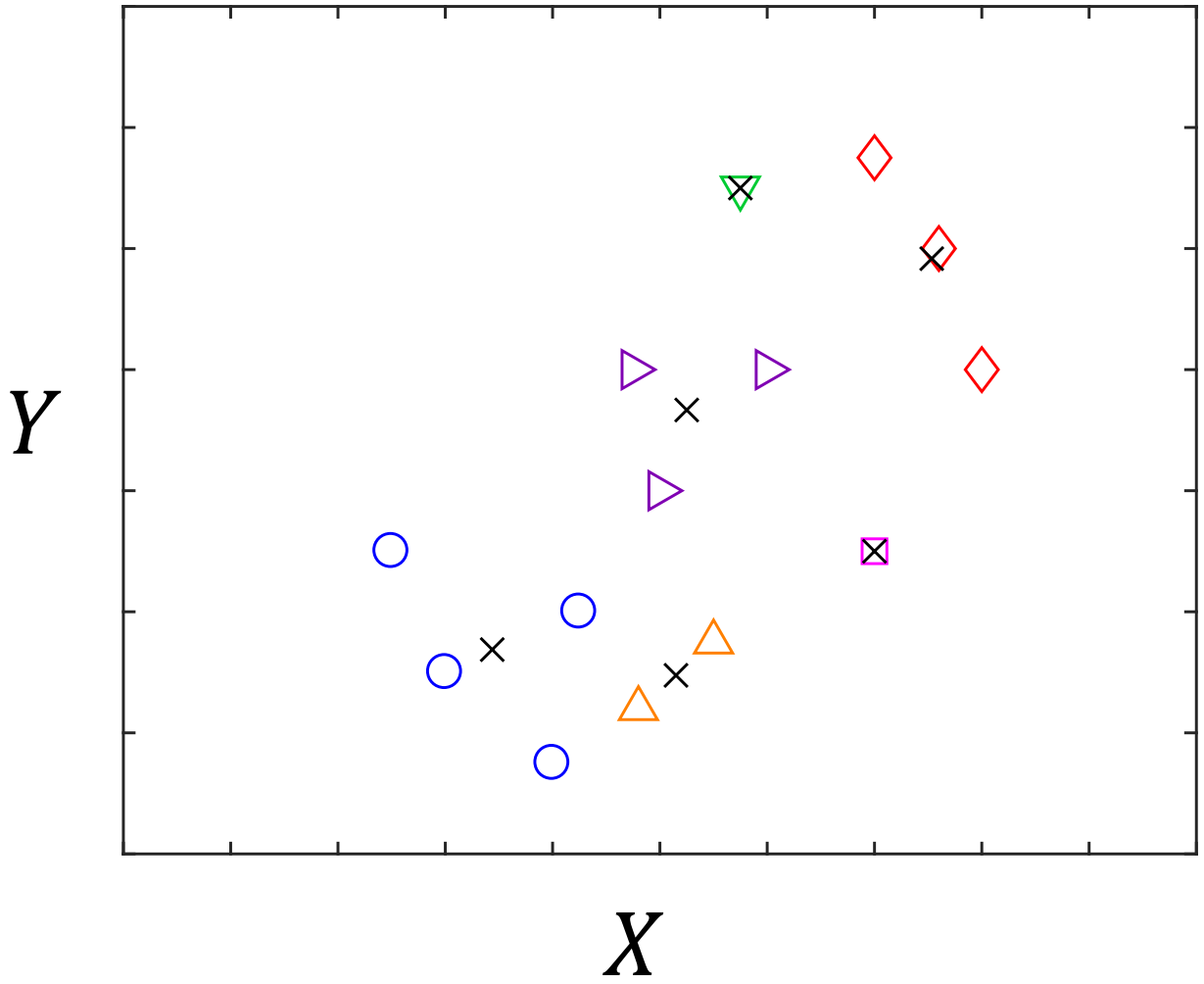
Traditional K-means Method (TKM)



Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Motivation

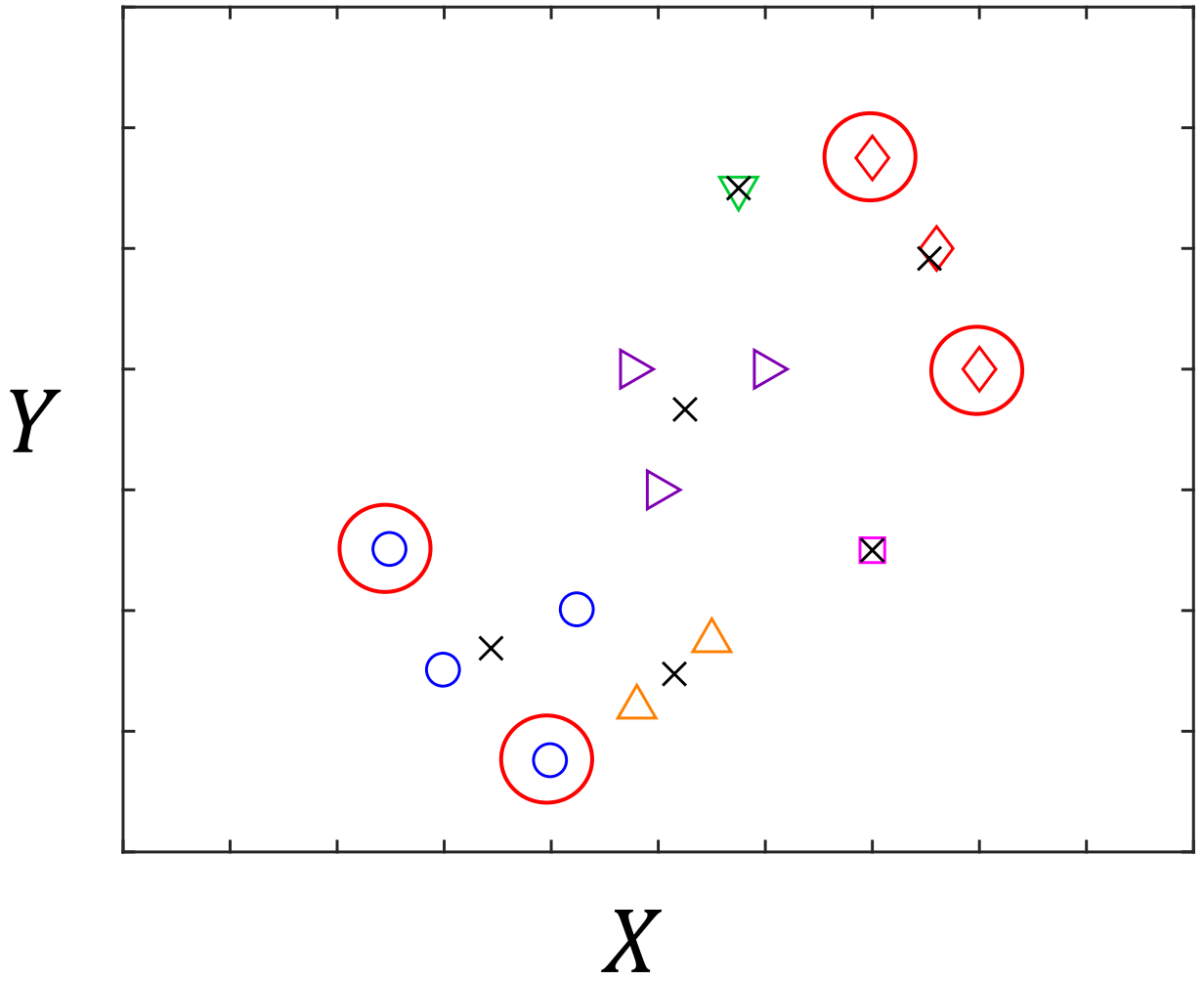
Traditional K-means Method (TKM)



Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

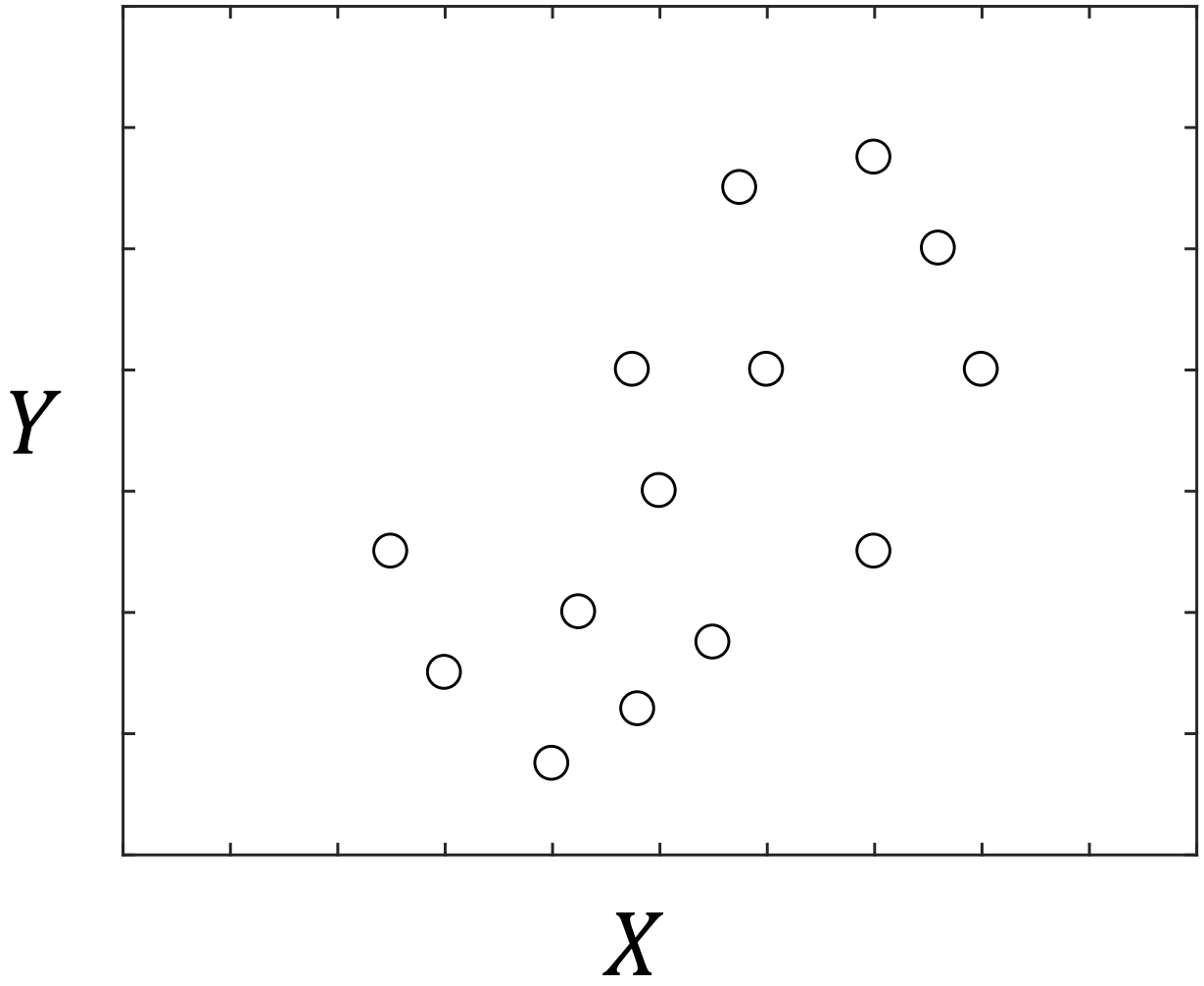
MKM – Motivation

Traditional K-means Method (TKM)



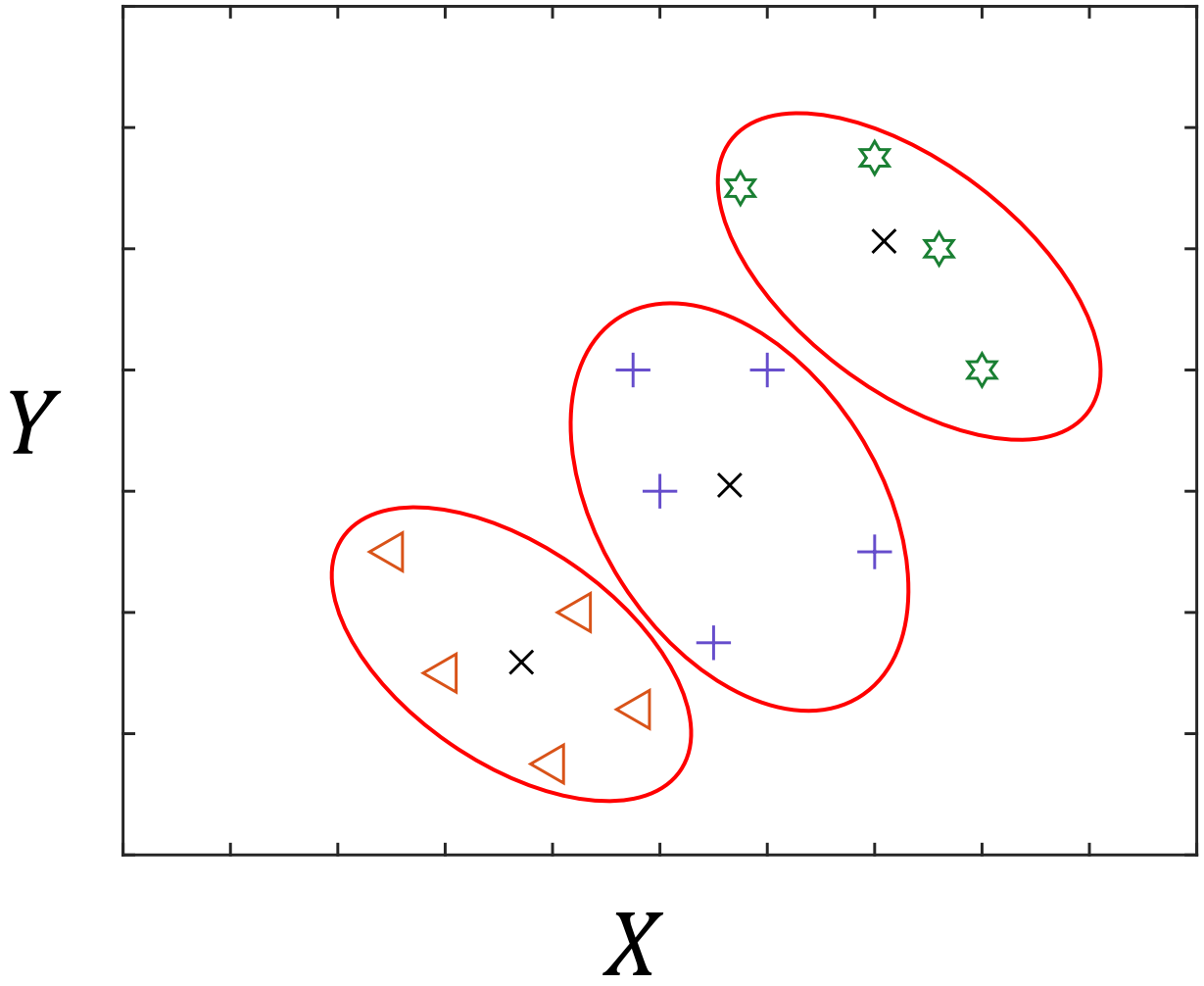
Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Illustrative example



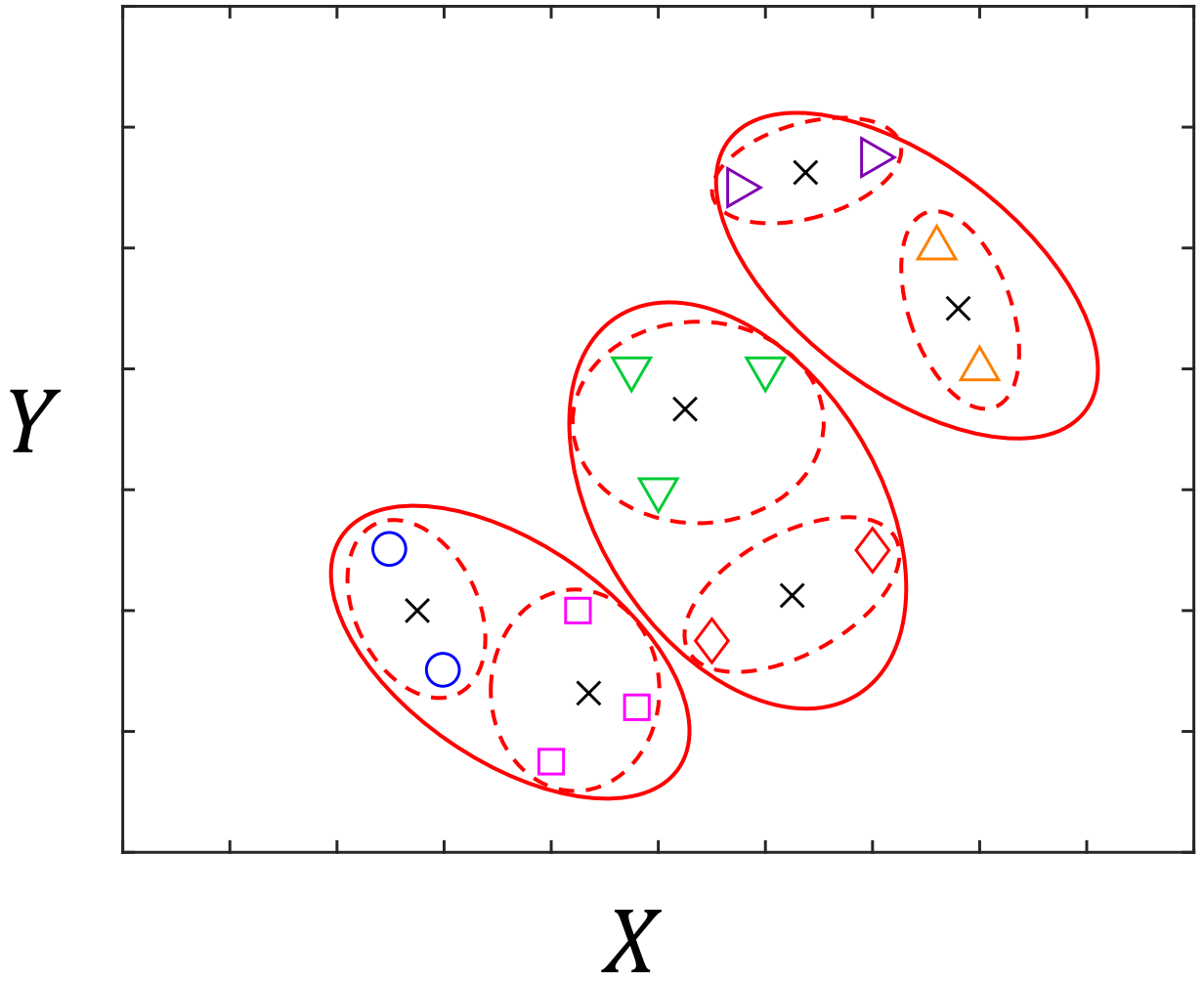
Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Illustrative example



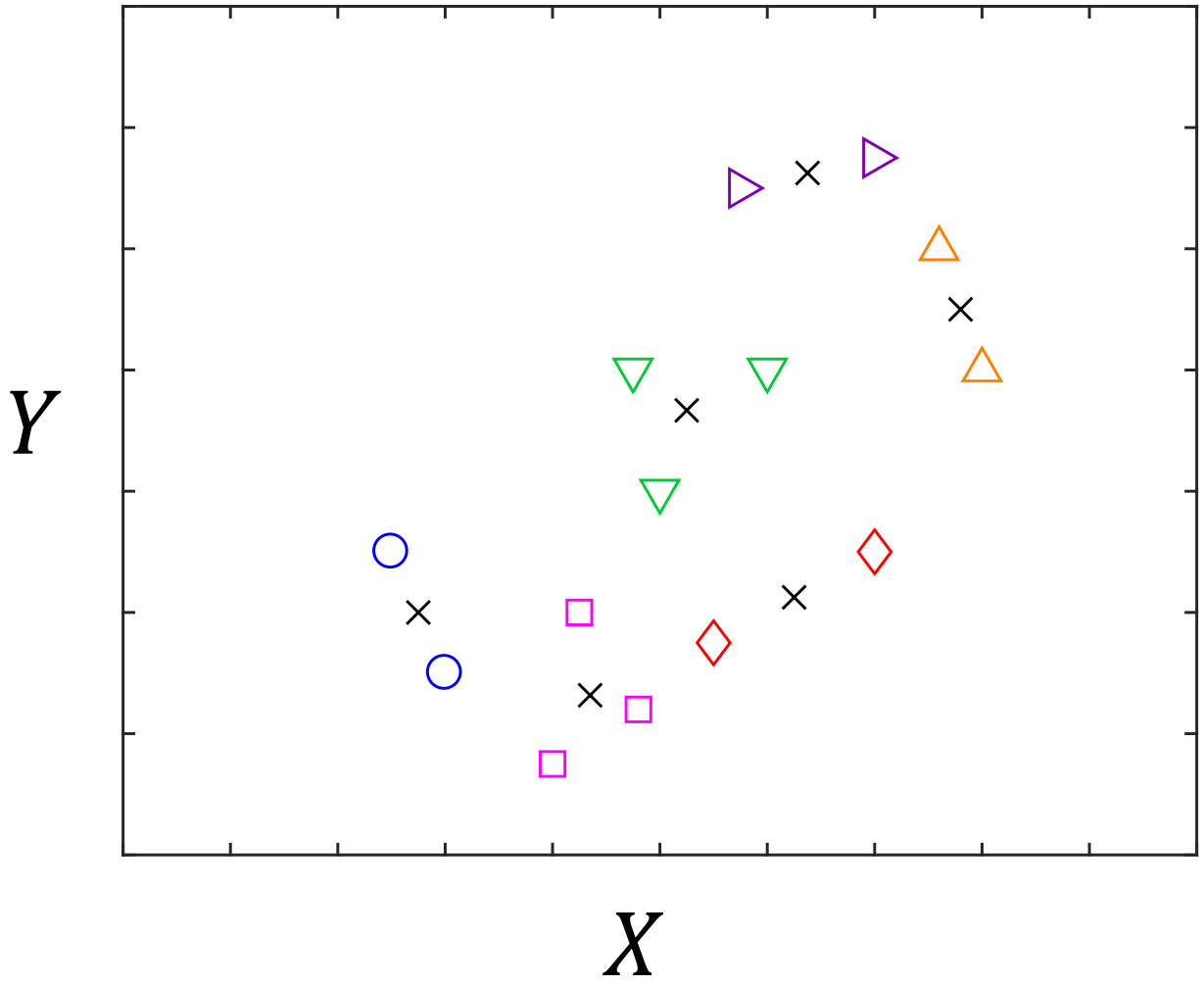
Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Illustrative example



Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Illustrative example



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MKM vs TKM

- G&TNEP problem.
- IEEE 24-bus Reliability Test System.
- 8760 hourly input data of electrical demand and wind power availability.
- Exact solution → 36 h 56 min.

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM vs TKM – Comparison procedure

Optimization model using all input data



Optimal objective function value / Exact solution $\left(OF^{\mathrm{E}}\right)$

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM vs TKM – Comparison procedure

Optimization model using RTPs/RDs

Investment plan

Optimization model using all input data and fixed investment decisions

Objective function value $\left(OF^{\mathrm{K}}\right)$

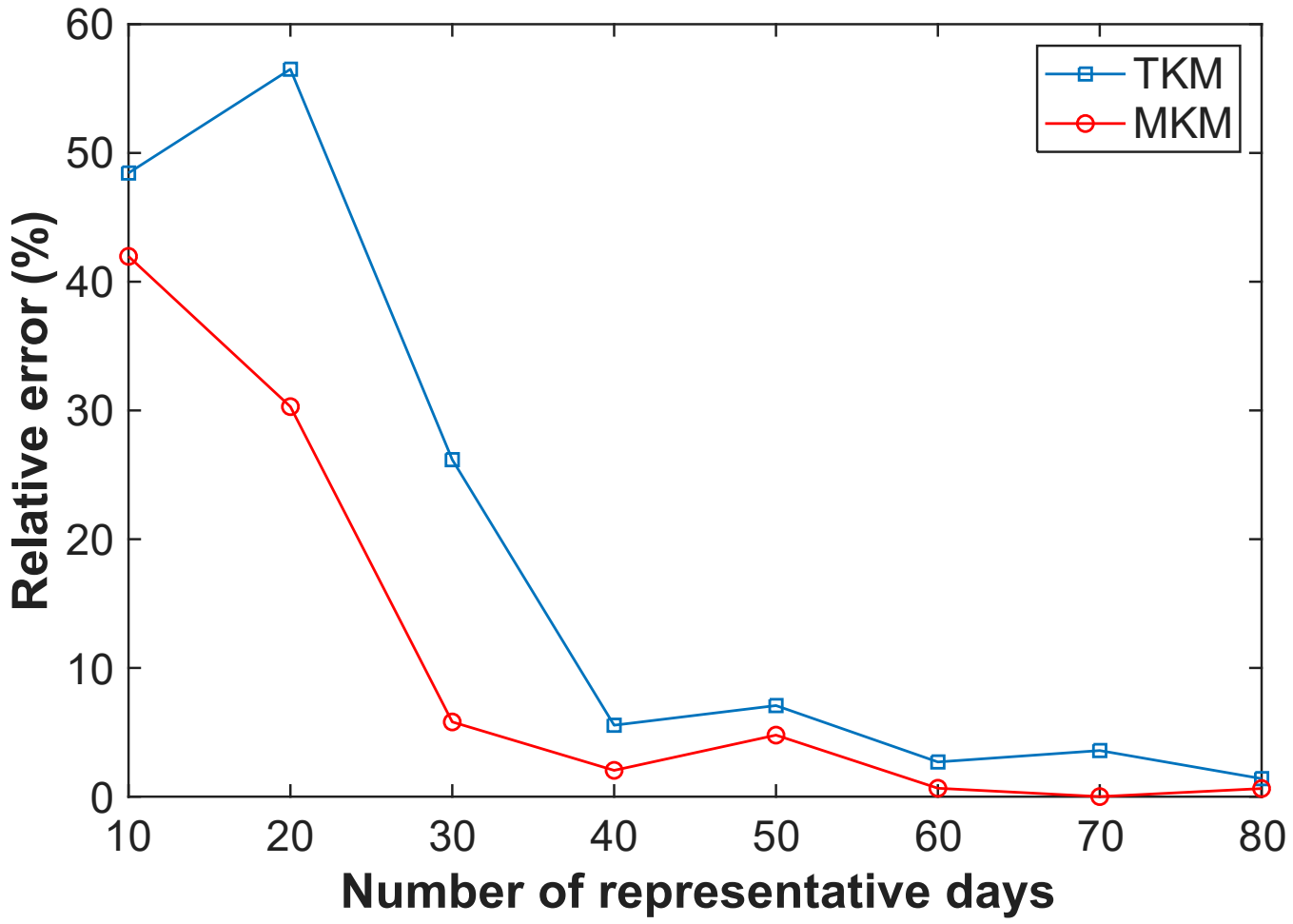
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MKM vs TKM – Comparison procedure

$$\text{Relative error} = \frac{\left|OF^{\mathrm{K}} - OF^{\mathrm{E}}\right|}{\left|OF^{\mathrm{E}}\right|} \cdot 100$$

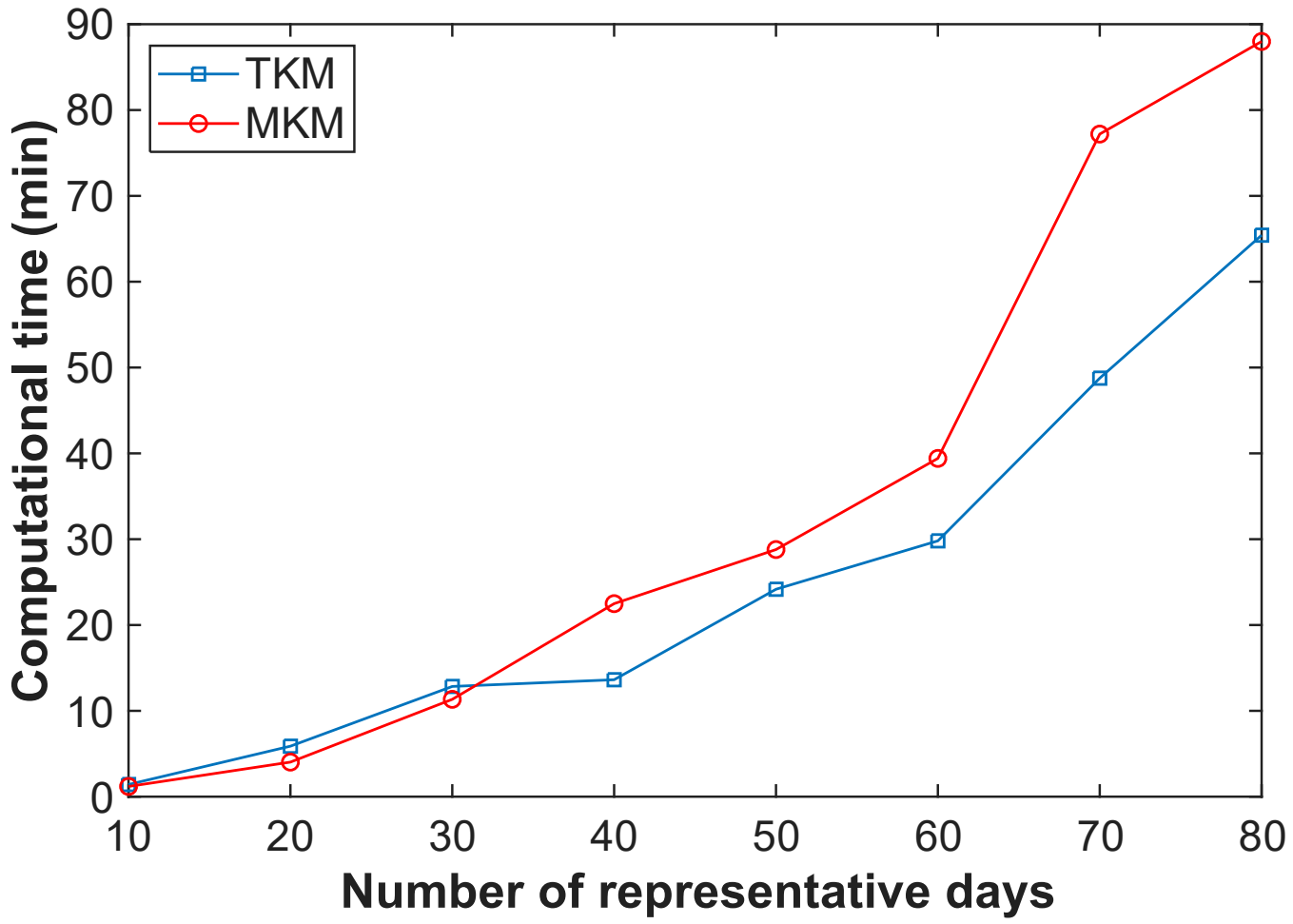
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MKM vs TKM



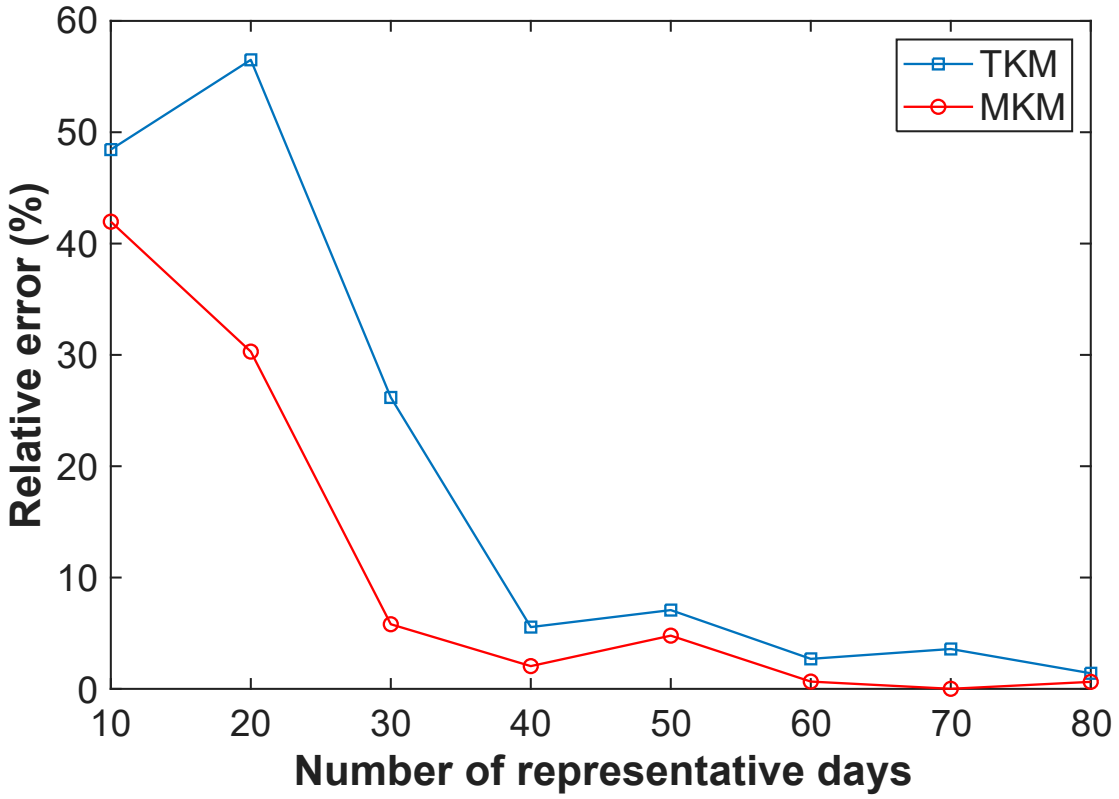
Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

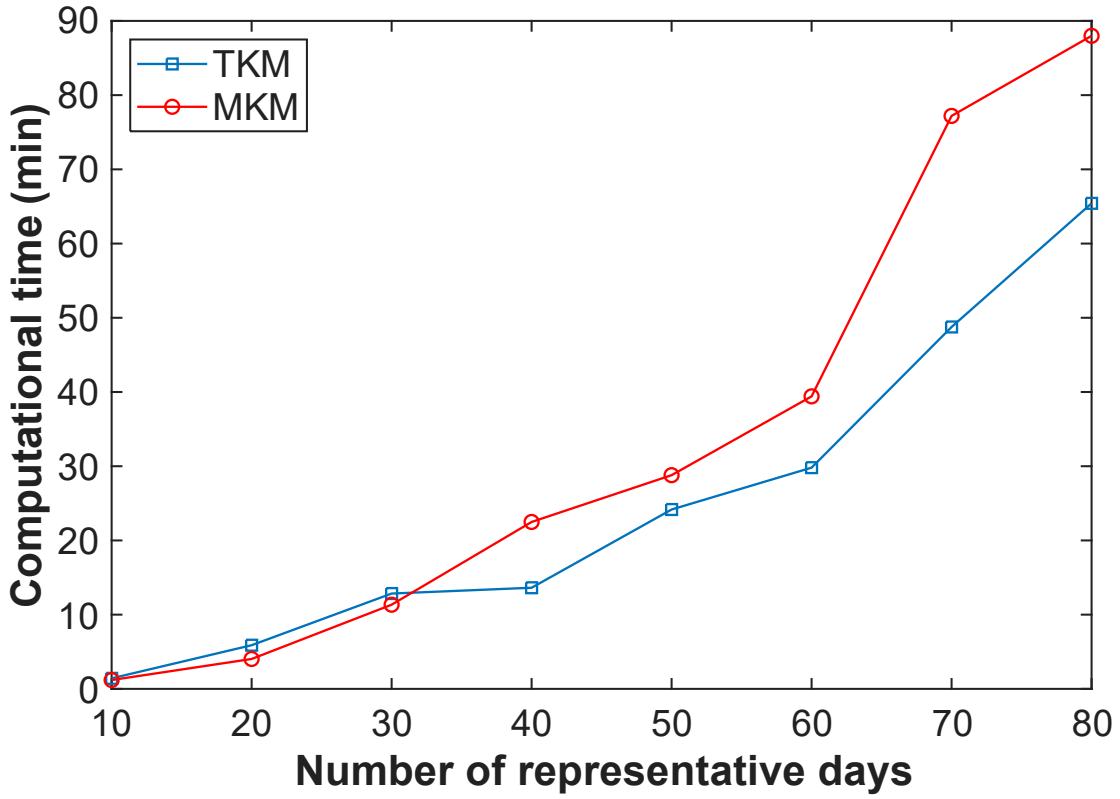
MKM vs TKM



Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

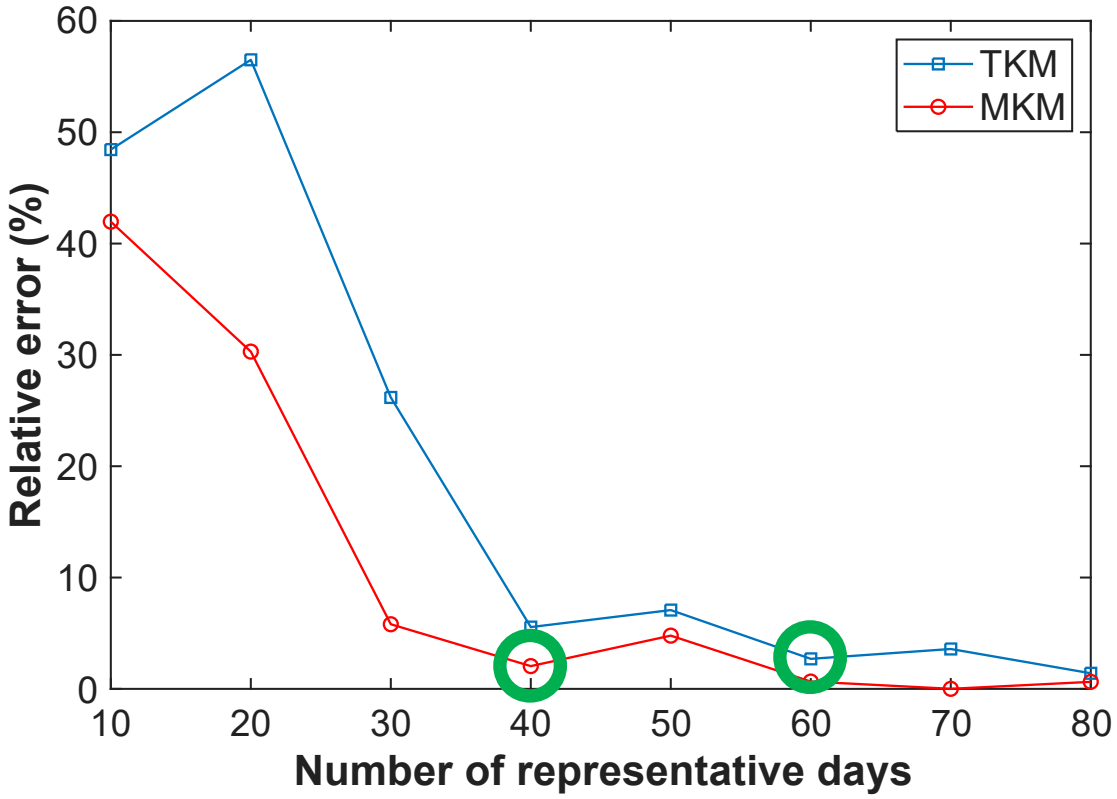
MKM vs TKM

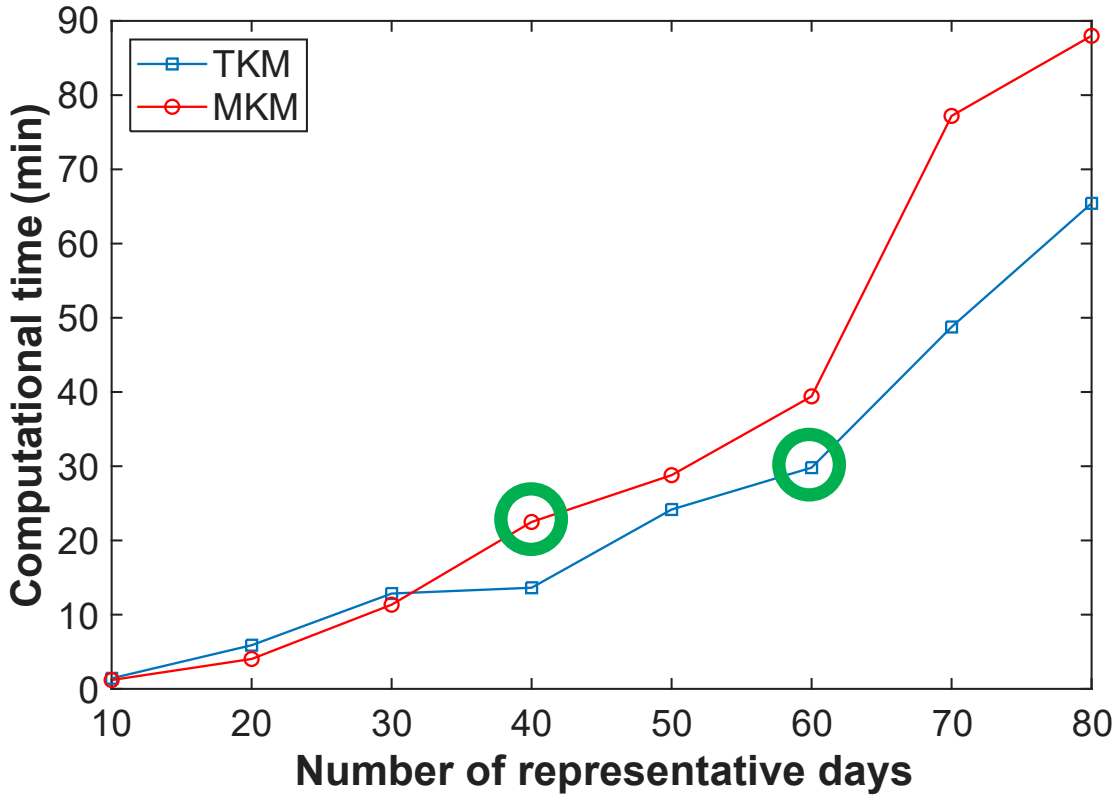




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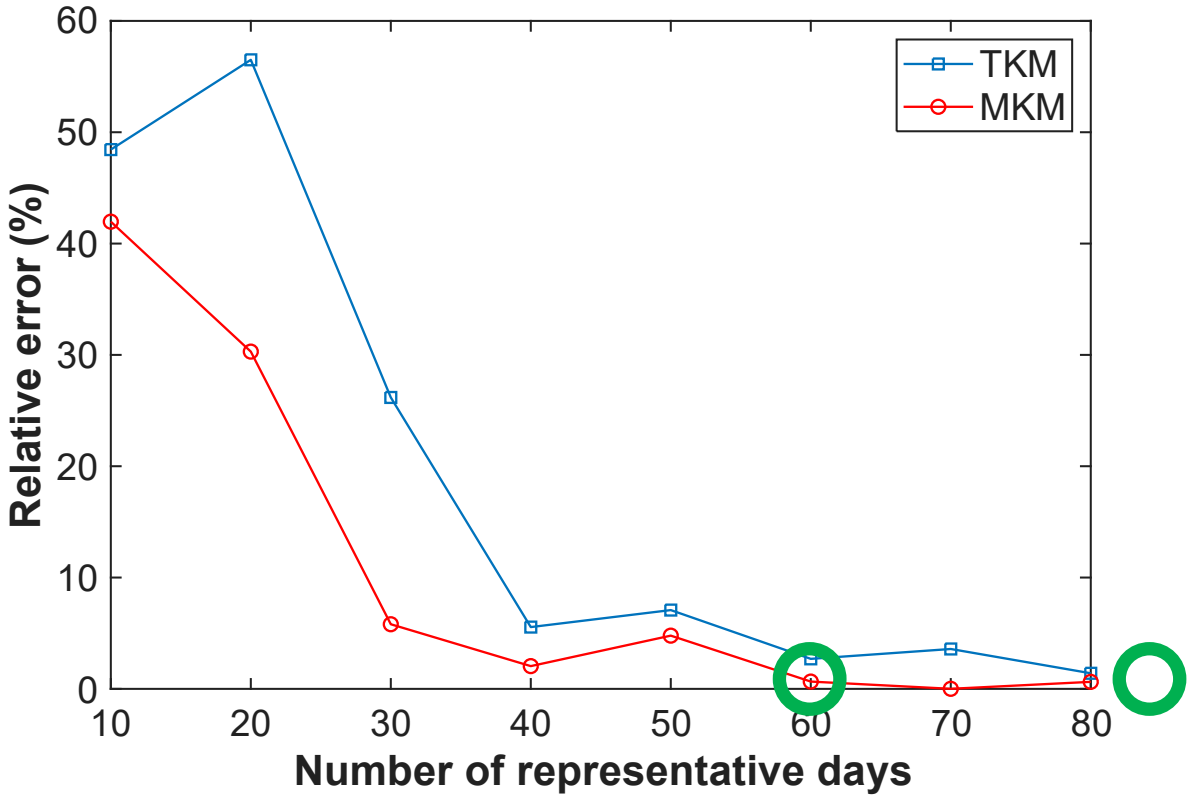
MKM vs TKM

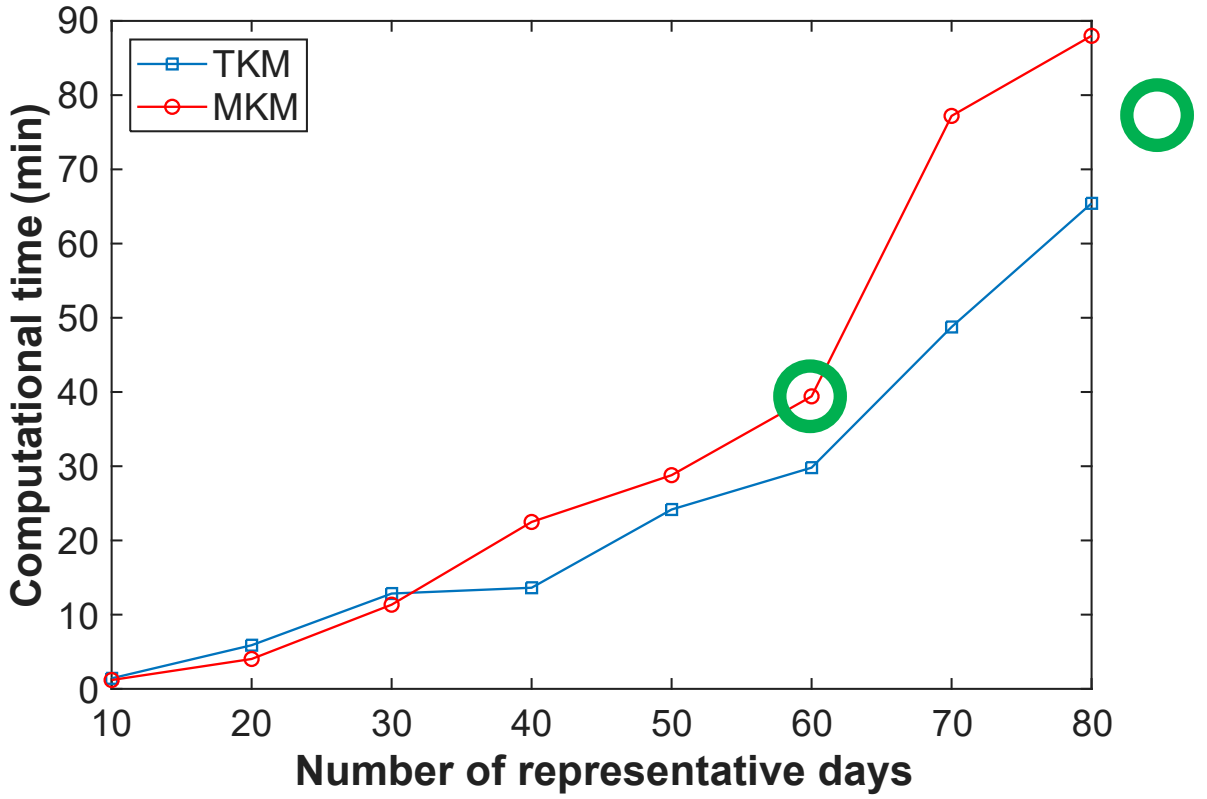




Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

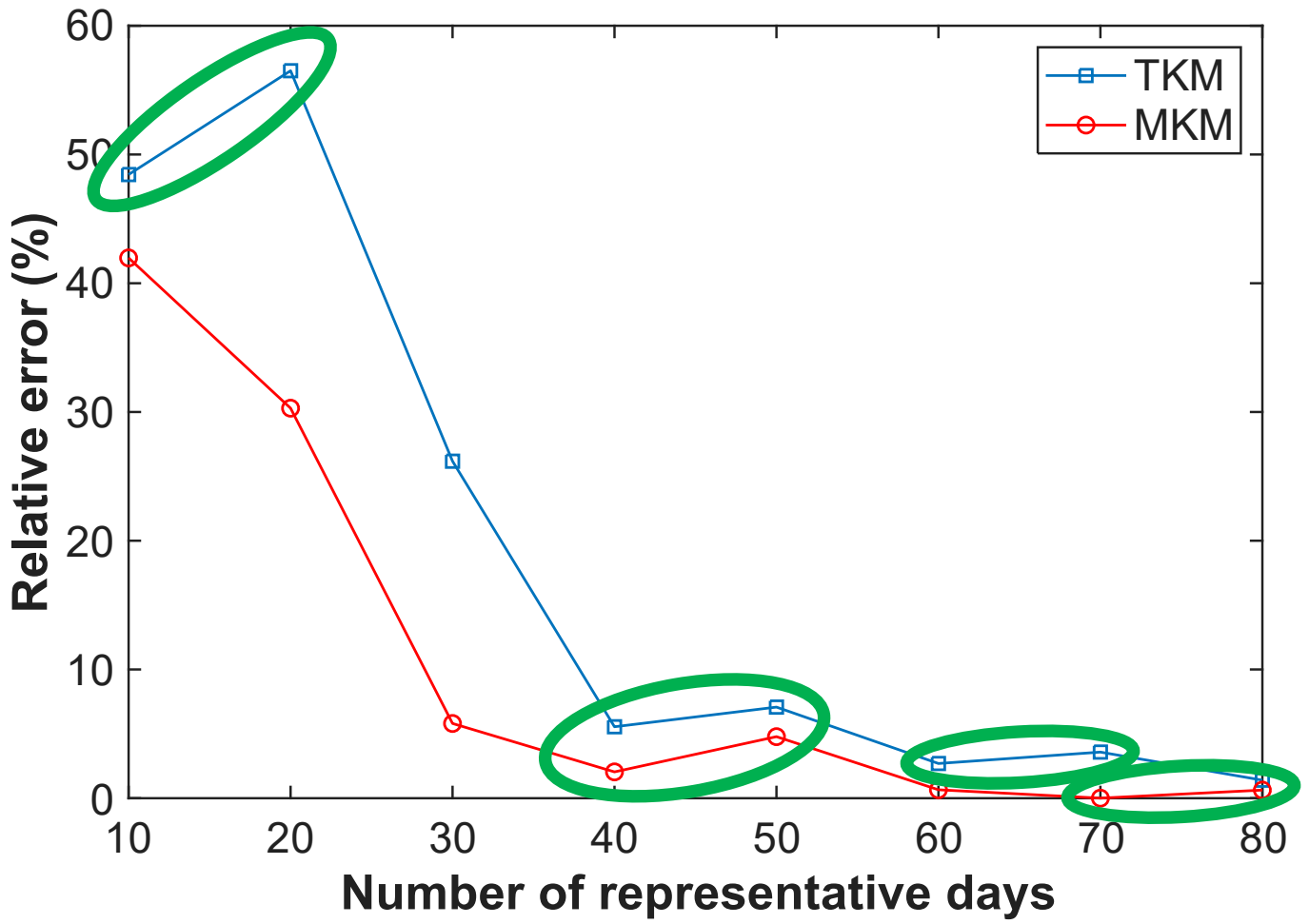
MKM vs TKM





Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM vs TKM



Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

MKM – Conclusions

- The MKM provides a solution closer to the exact solution than that provided by the TKM for a given number of RDs.

- The MKM also outperforms the TKM in terms of computational times needed to achieve the solution of the G&TEP problem for a given error with regard to the total annualized cost provided by the exact solution.

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Representative days for expansion decisions in power systems", *Energies*, vol. 13, no. 2, art. no. 335, Jan. 2020. DOI: 10.3390/en13020335.

Advanced aggregation techniques

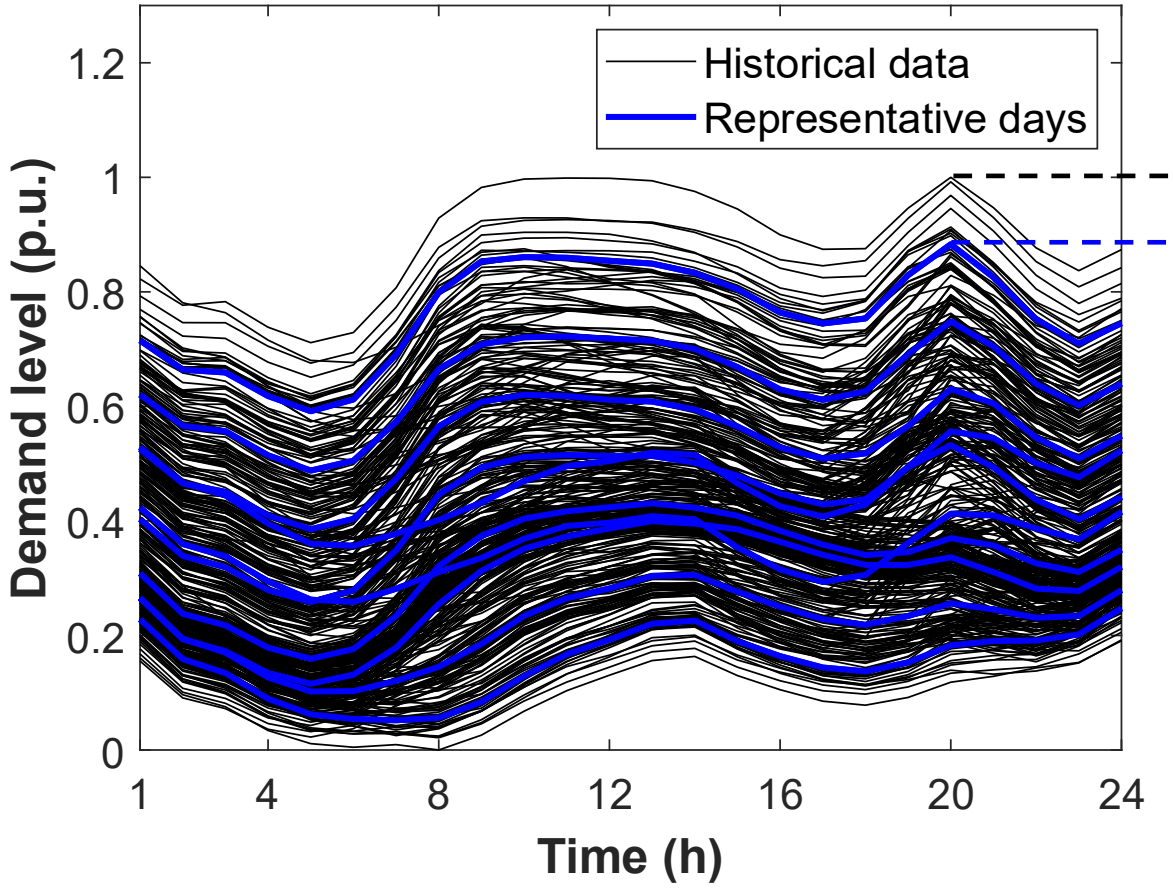
Works resulting from my Bachelor's Final Project and PhD Thesis:

- Modified K-means Method (MKM).
- Modified Maximum Dissimilarity Algorithm (MMDA).
- Priority Chronological Time-Period Clustering (PCTPC).
- Combination of previous techniques (MMDA + PCTPC).

Modified Maximum Dissimilarity Algorithm (MMDA)

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Motivation

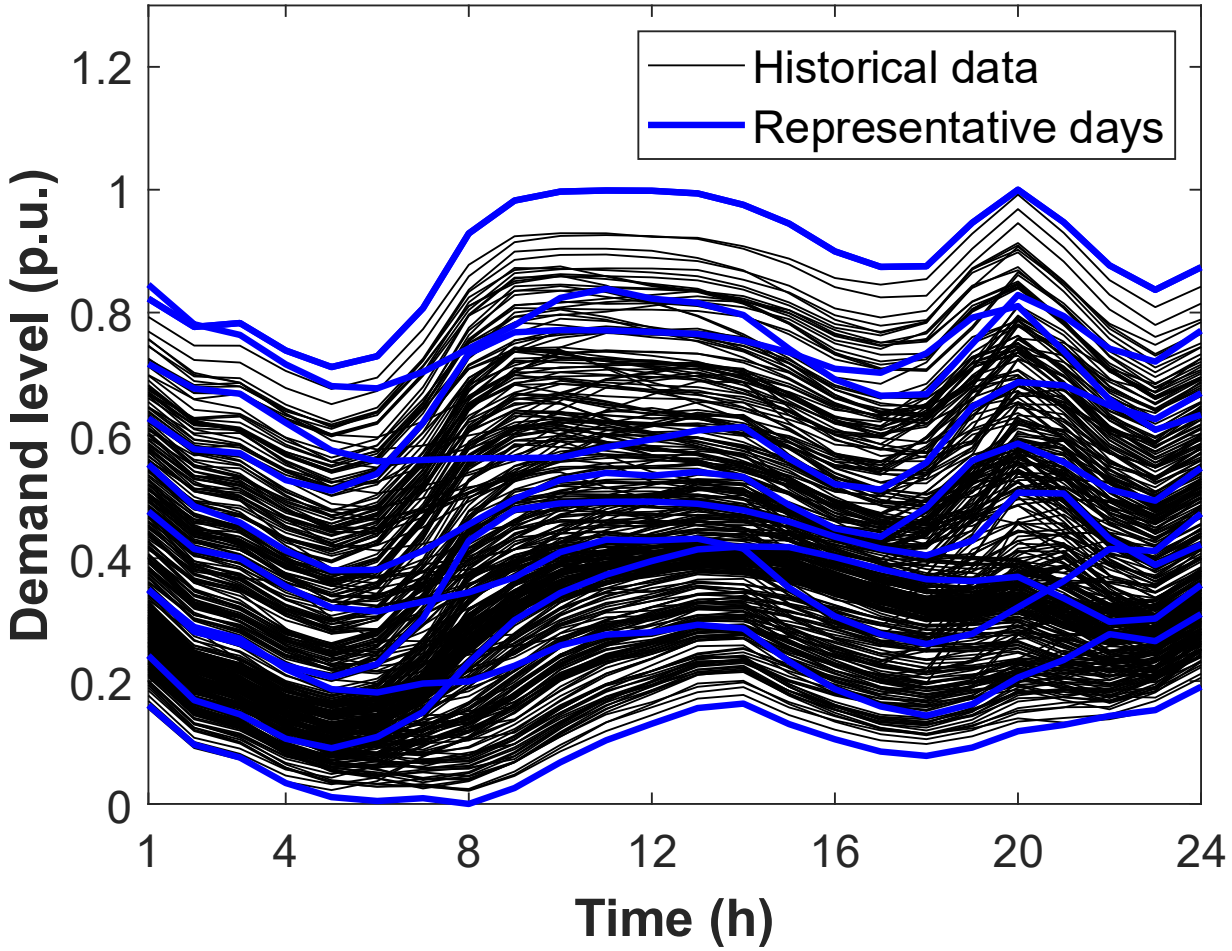


Peak demand level of the historical data

Peak demand level of the RDs

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

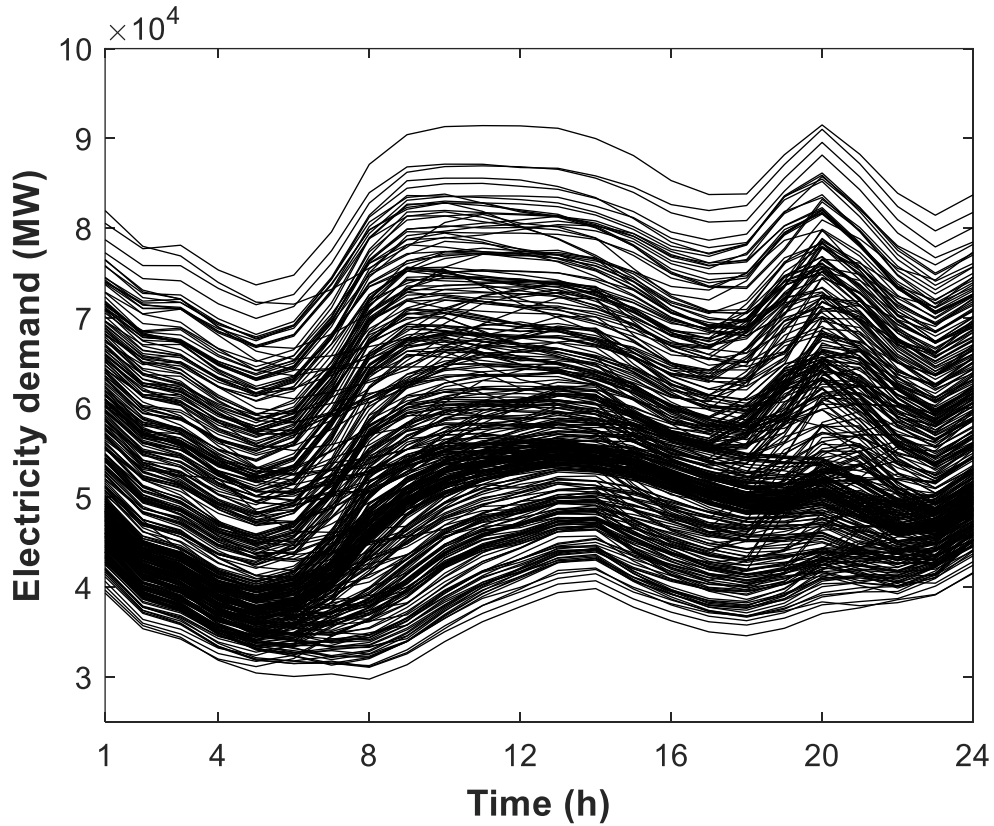
MMDA – Motivation



Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

Step 1) Select the number of representative vectors (K).



$$\left|\Omega^{J}\right| = 365$$

$$K = 10 \text{ RDs}$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

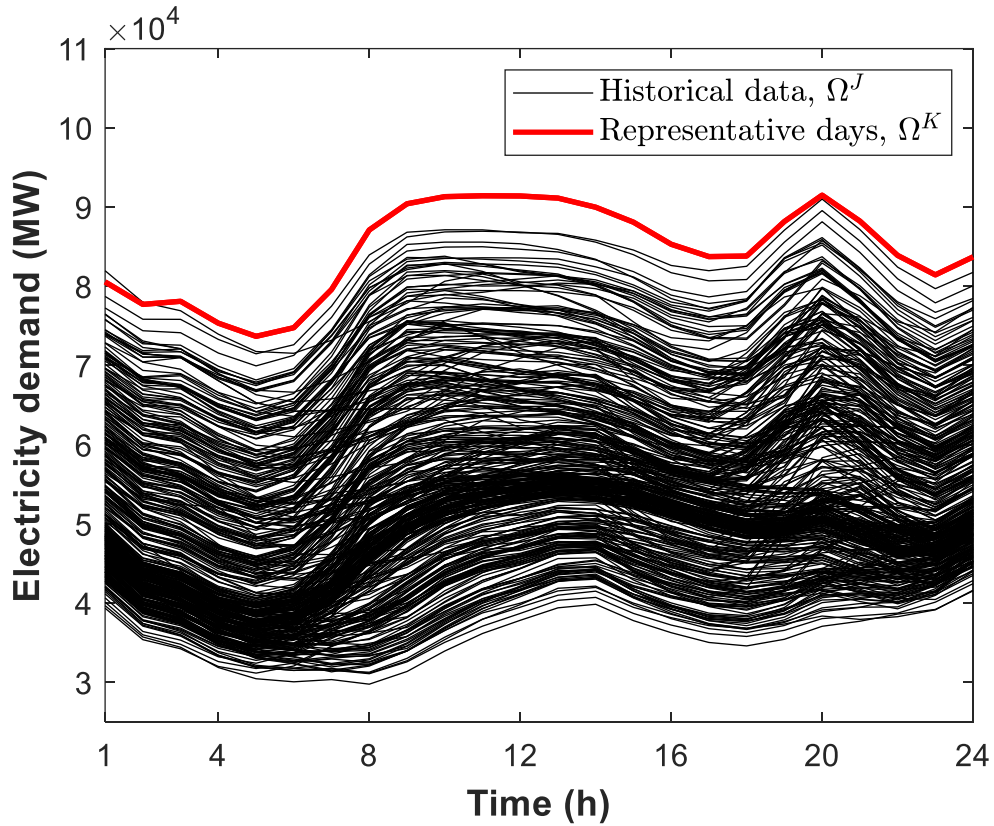
MMDA – Illustrative example

Step 2) Move the vectors of set Ω^{J} that contain the extreme conditions to set Ω^{K}, which is initially empty.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

Step 2)



$$|\Omega^J| = 364$$

$$|\Omega^K| = 1$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

Step 3) Compute the minimum Euclidean distances among each vector of set Ω^{J} and the vectors of set Ω^{K}.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

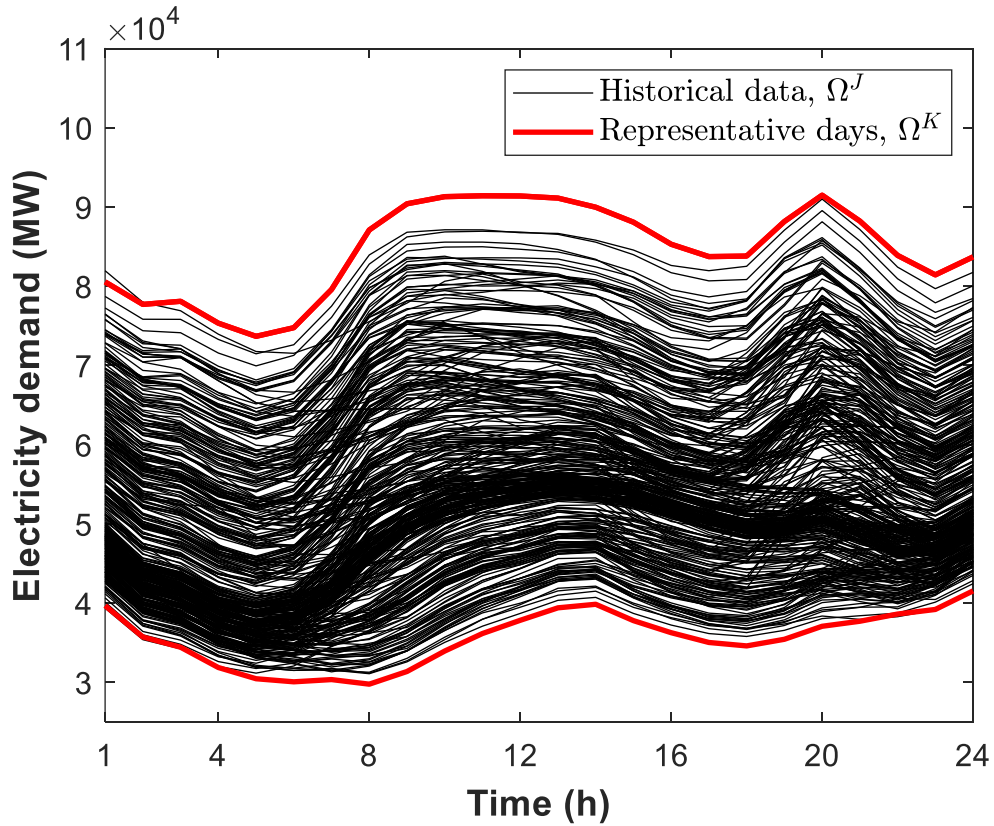
MMDA – Illustrative example

Step 4) Move the vector of set Ω^{J} that presents the largest of the distances computed in Step 3) to set Ω^{K}.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

Step 4)



$$|\Omega^J| = 363$$

$$|\Omega^K| = 2$$

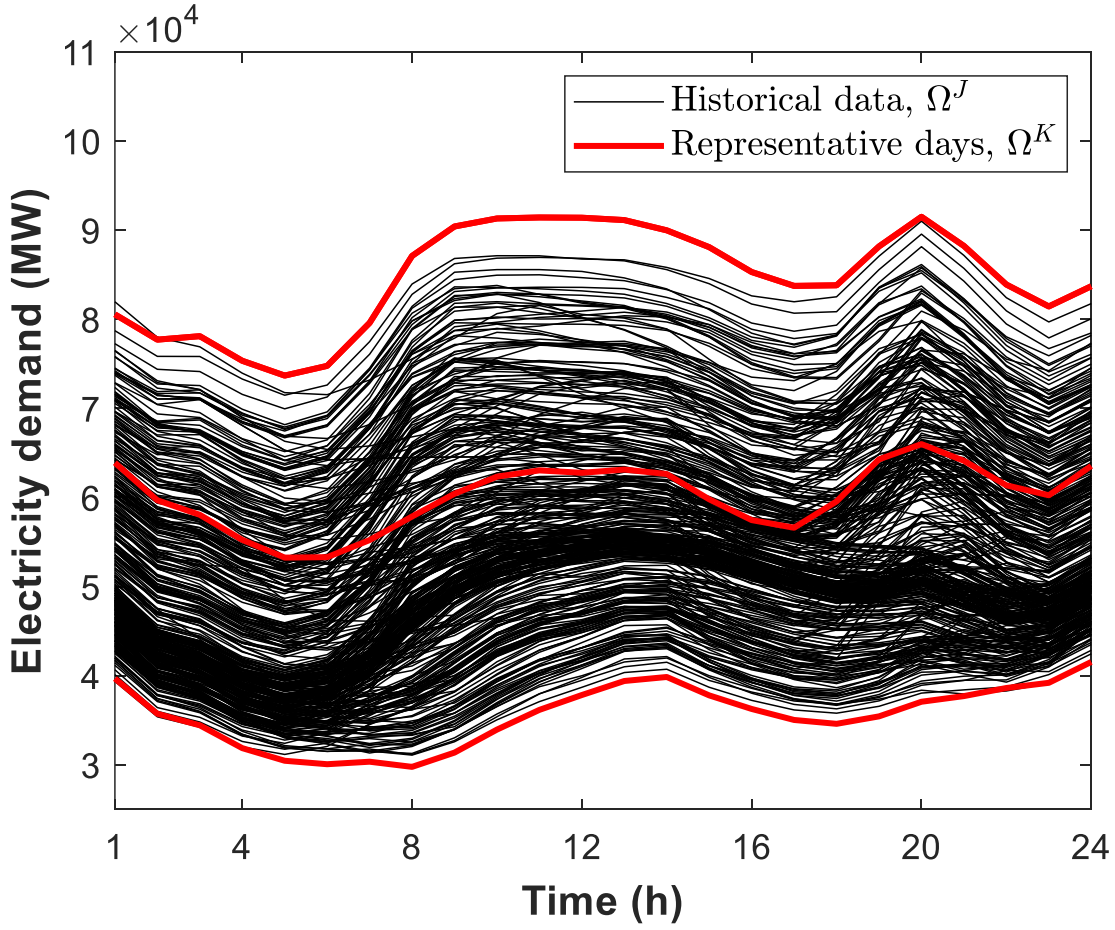
Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

Step 5) If $\left|\Omega^{\mathrm{K}}\right| = K$, stops. Otherwise, go to Step 3).

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

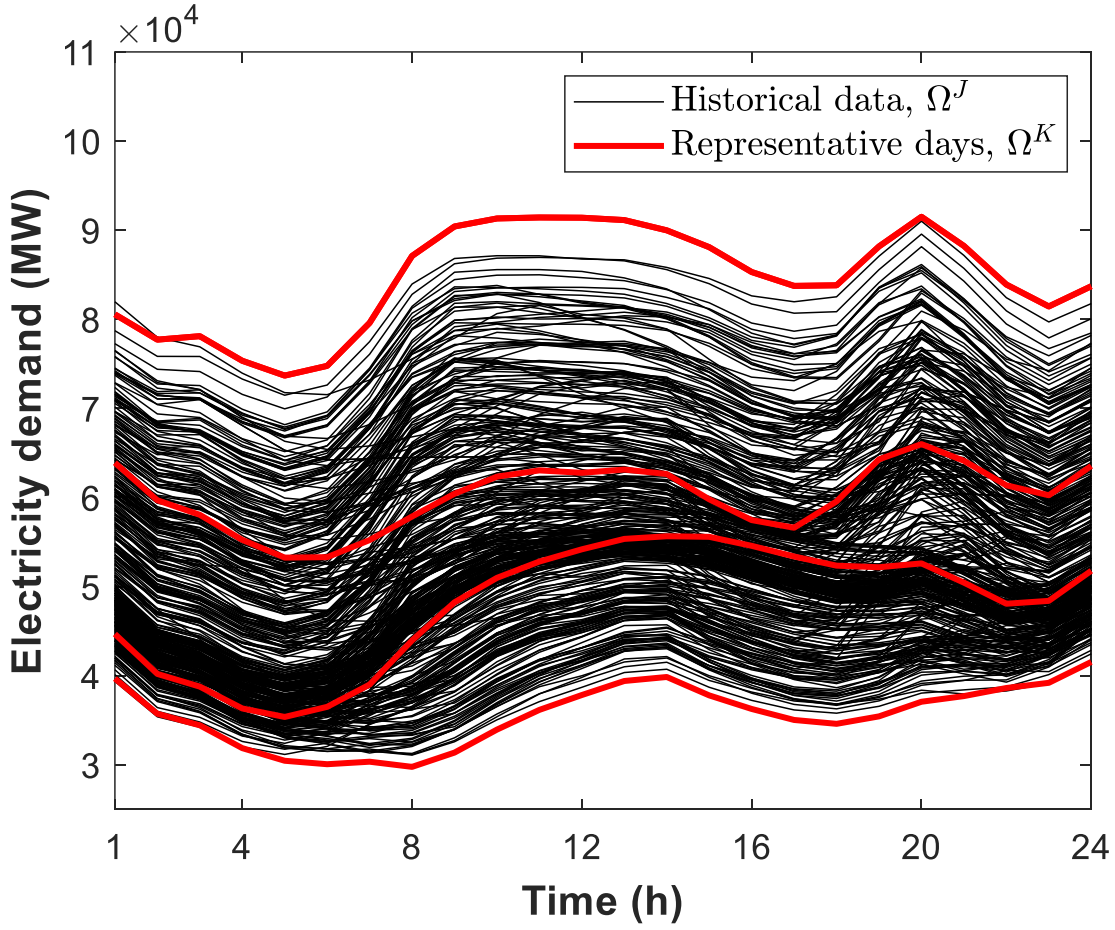


$$|\Omega^J| = 362$$

$$|\Omega^K| = 3$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

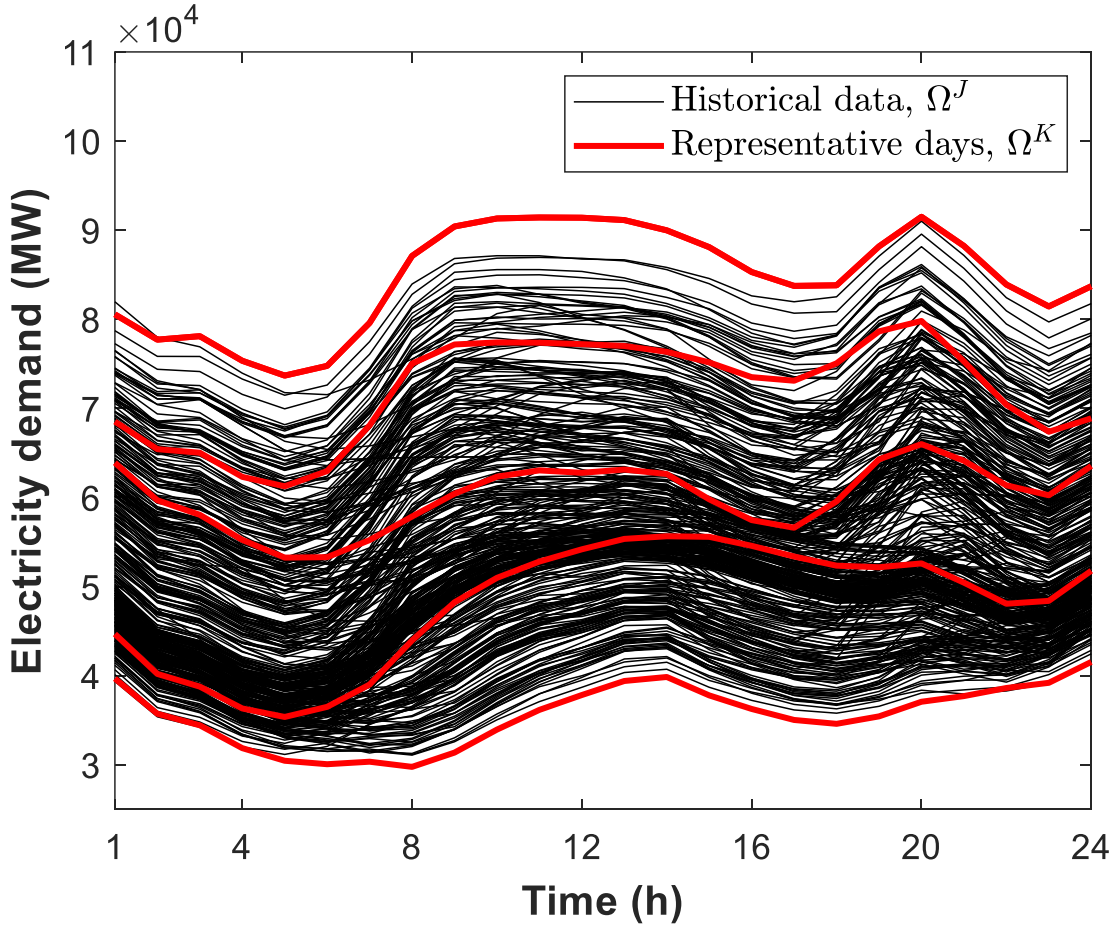


$$|\Omega^J| = 361$$

$$|\Omega^K| = 4$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

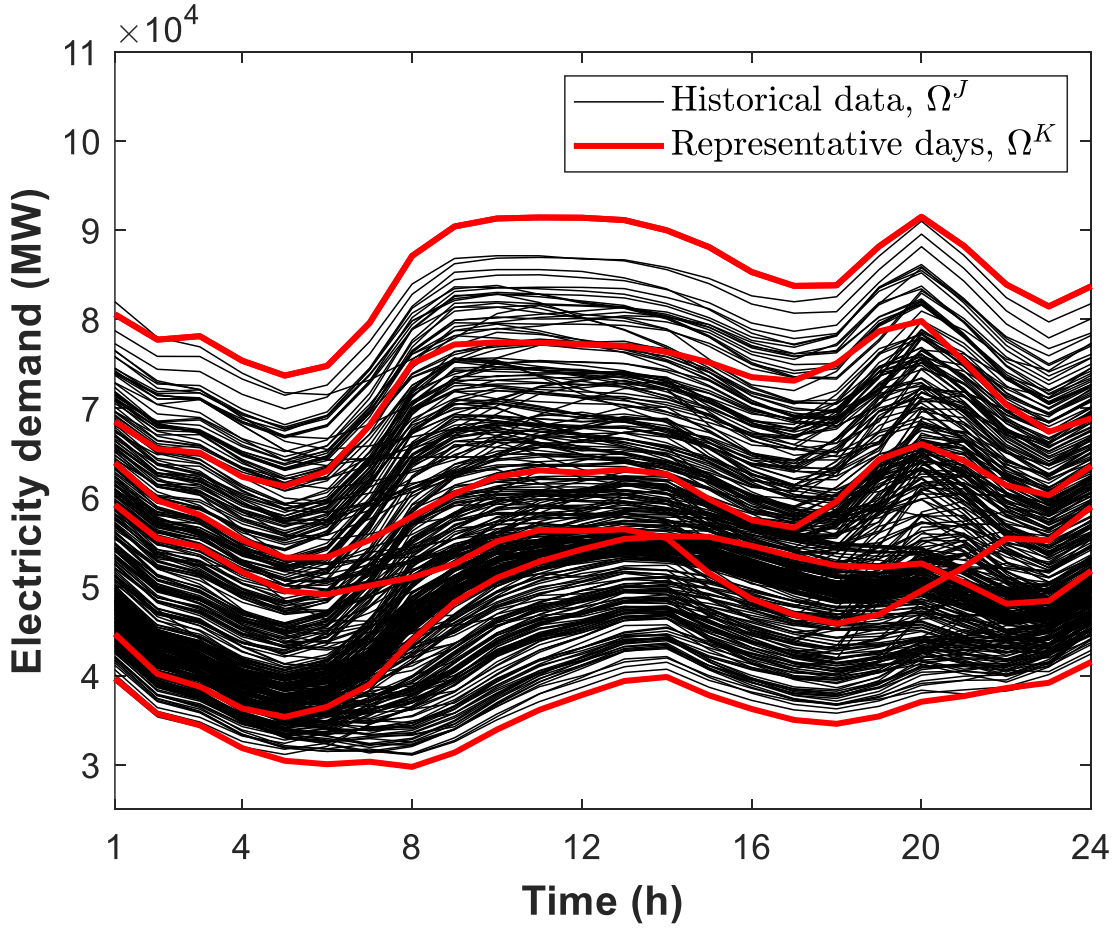


$$|\Omega^J| = 360$$

$$|\Omega^K| = 5$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

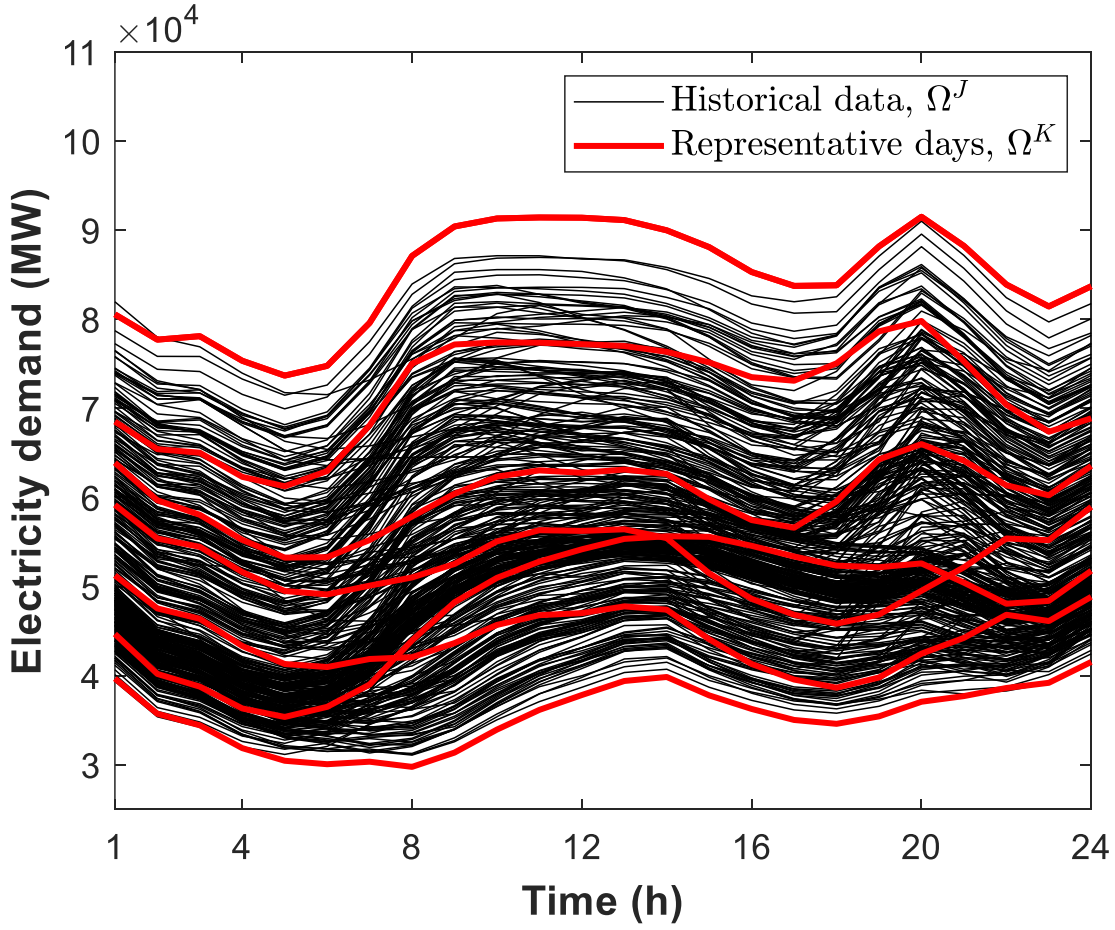


$$|\Omega^J| = 359$$

$$|\Omega^K| = 6$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

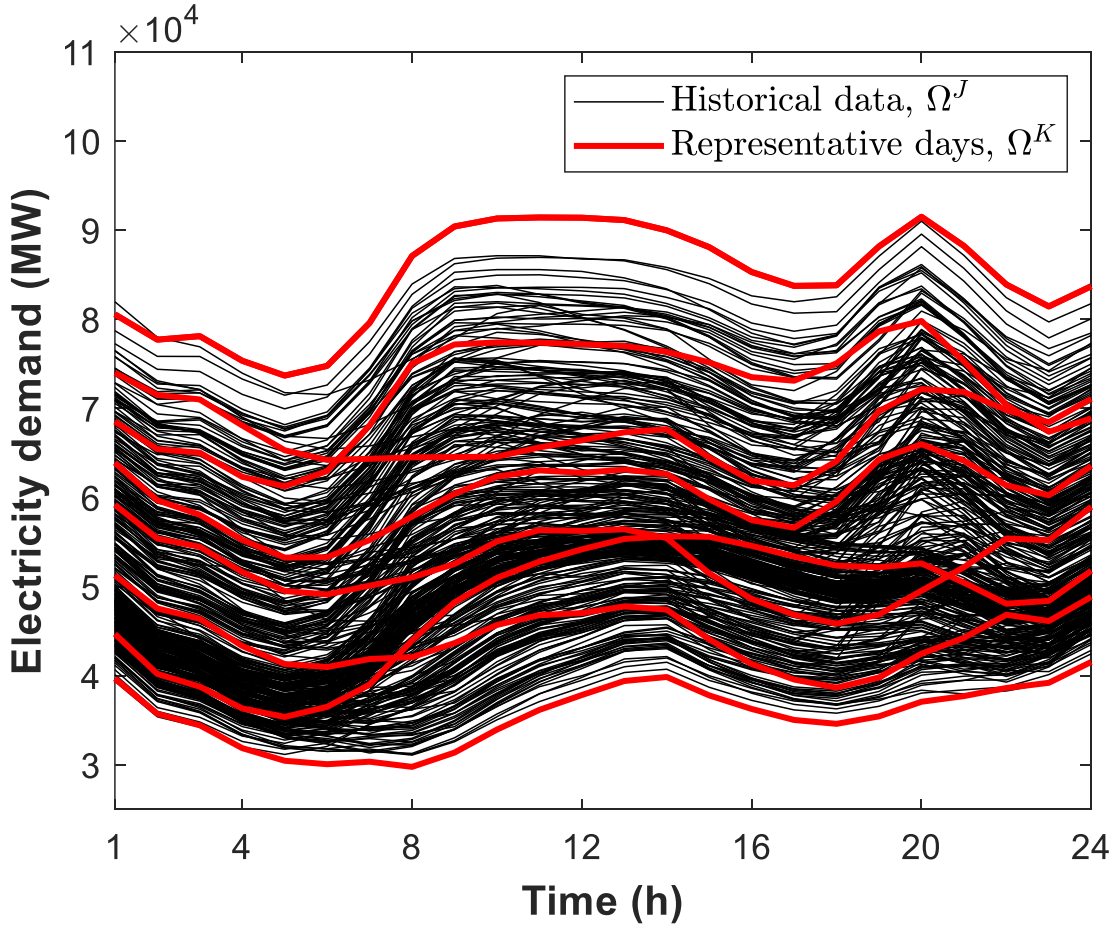


$$|\Omega^J| = 358$$

$$|\Omega^K| = 7$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

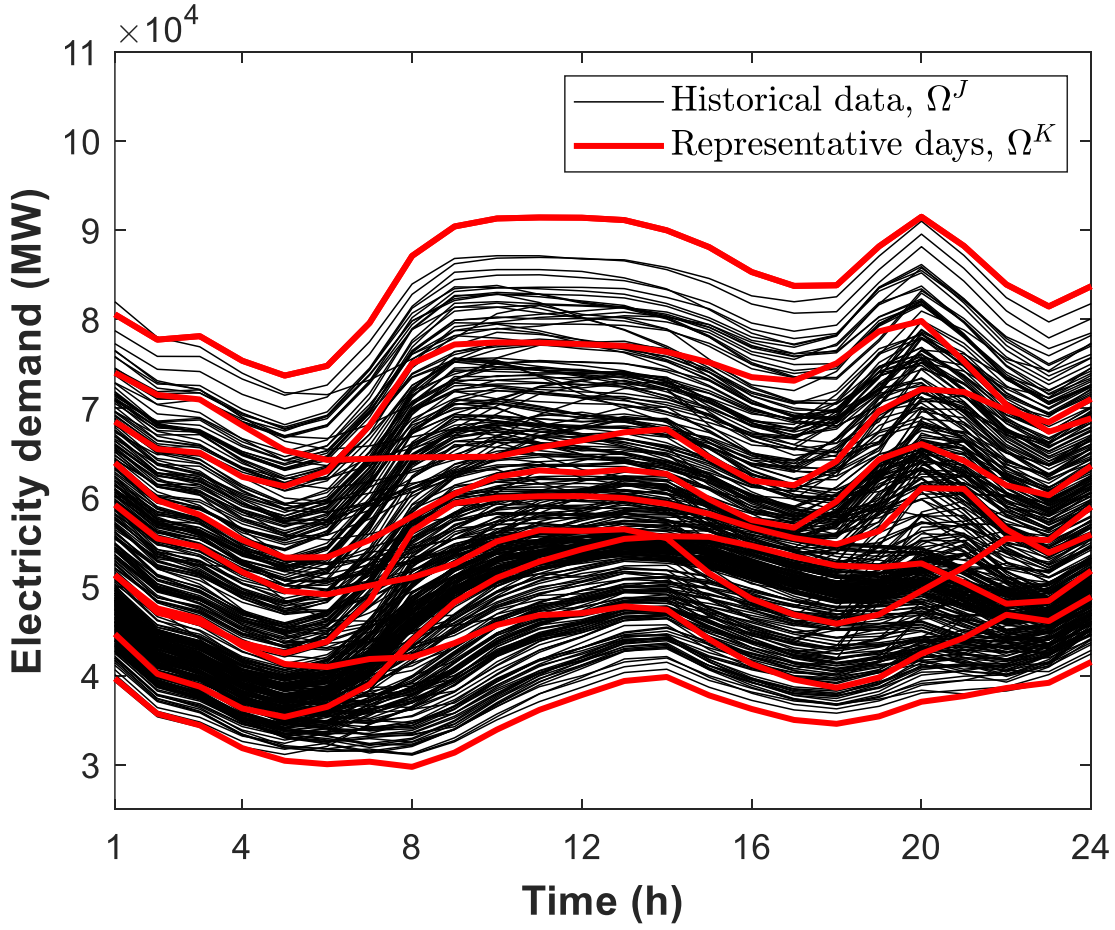


$$|\Omega^J| = 357$$

$$|\Omega^K| = 8$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Illustrative example

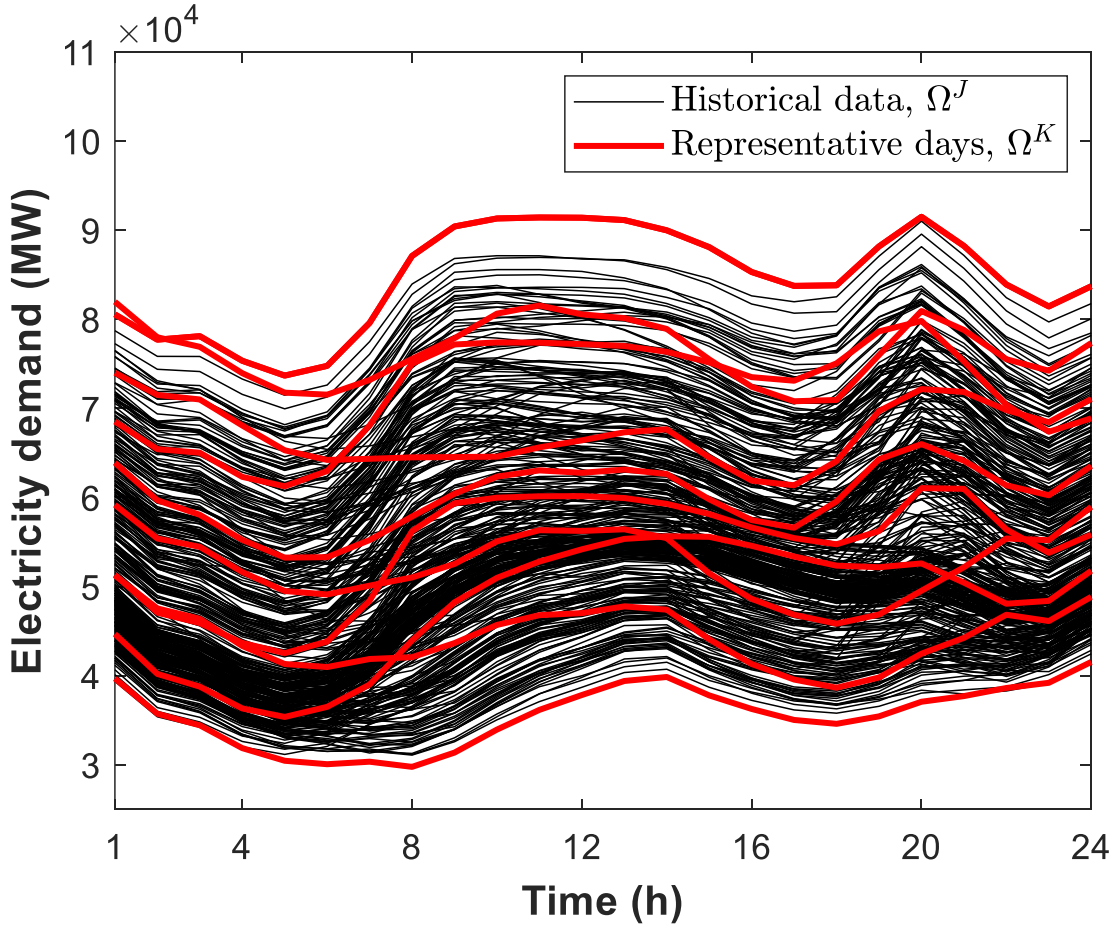


$$|\Omega^J| = 356$$

$$|\Omega^K| = 9$$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

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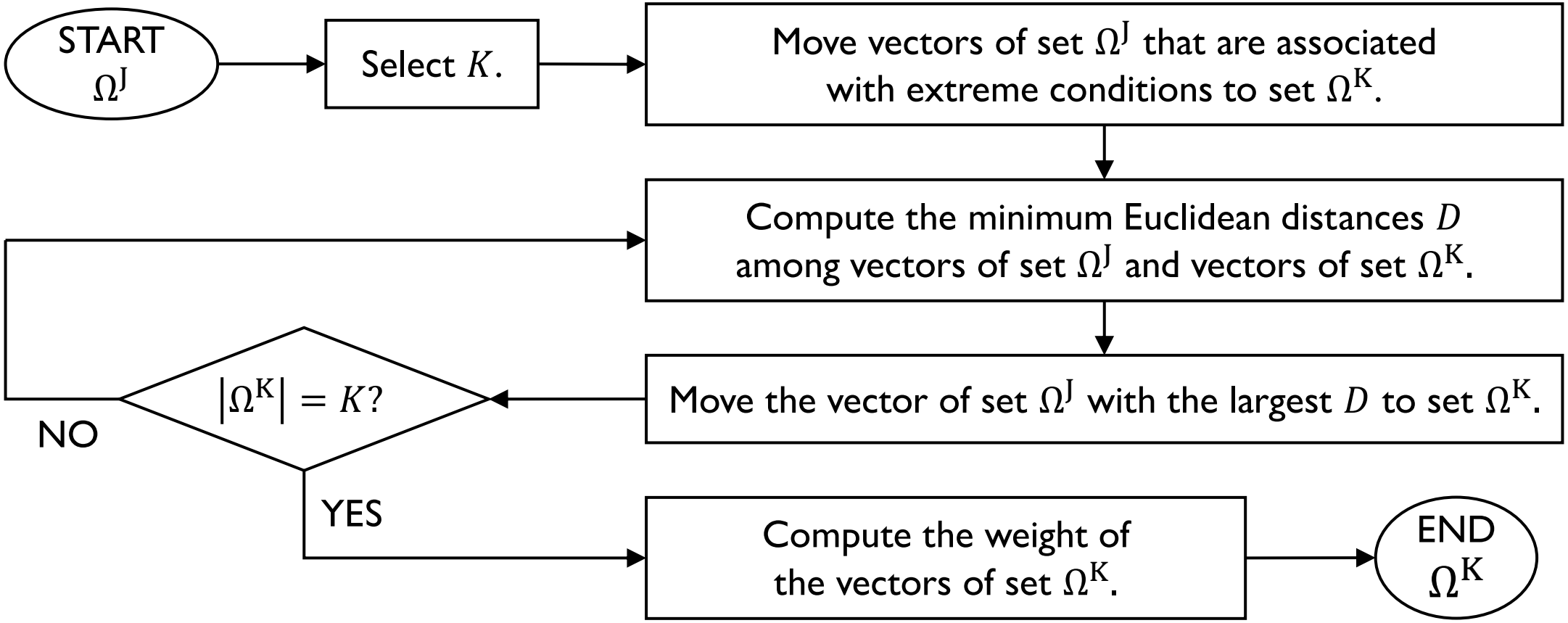


$$|\Omega^J| = 355$$

$$|\Omega^K| = 10$$

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MMDA – Flowchart



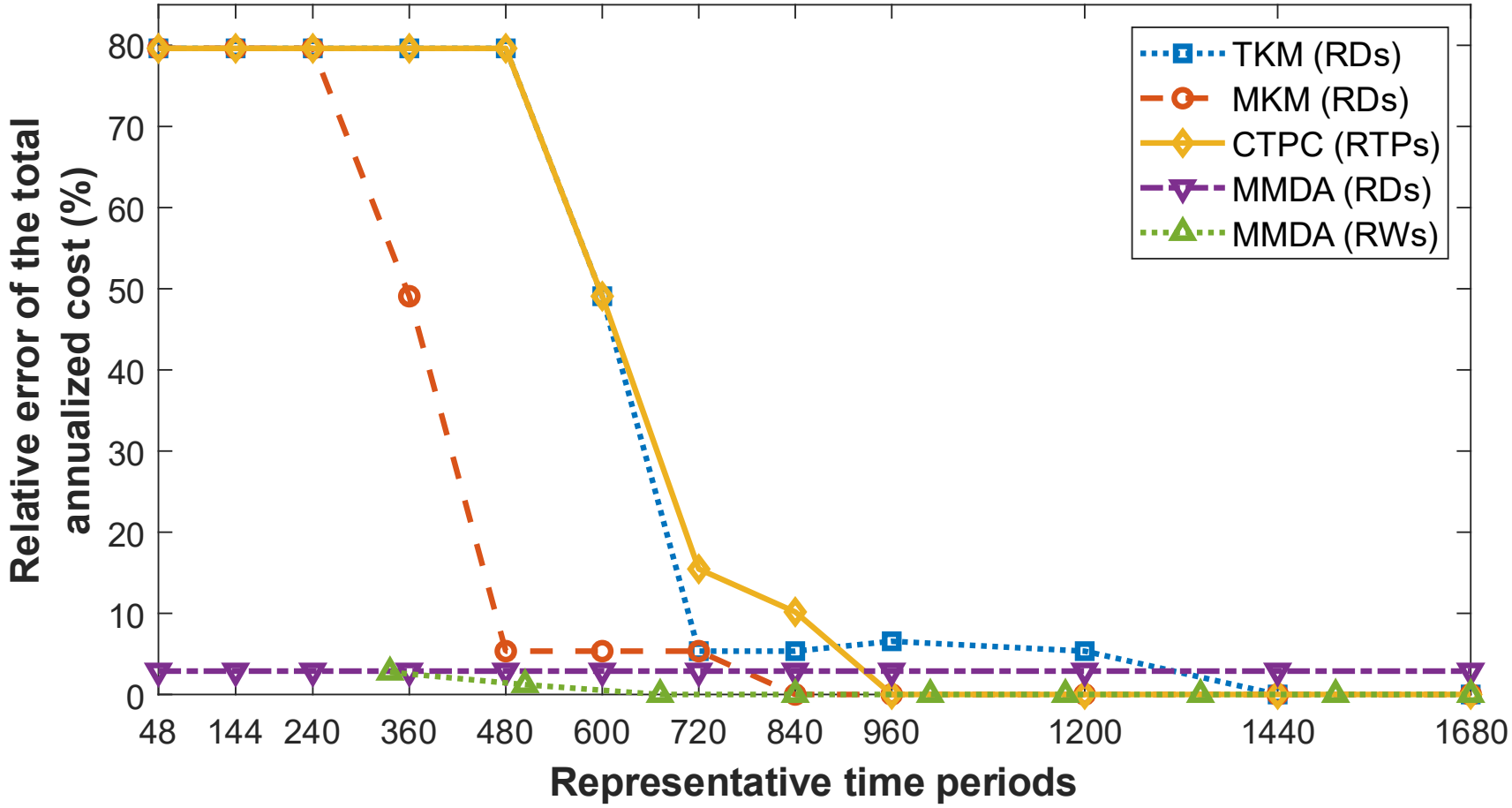
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MMDA vs other aggregation techniques

- TNEP problem.
- IEEE 24-bus Reliability Test System.
- 8760 hourly input data of electrical demand and renewable (wind and solar) power availability from Spain and France.

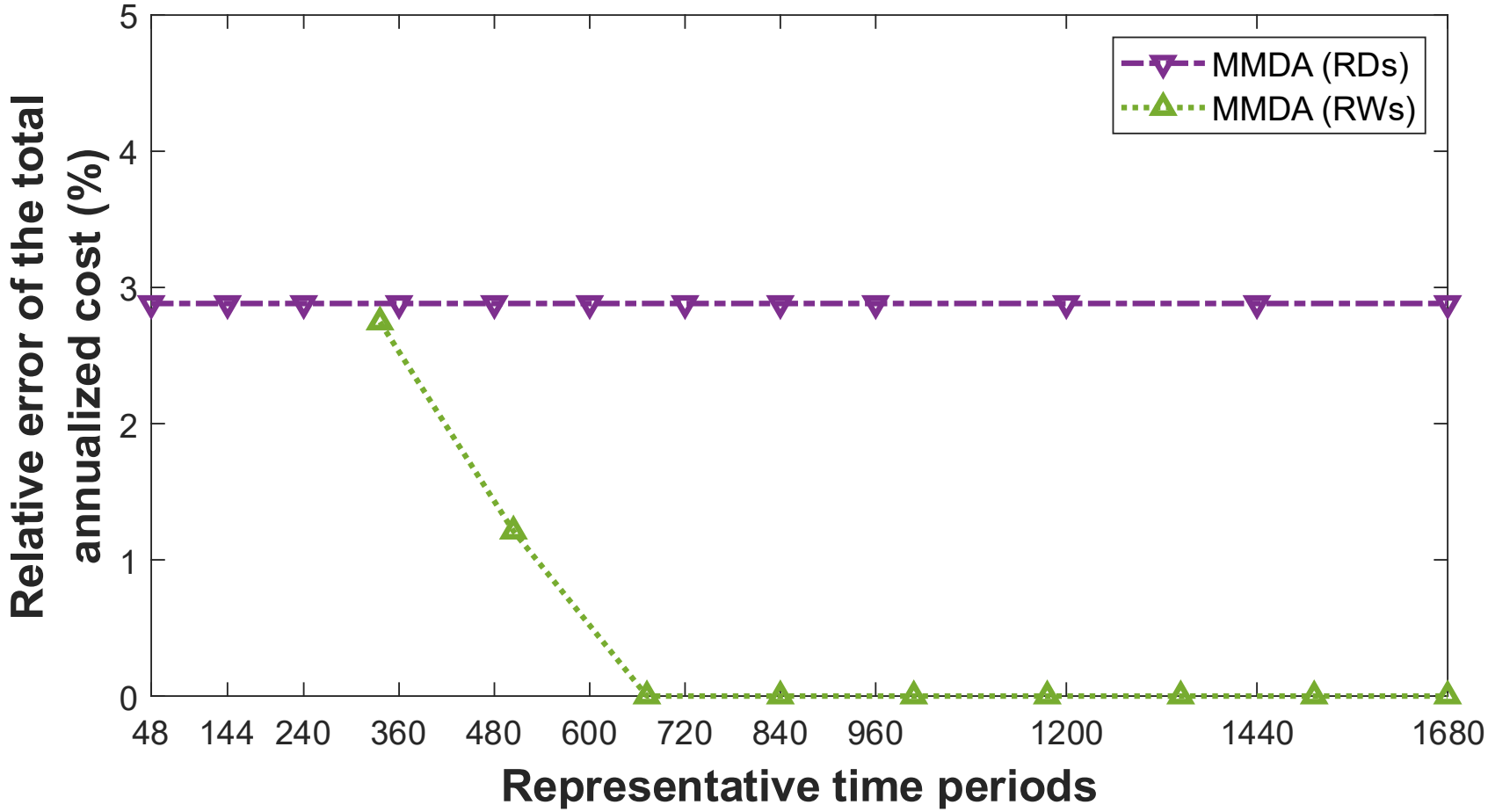
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MMDA vs other aggregation techniques



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MMDA vs other aggregation techniques



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MMDA vs other aggregation techniques

Requirements to prevent load shedding

Clustering Technique	RTPs	Computational time (s)	Relative error (%)
TKM (RDs)	1440	6316	0
MKM (RDs)	840	2015	0
CTPC (RTPs)	960	1832	0
MMDA (RDs)	48	28	2.88
MMDA (RWs)	336	2332	2.75

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

MMDA – Conclusions

- The use of the MMDA prevents load shedding in all cases analyzed.

- It leads to low relative errors when few representative time periods are considered. This motivates its use for more complex models.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Enhanced representative time periods for transmission expansion planning problems", *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 3802–3805, Jul. 2021. DOI: 10.1109/TPWRS.2021.3067148.

Advanced aggregation techniques

Works resulting from my Bachelor's Final Project and PhD Thesis:

- Modified K-means Method (MKM).
- Modified Maximum Dissimilarity Algorithm (MMDA).
- Priority Chronological Time-Period Clustering (PCTPC).
- Combination of previous techniques (MMDA + PCTPC).

Priority Chronological Time-Period Clustering (PCTPC)

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

PCTPC – Motivation

- Importance of long-term dynamics:
  - Key aspect when low renewable production takes place during a prolonged time → Load shedding and/or higher costs.
  - Inter-day storage → Increase the share of renewable generating units.

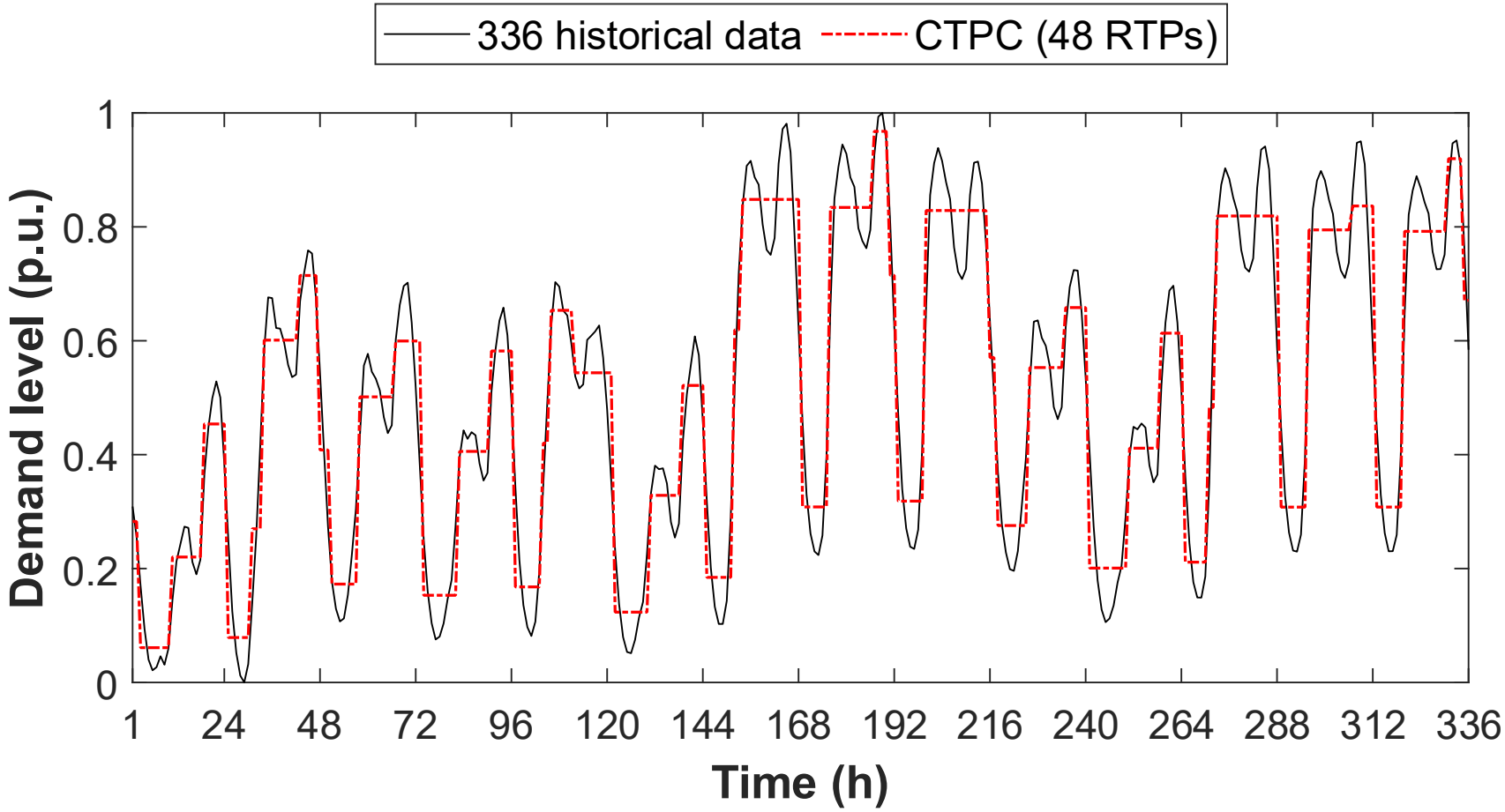
Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

PCTPC – Motivation

- Modeling long-term dynamic when representative vectors are used:
  - RWs → Limitation of one week.
  - RDs connected → Increase in the computational time.
  - CTPC → Poor representation of global and local maximum and minimum values.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

PCTPC – Motivation



Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

PCTPC – Rules

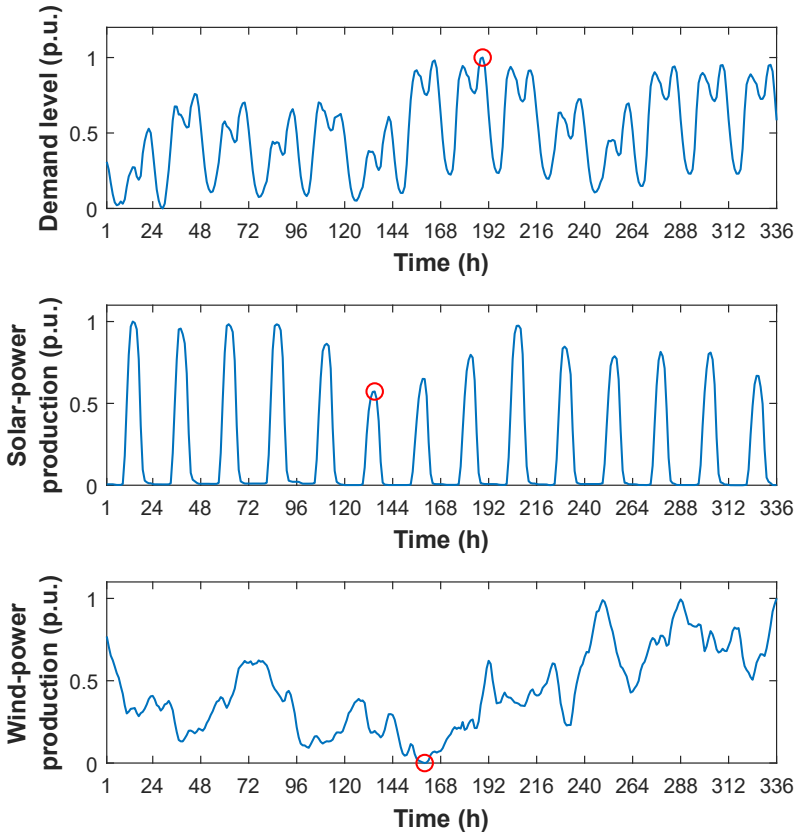
We proposed the **Priority** Chronological Time-Period Clustering (PCTPC). Input data are classified into:

- High priority (**H**): those values that must be represented independently of the number of RTPs considered.
- Medium priority (**M**): those values that must be represented whenever possible.
- Low priority (**L**): those values whose representation is assumed to be not so important.

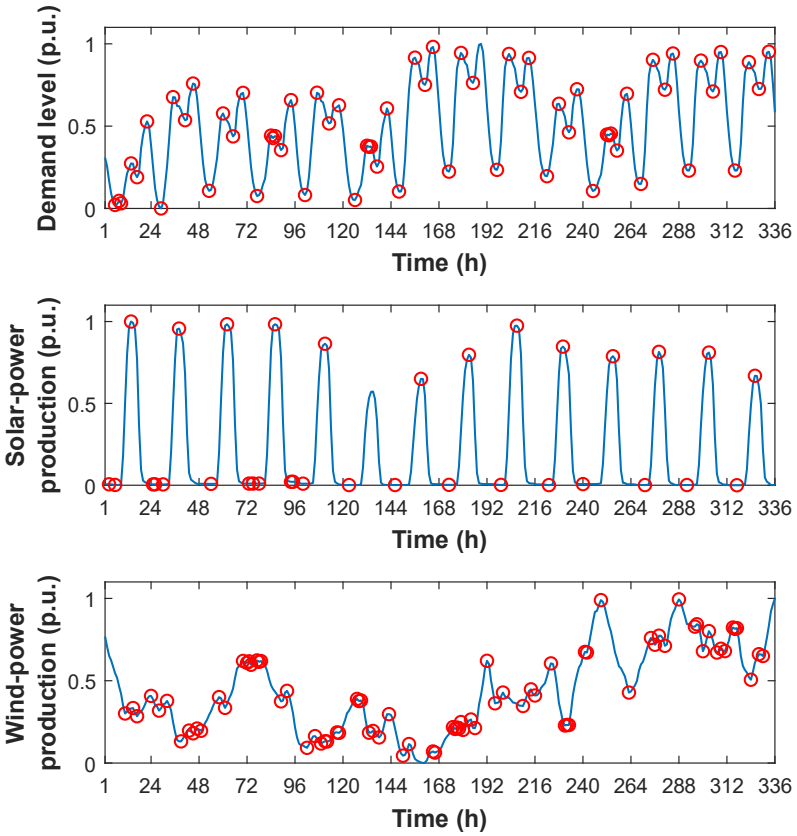
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PCTPC – Rules

High priority



Medium priority



Low priority:

Remaining data

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

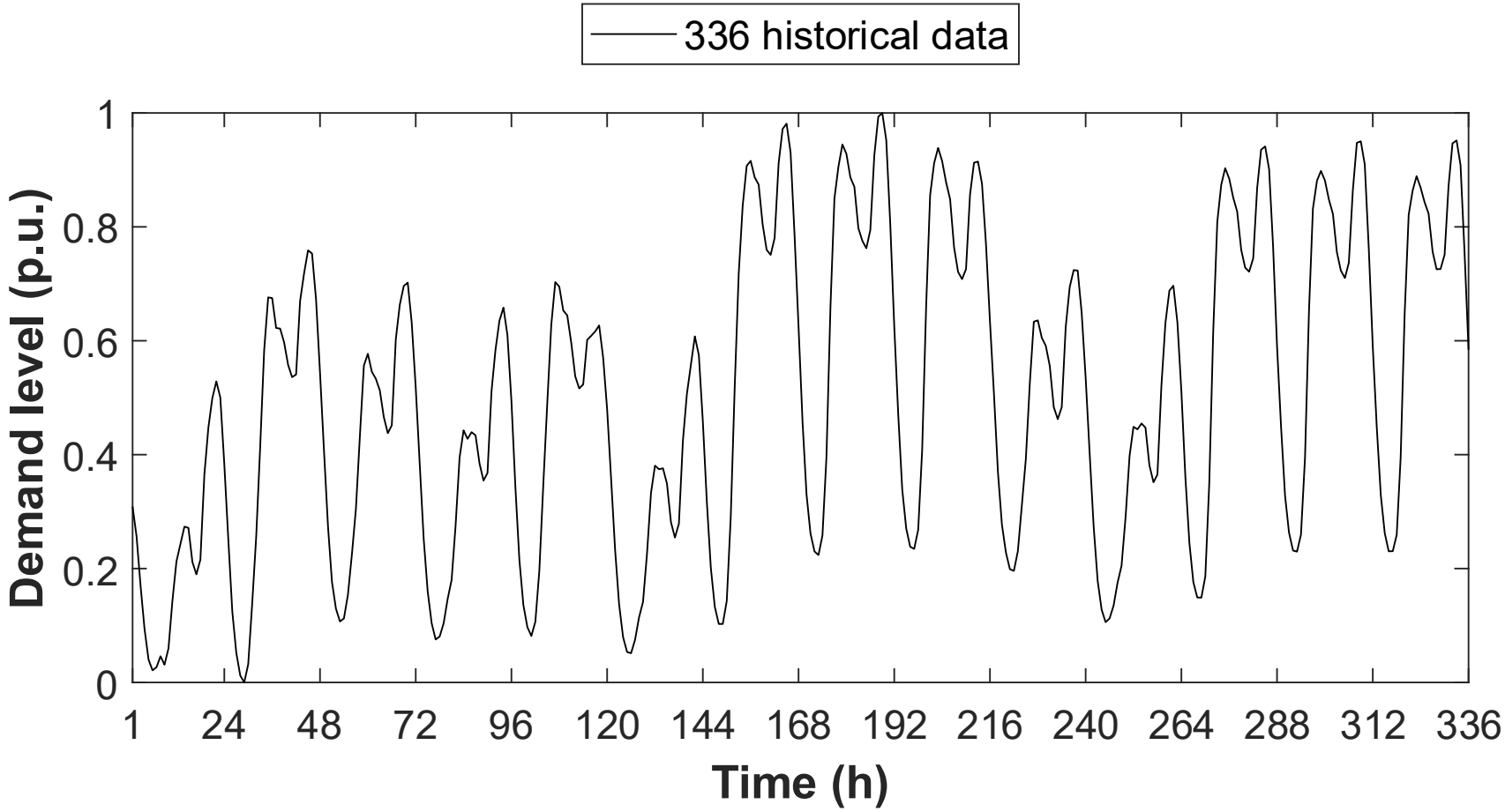
PCTPC – Rules

Merging rules:

- H & H → Find the next pair of adjacent clusters with minimum dissimilarity between them.
- H & M/L → Value of the centroid associated with H.
- M & L → Value of the centroid associated with M.
- M & M → Average value of the centroids.
- L & L → Average value of the centroids.

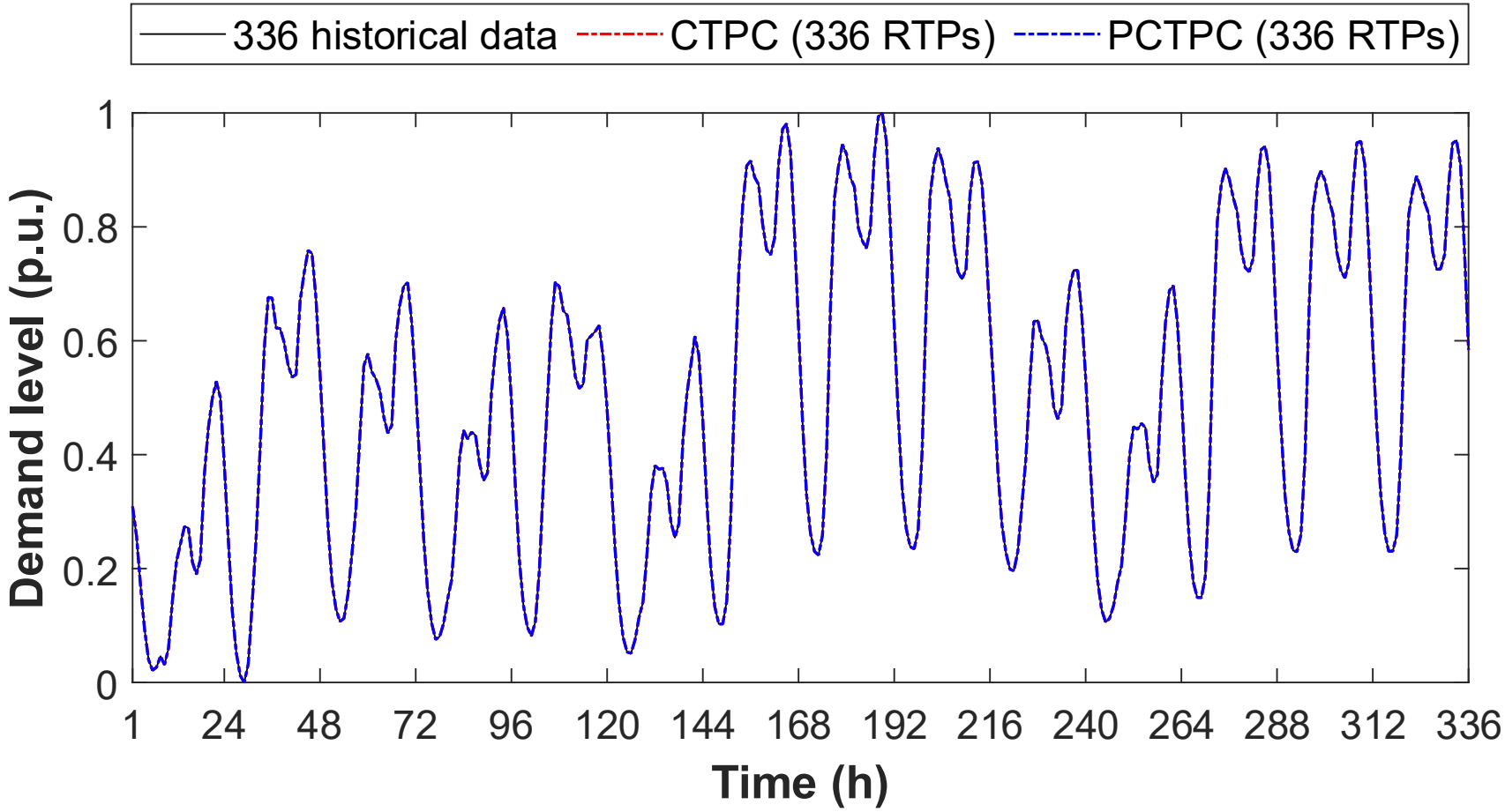
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PCTPC – Illustrative example



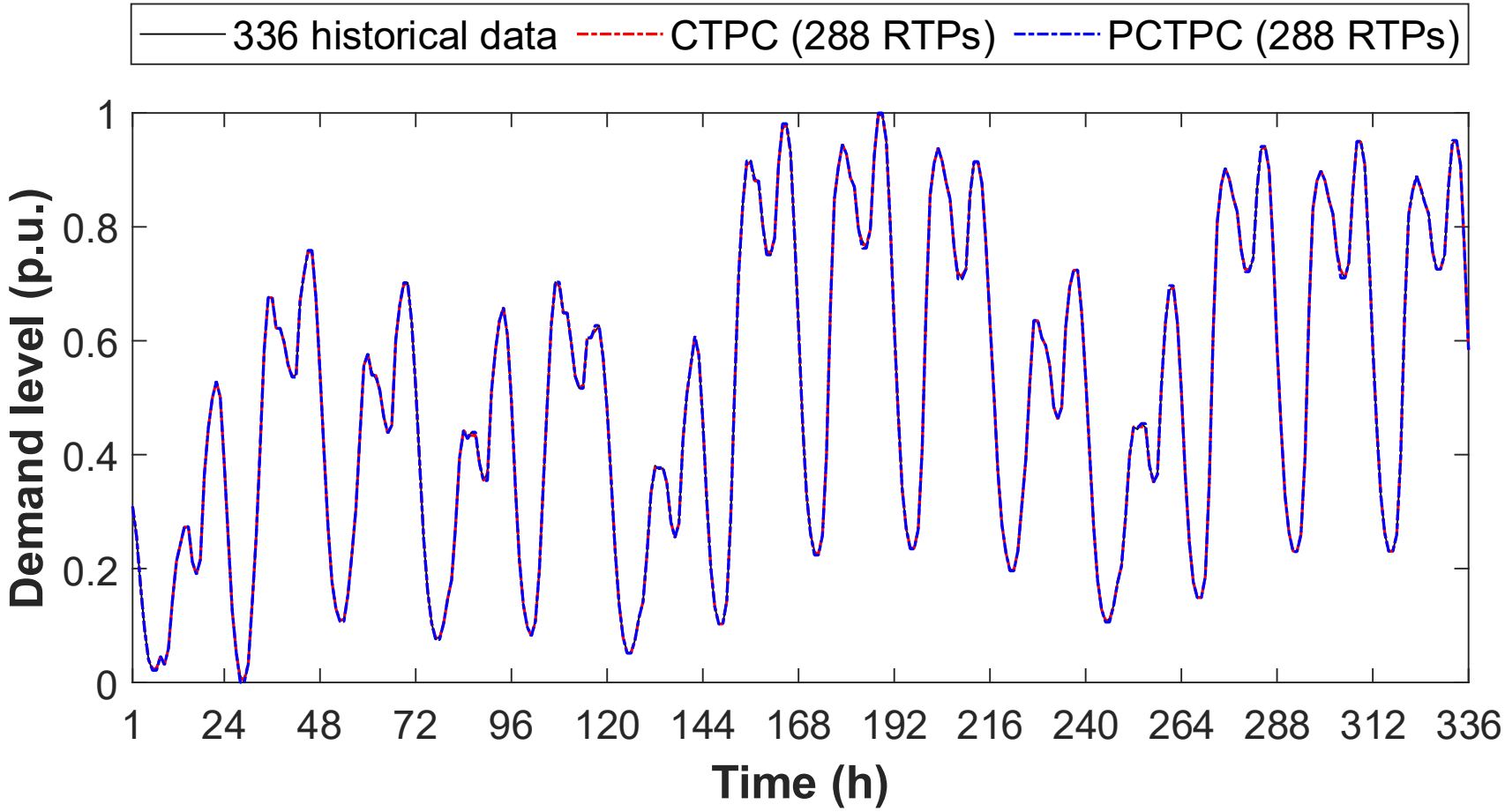
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PCTPC – Illustrative example



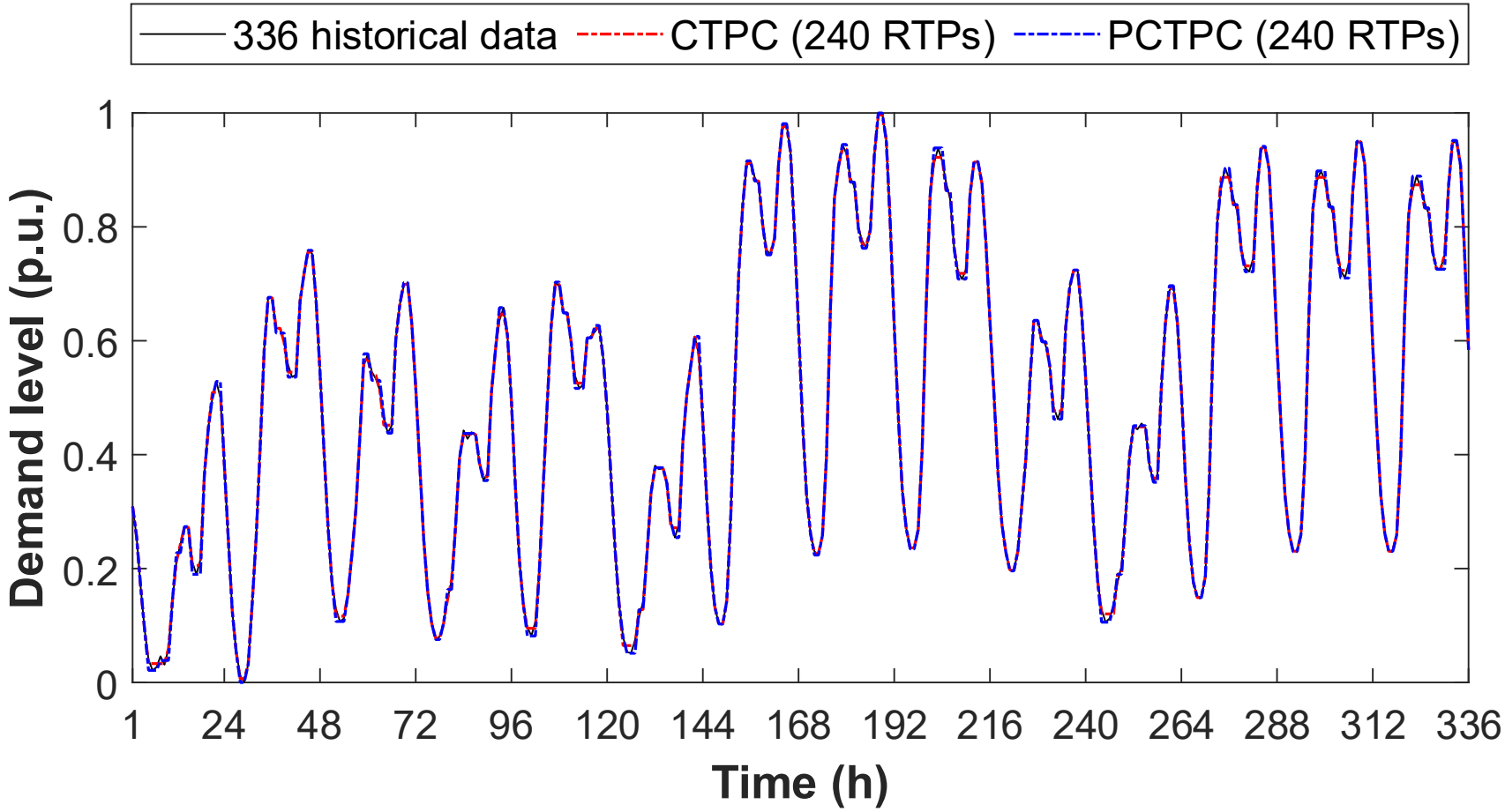
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PCTPC – Illustrative example



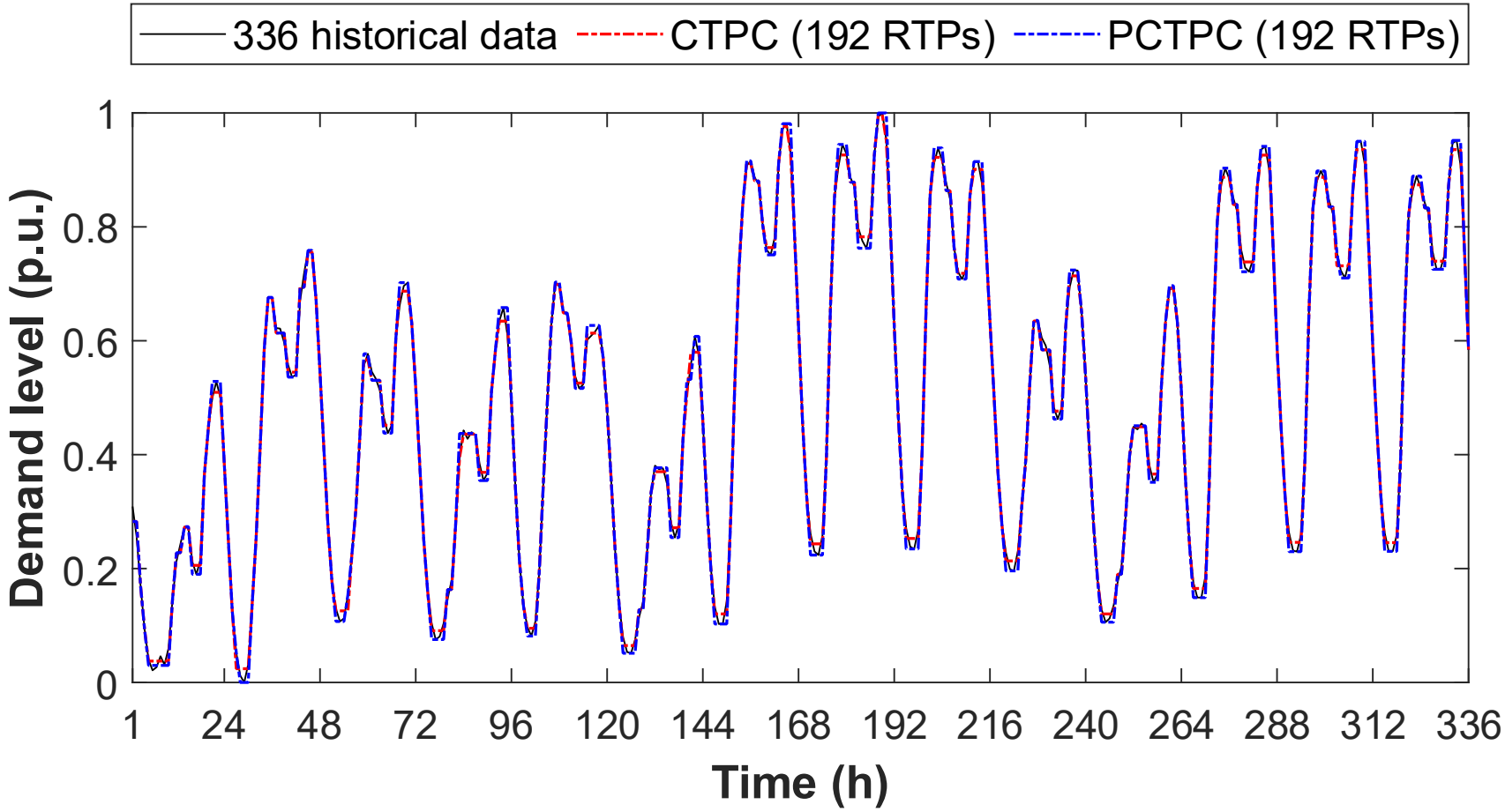
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PCTPC – Illustrative example



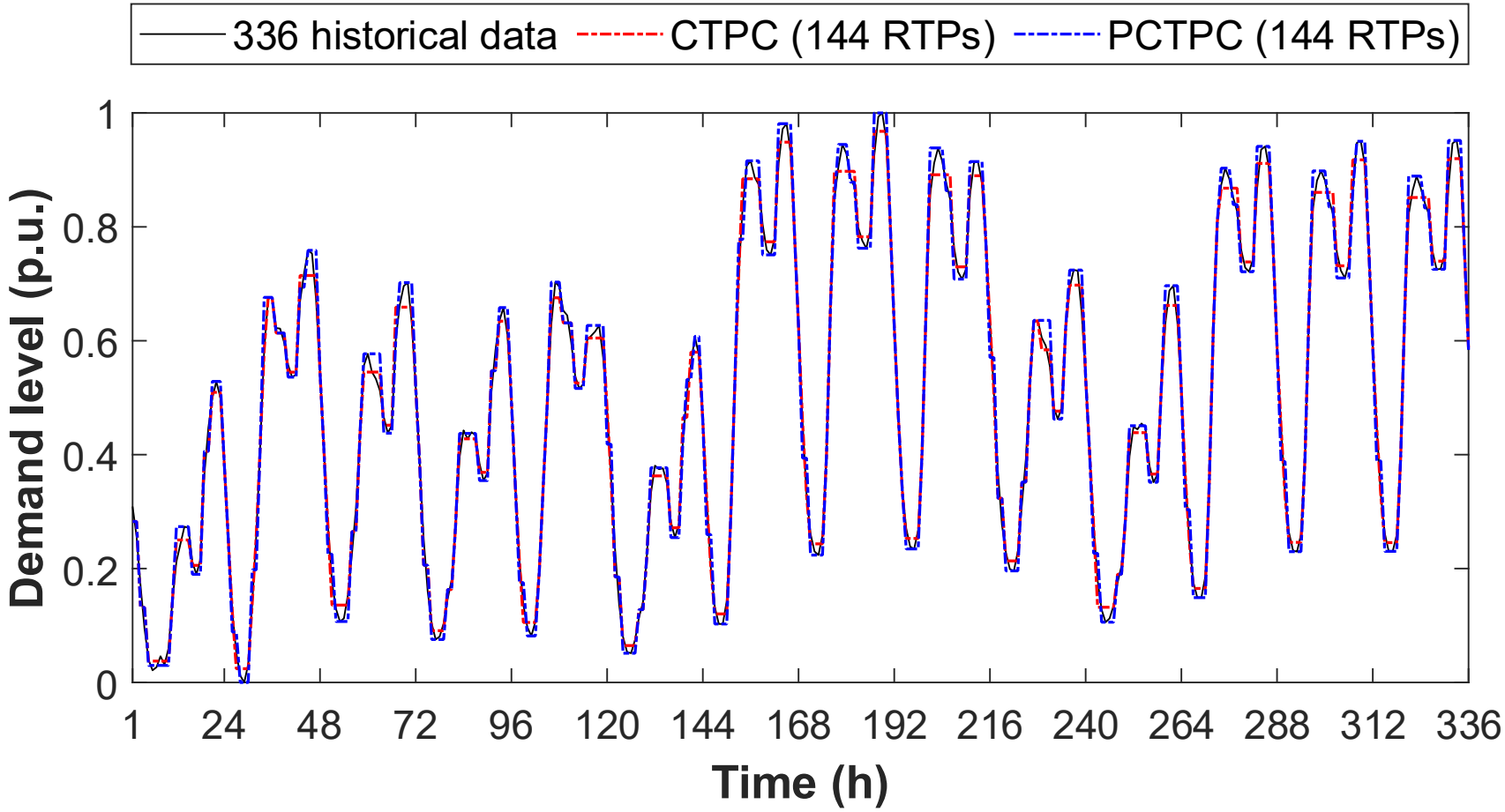
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PCTPC – Illustrative example



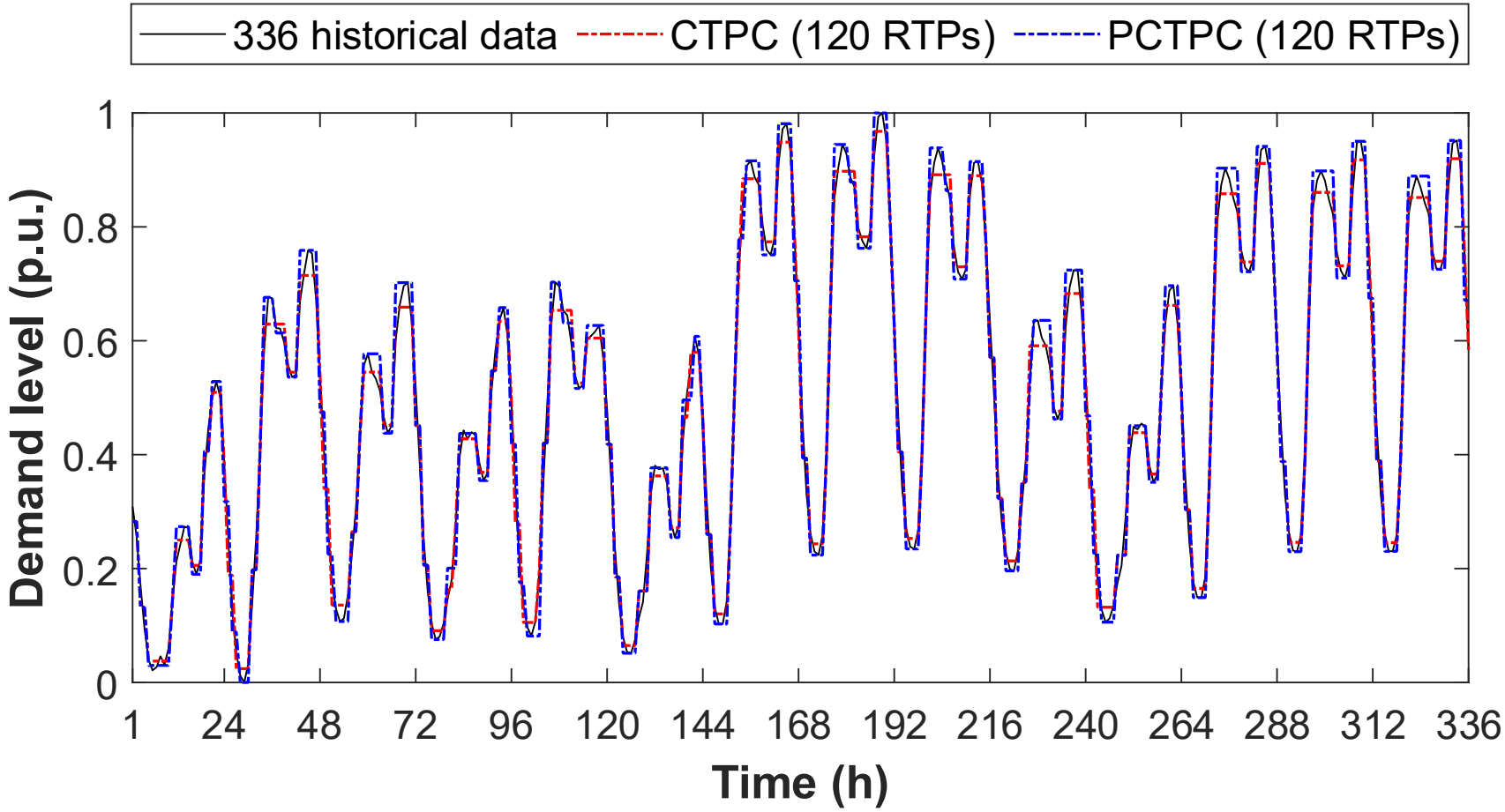
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PCTPC – Illustrative example



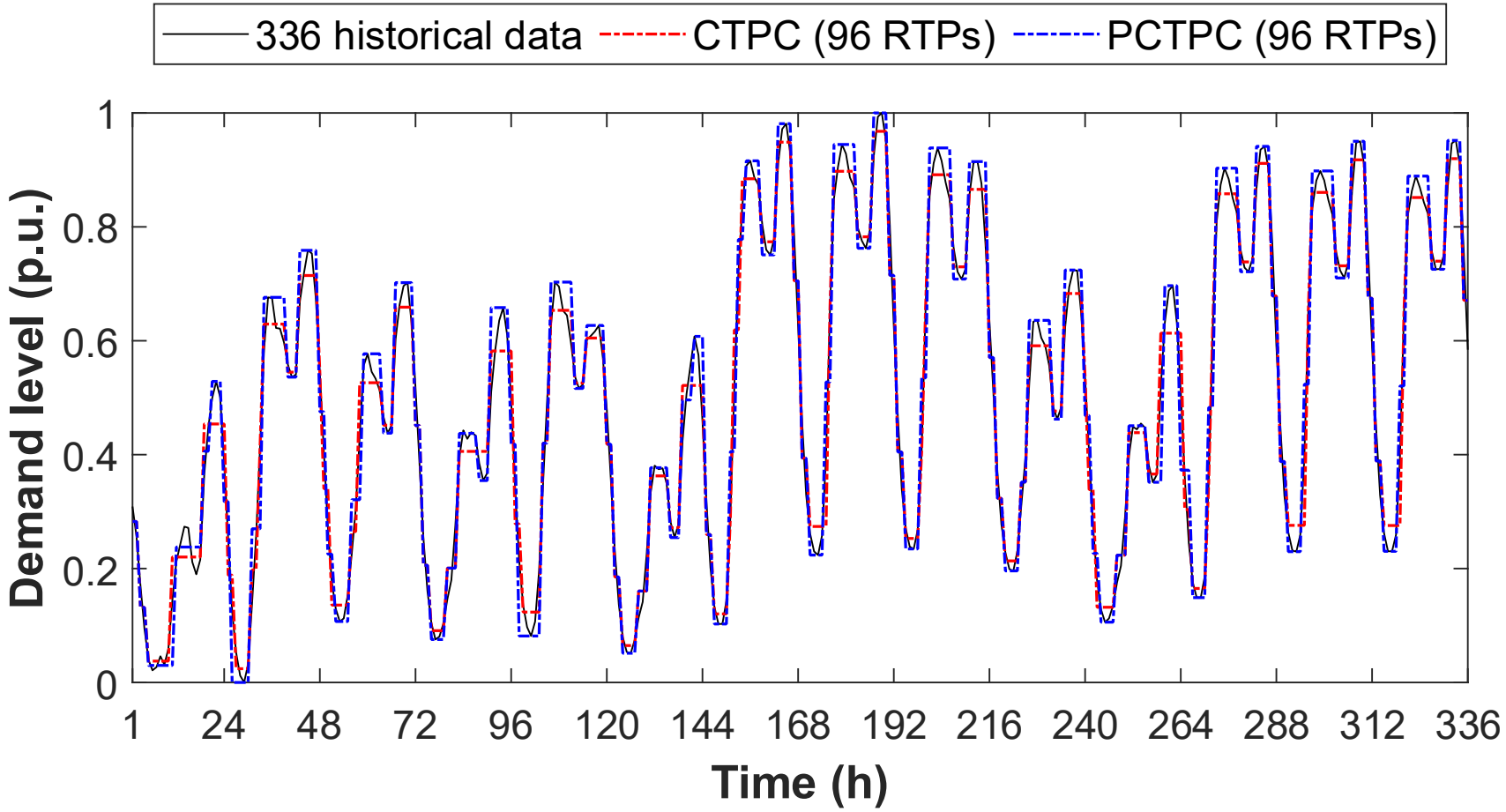
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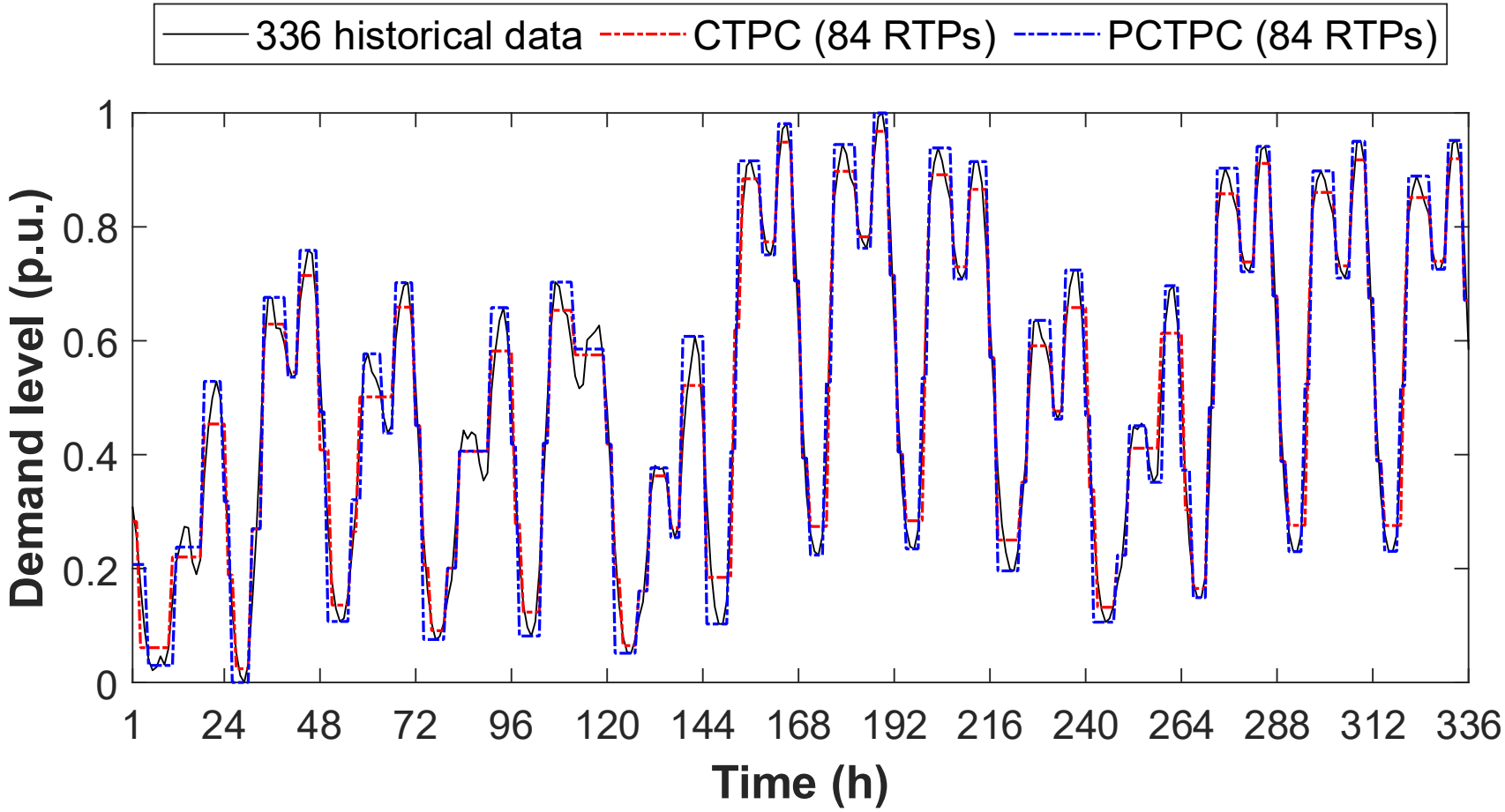
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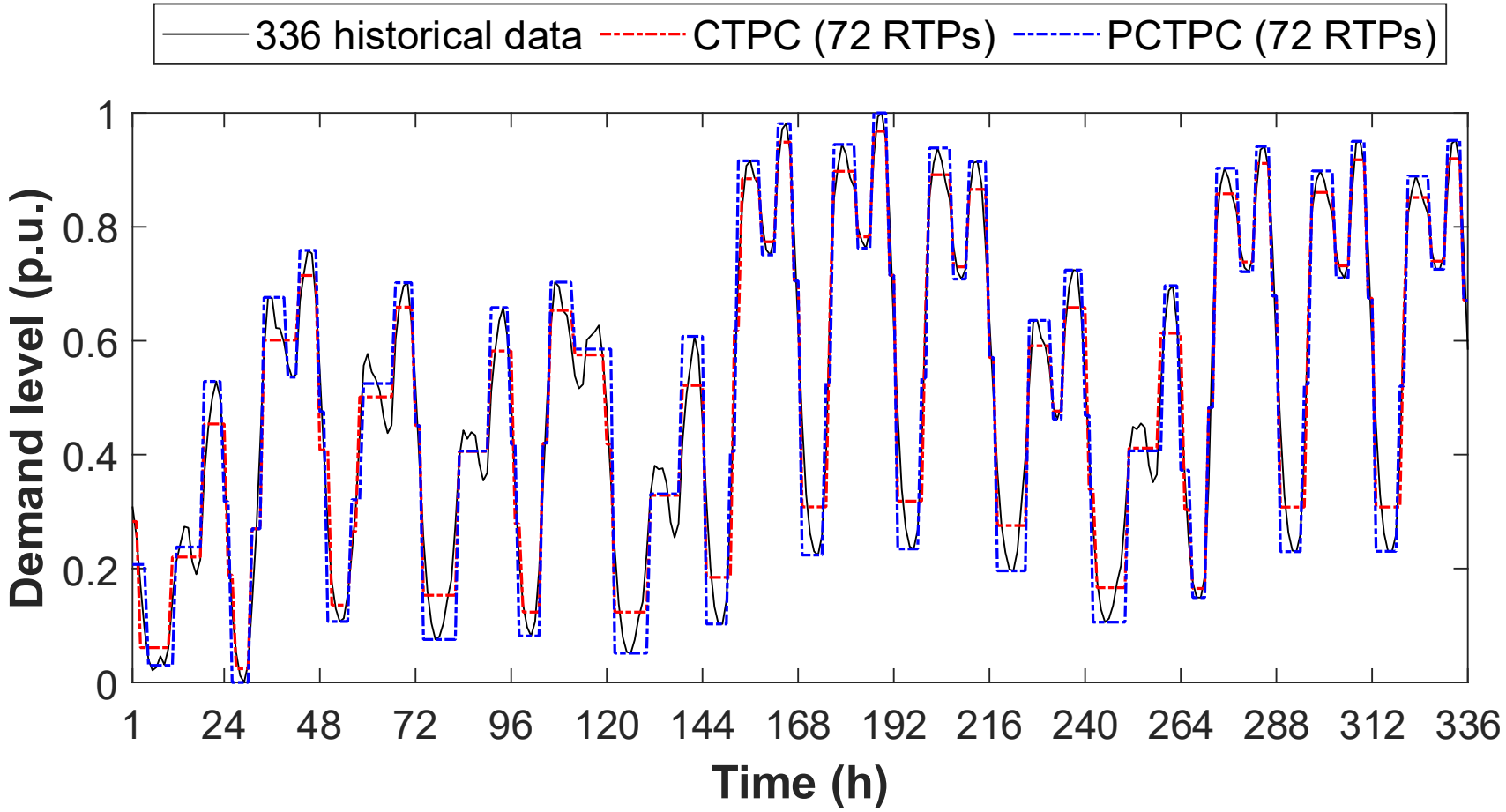
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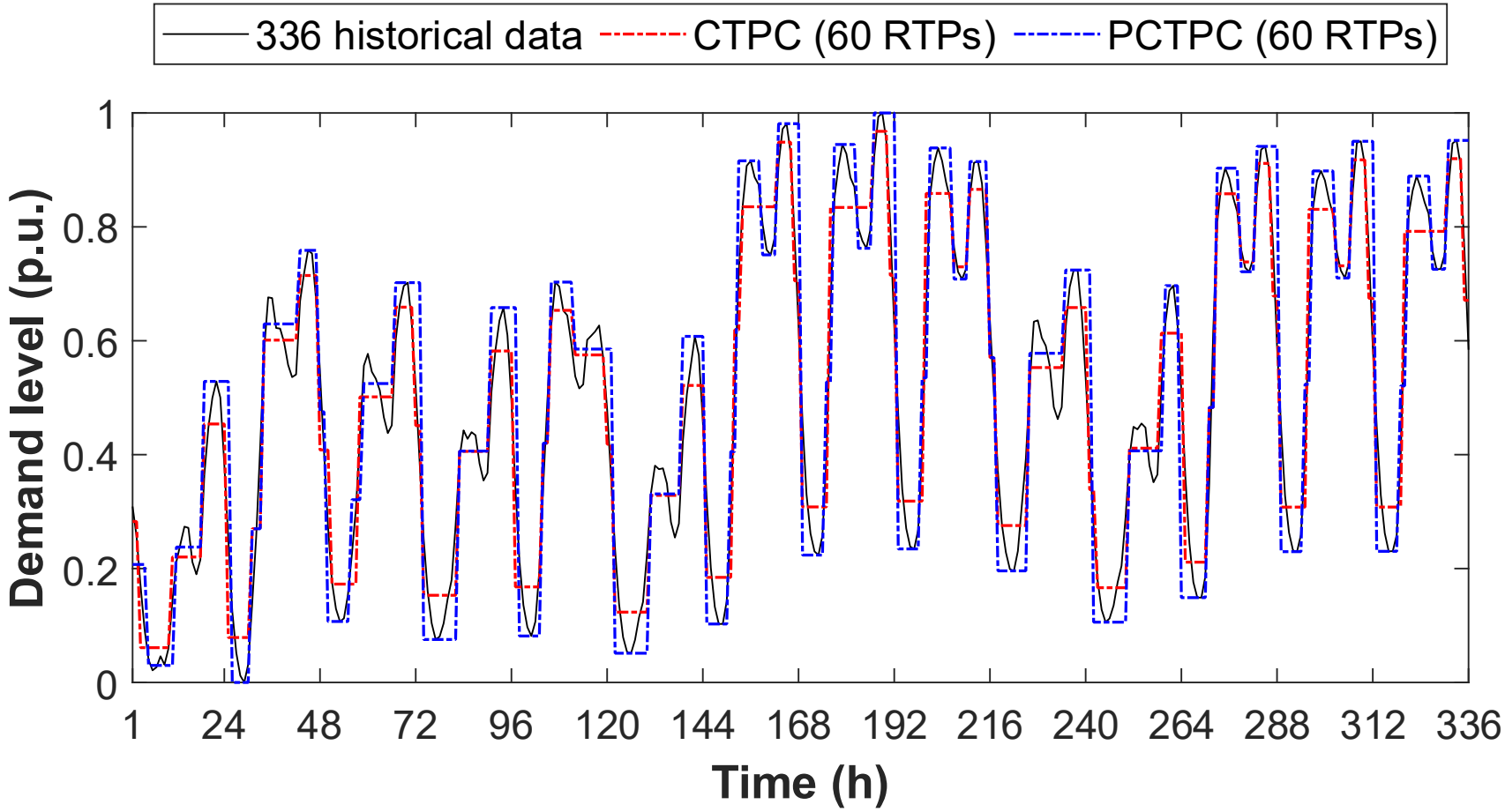
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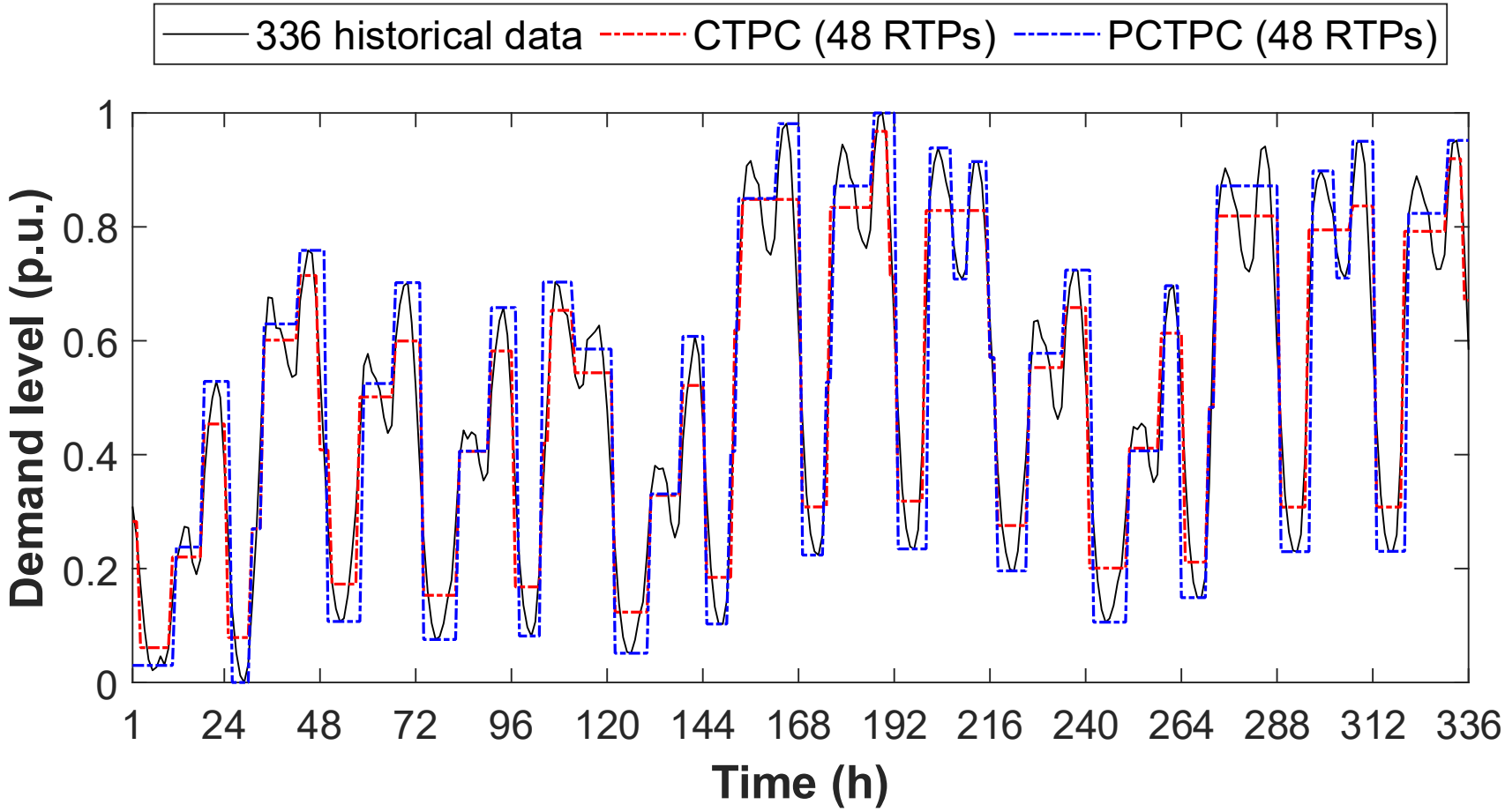
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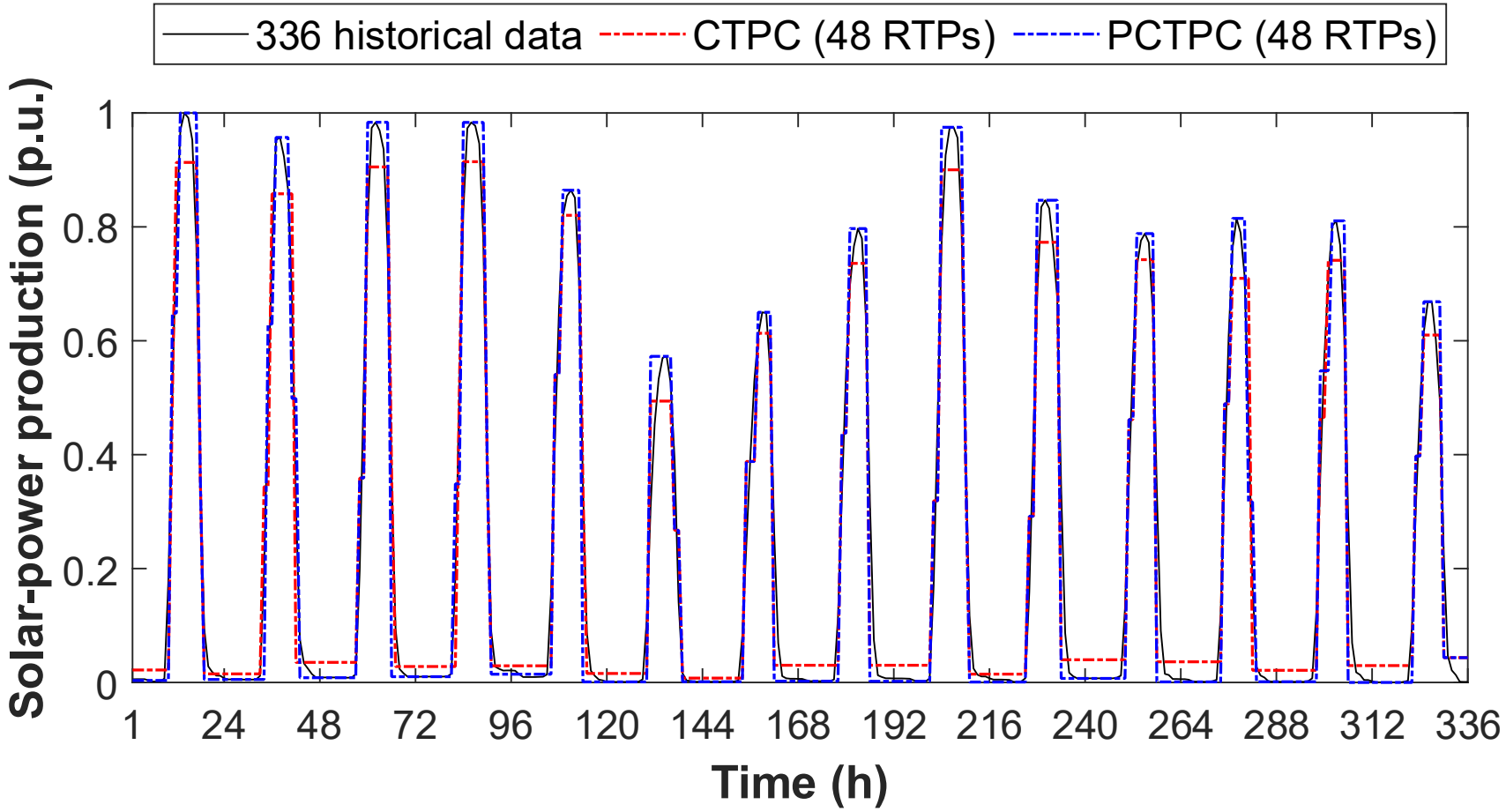
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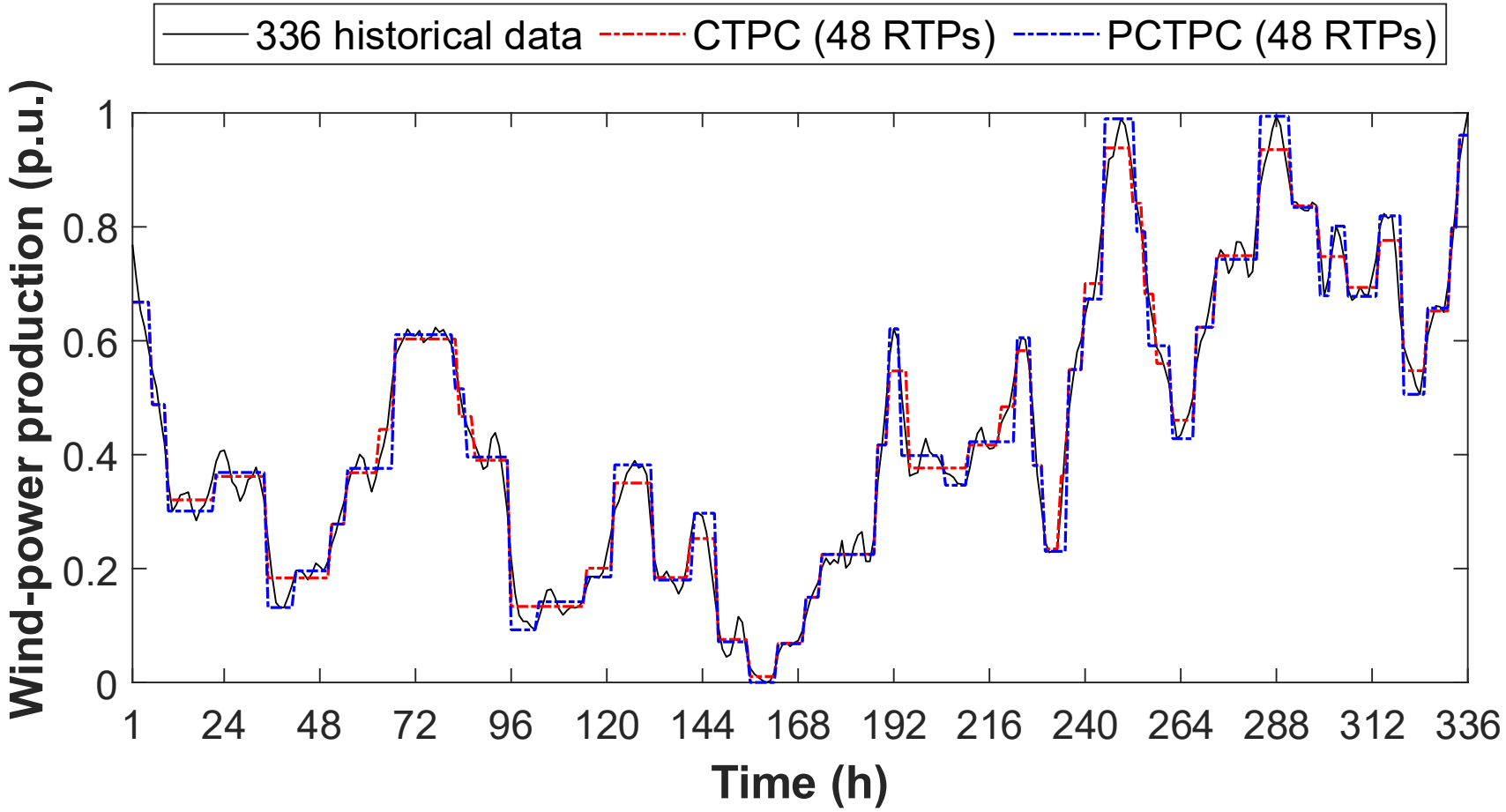
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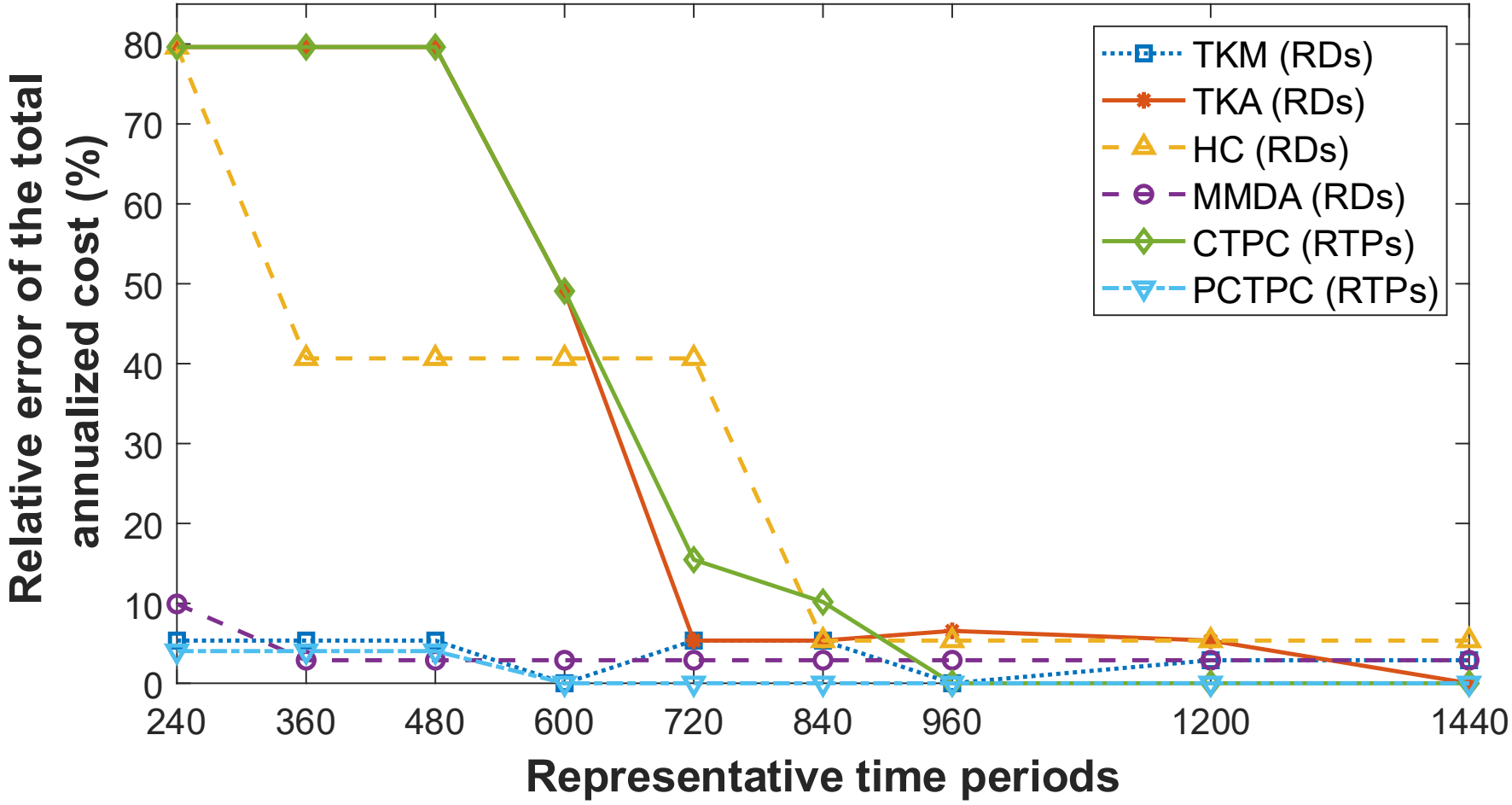
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PCTPC vs other aggregation techniques

- TNEP problem, GEP problem, and G&TNEP problem.
- IEEE 24-bus Reliability Test System.
- 8760 hourly input data of electrical demand and renewable (wind and solar) power availability from Spain and France.

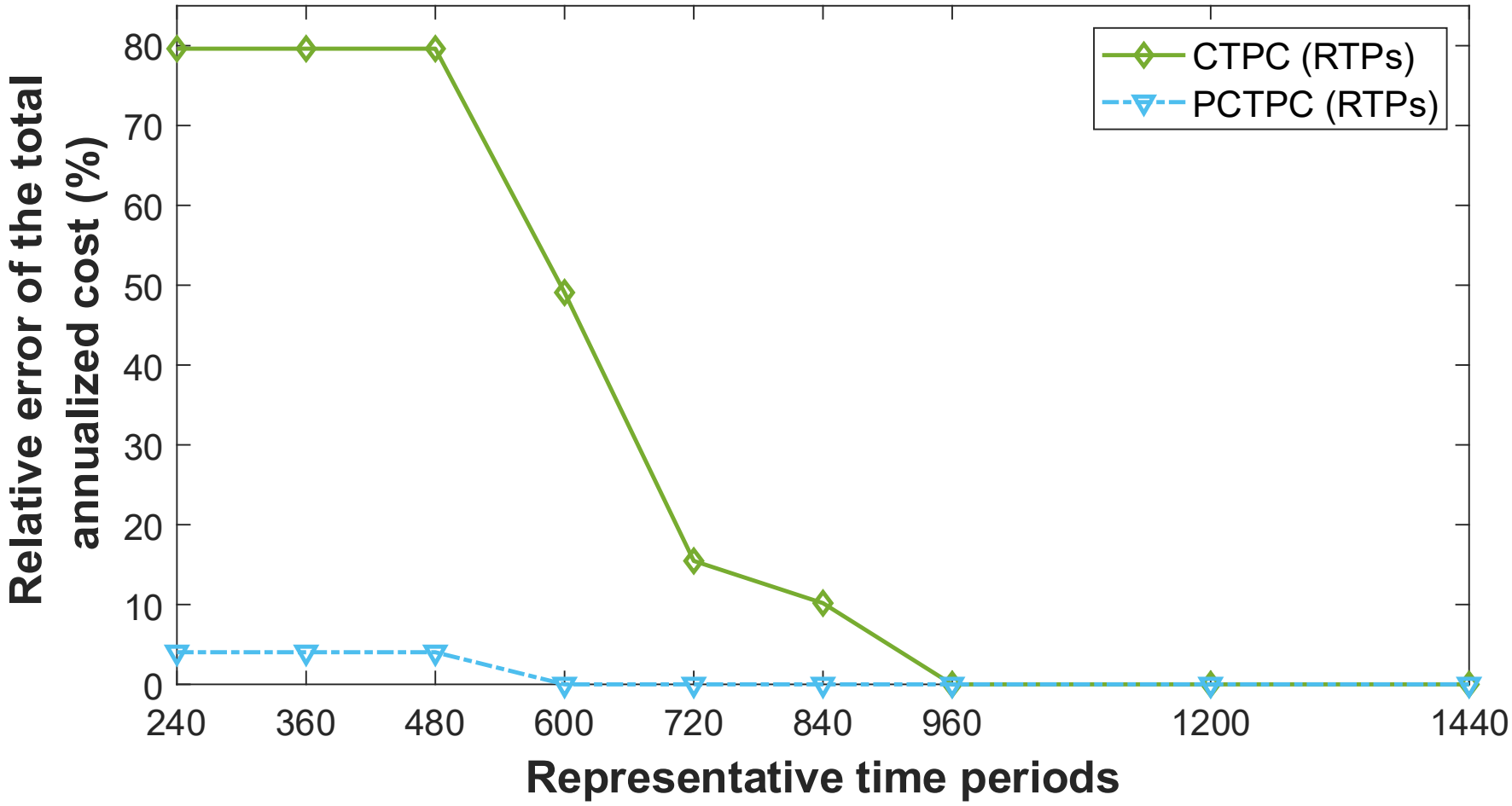
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PCTPC vs other aggregation techniques (TNEP)



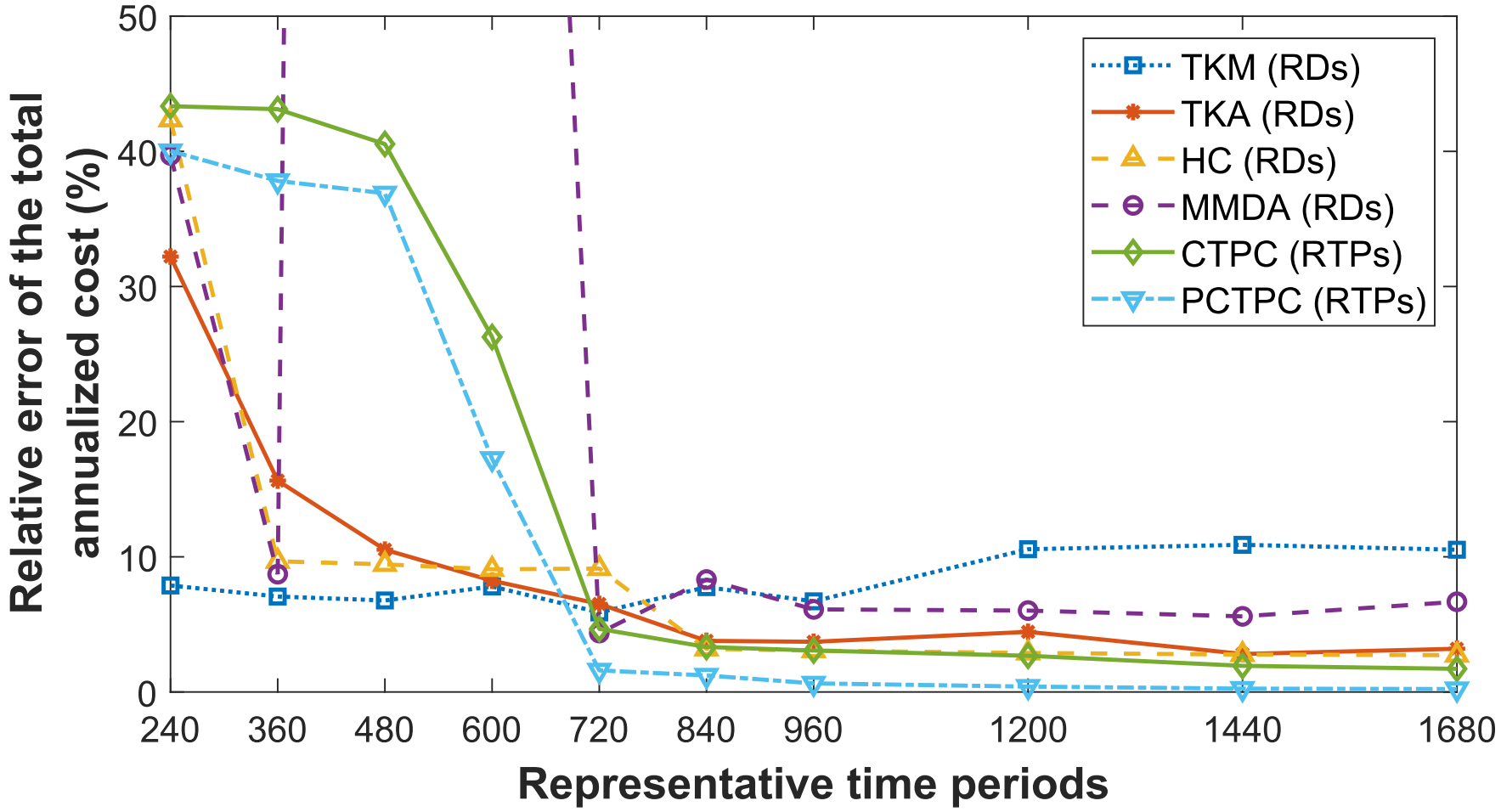
Reprinted with permission from Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

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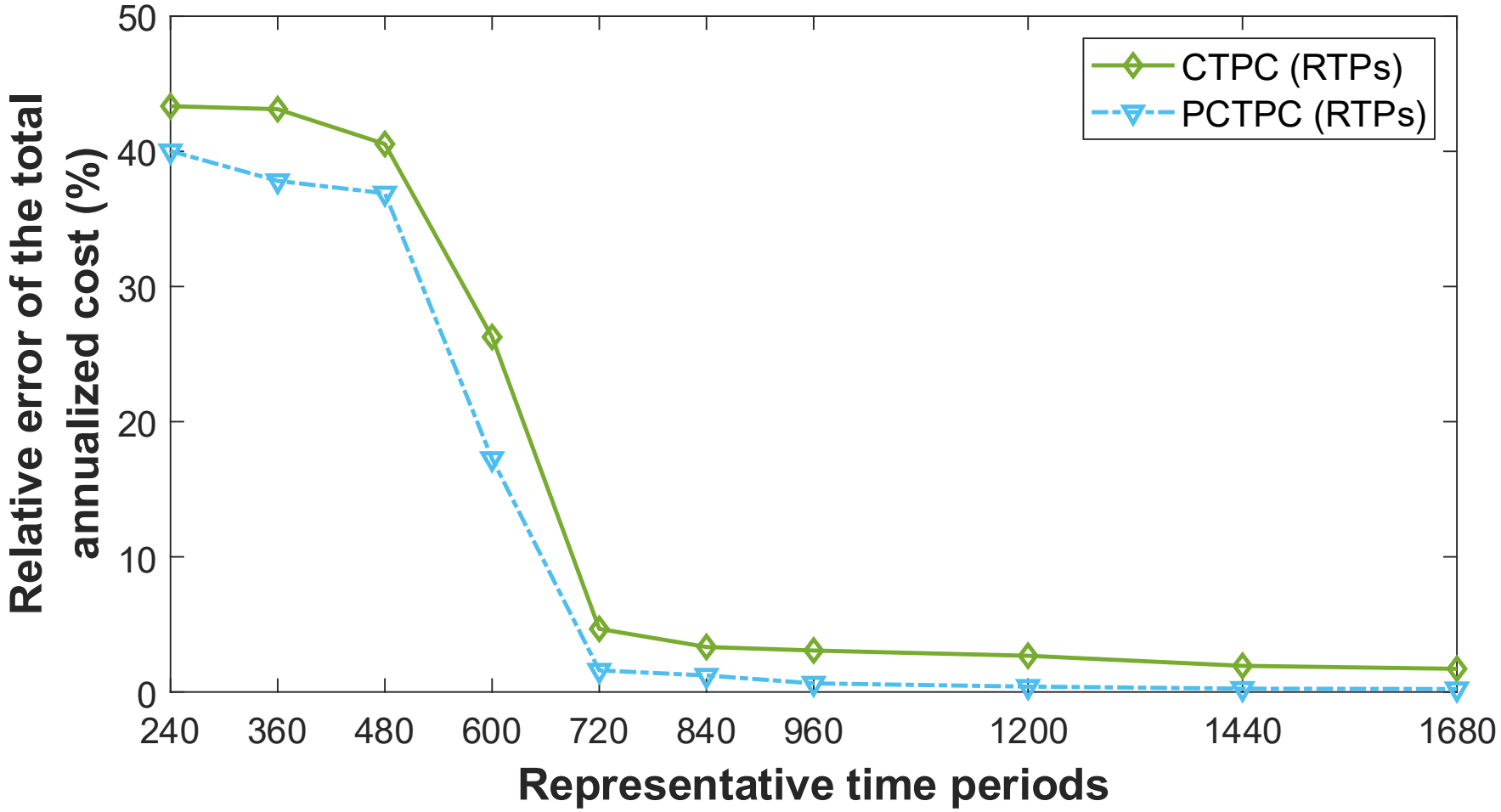
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PCTPC vs other aggregation techniques (GEP)



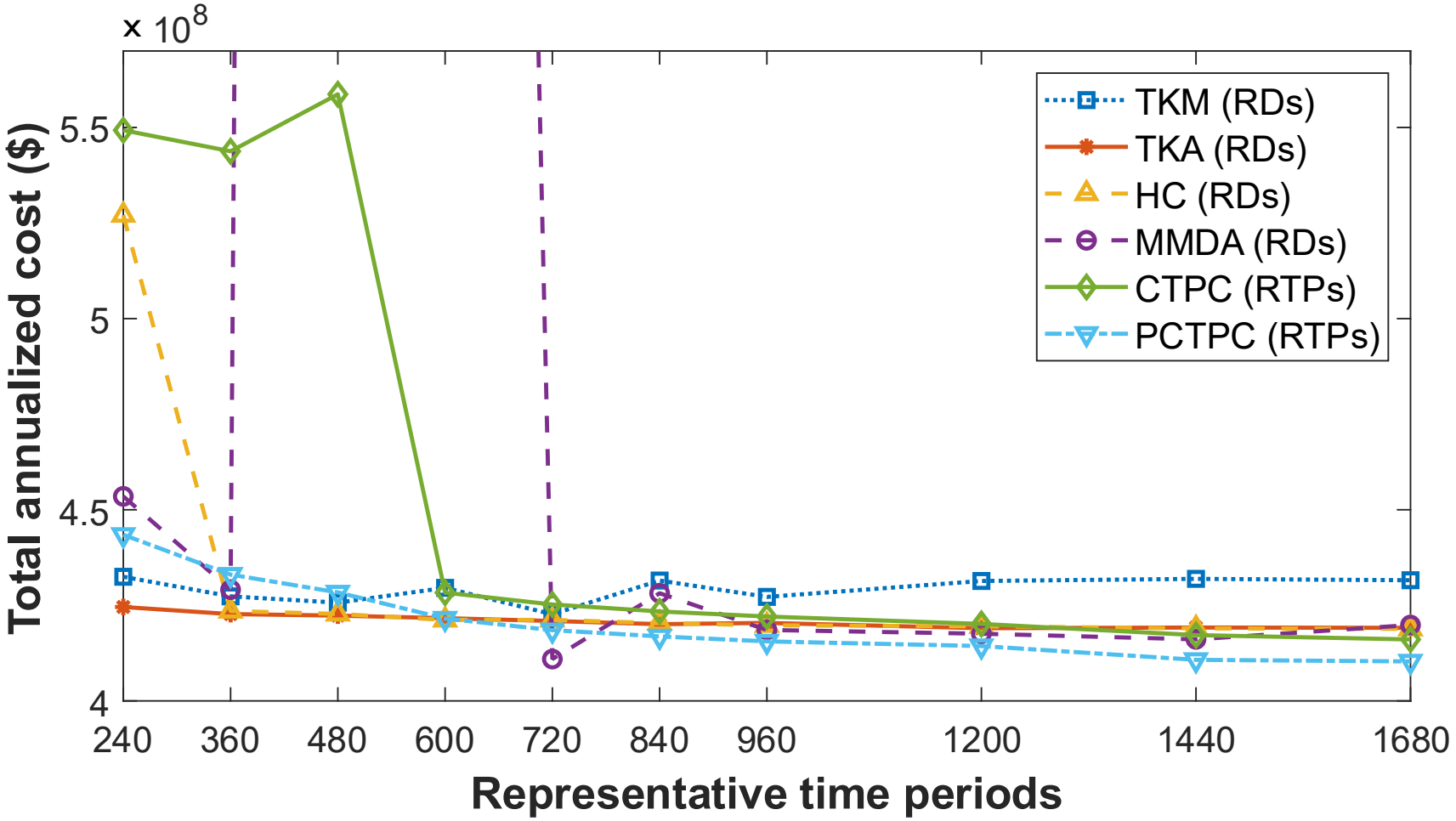
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PCTPC vs other aggregation techniques (GEP)



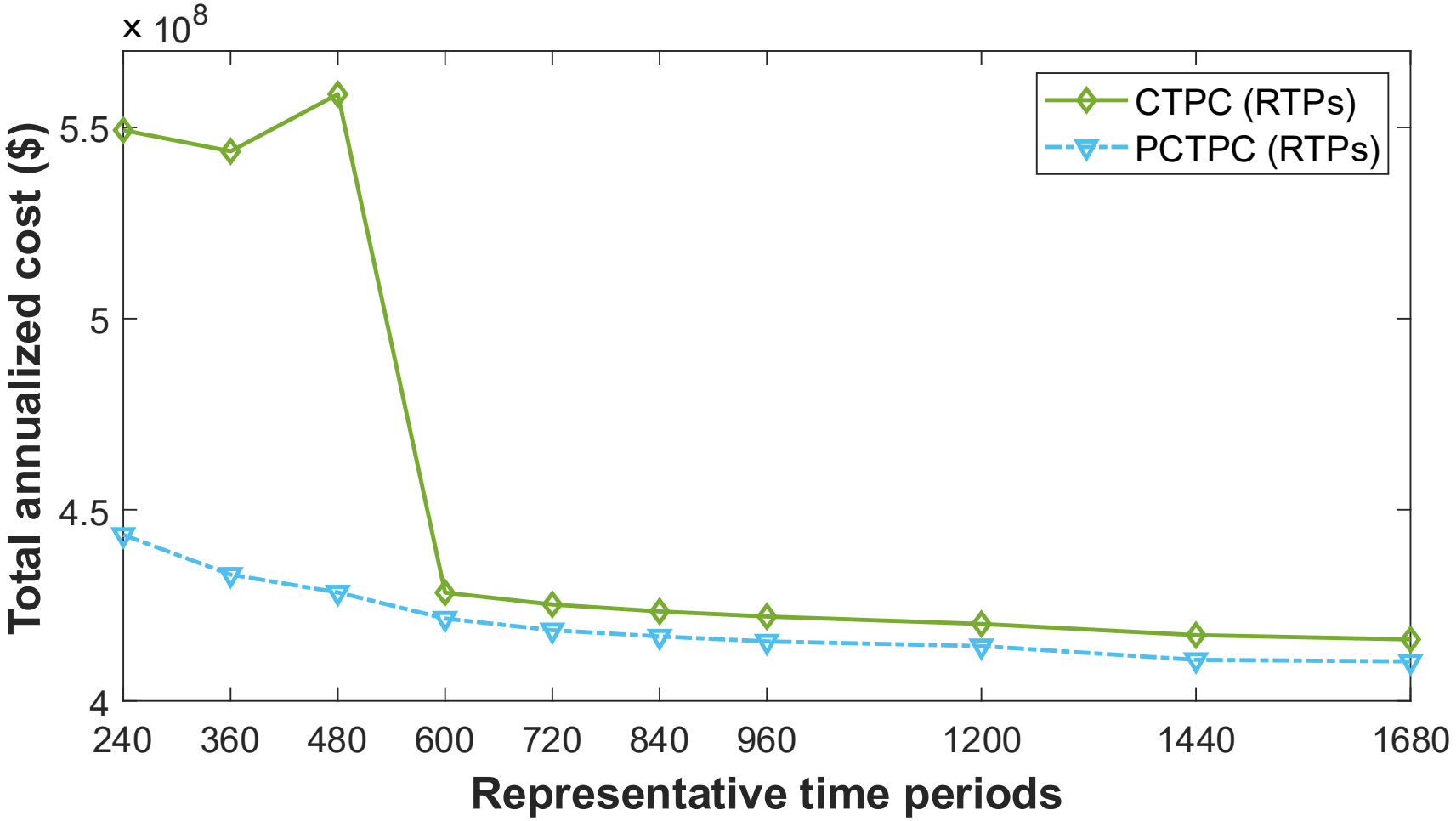
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PCTPC vs other aggregation techniques (G&TNEP)



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PCTPC vs other aggregation techniques (G&TNEP)



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PCTPC – Conclusions

- The PCTPC improves the representation of local maximum and minimum values of the historical data in comparison with the CTPC.

- The PCTPC leads to investment plans that involve lower relative errors of the total annualized cost than when the CTPC is applied.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Priority chronological time-period clustering for generation and transmission expansion planning problems with long-term dynamics", *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4325–4339, Nov. 2022. DOI: 10.1109/TPWRS.2022.3151062.

Advanced aggregation techniques

Works resulting from my Bachelor's Final Project and PhD Thesis:

- Modified K-means Method (MKM).
- Modified Maximum Dissimilarity Algorithm (MMDA).
- Priority Chronological Time-Period Clustering (PCTPC).
- Combination of previous techniques (MMDA + PCTPC).

MMDA + PCTPC

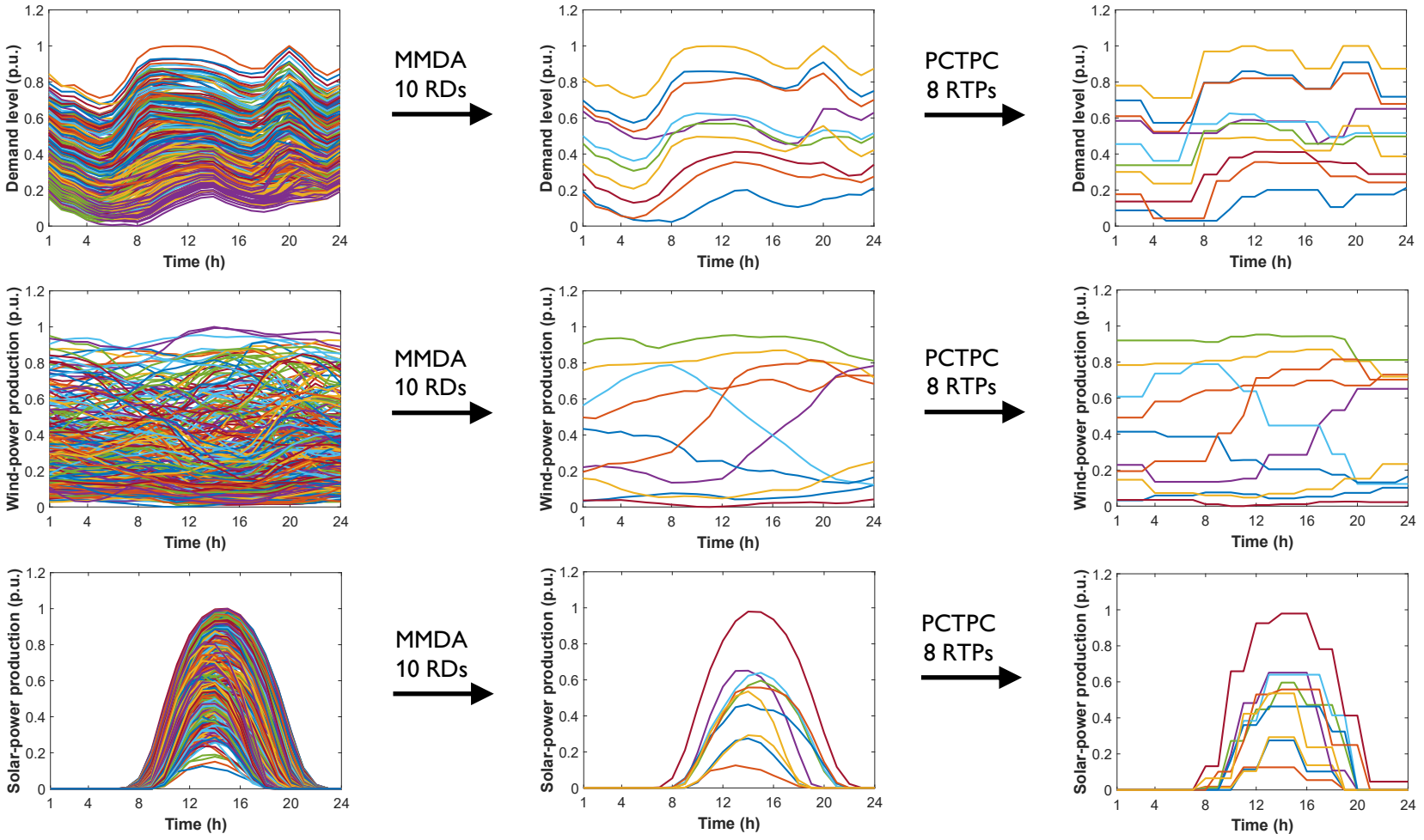
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MMDA + PCTPC

- Obtain K^{T} RDs/RWs using the MMDA.
- Then, apply the PCTPC $\left(K^{\mathrm{H}}\right)$ to each RD/RW independently.
- Total number of RTPs $= K^{\mathrm{T}} K^{\mathrm{H}}$.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Illustrative example



Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Case study

- G&TNEP problem.
- IEEE 24-bus Reliability Test System.
- 8760 hourly input data of electrical demand and renewable (wind and solar) power availability from Spain and France.
- Exact solution → Not solved after 168 h (1 week).

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Case study

Analysis carried out under three cases (different confidence levels, see lecture 4):

- MMDA + downsampling $\left(\boldsymbol{K^{\mathrm{T}} = 10, K^{\mathrm{H}} = 8}\right)$.
- MMDA + CTPC $\left(\boldsymbol{K^{\mathrm{T}} = 10, K^{\mathrm{H}} = 8}\right)$.
- MMDA + PCTPC $\left(\boldsymbol{K^{\mathrm{T}} = 10, K^{\mathrm{H}} = 8}\right)$.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Comparison procedure

Optimization model using one of the methods

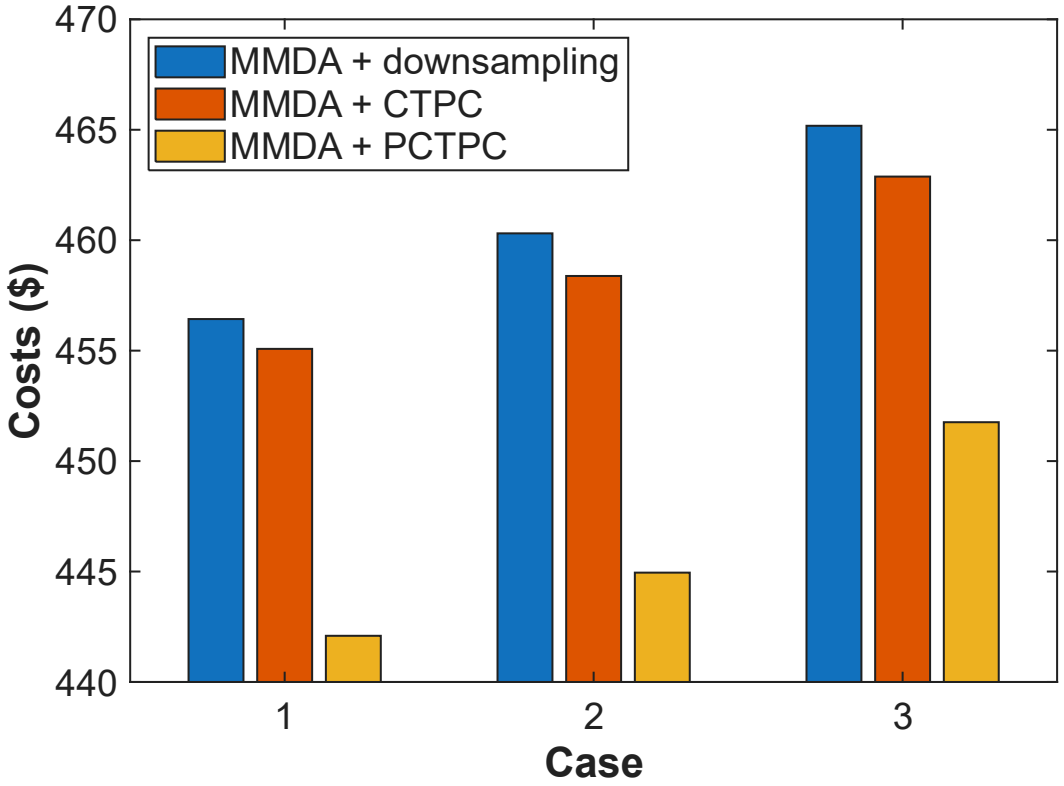
Investment plan

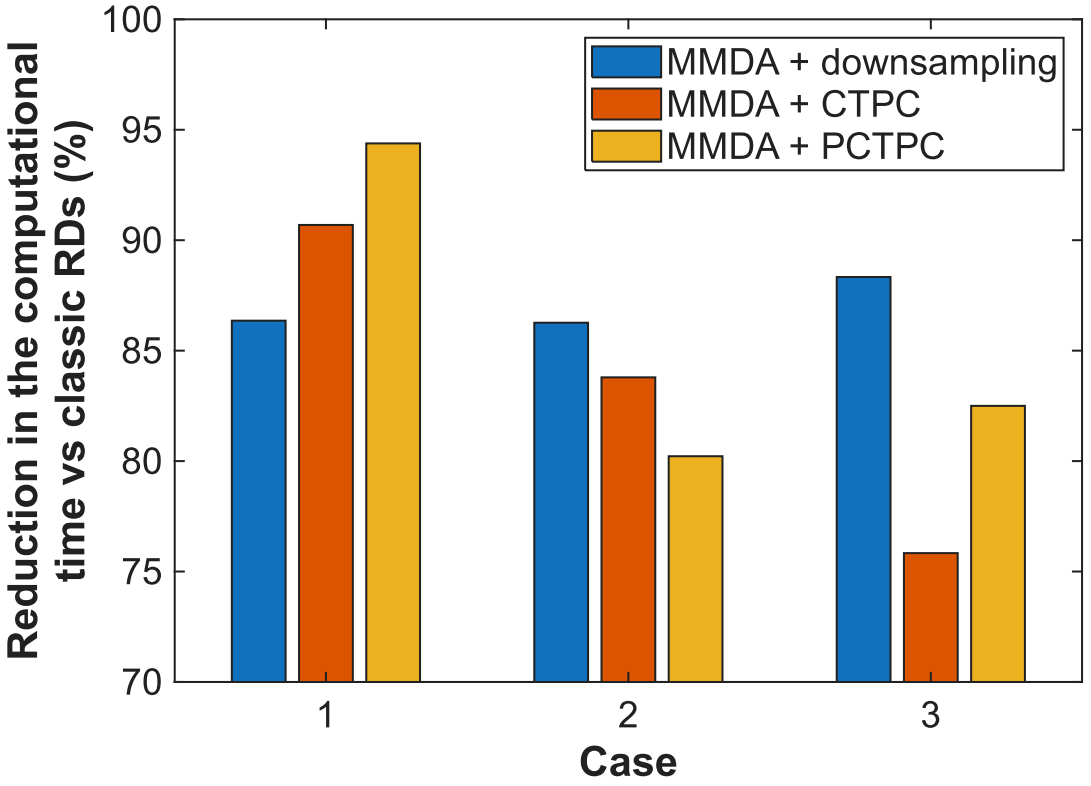
Optimization model using the RDs composed by 24 hourly time periods and fixed investment decisions

Objective function value $\left(OF^{\mathrm{K}}\right)$

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Case study





Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

MMDA + PCTPC – Conclusions

MMDA + PCTPC provides:

- A significant reduction in the computational time of over 94% with respect to the case in which RDs composed by 24 hourly RTPs are considered.

- Lower values of the objective function value (costs) than those attained when the MMDA is combined with the downsampling or the CTPC.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

7. Alternative aggregation techniques

Alternative aggregation techniques

Similar input data → Similar output data

Critical situations → Extreme values of input data

Alternative aggregation techniques

Similar input data → Similar output data

Critical situations → Extreme values of input data

This may not always be true.

Alternative aggregation techniques

Output-based clustering techniques:

- M. Sun, F. Teng, X. Zhang, G. Strbac, and D. Pudjianto, "Data-driven representative day selection for investment decisions: a cost-oriented approach", *IEEE Transactions on Power Systems*, vol. 34, no. 4, pp. 2925–2936, Jul. 2019.
- H. Teichgraeber, L. E. Küpper, and A. R. Brandt, "Designing reliable future energy systems by iteratively including extreme periods in time-series aggregation", *Applied Energy*, vol. 304, art. no. 117696, Dec. 2021.
- C. Li, A. J. Conejo, J. D. Siirola, and I. E. Grossmann, "On representative day selection for capacity expansion planning of power systems under extreme operating conditions", *International Journal of Electrical Power & Energy Systems*, vol. 137, art. no. 107697, May 2022.

Alternative aggregation techniques

Sonja Wogrin's team:

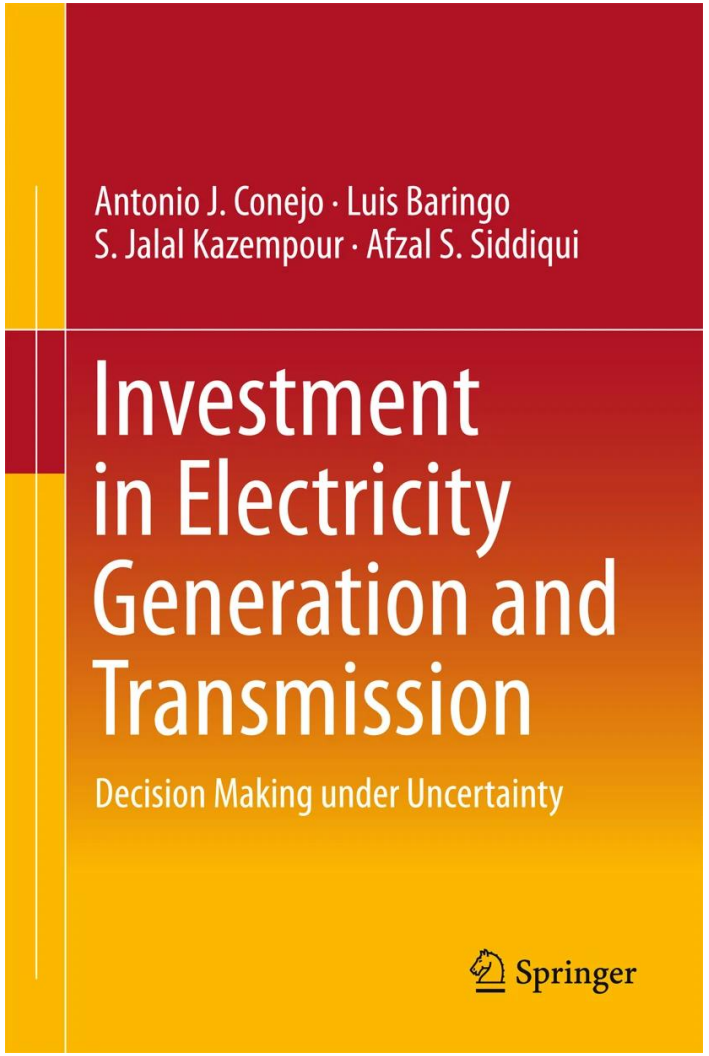
- S. Wogrin, "Time series aggregation for optimization: One-size-fits-all?," *IEEE Transactions on Smart Grid*, vol. 137, no. 3, pp. 2489–2492, May 2023.
- D. Cadona-Vasquez, T. Klatzer, B. Klinz, and S. Wogrin, "Enhancing time series aggregation for power system optimization models: Incorporating network and ramping constraints", *Electric Power Systems Research*, vol. 230, art. no. 110267, May 2024.
- T. Klatzer, D. Cadona-Vasquez, L. Santosuosso, and S. Wogrin, "Towards exact temporal aggregation of time-coupled energy storage models via active constraint set identification and machine learning", *Electric Power Systems Research*, vol. 263, art. no. 113737, Feb 2027.

Next lecture

- ~~Lecture 1: Introduction to Operations Research.~~
- ~~Lecture 2: Operation and Planning Problems.~~
- ~~Lecture 3: Operational Variability.~~
- **Lecture 4: Uncertainty and Risk.**
- Lecture 5: Further Applications.

References

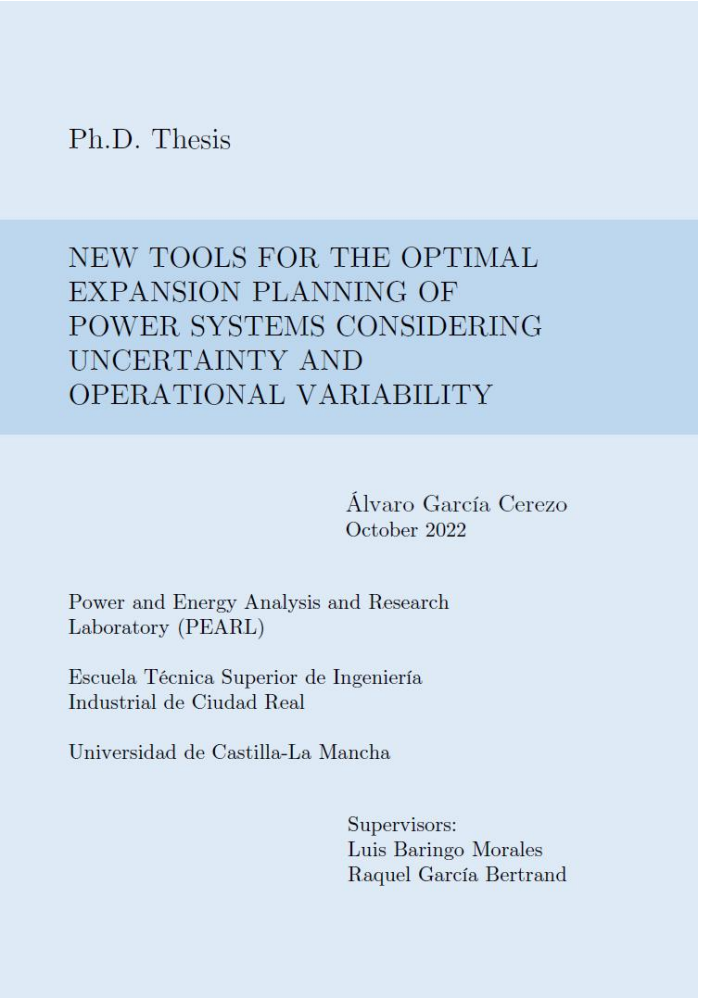
A. J. Conejo, L. Baringo, S. J. Kazempour, and A. S. Siddiqui, *Investment in Electricity Generation and Transmission: Decision Making under Uncertainty*. Springer, Switzerland, 2016. ISBN: 978-3-319-29499-5. DOI: 10.1007/978-3-319-29501-5.



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Á. García Cerezo, *New tools for the optimal expansion planning of power systems considering uncertainty and operational variability*. PhD Thesis, Universidad de Castilla-La Mancha, Oct. 2022. Permanent link: https://hdl.handle.net/10578/30715.

See Section 1.3.2: *Modeling Operational Variability through Aggregation Techniques*.



Ph.D. Thesis

NEW TOOLS FOR THE OPTIMAL EXPANSION PLANNING OF POWER SYSTEMS CONSIDERING UNCERTAINTY AND OPERATIONAL VARIABILITY

Álvaro García Cerezo
October 2022

Power and Energy Analysis and Research Laboratory (PEARL)

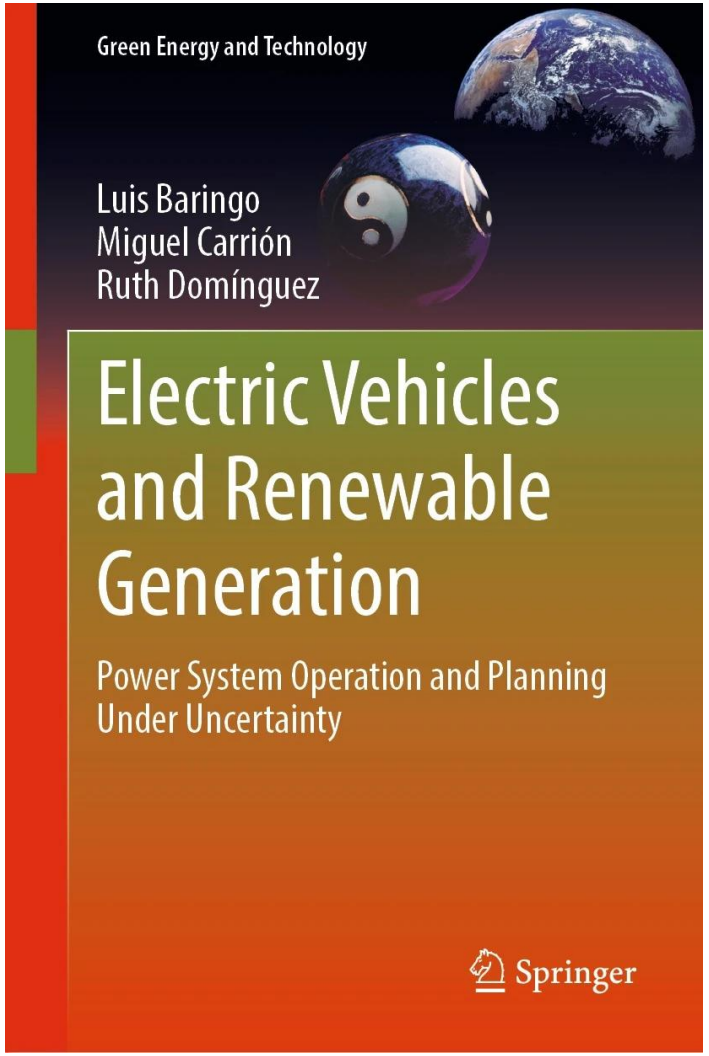
Escuela Técnica Superior de Ingeniería Industrial de Ciudad Real

Universidad de Castilla-La Mancha

Supervisors:
Luis Baringo Morales
Raquel García Bertrand

References

L. Baringo, M. Carrión, and R. Domínguez, *Electric Vehicles and Renewable Generation: Power System Operation and Planning under Uncertainty*. Springer, Cham, 2023. ISBN: 978-3-031-09078-3. DOI: 10.1007/978-3-031-09079-0.



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Thank you very much for your attention!

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